Four-dimensional quantum Yang–Mills theory: the proof, outlined

A natural-language account of the Lean proof (OpenAI-YM-20261028, commit 3456e5e)

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1 The proof in one reading

This is a substantial overview, about 10,000 words. The contents links locate individual topics; the exact statement fixes the scope of every claim.

1.0.1 The theorem

Let $G \subset U(m)$ be a closed subgroup whose Lie algebra $\mathfrak{g}$ is simple and nonabelian, for instance $SU(N)$ with $N \geq 2$; G need not be connected. The main theorem is `YangMills.jointExistence`; its precise statement is in statement.md. It asserts that there is one family of Wilson lattice gauge theories for G on four-dimensional tori, consisting of:

- spacings $a_N = 1/D_N \downarrow 0$ and physical side lengths $L_p \uparrow \infty$;
- bare couplings $\beta_{N,p}$ with $\beta_{N,p}/\log(1/a_N) \to b_* > 0$ uniformly in p;
- insertions $I_{N,p}(v, y)$ indexed by tokens v, each a continuous gauge-invariant function of the links in a stencil of fixed size in lattice units around the site y.

The family has the following property. For every strictly increasing sequence $p(k)$ of volume indices there are a further subsequence $p(q(k))$ and a continuum hierarchy satisfying (1)–(5) below.

(1) Lattice origin. Let $\mathrm{Conf}_n \subset (\mathbb{R}^4)^n$ be the open set of configurations of distinct points. For every n, every list of tokens v and every $f \in C_c^\infty(\mathrm{Conf}_n)$,

$$\lim_{k\to\infty} \lim_{N\to\infty} a_N^{4n} \sum_y f(\hat{y}) \left\langle \prod_{i=1}^{n} I_{N,p(q(k))}(v_i, y_i) \right\rangle_{N,p(q(k))} = \int f(x)\, S^{\mathrm{mix}}(\pi v; x)\, dx.$$

Here the sum runs over n-tuples of sites, $\hat{y}$ are their physical positions, and $\langle\cdot\rangle_{N,p}$ is the Wilson measure. - The inner limit runs over the whole sequence of spacings, at every volume whose fundamental box contains the support of f; this holds for all large k. - The test functions are joint functions of the n points and vanish near all coincidences. - One subsequence serves all n, v and f.

(2) Phases. On Conf_n, $S^{\mathrm{mix}} = \int_X S^\xi \, d\nu(\xi)$. Here X is a compact metric space, ν is a probability measure of full support, and $\xi \mapsto S^\xi(a; x)$ is continuous. All correlations S^s, $s \in X \sqcup \{\mathrm{mix}\}$, are smooth and permutation symmetric on Conf_n.

(3) Reconstruction. A selected phase ξ_* carries a Wightman theory: - a strongly continuous unitary representation of the proper orthochronous Poincaré group; - spectrum in the closed forward cone; - a vacuum that is unique up to scalars among translation-invariant vectors; - fields that are operator-valued tempered distributions on the invariant domain spanned by field monomials applied to the vacuum; - local commutativity and faithful field labels.

Its Wightman functions are boundary values, on product test functions, of the continuation of S^{ξ_*} that is holomorphic in the ordered complex times.

(4) Operator product expansion. One system of coefficients, with finitely many local fields of each degree, satisfies finite-order OPE bounds in the mixed state and in every phase.

(5) Nonfreeness. Let e_4 be the time direction. A field Θ of degree 4 with vanishing vacuum expectation satisfies

$$\lim_{r\downarrow 0} r^8 \log^2(1/r)\, G_\Theta(r) = \frac{3\dim_{\mathbb{R}} \mathfrak{g}}{\pi^4}, \qquad G_\Theta(r) = S^{\xi_*}(\Theta(-\tfrac{r}{2}e_4)\,\Theta(\tfrac{r}{2}e_4)).$$

Moreover, G_Θ agrees on no interval $(0,\delta)$ with a finite Wick profile, that is, a finite sum of products of free mass-shell kernels $\int E^q h(\mathbf{p})(2E)^{-1}e^{-rE}\,d^3\mathbf{p}$.

The lattice data are fixed before the volume sequence. The refinement and all continuum objects may depend on that sequence.

The stronger statement. The theorem is a projection of a longer statement, `YangMills.constructed`. That statement holds for every value $b > 0$ of the penalty parameter of the block-spin kernel; the public theorem takes $b = 1$. The construction supplies the following facts; the slope value is established in the proof of regularity, rather than stated as a separate conjunct of `YangMills.constructed`: - The slope is $b_* = \kappa_G = 11C_{\mathfrak{g}}/(24\pi^2)$. - A token is a pair (d, c), with c in the space V_d of gauge-invariant polynomials of weight at most d in the curvature and its covariant derivatives (curvature has weight 2, and each derivative adds 1). - The field spaces are $F_d = V_d/N_d$, where N_d consists of the labels whose mixed correlations with all separated spectators vanish. The label space of the relativistic theory is the directed union of the F_d, so there is no finite field multiplet. - Θ is the centred trace of a constructed conserved stress tensor. - Every phase carries an irreducible Wightman theory, with low-energy bounds that are uniform in the phase. - The correlations extend to distributions across coincident points; uniqueness is not asserted. - After finite-dimensional changes of frame, the rescaled correlations converge at short distance to those of $\dim \mathfrak{g}$ free Maxwell fields. - Nested and integrated forms of the OPE hold, as do further nonfreeness tests.

This overview describes the proof of the stronger statement.

What is not claimed. - No mass gap is proved; the available spectral bounds in every phase are polynomial. - The infinite-volume limit is not shown to be unique or independent of the subsequence. - The phases are not shown to agree, and X is not shown to be a point. - The mixed state is not shown to cluster. - The limit is not shown to be independent of the lattice data. - The two limits are taken in a fixed order. No diagonal limit, exchange of limits or rate of convergence is asserted. - No phase is shown to be a limit of lattice states; the lattice limit is the mixture. - The public theorem asserts nothing at coincident points and no Euclidean axioms. The construction proves those axioms internally. - There is no continuum measure on connections, no Wilson-loop observable or area law, and no particle or scattering theory. - Only finite Wick profiles are excluded. Generalized free fields with a continuous mass distribution, infinite sums and general composites of Gaussian fields are not excluded. - The OPE is a finite-order bound and is not compared with perturbation theory.

Normalizations. On $\mathfrak{g}$, put $\langle X, Y\rangle_{\mathfrak{g}} = -\frac{1}{m}\mathrm{Re}\,\mathrm{tr}(XY)$, and define $C_{\mathfrak{g}} > 0$ by $\mathrm{tr}(\mathrm{ad}X\,\mathrm{ad}Y) = -C_{\mathfrak{g}}\langle X, Y\rangle_{\mathfrak{g}}$. Two facts fix the coupling convention: - $1 - \frac{1}{m}\mathrm{Re}\,\mathrm{tr}\,e^X = \frac{1}{2}|X|^2_{\mathfrak{g}} + O(|X|^4)$; - each

unoriented plaquette is counted once in the Wilson action.

Hence $\beta = 1/g_0^2$ for the continuum action $\frac{1}{2g_0^2}\int \sum_{\mu<\nu} |F_{\mu\nu}|^2_{\mathfrak{g}}$. The constructed couplings satisfy, uniformly in p,

$$\frac{\beta_{N,p}}{\log(1/a_N)} \longrightarrow \kappa_G = \frac{11C_{\mathfrak{g}}}{24\pi^2} = 2b_0, \qquad b_0 = \frac{11C_{\mathfrak{g}}}{48\pi^2}.$$

For $SU(N)$ in the defining representation, $C_{\mathfrak{g}} = 2N^2$ and $\kappa_G = 11N^2/(12\pi^2)$. In that case β is the usual Wilson coupling $2N/g^2$, and the law above is the one-loop running $1/g^2 \simeq (11N/24\pi^2)\log(1/a)$ of Gross–Wilczek and Politzer.

1.0.2 The strategy

Order of limits. One lattice family is fixed first. Correlations of renormalized insertions are taken to the limit $a_N \to 0$ at fixed volume, along the whole sequence of spacings. The volume then tends to infinity along a subsequence of the prescribed one. The renormalization group (RG) is used only for the first limit. Its levels run from the Wilson lattice (level N) to the torus of side L_p at unit spacing (level 0). The bare coupling is tuned so that the effective coupling at level 0 is one fixed number $\bar{\beta}$, the same for all N and p. No RG step is taken above the unit scale; infinite volume is reached by compactness, from local bounds that are uniform in the volume.

Ultraviolet construction (Parts I–III). At each level the density is written as two pieces: - an explicit gauge-invariant action, whose normalized curvature-squared coefficient is the running coupling $\beta_{N,p,m}$; - polymer activities: regular ones, controlled where the curvature is small, and forced ones, which carry the large-field regions.

One block-spin step integrates the fine links against an exactly normalized, gauge-covariant probability kernel: - In the small-field region this is a Gaussian integral around a stationary background field, organized by a cluster expansion. - Large-field regions are kept and paid for by a lower bound on the quadratic energy of the step. - The marginal term is absorbed into the coupling. The coupling changes by the one-loop amount $\kappa_G \log L_m$ up to an error, and block ratios that grow slowly along the levels make the accumulated error $o(\log(1/a_m))$.

The same steps transport inserted observables. This second layer produces what later parts use: - chessboard bounds on renormalized insertions and on their difference quotients, uniform in the volume; - convergence at fixed volume, obtained from fixed-level convergence and uniform observable tails. The construction also supplies analytic estimates in a tuning parameter.

Euclidean limit (Part IV). - Chessboard bounds make the correlations locally bounded, uniformly in the volume. - A transfer-matrix representation and complex interpolation make them equicontinuous, and Arzelà–Ascoli gives the subsequence. - Difference quotients of insertions are close to insertions of higher degree. This yields derivative maps on the field spaces and kernels that are smooth on Conf_n. - The limit inherits translations, the signed permutations of the axes and reflection positivity. It does not inherit continuous rotations or clustering.

Symmetry (Part VI). - Lattice integration by parts gives shell identities for a conserved tensor. Its normalization is fixed by short-distance comparison with free Maxwell theory. - Reflection positivity controls the flux at infinity, and analyticity extends the Ward identities to all separated

configurations. - The infinitesimal rotations integrate to an $O(4)$ action, again by comparison with free Maxwell theory.

Phases and reconstruction (Part V). - The zero-energy subspace of the Osterwalder–Schrader (OS) space of the mixed state carries a commutative C^*-algebra; its proof of commutativity uses the $O(4)$ action. The characters of this algebra are the phases. - Each phase is reconstructed as a Wightman theory, from bounds on operators inserted in thin time slabs. Lorentz covariance comes from the Ward identities. - A low-energy spectral bound, uniform in the phase and derived from locality, upgrades vacuum uniqueness from almost every phase to every phase.

Identification (Part VII). - Along sequences of scales tending to zero, the theory converges, after changes of frame, to free Maxwell theory. - The one-loop coefficient produces an inverse-logarithmic law for the centered stress-tensor trace two-point function, and that law excludes the specified finite Wick profiles. - The RG also yields the OPE.

The logical order is not the order of the parts. Part VI uses the OS spaces of the mixed state from Part V and the free comparison from Part VII. Part V uses the $O(4)$ action from Part VI. Part VIII assembles everything for one family and one limit table.

1.0.3 Part I. The lattice theory

Unit: I.1 — Lattice, symmetries, gauge groups and the Wilson measure.

On $\Lambda_\ell = (\mathbb{Z}/\ell\mathbb{Z})^4$, a configuration assigns $U(y,\mu) \in G$ to each positively oriented link. With h the normalized Haar measure, the Wilson measure is

$$d\mu_{\ell,\beta}(U) = Z^{-1} \exp\Big[-\beta \sum_{y} \sum_{\mu<\nu} (1 - \tfrac{1}{m} \mathrm{Re}\,\mathrm{tr}\, U_{y;\mu\nu}) \Big] \prod_{(y,\mu)} dh(U(y,\mu)).$$

The energy lies in $[0, 12\ell^4]$, so this is a probability measure for every real β. It is invariant under gauge transformations, translations and the signed permutations of the axes.

Lattice schedule. - Block ratios are odd integers $L_m \geq 3$, with $D_{m+1} = D_m L_m$ and $a_m = 1/D_m$. - The level-m torus at volume p has $L_p D_m$ sites per side, with $4 \mid L_p$. - Odd ratios make the coarse lattice compatible with the signed permutations. - Supports are measured by cover numbers by balls of radius B, which obey a coarsening inequality uniform in the volume.

Group facts. On G the unit proves: - a real-analytic logarithm near 1; - an equivariant nearest-point retraction onto G, whose derivative is the orthogonal projection onto $\mathfrak{g}$; - a curvature map $F(u) \in \mathfrak{g}$ with $|F(u)| \leq \|u-1\|/\sqrt{m}$ and $F(vuv^{-1}) = \mathrm{Ad}(v)F(u)$.

These are qualitative statements about single group elements; quantitative control comes in Part II.

Role of simplicity. Simplicity enters through one fact: every Ad-invariant symmetric bilinear form on $\mathfrak{g}$ is a multiple of $\langle \cdot, \cdot \rangle_{\mathfrak{g}}$. Together with lattice symmetry, this leaves a single marginal coupling. The same fact is used in the one-loop computation and in the uniqueness of the stress tensor.

Reflection positivity. The Wilson measure is reflection positive for every $\beta \geq 0$, for reflections through planes of sites and through planes between sites (the argument of Osterwalder and Seiler).

This is proved in Part IV for the original lattice measure, not for an effective action, and it is used only through the chessboard inequality.

1.0.4 Part II. Fields, insertions and polymer activities

Units: II.1 — Field labels and their lattice representatives; II.2 — Polymer activities, norms, the Wilson family, the construction record, reflection-positive cells.

Labels. In radial gauge, a connection near a point is described by its Taylor coefficients, and the coefficient of order j has weight $j+2$. V_d is the space of $\mathrm{Ad}(G)$-invariant complex polynomials of weight at most d in these coefficients. Equivalently, it consists of the gauge-invariant polynomials of weight at most d in the curvature and its covariant derivatives. It is finite-dimensional, and $1 \in V_d \subset V_{d+1}$.

Lattice representatives. The basic lattice variables are "slots": plaquette curvatures transported to a base point along walks. - For each d, finitely many slots have linearizations with no common kernel on the jets of weight at most d. - The slot map then has an analytic Ad-equivariant left inverse near 0; it is chosen non-constructively. - Compose a basis of V_d with the order-d Taylor polynomial of this inverse, and average over the signed permutations. For each label c this gives a gauge-invariant polynomial Φ_c in the link entries of a fixed box stencil.

Two properties follow: - On links obtained by parallel transport of polynomial radial-gauge potentials at spacing h, Φ_c reproduces the classical label to order h^d. - Suppose the plaquettes near the stencil have chord at most t. Then a component with $f \geq 1$ curvature factors, and its derivatives, are $O(t^f)$. Conversely, a function with such bounds determines a label with uniformly bounded coefficients.

Renormalized insertions. The insertion of $v \in V_d$ at a site o is

$$O_{N,p}(v,o) = a_N^{-d}\, \Phi^{(d)}_{S_{1/\theta_N}\tau^{-1}v}(o).$$

- S_t multiplies the coefficient of a component with f curvature factors by t^f.
- θ_N is the small-field threshold of the finest level.
- τ is a linear automorphism of V_d. It commutes with conjugation and with the signed permutations, and it fixes the unit.

The RG defines τ. The running label coordinates of an insertion obey a transfer relation from level N down to level m, and τ is their value at a reference level $r(d)$ fixed in advance. So each insertion is normalized to have running coordinates exactly v at level $r(d)$. All labels of weight at most d, including the unit, may mix; no triangular structure is imposed. The large-field part of the inserted density goes into reserved forced polymers, so the construction never divides by the vacuum density.

Equation relations. Haar invariance gives the lattice Schwinger–Dyson identity $\langle\langle A, \nabla_e S_W\rangle\, S\rangle_\beta = 0$ for S independent of the link e. Products of such forces with slot polynomials span the linearized free-Maxwell equation ideal, up to terms of higher weight. They give lattice observables whose correlations with distant spectators vanish exactly. Two facts are established later, through the comparison with free theory: these relations become null in the limit, and they exhaust N_d.

Polymers, budgets and marks. - A polymer activity has a finite support of sites. - Regular activities are anchored at a site and measured by suprema of derivatives over small-field configurations. - Forced activities carry the large-field regions, together with an inventory of flagged sites. - Activities are controlled by budgets summed through each site, $\sum_{X\ni x} e^{a\,\mathrm{cov}_B(X)}\|K_X\| \le \text{budget}$. - Insertions carry marks: commuting variables λ with $\lambda^2 = 0$, one for each label or site that may occur. Exponentials of marked quantities are finite sums, and logarithms are finite recursions in the number of marks. - The weighted norm $\|a\|_w = \sum_x \prod_{\lambda\in x} w_\lambda\,|a(x)|$ is submultiplicative, so bounds do not grow with the number of marks. - A tree-graph bound without factorials controls connected clusters of overlapping supports.

1.0.5 Part III. The ultraviolet construction

1.0.5.1 The block-spin step and the background field

Units: III.1 — The block-spin step and the symmetric gauge fixing; III.6 — The stationary branch: background field of the effective action; III.3 — Free lattice kernels and covariances.

At level m, relative to product Haar measure, the density is

$$\rho_m(U) = \exp\big(v|\Lambda_m| - \beta\,\mathcal{K}(U) - \mathcal{L}(U) - \mathcal{E}(U)\big)\,\mathcal{G}(U).$$

- $\mathcal{K}$ is built from lattice paths: a plaquette term, smeared curvature squares, and cubic and quartic polynomials in transported curvatures. It is averaged over the signed permutations and normalized so that its quadratic symbol at zero momentum is the identity; this fixes the meaning of β.
- $\mathcal{L}$ is a loop term independent of β.
- $\mathcal{E}$ is the sum of the regular activities times small-field indicators.
- $\mathcal{G}$ is the gas of forced activities.
- At the finest level, $\rho_N = e^{-\beta_{N,p,N} S_W}$.

The kernel. One step is $\rho_m(V) = \int k_\beta(V,U)\,\rho_{m+1}(U)\,dU$. Given the fine configuration U, a finite tag t is drawn with probability $\lambda(U,t)$. Each coarse link is then drawn independently, with Haar density

$$Z_\beta^{-1}\exp\Big(-\beta\kappa(1-\tfrac{1}{m}\mathrm{Re}\,\mathrm{tr}\,V_b Q_b(U,t)^{-1})\Big),$$

where $\kappa > 0$ is the penalty. The predictor Q_b is built in two steps: 1. On a patch of radius $(J+3)L$ around each coarse site, a gauge frame is chosen. - If all route holonomies on the patch are close to 1, the frame is the unique minimizer of an averaged holonomy discrepancy. - Otherwise it is a uniformly drawn eligible axial gauge; such gauges exist for every configuration. 2. Fine transports of length L, read in these frames, are combined by a nonlinear (minimizing) average with a J-fold convolution weight, with a random fallback.

Properties of the kernel: - Haar invariance gives $\int k_\beta(V,U)\,dV = 1$ exactly. - The kernel is exactly covariant under fine gauge transformations and the lattice symmetries. - Hence $\int \rho_m$ equals the Wilson partition function at every level, and the step is defined without any small-field assumption.

Background field. For the integration, the links of a spanning forest are set to 1, and each remaining link is $\exp(a)$ times a fixed lift of the coarse field x. The background is the critical branch $a = \rho(x)$ of the classical Taylor model of the action (quadratic, cubic and quartic terms)

plus the blocking penalty, at fixed x. The steps are: - The Hessian at the flat field is a positive "bed" form plus the penalty. - Each non-tree link is bounded by the curvature in its block plus the response of the predictor. With this, the Hessian is uniformly positive in the vertical directions, with a constant independent of the volume and the forest. - The analytic implicit function theorem then gives ρ as a locally unique analytic germ with explicit linear part. Statements with a uniform radius and localized jets give control independent of the volume. - The branch is not shown to be a global minimizer, and this is not used. - Changing the forest multiplies the Hessian determinant by the square of a Jacobian and the Haar measure by the Jacobian itself. So the critical value and the one-loop term do not depend on the forest.

Free kernels. The free covariances are written in Fourier space through the actual averaging filters, which are entire and multiplicative under composition of blockings. Aliased momenta carry extra powers of $1/L$. - All constants (decay rates, strip widths, lower bounds) are fixed before the block ratio L. - The quadratic forms decay exponentially and are bounded below uniformly in the volume. - An exact identity on each pair of tori splits a quadratic form into a coarse part and a fluctuation part (a Schur complement).

1.0.5.2 Gaussian integration and the cluster expansion

Units: III.4 — Gaussian integration: change of Gaussian sector, determinants, window law; III.5 — Cluster expansion of the polymer gas; III.8 — Connected activities and backward dependence cones.

Change of variables. Active links are written in exponential coordinates around the background. Inactive links stay in Haar measure. - The change of variables to a Gaussian integral with covariance C and mean $m(Y)$ is an exact identity. It holds for any positive-definite C, keeps all Jacobians, and uses no smallness assumption. - The covariance actually used is the inverse of the gauge-fixed Gram form on the active links, not the inverse of the Wilson quadratic form. Its precision is bounded below uniformly in the volume. - Window law: the global inverse is approximated by inverses on windows of radius r, with error Ce^{-vr} uniform in the volume. - $\log\det$ separates into local terms and far-field terms with summable errors.

Cluster expansion. The gas is $\Xi(w) = \sum_s \prod_{i\in s} w_i$, summed over families of polymers with pairwise disjoint supports and real weights. With $B = \lceil (\log\widehat{\beta})^2 \rceil$, the smallness condition is

$$\sum_{i:\, x\in X_i} |w_i|\, e^{(a+\delta)\operatorname{cov}_B(X_i)} \le \gamma \quad \text{for every site } x, \qquad 4(2B+1)^4\gamma/\delta \le q < 1.$$

Under this condition: - $\log\Xi$ is an absolutely convergent sum of connected clusters, bounded through spanning trees (Penrose's partition scheme); - removing every polymer that meets a set Y changes $\log\Xi$ by at most $|Y|\gamma/(1-q)$, uniformly in the volume.

So the vacuum logarithm is extensive, but a local change costs only its size. A second, graph-free argument based on a deletion recursion gives bounds of the same type.

Connected activities and dependence cones. - Fixed-order connected coefficients come from the heat equation $\partial_t W = \frac{1}{2}(\Gamma : \nabla^2 W + \Gamma : \nabla W \nabla W)$ for $W_t = \log \mathbb{E}_{t\Gamma} e^V$. A rooted ℓ^1 norm with weight $e^{\sigma\,\mathrm{diam}}$ makes contraction and joining lossless, so these coefficients are polynomially bounded uniformly in the volume. There is no resummation. - Pulling a coarse observable back through the block kernel gives a function of the fine links in a window of fixed radius around its support.

The backward cone therefore grows by a bounded physical width per step. - For disjoint supports, the pull-back factorizes exactly.

1.0.5.3 Large fields and one nonlinear step

Unit: III.10 — One full nonlinear renormalization step: bounds on irrelevant terms.

At reference coupling $\hat{\beta}$ the small-field threshold is

$$t = \hat{\beta}^{-1/2} P(\hat{\beta}), \qquad P(\hat{\beta}) = \exp\left((\log\log\hat{\beta})^2\right),$$

well above the Gaussian fluctuation scale $\hat{\beta}^{-1/2}$.

Seeds. A seed is one of: - a fine plaquette with chord at least $t/4$; - a coarse link on which the coarse field departs from the predictor by more than a fixed multiple of t; - a flagged coarse site.

Energy lower bound. For every store A of seeds,

$$\kappa\, t^2\, \sharp A \;\le\; \text{quadratic energy of the step integrand, after subtracting the stationary predictor.}$$

The energy is a sum over five channels, one of which is the blocking penalty. The proof assigns charges to fine cells with bounded multiplicity, and controls the linearization defects on cells without seeds.

Why this suffices. Since $\beta t^2 \simeq P^2$, each seed gains a factor e^{-cP^2}. P^2 grows faster than any power of $\log\hat{\beta}$, so this gain beats: - the polylogarithmic number of connected seed sets through a site; - any per-seed loss of a fixed power of $\hat{\beta}$, or of order P^z with $z < 2$.

The decomposition into large-field sectors is exact: large-field sectors are integrated, not discarded.

One step. The step takes inputs at level $m+1$: regular norms at most $\hat{\beta}^{-11/10}$, forced norms at most $e^{-P^{1/5}}$, and the coupling in the band $[\hat{\beta}/2, 2\hat{\beta}]$. For L large, and then $\hat{\beta}$ large, it produces: - site-summable budgets for the new regular activities; - the bound $\exp\left((C\,\mathrm{bill}^5 - \beta\mu t^2)\sharp A\right)$ for each large-field sector, where $\mathrm{bill} = 1 + B + (\log\hat{\beta})^2$; - bounds on the probabilities of the remaining catastrophic events.

The marginal and vacuum terms are extracted, and the level-m bounds restored, by the reset below.

1.0.5.4 The marginal coupling and the schedule

Units: III.7 — Matching the Wilson energy to fourth order; III.9 — The marginal flow: the running coupling and its reset; III.2 — Choosing the constants: the order of choice and the RG schedule.

Matching and the one-loop coefficient. - At the finest level, a path representation is constructed whose jets of order at most 4 agree with those of the Wilson energy in exponential coordinates. - At later levels, the cubic and quartic parts are represented in curvature variables by quasi-local lifts (a cost-controlled Poincaré lemma). - In each step, the coefficient $\alpha_{N,m}$ of curvature squared produced by the one-loop term satisfies $|\alpha_{N,m} + \kappa_G \log L_m| \le D$, with D fixed before L.

The value of κ_G comes from an explicit momentum integral. The comparison symbol is

$$h(k) = 4C_{\mathfrak{g}}\Big(\frac{k_1^2}{|k|^6} - \frac{4k_1^2k_0^2}{|k|^8} - \frac{1}{|k|^4}\Big).$$

Its angular average is $4C_{\mathfrak{g}}(\frac{1}{4} - \frac{1}{6} - 1)|k|^{-4} = -\frac{11}{3}C_{\mathfrak{g}}|k|^{-4}$, and $\int_{1/L<|k|\le 1} |k|^{-4}\, d^4k = 2\pi^2 \log L$. Hence $(2\pi)^{-4}\int_{1/L<|k|\le 1} h = -\kappa_G \log L$. Comparing the step's Hessian with h up to $O(|k|^{-3})$ gives the bound on $\alpha_{N,m}$.

The reset. After the step, the blocked density is rewritten exactly, at every configuration including large fields: - a new coupling $\beta_{N,p,m} = \beta_{N,p,m+1} + \alpha_{N,m} + \delta$; - a new vacuum constant; - reorganized activities: flat values and isotropic quadratic parts are subtracted, compensating path terms are added, and mismatches of the small-field masks go into the forced gas.

The correction δ is the isotropic Hessian coefficient of the regular activities: - By simplicity and hyperoctahedral symmetry, their quadratic part at a point is δ times the identity on plaquette orientations. - $|\delta| \le 1$, and δ is fixed by a scalar equation with a unique root.

Altogether, with ω_j a drift allowance attached to regularity index j,

$$\big|\beta_{N,p,m+1} - \beta_{N,p,m} - \kappa_G \log L_m\big| \le \omega_{J_{m+1}}.$$

Endpoint. For real $|b| < r_*$ there is a trajectory with level-0 coupling $\bar\beta + b$. Its bare coupling is unique within the construction, and the trajectory continues analytically to complex $|b| < r_*$, with r_* independent of N and p. The family uses $b = 0$, so $\beta_{N,p,0} = \bar\beta$ for all N and p. An intermediate-value shooting argument also produces such trajectories; the analytic version is the one used for convergence at fixed volume below.

Running. Put $s_m = \log(1/a_m)$ and $\hat\beta_m = \bar\beta + \kappa_G s_m$. Induction from level 0 gives $|\beta_{N,p,m} - \hat\beta_m| \le \sum_{k\le m} \omega_{J_k}$, and the schedule makes this $o(s_m)$:

$$\beta_{N,p,m} = \bar\beta + \kappa_G \log(1/a_m) + o(\log(1/a_m)) \qquad \text{uniformly in } N \ge m \text{ and } p.$$

At $m = N$ this is the slope law with $b_* = \kappa_G$. Since $\bar\beta$ is chosen large, every level stays in the band $[\hat\beta_m/2, 2\hat\beta_m]$. The statement is leading-logarithmic only: there is no two-loop term.

Why the block ratios grow. The one-loop error per step is bounded but does not decrease with L. For the accumulated error to be $o(s_m)$, the allowance per unit of logarithmic scale must tend to zero. The schedule therefore imposes two conditions: - $\omega_{J_{m+1}} \le J_{m+1}^{-2} \log L_m$, which forces $\log L_m \to \infty$; - $\log L_m / s_m \to 0$, which keeps each step small compared with the total scale.

The ratios are constant while the regularity index is fixed, $L_m = \ell_{J_{m+1}}$. The regularity index J is nondecreasing, rises by at most one per level, tends to infinity toward the fine end, and stays constant on long windows of levels. One sequence (L_m) serves every cutoff N.

Order of choices. The constants are chosen in a fixed order: 1. the kernel with its penalty, and for every index j its regularity profile (derivative order, spatial power, support exponent, widths, path counts, drift allowance), together with the one-loop error constant D; 2. all block ratios ℓ_j, so that the allowances and the logarithms of the combinatorial costs are at most $j^{-2}\log \ell_j$; 3. the

smallness exponents; 4. the entry couplings and the volume conditions; 5. $\bar{\beta}$; 6. the assignment of regularity profiles to levels.

Each degree d has a reference level $r(d)$ with regularity index at least $d+3$. Beyond $r(d)$ the derivative order exceeds d and the spatial power is at least d. All choices precede any observable or extraction order, so every degree is served by the same family.

1.0.5.5 Renormalized observables, uniformly in the volume

Units: III.11 — Comparison with the free Gaussian measure: interacting rank and determinant budgets; III.12 — Bounds uniform in the volume and convergence as the spacing tends to zero; III.15 — Cost accounting for observables and the remainder comparison; III.16 — Analytic toolkit.

The chessboard root. The quantity propagated for observables is the chessboard root. For a function F of the links in one cell of a tiling of the torus by cells of side a_m,

$$\|F\|_T = \Big| \mathbb{E}_{\beta_{N,p}} \Big[\prod_{\text{cells}} \text{reflected copies of } F \Big] \Big|^{1/\sharp\text{cells}}.$$

The main bound reads as follows. For every degree d, radius R and $\varepsilon > 0$ there is A such that, for every $m \geq r(d)$, there is N_0 independent of p with

$$\|O_{N,p}(v,o)\|_{T_m(o)} \leq A\, a_m^{-d-\varepsilon} \qquad (N \geq N_0,\ \text{all } p,\ \|v\| \leq R).$$

It is proved by running the insertion through the same RG steps; ε absorbs losses smaller than any power. Reflection positivity enters only afterwards, through the chessboard inequality. That inequality makes the root a seminorm and bounds correlations of separated observables by products of roots.

No loss of $\log\beta$ per site. A root over c copies tolerates a fixed cost per copy, but not a factor $\log\beta$ per site. The determinant budget avoids this by a gauge count on a terminal torus with $c = L^4$ sites: - A tree gauge leaves $4c-(c-1) = 3c+1$ free links. - In the numerator, $3c-3$ of them are integrated one at a time, each being the only new link of a chosen plaquette. Each integral gives at most $b\,E^{\dim G}$, with $E \propto g = \beta^{-1/2}$. The four links that wrap around the torus, one per axis, are left over. - The normalization is bounded below by confining all $3c+1$ free links to Haar balls of radius $g/4$. - The ratio is therefore at most $\exp(c\,\pi - 4\dim G\,\log g)$, where the cost π per site is independent of g. - Since the number of copies is c, the root carries only $g^{-4\dim G/c}$.

Field content. Free Gaussian detectors give uniform rank detection: $\|\pi x\| \leq Z\|\text{probe}(\pi x)\|$, where π is the projection onto quotient labels. The limit field spaces are isomorphic, degree by degree, to those of free Maxwell theory, compatibly with conjugation and reflections.

Difference quotients. - A normalized lattice difference of an insertion of degree d equals an insertion $O(x)$ of degree $d+|l|$, up to a term whose root tends to zero. - Write $x = \pi x + (x - \pi x)$. The part $x - \pi x$ lies in the Maxwell equation ideal and has small root by the lattice equations of motion. The part πx is bounded by the rank detector. - So difference quotients satisfy root bounds of the same form. By the chessboard inequality, lattice correlations with any number of differences are bounded at r-separated points, uniformly in p.

These bounds are propagated through the RG; they do not come from differentiating a sup norm.

A rank-free trace bound. Let $X \geq 0$ be a matrix and let the insertions b_a be separated by gaps of more than $2r$ factors of X. Moving $m \leq r$ factors from one gap to another changes $\operatorname{tr} \prod_a b_a X^{Y_a}$ by at most $(\mathrm{cost}/r)\, m$. The cost is a product of trace norms with no dimension factor. The proof is a Hölder inequality for complex exponents, obtained from Hadamard's three-lines theorem, followed by a Cauchy estimate. By quadrature it extends to continuous positive kernels. Part IV uses it for equicontinuity.

Bookkeeping. The numerical conditions of all steps can be met in one fixed order: constants before L, then L, then the later constants, then β large. This holds uniformly across regularity profiles.

1.0.5.6 Convergence at fixed volume and the construction run

Units: III.12; III.13 — Extension to all labels: one common Wilson family; III.14 — The construction run: flat representations and the ultraviolet rate.

The obstacle. At fixed volume, two cutoffs N and N' have different bare couplings, each fixed by the condition $\beta_{N,p,0} = \bar{\beta}$. Continuity in a fixed bare coupling therefore compares nothing.

Fixed-level convergence and tails. The construction has a one-parameter family of trajectories: - The level-0 coupling is $\bar{\beta} + ws$, with a real tuning parameter $s \in [-1, 1]$. For each s the bare coupling is unique within the recursion. - The bare coupling and effective data have analytic dependence on the tuning parameter. The estimates on that dependence are separate from the Cauchy argument below. - For fixed j, convergence of the trajectory and stability of the observable transfers give convergence of the effective insertions and their polymer parts. Uniform convergence passes through the Haar integrals, and a positive lower bound on the normalized partition function gives convergent correlation approximations. - The omitted part is bounded, uniformly in the cutoff, by $2F_0 e^{-B_j/64}$, where $B_j = \lceil (\log \hat{\beta}_j)^2 \rceil \to \infty$. - Choose j to make this tail small, then take the cutoffs large enough that their level-j approximations are close. This proves the Cauchy property first at lattice-fitted configurations; equicontinuity extends it to all configurations distinct modulo the period. See III.12, §§4.5–4.6.

Combined with the difference bounds, this gives at each fixed volume: - limit kernels smooth away from periodic coincidences; - convergence of difference quotients to derivatives; - convergence of smeared moments along the whole sequence of spacings.

The family used by the public theorem. VIII.1 assembles the construction for one family, using the bounds and convergence results described in III.9–III.12. III.13 describes an alternative coordinated family with root bounds, an analytic shooting family and Maxwell rank comparisons. Its existence statements do not identify that family with the one used by `YangMills.constructed`; properties from the two constructions should not be combined without such an identification.

The construction run (III.14). III.14 packages the construction run. It also proves quantitative convergence of the order-zero quadratic part of the effective action (its position-space Hessian kernel) at each fixed level i, at rate $(a_N/a_i)^c$. - The difference between two cutoffs is written in curvature variables as $d^{\mathsf{T}} A d$, with A defined modulo kernels annihilated by this map. - So the curvature precision, which has zero modes, is never inverted. - This rate concerns the quadratic germ only, not the full effective action or the final limits.

1.0.6 Part IV. Infinite volume

Unit: IV.1 — Infinite volume: extraction of a subsequence and the Euclidean axioms of the limit.

Local bounds. - Reflection positivity gives the chessboard inequality $|\langle\prod_c F_c\rangle| \le \prod_c \|F_c\|_T$ for every $\beta \ge 0$. Log-convexity of roots under translation handles cells not centred at the insertion. - Put each insertion in its own cell of side $a_m < r$. Then $|C_{N,p}(v;x)| \le A^n$ whenever each pair of points has some coordinate at distance at least r, uniformly in p and for large N. - Near collisions, the same argument gives $|S(v;x)| \le Cr^{-4n(1+\max d_i)}$ when all points are at mutual distance at least r.

Equicontinuity. Let Gen_n be the set of configurations whose points have pairwise distinct coordinates on every axis. 1. Integrating out the links along one axis gives a positive transfer kernel K. A correlation becomes $\mathrm{Tr}\prod_k B_k K^{Y_k}/\mathrm{Tr}\,K^M$, with slab kernels B_k and gaps Y_k. 2. Moving one point shifts units between two gaps. The trace is entire in the size of the shift. 3. The rank-free Hölder bound controls the trace on a disc, and a Cauchy estimate gives a Lipschitz bound. 4. The trace norms of the slab kernels are themselves chessboard roots of $2q$ alternating copies, so the bound is uniform in the volume.

The result is $|C(x) - C(y)| \le 16nA^n|x-y|/R$ on compact subsets of Gen_n whose coordinate gaps exceed R. No derivative bound is used.

Extraction. Arzelà–Ascoli is applied twice, over countably many degrees, basis labels and compact sets: first to extract a spacing subsequence that works for all volumes, then a volume subsequence. This gives limit kernels on Gen_n. The complement $\mathrm{Conf}_n \setminus \mathrm{Gen}_n$ is a null set, and the kernels are locally bounded on Conf_n, so dominated convergence gives the smeared limits.

Derivatives and smoothness. - Exact summation by parts in the lattice positions moves a difference quotient onto the test function. - The ultraviolet estimates show that the lattice difference quotient of an insertion is chessboard-close to an insertion of one degree higher. - A dimension count, from the rank comparison, identifies the kernel of the limiting projections with the null labels. - Finite-dimensional compactness then gives fixed derivative maps $\partial_a : F_d \to F_{d+1}$ with

$$\partial_{x_i^a} S(\dots, v_i, \dots; x) = S(\dots, \partial_a v_i, \dots; x) \qquad \text{in } \mathcal{D}'(\mathrm{Conf}_n).$$

Iteration and a Fourier argument give $S \in C^\infty(\mathrm{Conf}_n)$, with derivatives polynomially bounded near collisions.

What the limit inherits. The limit inherits: - translations; - the group of signed permutations of the axes, acting on points and labels; - permutation symmetry, reality and the unit label; - reflection positivity at coordinate planes, for point and smeared tests.

It does not inherit continuous rotations, clustering or uniqueness.

Removing the spacing refinement. The extraction refines both the spacing and the volume sequences. At each admissible fixed volume, the smeared limit along the whole spacing sequence exists (Part III) and agrees with the subsequential one. This removes the spacing refinement from the final statement.

1.0.7 Part V. Phases and relativistic reconstruction

Units: V.1 — Decomposition of the limit into phases; V.2 — Osterwalder–Schrader reconstruction of a Wightman theory; V.3 — Reconstruction for every phase and the reconstructed theory.

The OS space of the mixture. Positive-time field words, modulo the null space of $\langle \theta w, w' \rangle_S$ and completed, form a Hilbert space. It carries: - a contraction semigroup e^{-tH} with $H \geq 0$; - a unitary representation of the spatial translations.

The joint spectrum lies in a half-space. No positive spectral gap is asserted. The zero-energy space may be large. Its projection Π is the strong limit of e^{-tH} as $t \to \infty$.

Thin slabs. A word w placed in a thin time slab defines an operator B on the OS space. To bound it, consider the train $v_{n+1} = B^* B v_n$, which contains n reflected copies of w and an arbitrary exterior word. - The lattice bounds give $\|v_n\|^2 \leq B_v A^{4n}$. Here A is polynomial in the word data and in the inverse buffer width, and B_v depends on the exterior word but not on n. - A power lemma removes B_v: if $\|(BA)^n v\| \leq K_v D^{2n}$ for all n, then $\|Av\| \leq D\|v\|$.

So the exterior word is paid once and disappears under the n-th root.

Phases. - The compressions $M(w) = \Pi\, B_w|_{\ker H}$ are independent of the width and placement of the slab. - $\langle \Omega, M(w_1) \cdots M(w_k) \Omega \rangle$ is the limit of S as the clusters are moved apart in time order.

The compressions commute: 1. Place four clusters on the rays $n(ue_4 + ve_1)$, with $(u, v) = (0,0), (1,2), (2,1), (3,3)$. 2. Exchanging the time axis with a spatial axis reverses the order of the middle pair. 3. $O(4)$ covariance and permutation symmetry of S then give $\langle \Omega, M_x M_y M_w M_z \Omega \rangle = \langle \Omega, M_x M_w M_y M_z \Omega \rangle$. 4. By density, $[M_y, M_w] = 0$.

The resulting phase decomposition: - The $M(w)$ generate a commutative unital C^*-algebra. Its Gelfand spectrum is X, which is compact and metrizable. - ν represents the vacuum state, and it has full support because the vacuum is separating. - The phase correlations are $S^\xi(w) = \xi(M(w))$. They are continuous in ξ, and $\int S^\xi \, d\nu = S^{\mathrm{mix}}$. - Reflection positivity, $O(4)$ invariance, the derivative and stress identities and uniform growth bounds hold for every ξ. They are proved as operator identities before any character is applied. - Every phase has the same null spaces as the mixture. - A variance argument over a countable total family of words gives a one-dimensional zero-energy space for ν-almost every phase. - The selected phase ξ_* is one such phase, chosen by this existence argument; no further condition is imposed. The trace law of Part VII is proved for the phase so chosen.

Reconstruction. For each phase, slab operators bounded by $C p^{-A}$ (with p the slab half-width) define

$$F_n(z; \mathbf{x}) = \big\langle \Omega,\, B(v_1)\, e^{i(z_1 - z_2)H} U(\mathbf{x}_1 - \mathbf{x}_2)\, B(v_2) \cdots \Omega \big\rangle.$$

This function is holomorphic for $\operatorname{Im} z_1 < \cdots < \operatorname{Im} z_n$, and it equals the Euclidean correlation at imaginary times. The Wightman theory is obtained in five steps: 1. **Boundary values.** Bounds of order k^{-m}, when all height gaps exceed k, give tempered boundary values W_n. They converge against every joint Schwartz test as the vector of heights tends to zero through all ordered vectors. 2. **Positivity and fields.** This mode of convergence allows two independently regularized vectors to be compared. So W is a limit of Gram matrices of OS vectors: it is positive, and the Wightman space embeds isometrically into the OS space. Fields act on finite words modulo the null space by

prepending a letter. This gives the invariant domain, the formal adjoints and the cyclic vacuum. 3. **Covariance and spectrum.** The rotation Ward identities of Part VI, in planes containing the time axis, continue to infinitesimal boost identities. These integrate to Lorentz covariance without any spectral assumption. The spectral measure of a Wightman vector equals that of its OS image, so it is supported in $p^0 \geq 0$; covariance then moves the support into the forward cone. 4. **Locality.** Only adjacent transpositions are needed. For a spacelike adjacent pair, both orderings have holomorphic germs, constructed with complex boosts. The germs agree to all orders on the Euclidean slice, because the Euclidean kernel is symmetric and smooth away from coincidences. 5. **Faithfulness.** A label whose Wightman functions vanish has a vanishing tube function, hence a vanishing Euclidean kernel.

Every phase. The low-energy bound uses only locality and the spectrum condition. Take A a finite sum of field words with compactly supported tests. - The commutator of time-translated copies of A vanishes for times at most 1 and spatial separations beyond a radius R independent of ξ. - Average over unit time, pair with a spatial-momentum bump of radius $2u$ (whose Fourier transform is $O(u^3)$ in three space dimensions), and use $|\mathbf{p}| \leq E$ on the forward cone. This bounds the spectral weight of each shell $u/2 < E \leq u$ by $O(u^2)$, and dyadic summation gives the first bound below.

$$\mu_{A\Omega_\xi}(0 < E \leq u) \leq M_A u^2 \quad (0 < u \leq 1), \qquad 0 \leq \langle A\Omega_\xi, (e^{-sH_\xi} - P_{0,\xi})A\Omega_\xi \rangle \leq M_A s^{-2} \quad (s \geq 1).$$

The constant M_A is independent of ξ, by compactness of X.

Vacuum uniqueness then extends from almost every phase to every phase: - The function $\xi \mapsto \langle A\Omega_\xi, e^{-sH_\xi} A\Omega_\xi \rangle - |\langle \Omega_\xi, A\Omega_\xi \rangle|^2$ is continuous. - For fixed A and s, the set of phases where this function lies in $[0, M_A s^{-2}]$ is closed by continuity. It contains the full-measure set of phases with unique vacuum; full support of ν makes this closed set all of X. - Letting $s \to \infty$ gives a unique vacuum in every phase.

The comparison map from the Wightman space to the OS space is surjective, proved independently by pushing time-ordered vectors apart. This transfers vacuum uniqueness to the OS space. Each phase theory is irreducible. Its Wightman distributions are continuous in ξ, and their ν-average is the boundary value of the continuation of S^{mix}. The bounds allow spectrum down to zero and give only polynomial decay s^{-2} in Euclidean time, not exponential clustering.

1.0.8 Part VI. Ward identities, rotations and the stress tensor

Units: VI.1 — Ward identities, the rotation action and the conserved stress tensor; VI.2 — The free Maxwell stress tensor and the local Ward charge.

The lattice has only the hypercubic symmetries. The proof obtains the generators of rotations from stress fluxes, and the group action by comparison with free Maxwell theory. It uses the OS spaces of the mixed state in the coordinate directions and the free comparison of Part VII. It uses no decay of correlations.

Free Maxwell theory. Consider $\dim \mathfrak{g}$ free Maxwell fields, with field spaces the G-invariant polynomials in on-shell curvature jets. - The centred, real, symmetric tensors of degree 4 that are covariant under signed permutations form a three-dimensional space. - For simple $\mathfrak{g}$, the divergence maps this space onto a two-dimensional space. So the conserved tensors form a line, spanned by

the Maxwell stress tensor. - For every Killing field v, the flux of $\sum_b v_b T_{ab}$ through a shell around an insertion equals the Lie derivative of the insertion. For elementary fields this follows from $-\Delta G_0 = \delta$; for polynomials, from the Leibniz rule and Wick contraction. - An explicit three-point function fixes the sign and normalization of the Ward constant.

Conserved tensor. There are scaling sequences $r_k \downarrow 0$ with frames U_k, isomorphisms for large k. Along them the rescaled interacting divergence converges to the free one. - Surjectivity is an open condition, so the interacting divergence is also onto, and its kernel is one-dimensional. - A nonzero conserved J in this kernel converges, after rescaling, to a multiple of the free stress tensor.

Shell identities. Lattice integration by parts in the link variables, with ultraviolet control of the error terms, gives shell identities. Take χ equal to 1 near x_0 and vanishing near the spectators:

$$-\sum_a \int \partial_a \chi(y)\, S(J_{ab}(y)\, B(x_0) \cdots)\, dy = S((p_b B)(x_0) \cdots).$$

With weight $K(y - x_0)$, for a rotation generator K, the same construction gives maps Y_K. Here p_b and Y_K are linear maps on labels, not yet identified with derivatives and rotations. Moving the centre of the shell gives $\partial p = p\, \partial$ and $\partial_b Y_K + \sum_c K_{cb}\, p_c = Y_K \partial_b$.

Flux at infinity. - For rotations, cut off the rotation field so that it is compactly supported and equal to the exact field on a box far larger than the configuration. - Conservation and symmetry cancel the cross terms. - The remaining exterior contribution is bounded through spectral measures of the OS semigroups along and transverse to a coordinate line, and it tends to zero. - This gives the angular identity for configurations on one coordinate line, together with a line-sum identity, and hence transversality of the label maps.

Normalization. In free theory, the finitely many relations satisfied by p and ∂ at a given degree have a one-dimensional solution space. Injectivity is an open condition, so the same holds along the scaling sequence. Hence $p_b = c\, \partial_b$, with c independent of the degree and $c \neq 0$. The tensor $T = J/c$ satisfies

$$-\sum_a \int \partial_a \chi(y)\, S^s(T_{ab}(y)\, B(x_0) \cdots)\, dy = \frac{\partial}{\partial x_0^b}\, S^s(B(x_0) \cdots)$$

in the mixed state. By the operator identities of Part V, it holds in every phase. Since $\partial\chi$ vanishes near every insertion, no contact terms arise.

All separated configurations. Positivity of Gram functions, together with the Ward values already obtained, gives analytic-vector bounds $\|f^{(m)}\| \leq C(\rho/2)^{-m} m!$. The angular Ward functional is therefore real-analytic along walks that move one point at a time. Starting from configurations on a line, it vanishes on all configurations whose points have pairwise distinct first coordinates, and by continuity on all of Conf_n.

Integration to $O(4)$. - The bracket defect $[Y'_u, Y'_v] - Y'_{[u,v]}$ (with $Y' = Y/c$) satisfies a summed identity. A rigidity statement, proved in free theory and transferred by the comparison, forces the defect to vanish. - The infinitesimal action converges to the Maxwell one: $U_k^{-1} Y'_K U_k \to \mathrm{Lie}_K^{\mathrm{Max}}$. - A perturbation of spectral projections of the Casimir operators gives exact intertwiners V_d. The action on F_d is $R_d(k) = V_d R_d^{\mathrm{Max}}(k) V_d^{-1}$. - Since $t \mapsto S(R(e^{tK})v; e^{tK}x)$ has zero derivative, and reflections act through the real structure, $S(R(k)v; kx) = S(v; x)$ for all $k \in O(4)$. - Averaging J/c

over $O(4)$ gives a conserved, real, symmetric, covariant tensor that satisfies both the translation and the rotation Ward identities.

1.0.9 Part VII. Identification as Yang–Mills theory

1.0.9.1 Free Maxwell fields at short distance

Units: VII.1 — Free Maxwell comparison theory; VI.2 — Interacting stress comparison; VII.6 — Scaling variants and fixed-field estimates. The comparison is assembled in VIII.1.

The reference theory. It consists of $\dim \mathfrak{g}$ free Maxwell fields built from $\Gamma(x) = 1/(4\pi^2|x|^2)$. - On the lattice, corrected covariances of finite-difference jets converge to $\partial_a \partial_b \Gamma$ away from coincidences. - Wick moments satisfy the same Gaussian recursion on the lattice and in the continuum, so correlations of quotient labels converge. - A triangular basis built from the Kelvin transform gives strict reflected Gram positivity for every label outside the ideal I_d generated by the free Maxwell equations and the Bianchi identities. - So the free null space is exactly I_d, and the free field space $M_d = P_d/I_d$, where P_d is the corresponding space of polynomials in free curvature jets, is detected by finitely many correlations.

The comparison. There are scales $r_k \downarrow 0$ and linear frames U_k on the field spaces with the following property. The correlations of the mixed state, rescaled by r_k and transformed by U_k, converge to the free Maxwell correlations: - with all derivatives; - uniformly on compact subsets of Conf_n; - for all degrees along one sequence.

The stress tensor converges to the Maxwell stress tensor. An upper bound on $\dim F_d$ makes the frames isomorphisms for large k, so $F_d \cong M_d$.

The frames are necessary: the claim is not that a fixed normalization of each composite field has a limit along every sequence of radii. The schedule contains a real parameter that moves the scale at which observables are read, and the estimates are uniform in it. Tuning this parameter, every sequence of radii tending to zero has a subsequence along which the comparison holds.

1.0.9.2 Running coupling

Unit: VII.5 — Asymptotic freedom of the running coupling.

With $g_{N,p}(m) = \beta_{N,p,m}^{-1/2}$, Part III gives

$$g_{N,p}(m)\,\sqrt{\log(1/a_m)} \longrightarrow \frac{1}{\sqrt{2b_0}} \qquad (m \to \infty),$$

uniformly in $N \geq m$ and p. This is the leading-logarithmic running of the coupling of the construction's effective actions. No two-loop term, no Λ parameter and no comparison with another renormalization scheme is claimed. For $SU(N)$, the conventionally normalized coupling $g_{\mathrm{std}} = \sqrt{2N}\,g$ satisfies $g_{\mathrm{std}}\sqrt{\log(1/a_m)} \to (11N/24\pi^2)^{-1/2}$.

1.0.9.3 The trace law and nonfreeness

Units: VII.2 — The trace of the stress tensor and the trace anomaly; VII.3 — The theory is not free: Gram positivity and exclusion of free-field profiles.

Leading form of the trace. The classical Maxwell and Yang–Mills stress tensors are traceless. The quantum trace comes from the scale dependence of the coefficient of curvature squared. - The proof pulls the trace of the lattice stress observable back through the RG steps. - Fourth-order

matching identifies the leading part of the trace as the derivative of the effective action with respect to scale. - The one-loop coefficient enters through a momentum-shell estimate of the kind used in Part III.

Along the scaling sequences, with β_k the running coupling at the RG level that reads scale r_k, and O_{F^2} the normalized curvature-squared label,

$$\beta_k\, r_k^4\, U_k^{-1}\Theta \longrightarrow 4d_2\, O_{F^2}, \qquad d_2 = -\frac{\kappa_G}{4}, \qquad \frac{\log(1/r_k)}{\beta_k} \longrightarrow \frac{1}{\kappa_G}.$$

In perturbative terms, this is the familiar $\Theta \simeq \frac{\beta(g)}{2g^3}F^2$, with $\beta(g)/(2g^3) = -b_0/2 = d_2$ at one loop. The trace is a curvature-squared field suppressed by $g^2(r) \simeq 1/(\kappa_G \log(1/r))$.

The two-point law. Here $\Theta = \sum_\mu T_{\mu\mu} - S^{\xi_*}(\sum_\mu T_{\mu\mu})\mathbf{1}$, in the selected phase, which has a unique vacuum. 1. The free two-point function of O_{F^2} is $3\dim\mathfrak{g}/(\pi^4|x|^8)$. 2. So at lattice-fitted configurations along the scaling sequence, $r_k^8 \log^2(1/r_k)\langle\Theta\Theta\rangle \to (4d_2/\kappa_G)^2 \cdot 3\dim\mathfrak{g}/\pi^4 = 3\dim\mathfrak{g}/\pi^4$. 3. Rotation invariance moves the pair onto the time axis. 4. The spectral representation in the phase gives $G_\Theta(r) = \int e^{-rE}\, d\mu(E)$ with a positive measure. So G_Θ is nonincreasing, which sandwiches $G_\Theta(r)$ between its values at fitted points. 5. If the limit failed along some sequence of radii, the flexible scaling above would give a subsequence along which it holds, a contradiction.

Hence the limit exists over all radii:

$$\lim_{r\downarrow 0} r^8 \log^2(1/r)\, G_\Theta(r) = \frac{3\dim_{\mathbb{R}}\mathfrak{g}}{\pi^4}.$$

The amplitude does not depend on the value of the one-loop coefficient. κ_G enters once through the trace and once through the running, and $(4d_2/\kappa_G)^2 = 1$ identically. So the test detects the inverse-logarithmic suppression and the free normalization of F^2, not the value of b_0; that value is established by the running law of Part III.

Exclusion of free profiles. Consider a finite Wick profile $W(r) = \sum_\alpha c_\alpha \prod_\beta K_{\alpha\beta}(r)$. Each factor $K(r) = \int E^q\, h(\mathbf{p})(2E)^{-1}e^{-rE}\, d^3\mathbf{p}$ has $E = \sqrt{\mathsf{m}^2 + |\mathbf{p}|^2}$ and h homogeneous. - W has an asymptotic expansion at $r = 0$ in the functions $r^a(\log r)^k$, with $a \in \mathbb{Z}$ and $k \geq 0$. - Along $r = 2^{-n}$, $r^8 \log^2(1/r)\, W(r)$ is then a finite sum of terms $c\,\rho^n n^{k+2}$, which can converge only to 0. - Since the limit for G_Θ is positive, G_Θ differs from W at points arbitrarily close to 0.

This excludes exactly the finite class of profiles in the statement, and it concerns only G_Θ on the time axis.

The stronger statement adds further tests: - a Wightman-side Wick test; - Maxwell–Fock tests. Consider any isometric embedding of the selected phase's Hilbert space into a Fock space of transverse photons that preserves the vacuum and translations and has transverse symmetric image. No such embedding makes Θ local with respect to the Weyl operators.

1.0.9.4 Operator product expansion and contact extensions

Unit: VII.4 — Operator product expansion and contact extensions.

The lattice statement. The product of insertions in one cell equals a single insertion of the same total degree, plus a remainder with small chessboard root. The proof is an induction over

the RG levels between the cluster scale and the observation scale: the part of degree at most D is kept, and the rest is paid into the remainder.

The continuum coefficients. They are read off rather than given by a formula. - A prescription fixes, degree by degree, biorthogonal systems of fields and detectors. The detectors are finite combinations of mixed-state correlations of the cluster with fixed, separated exterior words. - The prescription is averaged over $O(4)$ and symmetrized for reality. - The coefficient $c_{j,\kappa}(a;x,o)$ is a coordinate of the cluster's new component of degree j in a real basis.

Consider an (r,M,η)-cluster: points within Mr of o and at mutual distance at least ηr, carrying fields of total degree w. Take spectators at distance at least c from o and from each other. Then, for all sufficiently small r,

$$|c_{j,\kappa}(a;x,o)| \le r^{\,j-w-\varepsilon}, \qquad \Big|S^s(a,x;b,y) - \sum_{j\le D}\sum_{\kappa} c_{j,\kappa}(a;x,o)\, S^s(O_{j,\kappa},o;b,y)\Big| \le r^{\,D+1-w-\varepsilon}.$$

Uniformity and limits of the statement. - The coefficients do not depend on the spectators b and y, on the state s, or on D. - The threshold for r depends on the fields, on M,η,c,ε,D, and in the first bound on the channel, but it is uniform in s, x, o and y. - Uniformity in the spectators comes from the train argument of Part V, in which the exterior word is paid once. - Uniformity in the state comes from bounding the error in the zero-energy algebra, whose norm dominates every character and the mixture. - The loss ε absorbs logarithms. There is no logarithmic refinement, so the bound is weaker than the perturbative OPE. - Uniqueness of the coefficients is not asserted. Neither convergence as $D\to\infty$ nor agreement with perturbation theory is claimed. - Nested versions over finite trees of clusters, and integrated versions against measures of bounded variation, also hold.

Contact extensions. These use Taylor subtractions over forests of collapsing clusters on dyadic shells. The OPE remainder makes the shell sums converge. For every state and every list of labels this gives a distribution of finite order on $(\mathbb{R}^4)^n$ that agrees with the correlation on Conf_n. Not established: uniqueness, reflection positivity on overlapping supports, contact Ward identities, real-time products, and convergence of the lattice correlations on the diagonals.

1.0.10 Part VIII. Assembly

Unit: VIII.1 — Assembly of the main theorem.

The construction run depends only on G and the penalty b. It is chosen before any extraction order or observable. For each extraction order (a spacing sequence and a volume sequence), the assembly composes further refinements: - one refinement for the limits of the running coupling; - one that produces the limit table, with the Euclidean hierarchy, stress tensor, phases and reconstruction of Parts IV–VI.

To these it adds the reconstructions of all phases, the OPE prescription, the nested and contact results and the Maxwell comparison. Everything is transported to a common refinement, so all objects refer to one table.

The public theorem is the case $b=1$, with the full spacing sequence and the given volume sequence; its q is the volume part of the refinement.

No new estimate enters, but identifying all objects with one table is a real constraint. Separate existence results for a lattice limit, a Wightman theory, an OPE and a trace law would not show that they concern the same correlation functions. The formal construction supplies the hypotheses under which the limit theory is first proved. The outlines have not explained every supplier in full; the assembly unit records the limits of its source reading.

1.0.11 Relation to earlier work, and where the difficulty lies

Classical sources. - Balaban's program is the model for the ultraviolet analysis: block-spin transformations, gauge fixing around a background field, fluctuation covariances built from averaging operators, the small-field/large-field decomposition, and renormalization of the marginal coupling. - The classical cluster expansion (Glimm–Jaffe, Brydges, Kotecký–Preiss) organizes the polymer activities. - Reflection positivity (Osterwalder–Seiler) and chessboard estimates (Glimm–Jaffe–Spencer, Fröhlich–Israel–Lieb–Simon) connect lattice estimates with bounds on observables. - Osterwalder–Schrader and Glimm–Jaffe reconstruction turns Euclidean correlations into Wightman fields. - The slope is the one-loop coefficient of Gross–Wilczek and Politzer.

Departures from Balaban's scheme. - The block kernel is an exactly normalized, gauge-covariant Markov kernel with random tags, not a linear averaging operator in a fixed background gauge. - Integration uses a spanning-forest gauge. - The block ratios grow along the levels. - The threshold carries the factor $e^{(\log\log\widehat{\beta})^2}$. - The construction includes an analytic family in a tuning parameter. The unit outlines do not establish a comprehensive comparison with Balaban's papers.

Beyond the ultraviolet analysis, the proof does the following: - One schedule and one family serve composite fields of every degree. Each field has a fixed normalization, root bounds uniform in the volume, and derivatives controlled by rank detection, with no coupling-dependent loss per site. - Convergence at fixed volume is proved despite tuning that depends on the cutoff. - Equicontinuity in infinite volume comes from a rank-free trace inequality, without derivative bounds. - Commutativity of the zero-energy algebra is proved, not assumed. - Vacuum uniqueness in every phase follows from a uniform low-energy bound that replaces a mass gap or a cluster argument. - Lorentz covariance comes from Ward identities. - Rotation invariance is restored without decay of correlations, and the group action is obtained by comparison with free Maxwell theory. - The trace law holds as a limit over all radii. - The OPE is derived from the RG, with one system of coefficients for the mixture and every phase.

Where the difficulty lies. 1. **Gauge degeneracy and large fields** must be controlled with a bounded cost per site and per seed. This is done by the gauge count of the determinant budget, the energy lower bound for seeds, and the exactness of the kernel. 2. **Effective-action estimates must become estimates for observables**: for every composite field, its derivatives and its fixed-volume limit, inside one family. 3. **The passage from a lattice mixture to relativistic phases** has no continuous symmetry and no clustering to start from. It is handled by Ward identities with free comparison, the commutative zero-energy algebra, and spectral bounds based on locality.

The order of choice of constants and the transport of estimates through the construction account for a substantial part of the formal development. Its length alone does not measure mathematical novelty.

1.0.12 Steps whose informal account is incomplete

The theorem is proved in Lean, and the conditional statements of the limit theory are discharged within its proof. The unit outlines, however, describe the following steps only at the level of statements, or with readings marked tentative. A written proof will have to supply them.

- **One-loop identification (III.7, VII.2).** The slope bound is first proved for an auxiliary functional and then transferred to the completed Hessian of the actual step. The revised III.7 explains the determinant factorization and the transfer of the coefficient; several subsidiary real-quotient, complex-strip, tadpole, bubble and alias estimates remain inputs at statement level. The trace amplitude does not test this value.
- **Covariance and Hessian (III.4).** The Gaussian covariance is the inverse of the gauge-fixed Gram form on the active links. Whether it equals the inverse Hessian of the effective action at the background is left open.
- **Nonlinear step.**
  - The correspondence between the outputs of the one-step bounds (III.10) and the norms restored by the reset (III.9) is not written out.
  - Several hypotheses of the step were read only as statements.
  - The cluster expansion is proved for real weights (III.5), while the fixed-volume argument uses complex tuning.
- **Fixed-volume convergence (III.12, IV.1).** III.12 now explains fixed-level convergence, the uniform tail bound and the Cauchy argument; the underlying trajectory-convergence and multiscale estimates remain imported inputs. IV.1 locates the upgrade to the full spacing sequence, while leaving the upstream construction of its fixed-volume kernels outside that unit's review.
- **Shell identities and the flux at infinity (VI.1, VI.2).** The lattice derivation of the shell identities is given only at statement level. The exterior-flux estimate was not read. The descent of the rotation action to exact intertwiners is marked tentative.
- **Reconstruction (V.2).** The boundary values, the spectrum condition and the locality argument are described from statements; their mechanisms were confirmed in the Lean but not written out in the unit outlines.
- **Contact extensions and nested OPE (VII.4).** The shell estimates and the nested rates are described from docstrings.
- **Field relations (II.1).** It is not shown directly that the lattice equation relations exhaust the null space. The identification $F_d \cong M_d$ comes from the comparison with free theory.

1.0.13 Reading guide

The exact statement separates the public theorem from the stronger construction. The architecture describes what each part hands to the next. For the stated inputs and remaining gaps in this account, follow the relevant unit's "What it uses" and "What remains unexplained" sections. The assembly unit describes how the construction is put together.

Question	Units to read
What are the fields, and how are their normalizations fixed?	II.1, III.12, IV.1, VII.1
How does the ultraviolet analysis relate to Balaban's?	III.1, III.3, III.4, III.6, III.5, III.10

Question	Units to read
Where does the one-loop coefficient enter?	III.7, III.9, VII.5, VII.2
Why does one family serve every observable and every limit?	III.2, II.1, III.12, VIII.1; III.13 treats an alternative family
How are bounds made uniform in the volume?	III.11, III.12, III.16
How is infinite volume reached without an infrared RG?	III.16, IV.1
Why are there phases, and why does every phase reconstruct?	V.1, V.2, V.3; VII.4 for the train bounds
How are rotations and the stress tensor recovered?	VI.2, VI.1, with VII.1 and VII.6 for the free comparison
What makes the theory nonfree?	VII.2, VII.3, with VII.6 for arbitrary short-distance scales
What does the OPE say, and what is known at coincident points?	VII.4, then VIII.1 for the assembly

2 The architecture

This page describes how OpenAI's Lean proof of the construction of four-dimensional Yang–Mills theory is organised: its eight parts, what each part hands to the next, and the units that make up each part (detailed map).

It is read from the Lean statements and documentation and from the dependency graph. The mathematical framing follows the Yang–Mills group's assessment. Lean names link to the fixed source commit `3456e5e`.

2.0.1 The order of limits

The construction uses Wilson lattice theories on tori of side L_p, with lattice spacings a_N and couplings $\beta_{N,p}$ satisfying $\beta_{N,p}/\log(1/a_N) \to 11C_G/(24\pi^2)$. The public theorem leaves the positive limiting constant unspecified; see the exact statement.

Correlations of renormalized local observables are first taken to the limit $N \to \infty$ **at fixed volume**, along the whole sequence of spacings. They are then taken to the limit $p \to \infty$, along a subsequence that may be extracted from any prescribed sequence of volumes. The public form of this statement is `WilsonFamily.Converges`.

The renormalization group is used only for the first limit. The second is obtained by compactness.

2.0.2 The eight parts

part	name	units
I	Setting	I.1
II	Data of the construction	II.1, II.2
III	Ultraviolet construction	III.1–III.16
IV	Infinite volume	IV.1
V	Phases and reconstruction	V.1, V.2, V.3

part	name	units
VI	Symmetry	VI.1, VI.2
VII	Identification as Yang–Mills	VII.1, VII.2, VII.3, VII.4, VII.5, VII.6
VIII	Assembly	VIII.1

2.0.2.1 Ultraviolet units

- Blocking, constants and Gaussian estimates: III.1, III.2, III.3, III.4.
- Cluster expansion, stationary field, matching and dependence: III.5, III.6, III.7, III.8.
- Coupling reset, nonlinear step and observable limits: III.9, III.10, III.11, III.12.
- Alternative coordinated family, construction data and analytic tools: III.13, III.14, III.15, III.16.

The limit theory (Parts IV–VI) is written in the code around numbered work items; the units of those parts follow them.

2.0.3 What each part hands to the next

2.0.3.1 Ultraviolet → infinite volume Let G be a compact matrix group with simple nonabelian Lie algebra, and $b > 0$ a parameter of the block-spin kernel. The ultraviolet construction provides a Wilson family with the following properties.

1. **Specification.** The family satisfies its normalization, covariance and budget conditions, and its couplings obey the asymptotic-freedom law above.
2. **Chessboard bound, uniform in the volume.** For each degree d and $\varepsilon > 0$, every renormalized local observable at block scale m satisfies a chessboard bound of order $a_m^{-d-\varepsilon}$, uniformly in the volume, once the lattice is fine enough. A chessboard bound is a bound on the root of a product of reflected copies.
3. **Differences.** The same holds for finite-difference quotients, so correlation kernels at r-separated points are bounded uniformly in the volume.
4. **Fixed volume.** At each fixed volume the correlation kernels converge along the whole sequence of spacings, to smooth limits.
5. **Corrected fields.** The same estimates hold for the corrected fields.

For the subsequential extraction, the infinite-volume part uses 1–3 and 5, together with the reflection-positivity inequality on cells, which holds for every $\beta \geq 0$. Its conclusion: for every increasing sequence of volumes there is a subsequence along which all smeared correlations converge to kernels that are smooth away from coincident points. The fixed-volume convergence in item 4 then removes the spacing refinement; see IV.1. These two limits correspond to Theorems 4.1 and 5.1 of the group's assessment.

A caution about the code. About three quarters of the code filed under "full volume" (folder `YM/FullVolume`) belongs to the ultraviolet construction: it proves the volume-uniform bounds. The genuinely infinite-volume argument is short, about 7,000 lines.

2.0.3.2 Infinite volume → phases, reconstruction and symmetry The limit satisfies a Euclidean *hierarchy*:

- translation homogeneity;
- reflection positivity, including for smeared joint test functions;
- smoothness on separated configurations, with controlled derivatives;
- a compatible calculus of derivatives.

Reconstruction also needs a uniform bound on products of fields in thin time slabs.

2.0.3.3 Reconstruction → identification For every phase, a completed Wightman theory: unique vacuum, a light-cone bound, heat-kernel decay, and a trivial weak commutant. The heat bound is polynomial; no mass gap follows.

2.0.3.4 Symmetry → identification A conserved stress tensor and its Ward identities. Both are compared, along sequences of scales tending to zero, with the corresponding objects for free Maxwell fields.

2.0.3.5 Into the assembly A single Wilson family serves every part of the public theorem, from extraction to the trace test; VIII.1 describes the assembly. The construction run exists for every allowed G and every $b > 0$. III.13 describes an alternative coordinated family; its properties require a separate identification before they can be used for this family.

2.0.4 The work items of the limit theory

Parts IV–VI are written in the code as about seventy numbered work items. Each states one obligation, a "proof request" in the code's vocabulary, and proves it in its own file.

family	items	part	content
D	6	IV	the limit passes to derivatives of the correlation kernels; the Euclidean hierarchy
B	18	V	time-slab operators, decomposition into phases, reflection positivity and smoothness for each phase
C	29	V	Osterwalder–Schrader → Wightman: continuation in time, boost covariance, gluing of tubes, locality

family	items	part	content
A	17	VI	so(4) brackets, integration to the rotation action, the averaged conserved stress tensor, boosts, comparison with free Maxwell fields

These families interleave. D supplies the limit hierarchy; B begins with its OS spaces and slab operators. A uses those spaces and the free Maxwell comparison to obtain rotations and Ward identities. Those symmetry results enter both the phase decomposition in B and the boost and covariance arguments in C. The part numbers organize the subjects; they do not give a linear dependency order. See the strategy and the work-item introductions.

The same mechanism appears in the ultraviolet construction: families E–I cover the energy chain of one renormalization step, its stationary point and Hessian, label polynomials, and support bookkeeping.

See work-items.md for the complete list.

2.0.5 What does not follow the order of the parts

The Lean is not written in this order everywhere. Some definitions are needed before the part that explains their properties:

- the reflection-positive cells and chessboard roots;
- the record of the whole construction run;
- the basic types of the renormalization group and of the labels.

Some sections are filed far from their mathematical home. Two examples:

- the rotation-invariance estimate, in folder `Next/ROTATIONS`;
- some renormalization-group pieces filed with the trace-anomaly code.

None of this involves new mathematics. A written account should simply state these definitions early.

2.0.6 How reliable is this

- Reachable declaration sets for seventeen named results were compared with OpenAI's compiled record. This comparison does not verify every edge or the mathematical interpretation of the interfaces (method).
- The plain-language interfaces summarize statements and documentation. The unit index gives each outline's review status; its local limitations describe which arguments remain at statement level. The exact statement distinguishes the public theorem from additional construction conclusions.
- Report a correction with the page, heading and source reference.

3 The units

3.1 I.1. Lattice, symmetries, gauge groups and the Wilson measure

Part: I, Setting. **Lean:** `YM/Lattice` (1,215 lines), `YM/Gauge` (1,131), `YM/Space` (430), `YM/Corona` (405), `YM/Mink` (363), `YM/Wilson` (157). **Size:** 3,701 lines in 595 declarations used by the main theorem.

3.1.1 1. What this unit proves

Throughout, $G \subset U(N)$ is a closed subgroup, $N \geq 1$, $\ell \geq 1$ is integral, and $\Lambda_\ell = (\mathbb{Z}/\ell\mathbb{Z})^4$. Configurations assign $U(x,i) \in G$ to links $x \to x + e_i$; transport multiplies these variables, using inverses on backward steps.

Proposition I.1.1 (torus geometry). Every site has a unique lift $r_\ell(x) \in (-\ell/2, \ell/2]^4 \cap \mathbb{Z}^4$, and $|\Lambda_\ell| = \ell^4$. The metric $|x-y|_\ell = \max_i |r_\ell(x-y)_i|$ is translation and signed-permutation invariant; $|[y]_\ell|_\ell \leq |y|_\infty$ for every $y \in \mathbb{Z}^4$. *Status:* theorem. *Lean:* `lift_eq_iff`, `separation_triangle`.

Proposition I.1.2 (transport and symmetries). On Λ_ℓ or $\mathbb{Z}^4$, let $A = (\rho, b)$, $Ax = \rho x + b$, with $\rho = (\pi, \epsilon) \in B_4$ a signed permutation, $(\rho x)_{\pi(i)} = \epsilon_i x_i$, $b \in \Lambda$. Pullback assigns image-link transport and satisfies $1^* = \mathrm{id}$, $B^* A^* = (A \circ B)^*$ and $(A^* U)(x; w) = U(Ax; \rho w)$. For every $g : \Lambda \to G$, $U^g(x,i) = g(x) U(x,i) g(x + e_i)^{-1}$ gives

$$U^g(x; w) = g(x) U(x; w) g(x_{\mathrm{end}})^{-1}.$$

Transport depends only on links with endpoints visited by w. For $p = (x; i < j)$ put $U_p = U(x; e_i, e_j, -e_i, -e_j)$. Affine maps injectively send plaquettes to image squares $A_\sharp p$, bijectively on the torus; every conjugation- and inversion-invariant $\phi : G \to M$ satisfies $\phi((A^* U)_p) = \phi(U_{A_\sharp p})$. Gauge invariance means $F(U^g) = F(U)$ for all g; locality on X means dependence only on links with both endpoints in X. Periodic extension $E_b U(y,i) = U([y]_\ell + b, i)$ respects transport, gauge transformations and locality, and $E_b((\rho,a)^* U) = (\rho, 0)^* E_{\rho b + a} U$. *Status:* theorem. *Lean:* `travel_pull`, `travel_gauge`, `Face.send_price`, `expandAt_pull`.

Proposition I.1.3 (spacings and blocks). For $m, p \in \mathbb{N}$, odd integers $L_m \geq 3$ and strictly increasing positive integers P_p divisible by 4, set $D_0 = 1$, $D_{m+1} = D_m L_m$, $a_m = D_m^{-1}$, $\ell(m,p) = P_p D_m$, $s_m = \log(1/a_m)$. Then $a_m \downarrow 0$ strictly, $s_m \to \infty$, and $a_m \ell(m,p) = P_p$; placements $a_m r_{\ell(m,p)}$ lie in $(-P_p/2, P_p/2]^4$ and are preserved by subdivisions. If $\log L_m / s_m \to 0$ for $m \geq 1$, some $c > 0$ satisfies $a_{m+1} \geq c a_m^2$ for every m.

For every integer $L \geq 1$, $\ell' = \ell L$, the equivariant injective homomorphism $\iota x = [L r_\ell(x)]_{\ell'}$ satisfies $r_{\ell'} \iota = L r_\ell$. The bijection $(x,y) \mapsto \iota x + [r_L(y)]_{\ell'}$ defines parent and residue maps. For odd L, parent is B_4-equivariant with cubic fibres of radius $(L-1)/2$ centred at ιx. For odd L, $\mathrm{par}(z + e_i) = \mathrm{par}(z) + e_i$ if $r_L(\mathrm{res}\, z)_i = (L-1)/2$, and equals $\mathrm{par}(z)$ otherwise. For all fine sites z, z',

$$|z - \iota \mathrm{par}\, z|_{\ell'} \leq L/2, \qquad L|\mathrm{par}\, z - \mathrm{par}\, z'|_\ell \leq |z - z'|_{\ell'} + L.$$

Status: schedule definition, chosen in III.2; consequences theorems. *Lean:* `Schedule`, `mesh_inequality`, `Ratio.assemble_bijective`, `Corona.foreshorten`.

Proposition I.1.4 (cover numbers). Let $\mathrm{halo}_M(D) = \{y : \exists x \in D,\ |x-y|_\ell \leq M\}$, and let $\mathrm{cov}_M(X)$ minimize $|D|$ subject to $X \subset \mathrm{halo}_M(D)$ and halo connectivity by steps of sup-distance

at most 1; empty sets are allowed. For all sets and $M \in \mathbb{N}$, cover numbers are monotone and subadditive for intersecting sets,

$$|X| \le (2M+1)^4 \mathrm{cov}_M(X), \qquad |x-y|_\ell \le (4M+2)\mathrm{cov}_M(X) + 2M \quad (x, y \in X).$$

A halo is connected exactly when its centres are chained by steps at most $2M+1$, and $|\mathrm{halo}_M(D)| \le (2M+1)^4|D|$. For $b, k, m \in \mathbb{N}$, $b \ge 1$, $\mathrm{cov}_b(\mathrm{halo}_{kb}\{x\}) \le (2k+1)^4$; a scale-$mb$ cover D gives $\mathrm{cov}_b(X) \le (2m+3)^4|D|$. For subdivision ratio $L \ge 1$, $X \subset \Lambda_{\ell L}$, $M, B, s \in \mathbb{N}$, $B \ge 1$, $M \le 2B$,

$$L\mathrm{cov}_B(\mathrm{halo}_{sB}(\mathrm{par}\, X)) \le (L + 10\mathrm{cov}_M(X))(2s+13)^4.$$

Status: theorem. *Lean:* `coverNumber`, `Corona.collapse`.

Proposition I.1.5 (cost and Haar measure). G is compact and second countable. Put $\sigma(u) = 1 - N^{-1}\mathrm{Re}\,\mathrm{tr}\, u$, $\chi(u) = \|u-1\|_F/\sqrt{N}$. Both are continuous, conjugation- and inversion-invariant, vanish at 1, and satisfy $\chi^2 = 2\sigma$, $0 \le \sigma, \chi \le 2$, $\chi(uv) \le \chi(u) + \chi(v)$. Haar probability μ_G is invariant under inversion and both translations. *Status:* theorem. *Lean:* `chord_price`, `probHaar_rightInv`.

Proposition I.1.6 (Lie algebra and Casimir). $\mathfrak{g} = \{X \in M_N(\mathbb{C}) : e^{tX} \in G \ \forall t \in \mathbb{R}\}$ is a closed real Lie subalgebra of skew-Hermitian matrices, invariant under conjugation. Every curve $f : \mathbb{R} \to G$ differentiable at 0 with $f(0) = 1$ has $f'(0) \in \mathfrak{g}$. The invariant inner product $-N^{-1}\mathrm{tr}(XY)$ identifies $\mathfrak{g}$ with colours $\mathbb{R}^d$, $d = \dim \mathfrak{g}$, $c \mapsto X_c$, with $\|X_c\|_F = \sqrt{N}|c|$. Projection satisfies $X_{\Pi A} = P_{\mathfrak{g}} A$, $|\Pi A| \le \|A\|_F/\sqrt{N}$; conjugation induces orthogonal Ad, and $\mathrm{ad}(x)y$ represents $[X_x, X_y]$.

If $\mathfrak{g}$ is nonabelian with no nonzero proper ideals, every symmetric real bilinear form satisfying $B(\mathrm{ad}(x)y, z) = -B(y, \mathrm{ad}(x)z)$ for all x, y, z is a scalar multiple of the inner product. Writing $C_{\mathfrak{g}} = -d^{-1}\sum_i \mathrm{tr}(\mathrm{ad}(b_i)^2)$ in the orthonormal colour basis, for all colours x, y,

$$-\mathrm{tr}(\mathrm{ad}(x)\mathrm{ad}(y)) = C_{\mathfrak{g}}\langle x, y\rangle, \qquad C_{\mathfrak{g}} > 0, \qquad \kappa = 11C_{\mathfrak{g}}/(24\pi^2) > 0.$$

Status: theorems; constants definitions. *Lean:* `lie`, `invariant_scalar`, `trace_adjoint`, `casimir_pos`.

Proposition I.1.7 (small balls). For every $0 < r \le 1$, $\mu_G\{\chi \le r\} \ge (r/4)^{2N^2}$. *Status:* theorem. *Lean:* `shell_ge`.

Proposition I.1.8 (local group geometry). There exist open neighbourhoods $V \ni 1$, $W \ni 0$ and $\log_G : M_N(\mathbb{C}) \to \mathbb{R}^d$, real-analytic on V, with $\log_G(1) = 0$, $\log_G(e^{X_c}) = c$ for $c \in W$ and $e^{X_{\log_G u}} = u$ for $u \in G \cap V$. Every matrix X has a Frobenius-nearest point $u \in G$, satisfying $u^*X - 1 \in \mathfrak{g}^\perp$. Nearest points respect left and right translations by G and are unique in some tube about G; near 1 this retraction is analytic with strict derivative $P_{\mathfrak{g}}$. For every $\varepsilon > 0$, some $\delta > 0$ gives

$$\|u - v - P_{\mathfrak{g}}(X - Y)\|_F \le \varepsilon \|X - Y\|_F$$

whenever $\|X-1\|_F, \|Y-1\|_F < \delta$ and u, v are nearest to X, Y. Curvature $F(u) = \Pi((u - u^{-1})/2)$ is continuous, odd under inversion, satisfies $F(vuv^{-1}) = \mathrm{Ad}(v)F(u)$, $F(1) = 0$, $|F(u)| \le \chi(u)$, and has derivative c along e^{tX_c} at 0. *Status:* theorem. *Lean:* `plog_right`, `nearest_approximate`, `leading_curvature`.

Proposition I.1.9 (Wilson law). For product Haar μ_ℓ and plaquette holonomies U_p, set

$$E_\ell = \sum_{x,i<j} \sigma(U_{(x;i<j)}), \qquad Z_\ell(\beta) = \int e^{-\beta E_\ell} d\mu_\ell, \qquad d\mathbb{P}_{\ell,\beta} = Z_\ell(\beta)^{-1} e^{-\beta E_\ell} d\mu_\ell.$$

For every real β, E_ℓ is continuous, $0 \le E_\ell \le 12\ell^4$, $Z_\ell(\beta) > 0$, and $\mathbb{P}_{\ell,\beta}$ is a probability measure integrating continuous or bounded measurable complex functions. *Status:* definition with theorems. *Lean:* `law`, `partition_pos`.

Proposition I.1.10 (invariance). For every real β, affine lattice map A and $g : \Lambda_\ell \to G$, pullback and gauge transformation preserve μ_ℓ, E_ℓ and $\mathbb{P}_{\ell,\beta}$. *Status:* theorem. *Lean:* `law_pull`, `law_gauge`.

Proposition I.1.11 (continuum symmetries). $B_4 \to O(4)$ respects coordinate actions; skew-matrix exponentials lie in $O(4)$. With $x = (\vec{x}, x_t)$, set $\mathsf{m} = x_t^2 - |\vec{x}|^2$, $\overline{V}_+ = \{|\vec{x}| \le x_t\}$, $V_+ = \{|\vec{x}| < x_t\}$. The topological group $\mathcal{L} = \{g \in GL(4,\mathbb{R}) : \det g = 1,\ \mathsf{m} \circ g = \mathsf{m},\ g\overline{V}_+ = \overline{V}_+\}$ preserves Lebesgue measure and V_+ and contains time-fixing elements of $SO(4)$. For $R(\vec{x}, x_t) = (-\vec{x}, x_t)$, $g^\vee = RgR = (g^{-1})^T$ satisfies $\langle gx, g^\vee p\rangle = \langle x, p\rangle$. For every unit spatial vector e, the boost $B_e(r)$ is a continuous one-parameter subgroup of $\mathcal{L}$, with generator $(\vec{x}, x_t) \mapsto (x_t e, \vec{x} \cdot e)$. The closed cone is self-dual and equals $\{x : (g^\vee x)_t \ge 0\ \forall g \in \mathcal{L}\}$. For $x \in V_+$ and $p \in \overline{V}_+ \setminus \{0\}$, $\langle x, p\rangle > 0$. The Poincaré group $\mathcal{L} \ltimes \mathbb{R}^4$, acting by $x \mapsto \Lambda x + b$, preserves Lebesgue measure diagonally on every $(\mathbb{R}^4)^n$. Distinct-point configurations form an open set; pairwise distinct coordinates on every axis, or positive pairwise sup-separation, imply distinct points. *Status:* definitions and theorems. *Lean:* `Mink.Proper`, `Lorentz.from_half`, `Poincare`.

Proposition I.1.12 (cover boxes and routes). For every $r \in \mathbb{N}$, $B_r = \{-r, \dots, r\}^4$ is B_4-invariant with $(2r+1)^4$ points. For cover sites x, y, axis routes from x to y end at y, give the reverse route from y to x when axis order is reversed, and transform covariantly under affine maps. For $a \in \mathbb{Z}^4$ and an axis set J, slabs $\{y \in B_r : y_i = a_i\ (i \notin J)\}$ decompose into pinned layers when an axis $i \notin J$ is freed; their sums decompose accordingly for every function into a commutative additive monoid. Pinned layers $z, w \in \{-r, \dots, r\}$ correspond by translation $(w - z)e_i$. For every cover configuration and integer d, $U(x; \mathrm{straight}_j(d+1)) = U(x; \mathrm{straight}_j(d))U(x + de_j, j)$, where $\mathrm{straight}_j(d)$ has displacement de_j. *Status:* theorem. *Lean:* `route_end`, `slab_sum`.

3.1.2 2. What it uses

- No inputs from other units.
- Standard results: Haar uniqueness, product measures, matrix exponential, analytic inverse function theorem, spectral theorem, finite-dimensional volume scaling, Minkowski geometry.
- Local radii depend on G; the mesh constant depends on the schedule.

3.1.3 3. Strategy

The spacing schedule separates physical periods from mesh sizes; centred residues define the metric. Connected covers turn set size into path length, allowing coarsening by sparse sampling. Image-link transport reduces Wilson invariance to Haar invariance and conjugation/inversion invariance of the cost.

The exponential definition of the Lie algebra supplies tangent directions. Normal complements and the inverse function theorem give local coordinates and nearest points; simplicity forces invariant symmetric forms to be scalar.

3.1.4 4. Steps

1. Centred division gives the metric and subdivision; odd residue windows give parent equivariance. `Ratio.parent_signed`.
2. Traverse a fine cover by a path, sample parents every LB steps, then tile the resulting coarse halo; this proves the coarsening bound. `Corona.collapse`.
3. Walk induction gives transport covariance; gauge factors telescope. Reversal and change of starting corner give plaquette inversion and conjugation. `travel_gauge`, `Face.send_price`.
4. For the curve of I.1.6, approximate e^X by $f(1/n)^n$ to prove the tangent criterion. Isolation of normal exponentials gives the local logarithm. `tangent_generates`, `isolates`.
5. Minimize distance and use normality with local coordinates to identify nearest points. The symmetric operator associated with an invariant form has ideal eigenspaces; simplicity gives the Casimir identity. `nearest_is_focus`, `invariant_scalar`.
6. Pack ambient matrix balls for I.1.7; normalize the positive Wilson density and use link permutations, inversions and two-sided translations for invariance. `shell_ge`, `law_preserved`.
7. Boosts and Cauchy–Schwarz establish the cone characterizations. `cone_test`, `Lorentz.from_half`.

3.1.5 5. Where the difficulty lies

Local group geometry must handle disconnected groups. Coarsening is uniform in torus size. Real simplicity suffices without irreducibility of the defining representation. Reflections shift link base points; even-side lift ties prevent unrestricted placement covariance.

3.1.6 6. Notation

$[y]_\ell$: reduction; r_ℓ: centred lift; par: block parent; $\|\cdot\|_F$: Frobenius norm; $P_{\mathfrak{g}}$: real Frobenius orthogonal projection; ad: Lie bracket in colour coordinates. Time is coordinate 3.

3.1.7 7. What remains unexplained

Helpers outside the six principal folders remain uncovered. The condition $M \le 2B$ for the consecutive cover scales in III.5 and III.10 remains to be traced. Exact bibliographic theorem locations remain unchecked.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.2 II.1. Field labels and their lattice representatives

Part: II, Data of the construction. **Lean:** work items G01–G07: `YM/Labels/`, `YM/Source/`, `YM/Work/` (G01–G07), `YM/Next/FOUNDATIONS/`, `YM/Push/RELAY_MARK/`, `YM/Extra/SOURCE_HISTORIES/`, `YM/Push/LOCAL_SPLITTING/`, `YM/Push/RELAY_DIFFERENCE/`, `YM/Planned/ATOM_GUARDED_LAYOUT/`, `YM/Further/LOCAL_LOG_JETS/`, `YM/LocalH/`. **Size:** 10,805 lines in the main-theorem closure.

3.2.1 1. What this unit proves

Throughout, $G \subset U(n)$ is compact, $n \ge 1$.

Proposition II.1.1 (labels and frames). Let Tay_j consist of 4-tuples of symmetric $(j+1)$-linear maps $(\mathbb{R}^4)^{j+1} \to \mathfrak{g}$ satisfying $\sum_\mu z_\mu A_\mu(z,\dots,z) = 0$. A jet is $J = (J_j)_{j\ge 0}$; coordinates of J_j have weight $j+2$. For every $d \ge 0$, the space V_d of $\mathrm{Ad}(G)$-invariant complex jet polynomials of weight $\le d$ is finite-dimensional and determined by evaluation on real jets. It splits by weight and factor number (w, f); conjugation and $O(4)$ preserve grades. Its zero-factor part is $\mathbb{C}1$. There exist nested real pure-grade bases (b_i) of grades (w_i, f_i) with $b_0 = 1$. *Status:* definitions and theorems. *Lean:* `frames`.

Proposition II.1.2 (holonomies and sections). For every d, J, h, x, μ, the equation $u' = u\, hA_\mu^{(d)}(h(x+te_\mu))$, $u(0) = 1$, has a unique solution in G, where $A_\mu^{(d)}(z) = \sum_{j+2\le d} J_{j,\mu}(z,\dots,z)$. Its time-one endpoint $U_{J,h}^{(d)}$ satisfies $U_{J,h} = U_{\mathrm{sc}_h J,1}$ and $U_{J,0} = 1$. A section (S, Φ) maps coordinates linearly to link-entry polynomials on finite S. Soundness requires signed-permutation-invariant $S \ni 0$, gauge invariance, $\Phi_{e_0} = 1$, $\Phi_{\bar c} = \overline{\Phi_c}$, $\Phi_c(s^*U) = \Phi_{s\cdot c}(U)$, and equality of the C^d jets of $\Phi_c(U_{J,h})$ and $p_c(\mathrm{sc}_h J)$ at 0, for all c, J. *Status:* definitions; ODE theorems. *Lean:* `Hol.exists_ode`; `SectionData.Sound`.

Proposition II.1.3 (primary sections). For every holonomy datum and coherent frames there are sound sections Σ_d whose column i is a polynomial in finitely many transported plaquette curvatures, every monomial having at least f_i factors. For every $k, P \in \mathbb{N}$ there is $A \ge 0$ such that, for every budget b of order k, power P, side and placement, column i is b-regular with norm $\le A t_b^{f_i}$. For the fixed primary sections, nondecreasing $\pi(D) \ge 1$ bounds these costs for $d, k, P \le D$, and $c_d \ge 1$ bounds all column values globally. *Status:* theorem. *Lean:* `Integrated.source_sections`.

Proposition II.1.4 (readout). Fix holonomies, d and a frame. Every b-regular f on $X \subset \Lambda_\ell$, with $k_b \ge d$, has a unique read label $p \in V_d$ at every o: test jets agree through order d if $10\mathrm{dist}(x, o) < \ell$ for all $x \in X$; otherwise $p = f(1)1$. If also $P_b \ge d$ and $\|f\|_b \le c$, then $\|S_{t_b}\mathrm{coord}(p)\|_1 \le Cc$, with $C \ge 1$ depending only on holonomies and frame. A sound placed section reads p_c if $20(|x_\mu|+1) \le \ell$ throughout its stencil. *Status:* theorem. *Lean:* `exists_read`; `price_of_readout`.

Proposition II.1.5 (equation sections). Let $\mathcal{E}_d$ be the labels vanishing on all jets of radial potentials whose curvatures satisfy the free Bianchi and divergence equations, degree by degree. For every holonomy datum, d and frame, there exist a sound section with the slot structure and cost of II.1.3 and an automorphism R commuting with conjugation and signed permutations, such that $\Phi_{\mathrm{coord}(Rp)}$ is micro for every $p \in \mathcal{E}_d$. Micro means continuous and, for some r, M, orthogonal under every Wilson expectation with $\ell \ge M$, every placement, and $\beta \ne 0$, to every bounded measurable spectator using only links with both endpoints farther than r from the placement. The choices exist simultaneously along coherent frames. *Status:* unconditional theorem. *Lean:* `Portfolio.build`.

Proposition II.1.6 (observables). A record carries a history, holonomies, frames, sections, reference levels and transfer families. Define

$$O_{N,p}(v, o) = a_N^{-d}\Phi^{(d)}_{S_{1/\theta_N}\tau^{-1}v}(o).$$

For all records this is linear, continuous, local and literally translation covariant; soundness gives gauge invariance. Under Makes and $r(d) \le N$, it respects reality and affine hypercubic pull-backs, satisfies $O_{N,p}(a^d_{r(d)}e_0, o) = 1$, and has reference-level readout v. Affine covariance is $O(v, A_{s,b}o; U) = O(s^{-1}\cdot v, o; A^*_{s,b}U)$. Here Makes requires selection, soundness, $d + 3 \le J(r(d))$, $k_{b_m} \ge d+1$, $P_{b_m} \ge d$ for $m \ge r(d)$, $20(|x_\mu|+1) \le \ell_{N,p}$ on S_d when $r(d) \le N$, column bounds

$\kappa_N \theta_N^{f_i}$ and based transfers. *Status:* theorems; Makes holds for VIII.1's family. *Lean:* `Family.Makes`; `Family.O`.

Proposition II.1.7 (transfers). With terminal linear automorphism τ, a based family for (Σ, r) requires $r \le N$ and, for every v and $r \le m \le N$, linear coordinates and remainders on fixed supports, regular remainders on $X_i \ni 0$ with $g_i(1) = 0$ and vanishing test derivatives through d when nonwrapping, and measurable gauge-invariant forced functions local in their supports and flags, vanishing unless eligible and carrying 0, and remainder bound $(\sqrt{w_m}, \kappa_m \|x_m^\circ\|_1)$. A bound (κ, c) means summable regular and forced weighted masses $\le a$ and $\le \kappa b$, respectively, for $a, b \ge 0$, $a + b \le c$, dominating regular norms and forced absolute values pointwise. It requires $x_N = v$, zero terminal regular remainder, $\mathcal{D}_N = \rho_N \Phi_{S_{1/\theta_N} v}$, and $\mathcal{D}_m = L_m^d T^{(m)}_{\beta_{N,p,m+1}} \mathcal{D}_{m+1}$. Also $\|x_{m+1}\|_1 \le \kappa_{m+1} \|x_m\|_1$, $\|x_m\|_1 \le \kappa_{m+1} L_m^d \|x_{m+1}\|_1$, $x_r = \tau v$, τ respects reality/symmetry, and $x_m(a_N^d e_0) = a_m^d e_0$. For selected histories, $\int \mathcal{D}_m(E_m(\tau^{-1} v)) = a_m^d Z_{N,p} \langle O_{N,p}(v, 0) \rangle$. *Status:* specification, supplied by III.10; integral identity is a theorem. *Lean:* `TransferData.Based`.

Proposition II.1.8 (birth). Let ρ be lawful on Λ_ℓ with $w > 0$, threshold θ, $o \in \Lambda_\ell$, $Y = Y_o$, and Σ sound with $|\Phi_{e_i}(o; \cdot)| \le B$, $B \ge 1$. Suppose distinct reserved zero-activity forced addresses exist for every finite $\emptyset \ne I \subseteq X$ and auxiliary index, with support X, inventory I and everywhere-satisfied auxiliary test. If

$$B \theta^{-f_{\max}} \mathrm{mass}(Y, 1)(e^{|Y| w} - 1) \le \sqrt{w},$$

then, for every reservation index, there is linear $E(v)$ with fixed supports, $x = v$, $g = 0$, Format and bound $(\sqrt{w}, \|v^\circ\|_1)$, satisfying $\mathcal{D}(E(v)) = \rho \Phi_{S_{1/\theta} v}$ pointwise. The forced part is precisely the large-field supplement. The guarded variant (`Guard.summon`, used at level N with w=w_N) additionally requires a selected history. It requires the auxiliary tests only when the support has exactly the prescribed flagged inventory, together with a separate zero-activity assumption. The section bound and fee condition are unchanged. *Status:* theorem. *Lean:* `launched`.

Proposition II.1.9 (numerical conditions). Let $C_j \ge 1$, let (K_j) be nondecreasing with $\mathsf{K}_j \ge 1$, and a stage profile satisfy $j \le k_j, P_j$, $\pi(\max(k_j, P_j)) \le \bar{\kappa}_j$, $(C_j + 1)\Omega_d \le \mathsf{K}_j$ for $d \le j$, and $64 \mathsf{K}_{j+1}^2 \le \bar{\kappa}_j$. There exist n_j, β_j^* such that every admissible level system of this profile with stage block factors $\ge n_j$ and $\hat{\beta}_m \ge \beta^*_{J(m+1)}$ satisfies $J(m) \le k_{b_m}, P_{b_m}$, $\pi(\max(k_{b_m}, P_{b_m})) \le \kappa_m$, latch for $d \le J(m)$, and

$$64 \mathsf{K}_{J(m)+1}^2 \le \kappa_m, \qquad 10^5 \mathsf{K}_{j+1}^6 (1 + \kappa_{m+1})^2 \le L_m^{1/8},$$

$$\eta(1 + \kappa_{m+1})\Omega_d \le L_m^{-3/8}, \qquad (C_j + \eta \kappa_{m+1})\Omega_d \le \mathsf{K}_j,$$

where $j = J(m + 1)$, $\eta = L_m^{-2/5}$, $d \le J(m)$. Latch requires sufficient side length and II.1.8's fee bound at every placement, with $B = c_d$, $\theta = \theta_m$, $w = w_m$. Such plans exist by enlarging only stage constants whenever $j \le k_j, P_j$. For fixed d and support exponent, latch holds at sufficiently large coupling. These conditions and $d + 3 \le J(r(d))$ imply the section clauses of Makes. *Status:* theorems. *Lean:* `Doors.afford`; `Issue.refer`.

Proposition II.1.10 (comparison). Diagonal sign changes, translations and conjugation preserve centred root tests of interior functions. If F_0 is interior to a centred level-k cell with root $\le A$, and T has side divisible by that cell side and satisfies the seam inequality, every translate at distance at least three cell sides from its boundary has root $\le A$. Under Makes, fix $\gamma > 0$ and an extraction order. If finitely many eventually literal-translate families have centred roots $\le e^{-\gamma s_N}$ eventually at arbitrarily large fixed levels $k \le N$, with interior central support, then combinations

with placement-independent coefficients eventually bounded by $e^{\varepsilon s_N}$ for every $\varepsilon > 0$ are close with bound $y = e^{-\gamma s_N/2} \to 0$: for every $j, m \in \mathbb{N}$ there is $k \geq m$ such that eventually $k \leq N$ and, for every level-k link tiling T and $o \in T.\mathrm{margin}(0, \lfloor T.\mathrm{side}/(8(j+1)) \rfloor)$, the combination is interior to cell 0 with root $\leq y$. *Status:* theorems. *Lean:* `offset`; `gather_close`.

3.2.2 2. What it uses

- I.1: Wilson/Haar measure and lattice symmetries.
- II.2: regularity, norms, lawful polymers, tilings.
- III.1: mass-preserving blocking.
- III.2/III.9: profiles and selected histories.
- III.10–III.11: transfer estimates.
- III.16: Taylor calculus.
- IV.1: fused extraction order.

Standard inputs: matrix ODEs, finite-group averaging, analytic inverse function theorem, polarization, norm equivalence, Haar integration by parts, reflection positivity.

3.2.3 3. Strategy

Invert a finite sampling of transported plaquette curvatures to realize jet labels. Readout reverses this construction. Link integration by parts realizes equation labels; marking reserved polymers restores the large-field complement of the insertion.

3.2.4 4. Steps

1. Slots detect truncated jets: shrinking plaquette circulations recover curvature, hence radial potentials. *Lean:* `Work.G03.answer`.
2. Taylor expansion of an equivariant inverse gives sections; each slot factor costs one threshold factor. *Lean:* `Work.G05.answer`; `Work.G07.answer`.
3. Rescaled test links, polarization and finite-dimensional norm equivalence give uniform readout. *Lean:* `coeff_bound`.
4. Force lifts avoid the varied link; a unitriangular change absorbs higher-grade errors. *Lean:* `weaveBuild`.
5. Splicing the large-field supplement into reserved addresses gives exact birth. *Lean:* `Attach.paid`.
6. Seam log-convexity extends centred bounds to offsets. *Lean:* `offset`.

3.2.5 5. Where the difficulty lies

Readout constants must avoid volume dependence. Equation lifts must retain independence of the varied link. Lawfulness alone supplies no reserved addresses. Higher grade orders factor number first, then weight. Stencils are finite but nonexplicit.

3.2.6 6. Notation

$\mathrm{sc}_h J = (h^{j+2} J_j)_j$; $(S_t c)_i = t^{f_i} c_i$; $p_c = \sum_i c_i b_i$; $c^\circ = c - c_0 e_0$; $Y = o + S$. $a_m, \theta_m, w_m, \kappa_m, J(m)$ are history parameters; $\Omega_d = 1 + 2\sum_{s,i} \|s \cdot e_i\|_1$ sums over signed permutations and frame indices. $\mathcal{D}$ is the first-order inserted density. Budget regularity means locality, measurability,

gauge invariance and C^{k_b} exponential-coordinate regularity on the small-field set. The threshold satisfies $t_b \in (0,1]$. Its norm bounds all ordered left derivatives, including values, in directions $|A_e| \le t_b(1+\mathrm{dist}(e_-,o))^{P_b}$. Bounds include the small-field boundary. A slot is $\mathrm{Ad}(U_\omega)F_P$, where F_P projects $(U_P - U_P^{-1})/2$ onto $\mathfrak{g}$. Eventually means volume first, then mesh.

3.2.7 7. What remains unexplained

Identification of the auxiliary equation sections with limit null labels remains tentative. Compatibility with all later transfer estimates belongs to III.2 and III.10–III.15.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.3 II.2. Marker algebra, polymer combinatorics and Gaussian estimates

Part: II, Data of the construction. **Lean:** `YM/Marker/`, `YM/Ac/`, `YM/Ovens/Coffee.lean`, `YM/Further/SHELL_BIRTH_49/Fusing/Coal.lean`, `YM/Extra/MULTISOURCE_TRAJECTORIES/`, `YM/Push/GRID_APPROXIMATION/Nil.lean`, `YM/MergeProof/`, `YM/Next/RANK/`, `YM/Vac/Treasury.lean`. **Size:** 3,691 lines in 793 declarations used by the main theorem.

3.3.1 1. What this unit proves

Proposition II.2.1 (marker algebra). For finite V and a downward-closed family $R \subseteq 2^V$ containing $\emptyset$ and singletons, $\mathcal{A}_R = \mathbb{C}^R$ is a commutative unital algebra: $\delta_x\delta_y = \delta_{x\cup y}$ for disjoint x, y with $x \cup y \in R$, and zero otherwise; $1 = \delta_\emptyset$. Constant coefficients multiply. If $a(\emptyset) = 0$, then $a^k(x) = 0$ for $|x| < k$, hence $a^k = 0$ for $k > |V|$. *Status:* theorem. *Lean:* ring structure, `Poly.stops`.

Proposition II.2.2 (norm). For positive w_v, set $w^x = \prod_{v\in x} w_v$, $\|a\|_w = \sum_{x\in R} w^x|a(x)|$ and $\mathrm{Exp}(a) = \sum_{k=0}^{|V|} a^k/k!$. Then $\|1\|_w = 1$, $\|c\delta_x\|_w = w^x|c|$, $\|ab\|_w \le \|a\|_w\|b\|_w$, and $\|\mathrm{Exp}(a)\|_w \le e^{\|a\|_w}$. A countable family with $\|f_i\|_w \le b_i$ and $\sum_i b_i < \infty$ has a coefficientwise sum of norm at most $\sum_i b_i$. *Status:* theorem. *Lean:* `multiplication`, `budgeted`.

Proposition II.2.3 (connected polynomial). For zero-constant a, b, $\mathrm{Exp}(a+b) = \mathrm{Exp}(a)\mathrm{Exp}(b)$ and $\mathrm{Exp}(-a)\mathrm{Exp}(a) = 1$. For nonempty $x \in R$, $\mathrm{Exp}(c\delta_x) = 1 + c\delta_x$. Every $F(\emptyset) = 1$ has a unique zero-constant C with $\mathrm{Exp}(C) = F$; C depends continuously on F. *Status:* theorem. *Lean:* `Exponential.pass_add`, `connected_spec`.

Proposition II.2.4 (tagging; repacking). For finite s and $\tau : \iota \to R$, a nonzero tagged product requires pairwise disjoint tags with union in R; it equals $(\prod_{i\in s} c_i)\delta_{\bigcup_i \tau(i)}$, and weights factor. If $\sigma : \iota \to \kappa$, admissibility is any property of finite slot sets, and $\sum_i \|F_i\|_w < \infty$, then

$$\sum_{\substack{s \text{ finite: } \sigma|_s \text{ injective} \\ \sigma(s) \text{ admissible}}} \prod_{i\in s} F_i = \sum_{\substack{S \text{ finite} \\ S \text{ admissible}}} \prod_{k\in S} \sum_{\sigma(i)=k} F_i,$$

with summable terms. *Status:* theorem. *Lean:* `tagged_prod`, `Packets.repack`.

Proposition II.2.5 (components). For arbitrary finite supports X_i and finite label set s, overlap-chain classes partition s into nonempty connected sets, called sparks, with disjoint support-unions. Conversely, disjointly supported sparks recover exactly these classes. Removing $i \in s$ from a spark s leaves components each meeting X_i. *Status:* theorem. *Lean:* `separates`, `excise`.

Proposition II.2.6 (tree bound). Let $M \in \mathbb{N}$, $a, \gamma, f_i \geq 0$ and $(2M+1)^4\gamma \leq 1$. If, at every site x, $\sum_{i:x\in X_i} f_i e^{(a+1)\mathrm{cov}_M(X_i)}$ is summable and at most γ, then

$$\sum_{\substack{s \text{ spark} \\ x\in X(s)}} e^{a\mathrm{cov}_M(X(s))} \prod_{i\in s} f_i \leq e^\gamma - 1,$$

and this family is summable; labels are not repeated. *Status:* theorem. *Lean:* `Ac.forest`.

Proposition II.2.7 (two overlaps). Give each label finite mentions and walls, and mark some labels dirty. Within a selection, a label is hot when mention-connected to dirt; its fuel comprises mentions and, when hot, walls in disjoint copies of their ambient sets. Fuel components retain their internal hotness. They bijectively decompose finite selections into nonempty fuel-connected sets with disjoint mention-unions and disjoint wall-unions among dirty-containing components. If walls are lattice supports and mention overlap implies wall overlap inside such a connected set, it is wall-connected. *Status:* theorem. *Lean:* `Kiln.undo`, `Kiln.attends`.

Proposition II.2.8 (fusion). Fix a gauge group G, four-dimensional periodic lattice Λ, activity parameters and regular/forced addresses (forced inventories nonempty and contained in their supports, with extra tests) and coefficients E_i, K_i, plus logarithms ℓ_i on nonempty supports Y_i, indexed by $\mathbb{N}$. Patches are forced supports or Y_i. Pellets are finite patch-connected selections containing a forced label, with disjoint forced supports; their reaches are K_i and $e^{\ell_i} - 1$. Pellets admit a bijective enumeration. Fusion preserves regular activities and assigns each pellet its union support, union forced inventory, conjunction of forced eligibility tests, and eligibility indicator times its reach-product. *Status:* definitions; enumeration theorem. The density, bound, covariance and parametric continuity results belong to III.10. *Lean:* `fused`, `encoding`.

The digest condition requires unfused validity with bounds δ, b; logarithms supported on patches meeting the large-field set, uniformly summably bounded, measurable, gauge invariant, local in internal links and flags, and affine covariant. Fusion bounds require $\kappa, \gamma \geq 0$, $(2B+1)^4\gamma \leq 1$, and a configuration-independent $f \geq 0$ with $|\mathrm{reach}| \leq e^{-\kappa} f$ on forced labels and $|\mathrm{reach}| \leq f$ otherwise, satisfying II.2.6's hit condition with cover size B and exponent $e_P + 1$. *Lean:* `Digests`, `Buys`.

Proposition II.2.9 (switching). For finite ι, $\mathrm{val}_x(g)(r) = \sum_{i\subseteq r} x^{|i|} g(i)$ is invertible for $x > 0$. A model satisfies $\mathrm{val}_x(g_x) = f_x$ on $[0,1]$ and $g_x \to g_0$; its jet is g_0. A family is paid at cost d when models share a bound $|g_x^u(i)| \leq b(i)$ with $b \geq 0$, $\sum_i b(i) \leq d$. Costs add under finite sums and multiply under finite products; summable costs permit infinite sums and commuting jets with sums. If modeled $f_x(r) = f_x(r \setminus \{v\})$ for every r and $x \in [0,1]$, then $\mathrm{jet}(f)(i) = 0$ whenever $v \in i$; pointwise measurability of paid families passes to jets. If $\|\rho_v\|_w \leq 1$, then $\sum_i \mathrm{jet}(f^u)(i) \prod_{v\in i} \rho_v$ has a common coefficient majorant of weighted norm at most d. *Status:* theorem. *Lean:* `entire`, `Accent.crystal`.

Proposition II.2.10 (Gaussian calculus). For complex symmetric P, Q, define $\mathcal{G}_Q p = p(X + \partial_Q)1$ and $\mathbb{E}_{Q,m} p = (\mathcal{G}_Q p)(m)$. These satisfy Wick recursion, $\mathcal{G}_P \mathcal{G}_Q = \mathcal{G}_{P+Q}$, and covariance transport $Q \mapsto AQA^T$ under substitutions with finitely supported rows. A real random vector ξ under a probability law with all polynomial moments and $\mathbb{E}[\xi_v q(\xi)] = \mathbb{E}[(\partial_{Q_{v\cdot}} q)(\xi)]$ for every v, q has expectation $\mathbb{E}[q(m+\xi)] = \mathbb{E}_{Q,m} q$ for every real m. If $B \geq 1$, $|m_v|, |Q_{vw}| \leq B$, $\deg p \leq d$, then $|\mathbb{E}_{Q,m} p| \leq \|p\|_1 \pi(B,d)$. Two such pairs differing entrywise by at most $E \geq 0$ have expectation difference at most $\|p\|_1 E\pi(B,d)$. *Status:* theorem. *Lean:* `warms`, `payment`, `Common.differs`.

Proposition II.2.11 (polynomial steps). Let Q be symmetric and H_v linear. If $C \geq 1$, $\|H_v\|_1, |Q_{vw}| \leq C$, $\deg p \leq d$, then $T_{H,Q} p = (\mathcal{G}_Q p) \circ H$ has degree at most d and norm at

most $\|p\|_1(2(d+2)C)^d$. Two such pairs differing by at most $\varepsilon \ge 0$ in these norms have output difference bounded by $\|p\|_1\varepsilon(2(d+2)C)^d$. Zero cross-covariance between variables of P, R gives $T(PR) = T(P)T(R)$. *Status:* theorem. *Lean:* `crux`, `crossing`, `bridge`.

Weighted coefficient quantities $\sum_\beta r^{w\cdot\beta}|c_\beta|$, for $w : V \to \mathbb{N}$, $r \ge 0$, are submultiplicative, and norms for $r > 0$. If $0 \le \varepsilon \le 1$, $|Q_{vw}| \le \varepsilon$, $\deg p \le d$, then $\|\mathcal{G}_Q p - p\|_1 \le \|p\|_1\varepsilon(2(d+2))^d$. For linear f_v and polynomials A_v without terms below degree two, $\|f_v\|_1, \|A_v\|_1 \le B$, $B \ge 1$, $0 < t \le 1$, symmetric K with $0 \le \varepsilon \le 1$, $|K_{vw}| \le \varepsilon$, and $\deg P \le n$,

$$\|\Phi_t(P) - (\mathcal{G}_K P)\circ f\|_1 \le \|P\|_1(ntB(2B)^n + \varepsilon(2(n+2))^n B^n),$$

where $\Phi_t : X_v \mapsto f_v + t^{-1}A_v(tX)$. A measurable $g \in [0,1]$ with $\int(1-g)d\mu \le \eta$ is an η-guard. For probability μ and measurable f with $\int |f|^2 d\mu \le B < \infty$, $|\int gf\,d\mu - \int f\,d\mu|$ is at most $\varepsilon' B + \eta/\varepsilon'$ for every $\varepsilon' > 0$. Under II.2.10's probability-law hypotheses and entry bounds, $\int |p(m+\xi)|^2 \le \|p\|_1^2\pi(B, 2d)$. For finite V, $d \in \mathbb{N}$, $\delta > 0$, there exist finite nonempty $T \subseteq [-\delta,\delta]^V$ and $C \ge 0$ with $\|p\|_1 \le C\max_{x\in T}|p(x)|$ for every $\deg p \le d$. *Lean:* `LiftMark.stock`, `escaped`, `Spindle.molds`.

Proposition II.2.12 (scales). For $s > e$, put $y = \log\log s$, $\Pi = e^{y^2}$, $\theta = s^{-1/2}\Pi$, $h = e^{y^{3/2}}$, $j = h^{1/5}$, $M = \lceil(\log s)^2\rceil$, $W = M^2$. For every fixed $a, p > 0$,

$$y^2 \ll j \ll h \ll \Pi^a \ll s^p, \qquad W \sim (\log s)^4.$$

For fixed $a > 0$, $k \in \mathbb{N}$, $c \in \mathbb{R}$, eventually $W^k, h \le \Pi^a$, $c\log W \le ay^2$, $y(s)^2 \le aW(s/2), aj(s/2)$ and $j(s) \le h(s/2)$. Eventually $s\theta^2 = \Pi^2$, $s\theta^3, s\theta^4 \le 1$, and $2\Pi^v\theta^2 \le 1$ for every $v \le 1$. *Status:* theorem. *Lean:* `uniform_loop`, `rate_fire`.

Proposition II.2.13 (numerical feasibility). Fix $m, Q \in \mathbb{N}$, $e, U, V \ge 0$, $a > 0$, $\alpha \ge 1$, $L > 1$, and $G(u) = e^{-ah(u)^2}$. Eventually, uniformly for $u, v \in [s/2, s]$, $(2W(u)+1)^4\Pi(s)^{-1} \le 1$, $G(u) \le \Pi(s)^{-3/4}e^{-7j(v)/10}$ and $(\alpha+1)^{\lceil 10\log M(s)/\log L\rceil}\Pi(s)^{-3/4} \le \Pi(s)^{-1/2}$. For fixed $k, n \in \mathbb{N}$, $A \ge 1$, $\phi = 1/(4(nk+k+1))$, eventually

$$n\Pi^{-2}(A+\Pi^{-2})^n\pi(\Pi^\phi, nk) + A^n\Pi^{-2}\pi(\Pi^\phi, nk) \le \Pi^{-3/2}.$$

Status: theorems and numerical definitions. *Lean:* `on_time`, `collected`.

For numerical records with $t, g, H_0 \in \mathbb{N}$, $b, c, f, F \in \mathbb{R}$, $K \ge 1$, $c \ge 1$, $F \ge 0$, $f + c \le -8$, $g \ge 6$, $U = V$, $f = 1/16 - U$, $2V + 2 \le g - 1$, put $C_d = (2(d+2)K)^d$. If $b \ge 0$, $L > 1$ and $\log L \ge 2(b+2)\log q$ for $q \in \{\alpha+2, C_t, C_{2t}\}$, then at $H = \lceil(b+2)y^2/\log L\rceil$ eventually $(\alpha+2)^H, C_t^H, C_{2t}^H, H \le \Pi$ and $\Pi^{2c+1} + \Pi^{2c+2} \le \Pi^{2c+3}$. Consequently $(\alpha+2)^r\Pi^{-g} \le \Pi^{-4}$ for $0 \le r \le H$. Such block-size bounds hold for all sufficiently large L; parameters precede L, which precedes s. *Lean:* `racetrack`.

Put $A_Q = (2Q+1)^4$ and $R_Q(z) = e^{zA_Q}$. For $B, \varepsilon, w \ge 0$, $D \in \mathbb{N}$, define

$$q_0 = w\pi(B, D), \quad d_0 = wDB^D\varepsilon,$$

$$q_1 = d_0 + \varepsilon w^2\pi(B, 2D) + \varepsilon + w\varepsilon\pi(B, D),$$

$$q_2 = 2q_0d_0 + \varepsilon w^4\pi(B, 4D) + 2\varepsilon + w^2\varepsilon\pi(B, 2D), \quad \mathcal{Q} = q_1 + q_2 + (q_0+q_1)q_1 + q_1q_0.$$

For these records, if $F \ge g + 10Z + 10$, $Z = 2c + 8t + 13$, eventually $4R_Q(e+2)\mathcal{Q}(\Pi, \Pi^{-F}, \Pi^{2c+3}, 2t) \le \Pi^{-g}$. *Lean:* `funding`.

3.3.2 2. What it uses

- I.1: lattice supports, connected cover numbers, cardinality and intersecting-union bounds.
- III.1: activity format, eligibility, weighted bounds and gas.
- Kotecký–Preiss induction; nilpotent exponentials; Boolean Möbius inversion; Gaussian integration by parts and Wick recursion; finite-dimensional norm equivalence.

3.3.3 3. Strategy

Square-zero markers encode allowed products; their exponential separates connected contributions. Connected-component decomposition and a Kotecký–Preiss induction control polymer sums. Switching transfers scalar coefficient bounds to markers. Algebraic Gaussian convolution controls polynomial comparisons, and separated scales absorb their fixed-degree losses.

3.3.4 4. Steps

1. Construct the connected polynomial by $C_{k+1} = C_k + F - \mathrm{Exp}(C_k)$: each correction removes another marker order. `rolls`.
2. Remove one root per component. The extra unit in the hit exponent pays each attachment; the cardinality bound closes induction. `lookup`, `Ac.forest`.
3. Restrict hotness to fuel components to recover the two-overlap decomposition. `hot_comp`, `Kiln.undo`.
4. Invert switching triangularly and pass convergent models to zero. `Models.exact`.
5. Apply Wick recursion inductively to moments and differences. `width`, `differs`.
6. Compare logarithms of the scale functions and intersect the eventual inequalities, uniformly on $[s/2, s]$. `mint`.

3.3.5 5. Where the difficulty lies

Cover subadditivity requires intersecting unions. Hotness depends on the selection. Switching needs convergence, beyond boundedness. Numerical feasibility supplies no analytic trajectory estimates.

3.3.6 6. Notation

$P = (r, q, e_P, B, t_P)$ has $r, q, B \in \mathbb{N}$, $r \geq 6$, $q \geq 2$, $B \geq 1$, $e_P > 0$, $0 < t_P \leq 1$; e_P is its support exponent. Polynomials are complex; degrees are total and lie in $\mathbb{N}$. $\partial_b X_v = b_v$. $X(s) = \bigcup_{i \in s} X_i$; cov_M is the least number of radius-M balls whose union is connected and contains the support. $\|p\|_1$ sums absolute coefficients; $\pi(B, d) = (2B(d+1))^d$. $f \ll g$ means $f/g \to 0$.

3.3.7 7. What remains unexplained

III.11 must construct restarted representations and verify analytic hypotheses along trajectories. The auxiliary regrouping and switching-cost declarations' full role remains to be traced.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.4 III.1. The block-spin step and the symmetric gauge fixing

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/RG/` (Block, Patch, Measurable, Push, Masks, Activities, PathActions, Bounded, History), `YM/Branch/`, `YM/Pressure/`, `YM/PREC/`, `YM/PCalc/`, `YM/Tree/`, `YM/Exterior/`, and the small packages `YM/RG17/`, `YM/Push/CORRECTED_COLUMNS/`, `YM/Planned/CALIBRATED_MIXED_HISTORY/`, `YM/UVNext/ANNULUS_COMPARISON/`, `YM/FullVolume/Train/`, `YM/SectorTools/ACTUAL_SCALAR_LEDGER/`. **Size:** 4,219 lines (693 declarations) in the main-theorem closure.

3.4.1 1. What this unit proves

Throughout, $N \geq 1$, $G \subset U(N)$ is closed, $c \geq 1$, $f = cL$, $L \geq 3$ is odd, $J \geq 4$, $R = (J+3)L$, and $K = (\kappa, \delta, \varepsilon, a) > 0$. Branch conditions mean uniqueness of the centre-normalized minimizing frame when every route holonomy in every centre axial gauge has chord $< \delta$ (B1), uniqueness of the minimizing predictor when its minimum price is $< \varepsilon$ (B2), and nonempty eligible axial tags for every configuration (B3).

Proposition III.1.1 (Markov kernel). Under (B1)–(B3), for every real β, ρ_β is nonnegative, Borel, bounded and satisfies $\int \rho_\beta(U,V)d\mu_c(V) = 1$ for every U. For every Banach-valued $F \in L^1(\mu_f)$, $\mathcal{P}_\beta F(V) = \int \rho_\beta(U,V)F(U)d\mu_f(U)$ exists for every V, is integrable, depends linearly on F, and preserves its integral. Positivity and integrability in V need no branch conditions. *Status:* theorem. *Lean:* `isKernel`.

Proposition III.1.2 (covariance). Under (B1)–(B3), for every β, U, V, fine gauge k, signed axis permutation s and coarse site b,

$$\rho_\beta(k \cdot U, (k \circ \mathrm{embed}) \cdot V) = \rho_\beta(U,V) = \rho_\beta(s_{Lb}U, s_b V), \qquad \mathrm{embed}(x) = Lx.$$

A calendar satisfying branch conditions on every patch has block control and Markov kernels at every step. *Status:* theorem. *Lean:* `covary`; `channel`.

Proposition III.1.3 (uniform branch constants). Let d be as in III.1.5 and $r_G > 0$ a unique-nearest-point tubular radius of G. For every $\kappa > 0$, $0 < \delta \leq d/\sqrt{N}$, $0 < \varepsilon \leq r_G^2/(2N)$ and $a \geq 16$, (B1)–(B3) hold for every patch. Every configuration and axis ordering admit an eligible root. *Status:* theorem. *Lean:* `Axial.viable`; `RG.frame_unique`; `RG.predictor_unique`.

Proposition III.1.4 (axial gauge). For every list l of distinct axes, $r \in \mathbb{N}$, $o \in B_r$ and configuration U on $\mathbb{Z}^4$, put $S = \{y \in B_r : y_j = o_j \ (j \notin l)\}$. Some $x \in S$ gives an axial gauge $g_y = T_U(x \to y; l)$ with $E_S(g \cdot U) \leq 4|l|(2r+1)F_S(U)$. *Status:* theorem. *Lean:* `Axial.existence`.

Proposition III.1.5 (synchronization uniqueness). There exists $d(G) \in (0, 1/4]$ such that, for all finite nonempty n, Λ, data $W : n \times n \times \Lambda \to G$ with $W_{xxk} = 1$, $W_{yx\sigma(k)} = W_{xyk}^{-1}$ for a permutation σ, and $\|W_{xyk} - 1\| \leq d$ have this property: any two global minimizers over G^n of $\mathrm{avg}_{x,y,k}\mathrm{pr}(h_x W_{xyk} h_y^{-1})$ differ by one global left multiple. *Status:* theorem. *Lean:* `Branch.align`.

Proposition III.1.6 (frame comparison). There exists $A(G) > 0$ such that, for every patch, K, configuration W, positive-frame-mass tag, realized frame h, and gauge $H : \mathbb{Z}^4 \to G$, some $q \in G$ satisfies

$$\sum_{y\in B_R} \mathrm{ch}(q^{-1}h_y H_y^{-1}) \le A(2R+1)[E_{B_R}(H\cdot W) + a(2R+1)F_{B_R}(W)].$$

No branch conditions are needed. *Status:* theorem. *Lean:* `Pressure.synchronized`.

Proposition III.1.7 (common gauge). Fix G, K and $j \ge 4$; set $\ell = 2j+15$, $D = 16\ell$, $C = A\ell(D+a\ell)$ and $R' = (j+7)L$. For every step with $J = j$, fine U, coarse x, and tag usable at x, there exist $H : \mathbb{Z}^4 \to G$ and $p_y \in G$ for $y \in B_2$ such that, for every such y,

$$E_{B_{R'}}(H\cdot U^{(x)}) \le DL\,\mathrm{port}(x,U), \qquad \sum_{i\in B_R} \mathrm{ch}(p_y^{-1}h_i^{(x+y)}H_{i+Ly}^{-1}) \le CL^2\mathrm{port}(x,U).$$

Every positive-tag-mass tag is usable everywhere. *Status:* theorem. *Lean:* `Pressure.planting`.

Proposition III.1.8 (coarse curvature). With III.1.7's constants and $C_8 = 8\max(1,\varepsilon^{-1})(2C+D)$, for every step with $J = j$, configurations U, V, coarse site x, tag t usable at x, and plane p,

$$\mathrm{ch}(Q(U,t)_{(x,p)}) \le C_8L^{-2}\mathrm{port}(x,U), \qquad \mathrm{ch}(V_{(x,p)}) \le \mathrm{fuel}(Q,V)_{(x,p)} + C_8L^{-2}\mathrm{port}(x,U).$$

Status: theorem. *Lean:* `Pressure.qdc`.

Proposition III.1.9 (densities and histories). Representations evaluate to an exponential of vacuum minus action and masked regular activities, times the gas. Flags mark sites touched by plaquettes of chord above threshold. Reading links and flags in X requires only links in X and corners of plaquettes meeting X. The gas equals 1 without flags. Every lawful representation (path bounds and valid activities) has bounded Borel evaluation. Selected histories start at the bare Wilson density and obey exact pushes; every selected history satisfies $\int \mathrm{eval}_{N,p,m}\,d\mu = Z_{N,p}(\beta_{N,p,N})$ for all $m \le N$. *Status:* definitions, with four consequences proved; quantitative construction belongs to III.3–III.10. *Lean:* `UsesFinal.allReads`; `gas_one`; `Rep.envelope`; `History.partition_eq`.

Proposition III.1.10 (tube identity). For every tube, path cutoff, real thresholds and U, Q, V, the six works satisfy

$$W_F + W_I + W_C + W_N + W_L + W_D = C_f + P - C_c,$$

where C_f, C_c are truncated crown sums and $P = \sum_b \kappa\,\mathrm{ch}(V_bQ_b^{-1})^2/2$. *Status:* theorem. *Lean:* `Tube.locates`.

Proposition III.1.11 (energy per seed). Estimator data require: for every sufficiently large L and then sufficiently large W, some $\kappa_0 > 0$ gives $\kappa_0 t^2|A| \le \mathrm{Work}(A)$ at all sufficiently large reference couplings, uniformly over compatible tubes, forests and plots with trivial trunks and $c \ge 100(1+\varrho)$. For a directed history satisfying ringing, each $J_0 \le i \le j \le i+1$ admits $\mu > 0$ such that $\mu t_{m+1}^2|A| \le \mathrm{Work}(A)$ for all $N, m < N, p$ with $(J_{m+1}, J_m) = (j,i)$ and every plot at $(\hat\beta_m, F(j,i))$ with trivial trunks. *Status:* requirement, proved in III.10; transfer theorem here. *Lean:* `Stores.Plot`; `Teller.Grants`; `Testing.notation_ticket`.

Proposition III.1.12 (kinetic specification). Fix levels, a quadratic horizon, Haar-chart density and positive stage constants. Kinetic data assign shifts α, path data and free-link maps

ψ. Stages and factors agree with the fixed choices; rows attach to the horizon and satisfy stage bounds. At level N, the unmasked, unclipped action is analytic, matches Wilson derivatives through order four, and has zero loop coefficients. For every $m < N$, volume, forest and tag, let $\mathcal{J}_t$ be the fine unmasked, unclipped unit-coupling action without loops plus predictor penalty, on fields with trunks 1 and free links $\exp(b_e)(\iota \exp u)_e$, where ι copies coarse links across block boundaries. Requirements, for every free field v, are

$$D_b^2 \mathcal{J}_t(0,0)[v,v] \geq \eta \sum_{e \text{ free}} |v_e|^2, \qquad D_b \mathcal{J}_t(u, \psi(u)) = 0$$

near zero: $\psi(0) = 0$, both maps are analytic, $D\psi(0)$ is the prescribed background adjusted to tree gauge minus the linearized block copy, and the analytic coarse action matches $\mathcal{J}_t(u, \psi(u))$ through order four. Writing $H_t(u) = D_b^2 \mathcal{J}_t(u, \psi(u))$ in physical coordinates,

$$T_t = \mathcal{L}_f^\circ(U_F(u, \psi(u))) + \tfrac{1}{2} \log\det H_t - \log \prod_{e \text{ free}} J(\psi(u)_e), \qquad D = \alpha \mathcal{Q}_c^\circ(\exp u) + \mathcal{L}_c^\circ(\exp u).$$

$\mathcal{L}^\circ, \mathcal{Q}^\circ$ denote loop and quadratic actions; J is the Haar-chart density. Both T_t, D are analytic, $\det H_t > 0$ near zero, and

$$|D^2 T_t(0)[a,b] - D^2 D(0)[a,b]| \leq |\Lambda_c| \Theta \Lambda e^{-\gamma c}$$

for real plane waves with direction-colour amplitudes bounded by 1; Λ is the fine loop bound. Also $|\alpha + \kappa_{\text{slope}} \log L_m| + 1 \leq \text{drift}_{m+1}$. Directed histories require paid levels, a sufficient horizon, matching recipe and local Haar-chart formula. Moves require exact push, continuation, retention, lawful coarse data, forced-activity bound $\exp[-g_j \text{monitor}_{m+1}^2]$, and $|\beta_m - \beta_{m+1} - \alpha| \leq 1$. *Status:* definitions; conditional constructors for simple nonabelian $\mathfrak{g}$ belong to III.2. *Lean:* `Kin.Planned`; `Directed.Move`.

Proposition III.1.13 (tree gauge). For odd L, block forests exist. Inverse tree transport gives a gauge equal to 1 at roots, depending only on trunks, and trivializing them. Every real-Banach-valued integrable F invariant under gauges trivial at roots satisfies $\int F d\mu_f = \int F(\text{drape}\, s) d\mu_{\text{free}}(s)$. For a kernel satisfying III.1.2(a) and gauge-invariant integrable F, the same reduction applies to $\mathcal{P}_\beta F(V)$. *Status:* theorem. *Lean:* `Forest.pack`; `Forest.pushing`.

Proposition III.1.14 (linearization). Under (B1)–(B3), for every tag t and coarse link b, $v \mapsto Q_b((\exp v_e)_e, t)$ on fine $\mathfrak{g}$-valued 1-forms equals 1 at $v = 0$ and is real analytic nearby, using minimizing branches. Its tag-independent derivative is $\mathcal{Q}v = \mathcal{M}v + d_0 \mathcal{F} v$, and $\mathcal{Q} d_0 \varphi = d_0(\varphi \circ \text{embed})$ for every fine 0-form. Frames have derivative $h_y^{(x)\prime} = \tau_x(y) - \tau_x(0)$. The kernel is analytic near $(1,1)$ for fixed β; neighbourhoods may depend on the step. *Status:* theorem. *Lean:* `Push.derivative`; `Push.prices`.

3.4.2 2. What it uses

- I.1: lattice geometry, Haar invariance, Wilson energy and nearest-point projection.
- III.16: offset lengths; III.2: schedules and directed histories; III.10: energy estimates.
- Tubular neighbourhood theorem, analytic implicit function theorem, Fubini, Jensen, continuity of unique compact minimizers.

3.4.3 3. Strategy

Minimize the average route price to fix centre-normalized frames, transport across blocks, and average the connectors. The predictor Q is the nearest group point to their weighted matrix mean on its minimizing branch. The kernel averages $\prod_b Z_\beta^{-1} e^{-\beta\kappa \mathrm{pr}(V_b Q_b^{-1})}$ over tags, with $Z_\beta = \int_G e^{-\beta\kappa \mathrm{pr}(g)} dg$. Unique minimizers handle small fields; symmetric sampling of curvature-controlled axial frames and connectors handles the remaining configurations. A common macro-box gauge compares neighbouring frames.

3.4.4 4. Steps

1. **Uniform cutoffs.** Normalize minimizers by nearest centroid points; tangential cancellation makes feedback contract. Slab induction gives eligible axial roots. `align`; `existence`.
2. **Kernel.** Tag probabilities and Haar noise each have mass one; uniqueness and tag permutations give covariance. `density_total`; `channel`.
3. **Curvature.** Synchronization costs L^2, while $w \le L^{-4}$; four padded links give the plaquette bound. `indicated`; `qdc`.
4. **Local analysis.** The implicit function theorem gives derivatives; Haar invariance removes trunks. Adjointness cancels cross terms and squares in the tube identity. `scheme`; `pack`; `Tube.locates`.

3.4.5 5. Where the difficulty lies

Uniqueness must remain uniform in patch size. Both branches must preserve area scaling. Global regularity is Borel; analyticity is local.

3.4.6 6. Notation

$B_r = \{-r, \dots, r\}^4$; $\mathrm{ch}(u) = \|u-1\|/\sqrt{N}$, $\mathrm{pr}(u) = 1 - \mathrm{Re}\,\mathrm{tr}(u)/N$; $\|\cdot\|$ is Frobenius, $|v| = \|v\|/\sqrt{N}$. μ denotes product probability Haar measure; E_X, F_X sum link and plaquette chords. $U^{(x)}$ is the root-centred periodic lift; port $= F_{B_{R'}}(U^{(x)})$. Usable means positive frame masses on $x + B_2$; fuel sums four link discrepancies with multiplicity. w is the J-fold block-uniform convolution. τ averages negative route integrals, $\mathcal{M}$ averages straight length-L integrals, $\mathcal{F}v(x) = \tau_x(0) - \sum_y w_y \tau_x(y)$, and d_0 is forward difference. Here $\tau_x(y) = -\mathrm{avg}_{z\in B_R, o\in S_4} \int_{Lx+y\to Lx+z;o} v$.

$\mathrm{monitor}_m = e^{(\log\log\hat\beta_m)^2}$; $t_m = \mathrm{temp}(\hat\beta_m)$, $\mathrm{temp}(z) = z^{-1/2} e^{(\log\log z)^2}$; $t = \mathrm{temp}(\hat\beta + \kappa_{\mathrm{slope}} \log L)$, $\varrho = \lceil (\log\hat\beta)^2 \rceil$, $\kappa_{\mathrm{slope}} = 11C_A/(24\pi^2)$.

A store A comprises fine plaquettes, coarse links and coarse sites; $|A|$ sums their cardinalities. $\mathrm{Work}(A)$ sums local works whose fine bases' parents or coarse bases lie within distance 2ϱ of its locations, with cutoff $\lfloor \varrho/10^5 \rfloor$. $F(j,i) > 0$ is the assigned wire parameter.

3.4.7 7. What remains unexplained

Energy lower bounds and kinetic construction belong to later units. The uses of B_2 rather than B_1, and $J \ge 4$ rather than $J \ge 1$, remain unspecified here.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.5 III.2. Choosing the constants: the order of choice, the level schedule and the shooting argument

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/Price/` (the constants and the level schedule, 406 lines), `YM/Shoot/` (shooting, 228 lines), `YM/Integration/RG05Preconstruction/`, `YM/Integration/NativeSelection/` and other files of `YM/Integration/`, `YM/RG08/Calendar/`, and parts of `YM/RG08/`, `YM/RG12/`, `YM/FullVolume/` (in particular `Roll/`) and `YM/TraceExtension/`. **Size:** 18,156 lines in the main-theorem closure.

3.5.1 1. What this unit proves

Proposition III.2.1 (order of choice). For simple G, choose:

1. A sheet: kernel K with branches on every patch; first stage $j_0 \geq 4$; floors $f_j \in \mathbb{N}$; stage records with a clip of relaxation r_j, derivative orders $k_j \geq 6$, spatial powers $q_j \geq 2$, support exponents $a_j > 0$, widths $w_j > 0$, path counts $\varpi_j > 0$, drift allowances $\omega_j \geq 0$, column bounds $\gamma_j \geq 1$, $r_j w_j \geq 12$; nondecreasing k, q, a.
2. Nondecreasing odd $\ell_j \geq \max(3, f_j)$, with, for $j \geq j_0$,
$$\omega_j/\log \ell_j \leq \min(\kappa/4, j^{-2}), \qquad \log \mathcal{C}_j/\log \ell_j \leq j^{-2}, \qquad \mathcal{C}_j = 1 + \varpi_j + w_j^{-1} + \gamma_j + r_j + a_j.$$
3. Positive nonincreasing $\theta_j \leq 1/j$ for $j \geq j_0$.
4. Unrestricted delays: real entry couplings β_j^{ent}, real barrier-dependent entry scales $\sigma_j^{\mathrm{ent}}(\bar\beta)$, natural shells S_j and exponents g_j.
5. A barrier $\bar\beta > 1$, with $\bar\beta \geq \beta_{j_0}^{\mathrm{ent}}$, with $x^{-1/2} e^{(\log\log x)^2} \leq 1$ for every $x \geq \bar\beta$.
6. A wait: real readiness scales σ_j^{rdy}, windows $h_j : \mathbb{R} \to \mathbb{N}$ with $h_j(s)/s \to 0$ as $s \to \infty$ for each fixed j, and natural $n_{\min}$.

Delays and waits are ordered componentwise. Factors exist for every sheet; θ can lie below any prescribed positive sequence; barriers exist above any prescribed real. *Status:* definitions and three existence theorems. *Lean:* `Sheet`, `Rolls.exists_rolls`, `Protection.exists_margin`, `Bar.exists_above`.

Proposition III.2.2 (schedule). For simple G and any III.2.1 data, there are J_m, $L_m = \ell_{J_{m+1}}$, $n_p = 4(M + p + 1)$ with natural M, retaining the prescribed data, such that:

- $J_0 = J_1 = j_0$; J is nondecreasing, increases by at most one, and diverges; $k_{J_m}, q_{J_m}, a_{J_m}$ are nondecreasing.
- For every m, $\beta_{J_{m+1}}^{\mathrm{ent}} \leq \hat\beta_m$; also $\sigma_{J_{m+1}}^{\mathrm{ent}}(\bar\beta) \leq s_m$ if $J_{m+1} > j_0$.
- For every m, p, with $j = J_{m+1}$: $n_p \geq n_{\min}$, $S_j(1 + B_m + B_{m+1})^{g_j} \leq n_p e^{s_m}$, $100(B_m + 1) \leq n_p e^{s_m}$, $100(R_m + 1) \leq n_p e^{s_{m+1}}$.
- Each $j \geq j_0$ has some k with $\sigma_j^{\mathrm{rdy}} \leq s_k$, $h_j(s_k) \leq k$, and $J_i = j$ for $|i - k| \leq h_j(s_k)$.
- $\hat\beta_m > 1$, $t_m \leq 1$, $0 < \theta_{J_m} \leq 1/J_m$; $\omega_{J_{m+1}}/\log L_m$ and $\log \mathcal{C}_{J_{m+1}}/\log L_m$ are at most J_{m+1}^{-2}; $\log L_m/s_m \to 0$. Kernels are well posed, lattice symmetric and gauge covariant.

Every positive integer divides n_p infinitely often. *Status:* theorem; divisibility variant in III.10. *Lean:* `Wait.ready`, `Wait.engine`.

Proposition III.2.3 (shooting). Let G be simple; fix a level system with $\omega_{J_{m+1}} \leq (\kappa/4)\log L_m$ and a recipe prescribing paths $P_{N,m}^{\mathcal{R}}$ and start, retention and transition tests. Admissibility

requires these paths, the level budget point and clip, width w_{J_m}, path count ϖ_{J_m}, loop cap $2M_m^{\theta_{J_m}}$, allowances $\widehat{\beta}_m^{-11/10}, e^{-M_m^{1/5}}$, and retention. Continuity requires continuous β, v, fixed activity address sets, and differences tending to zero in summed budget masses of regular derivative bounds through budget order on the healthy domain and forced uniform bounds on all configurations.

Assume, for every N, p, a continuous admissible Wilson start family on $\mathfrak{B}_N$ with coupling equal to its parameter, $v = 0$, density equal to the Wilson weight at that coupling, passing the start test. Assume also that every continuous admissible level-$(m+1)$ family on any $[l, u]$ with $l \le u$, $m < N$, with couplings in $\mathfrak{B}_{m+1}$, admits a continuous admissible level-m family whose density is its block-spin push-forward at the input coupling, passing the transition test, with $|\beta' - \beta - \kappa \log L_m| \le \omega_{J_{m+1}}$.

Then for all N, p a selected history exists: admissible representations in $\mathfrak{B}_m$, Wilson top with $v = 0$, these transitions, $\beta_{N,p,0} = \bar{\beta}$, paths independent of p. For every $\varepsilon > 0$, some m_0 gives $|\beta_{N,p,m} - \widehat{\beta}_m| \le \varepsilon s_m$ for all $N \ge m \ge m_0$, all p. *Status:* conditional theorem; hypotheses discharged in III.10. *Lean:* `Recipe.races`.

Proposition III.2.4 (joint purchase). Assume simple G, kernel K branched on every patch, first stage ≥ 4, nondecreasing $k_j \ge \max(6, j)$, $q_j \ge \max(2, j)$, $a_j \ge 100$, nondecreasing $Q : \mathbb{N} \to \mathbb{N}$ with $Q_j \ge 1$, records Ω, Δ, degree-300 tableau at ξ and contraction condition `sphere` at $(\xi, \omega_v, \omega_t)$, with $\xi \ge 2$, $0 < \omega_v \le 1$, $\omega_t > 0$, and prediction π prepaid at every sheet and price map.

There exist a sheet s preserving the prescribed kernel, first stage, derivative orders, powers and support exponents, price map T, real sequence Y, nondecreasing $K' \ge 1$ and floor $\tau \ge f^s$. For every floor $k \ge \tau$, factors exist; for every block-factor sequence ℓ for s with floors k there is U; for every wage $w \ge U$, postage rate and exchange datum there are post-L costs; for every positive θ^0 there is protection $\theta \le \theta^0$; for every menu, auxiliary record and delay T_0 there is $V \ge T_0$; for every $V' \ge V$, barrier and wait W there is $Z \ge W$; for every $W' \ge Z$ and level system meeting III.2.2 with divisible periods there are a directed datum and factor X recording these choices, satisfying $\pi(X)$.

Wages, exchanges, menus and auxiliary records equal the prescribed data; postage rates do not exceed the prescribed rates. The same factor satisfies the III.12 conditions `Degree`, `Savvy`, `Roofing`, `PaysW`, `Spire`, `Wild`, `Tower.Sav` with `Canyon`, `Cadre`, `AccordV` with Q (III.11), top kinetic data (III.9), amplification (III.1), and a bandage with extras (Ω, Δ, b) (III.15). Support exponents are ≥ 100; $e^{(a_j+2)(2\,\mathrm{span}_{ij}+13)^4} \le \ell_i$ for $j \le i \le j+1$. The bandage also supplies the conditional family package `Unlatched` with a vacuous additional conclusion, under the family hypotheses of III.2.5. *Status:* theorem. *Lean:* `richFaresWithPrediction`.

Proposition III.2.5 (plentiful construction). Under III.2.4 and $Q_i \ge 2$, its conclusion additionally ensures: every compatible trajectory, reference levels with $d + 3 \le J_{t(d)}$, section transfers and consecutive gauge doors satisfying family consistency, door compatibility and prescribed transfer bounds from level $t(d)$ whenever $t(d) \le N$ give, for every extraction order, the complete weak-continuum requirement (III.11) and trace gallery (VII.2). Floors, delays and waits may increase; the prediction persists. *Status:* theorem. *Lean:* `plentifulWithPrediction`, `Unlatched`.

Proposition III.2.6 (prediction). Let π be a prediction. For simple G, a sheet with the perceived-estimate package of detailed Lemma 4.16 at fixed tariff and scope, and $\tau > 0$, suppose: for every choice of targets with the canonical chart, expenses with deficit $-7/16$, programs, initial

wages and baton there is a floor u; for every floor $l \geq u$, factors, wages, duration bounds, register, postage, scouts, protection, step records with $\varepsilon \leq 1/4$ on transitions with output stage at least the first stage, post-L costs and delay V, there is $W \geq V$; for every later delay, barrier, wait, pricing record, directed datum and remaining factor data, shelter and step conditions and divisible periods imply π on the assembled factor. Then the complete native purchase of detailed Lemma 4.17 retains π on the emitted factor, on level systems with divisible periods, maximizing its floor with u and inserting W before its booth and final delay. *Status:* theorem; used with vacuous prediction. *Lean:* `company`, `Prepaid`.

Proposition III.2.7 (complete run). Given simple G, penalty $b > 0$, records Ω, Δ, custody request $\mathcal{H}$, eventually paid Λ, and a demand assigning every sheet and price map nonempty choices c, floors $e_c : \mathbb{N} \to \mathbb{N}$ and real arrays $F_c(n,i), F_c'[\ell](https : //github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/n,i)$, there is a run with a campaign, penalty b, preselection s, q, c, and floor $n \geq \max(f^s, e_c)$. It uses s with floors n and the price map derived from q; both arrays are paid as in III.2.8(b), using actual ℓ, and $F_c'[\ell](https : //github.com/scottnarmstrong/yang - mills - informalization/blob/main/outline/units/0, J_{m+1}) \leq \widehat{\beta}_m$ for all m. Orders satisfy $k_j, q_j \geq j$; Λ, treasury funding, common-period funding, pillars, the funded, news and read-out records, custody and sky hold. Any universally prepaid prediction can persist. *Status:* theorem. *Lean:* `selectRunWithTreasuryPeriod_preselected_quote_orders_postnumeric`.

Proposition III.2.8 (persistent payments).

(a) Eventually paid means: for every sheet, factors, protection and delay V, some $X \geq V$ ensures the property for every delay $W \geq X$, every barrier and wait, and every system meeting their entrance conditions. Conjunction preserves it.

(b) For every real array $F(n,i)$, eventually paid are $1 \leq 1 + \log\log\widehat{\beta}_m$ at all levels and: for every n, eventually in m, $|F(n,i)| \leq 1 + \log\log\widehat{\beta}_m$ for all $i \leq J_{m+1}$. For every real sequence F, $F(J_{m+1}) \leq \widehat{\beta}_m$ at all levels is eventually paid.

(c) Every countably encoded family of requests (stage floors and costs depending on the transition, block geometry and stage record X_{J_m} at step m) has a common floor pre fixed before factors. If $\ell_j \geq \mathrm{pre}(j)$ at every stage, every delay can be enlarged so every later delay, any barrier and wait, and every admitted system satisfy, for $\mathrm{code}(\alpha) \leq J_n$, $m \geq n$, $\mathrm{pre}_\alpha(J_{m+1}) \leq L_m$ and step cost $\leq 1 + \log(2 + \log\widehat{\beta}_m)$.

(d) For simple G and every kernel, a plan fixes charge names and dictionaries of finite words and spectra for every request before factors. Above its floor, delays can be enlarged persistently to fund requests and the inverse charge from level zero. *Status:* definition in (a); theorems. *Lean:* `EventuallyPaid`, `eventuallyPaid`, `minDelay`, `buy`.

Proposition III.2.9 (analytic estimates). Norms are componentwise exponentially weighted displacement ℓ^1 norms, summing colour indices for multilinear kernels; colour spaces are fixed and finite, and constants may depend on them. Crosses map coarse fields to fine fields at block size L. Vanishing orders concern symbols at zero momentum.

- For $\sigma, \tau > 0$, real C, natural d, some $0 < \eta \leq \sigma$, $K \geq 0$ give $\|A^{\mathsf{T}} R B\|_\tau \leq K M L^{2-d}$ for positive odd L with $\tau \leq L\eta$, $\|A\|_\tau \leq C/L$, $\|\nabla^s B\|_\tau \leq C/L^{1+|s|}$ through order d, $M \geq 0$, $\|R\|_\sigma \leq M$, and vanishing order d. For one-form rows, $d = 3$ gives $ML^{-1/2}$ for all sufficiently large odd L. `pullback_gain`, `large_contraction`.

- For natural r, d, $\sigma, w > 0$, $C \geq 0$, some $0 < \eta \leq \sigma$, $K \geq 0$, fixed before L, give transfer bound KBL^4/L^{r+1+d} at width w. Assume a real $(r+1)$-linear kernel of width σ, bound B, supported on tuples with first site zero, vanishing order d in independent momenta; positive odd L, $(r+2)w \leq L\eta$; crosses with $\|\nabla^i Q\|_{(r+2)w} \leq C/L^{1+|i|}$ through order d; transfer through $r+1$ copies of Q, followed by symmetrization and recentering. `flow_soft`.
- For finite coordinates I, natural n, $\sigma > 0$, some $0 < \eta \leq \sigma$, $K \geq 0$ ensure: every real kernel $f : \mathbb{Z}^I \to \mathbb{R}$ of weighted bound $A \geq 0$, vanishing order n, has real kernels g_b of bound KA, width η, with $\hat{f}(p) = \sum_b \prod_{a=1}^{n} (e^{ip_{b(a)}} - 1)\hat{g}_b(p)$ for all real p, where $b : \{1, \dots, n\} \to I$. `real_difference_division`.
- Fix step scenes with input coupling tending to infinity, leaves, cutoff parameter, radii satisfying the logarithm-chart conditions and fluctuations z_i; put $H_i = \frac{1}{2} \log \beta_i^{\mathrm{out}}$, $\eta_{*,i} = e^{-H_i}$. Assume fluctuation bounds, leaf weights and third-derivative bounds D_i have upper exponential rate zero relative to H_i; eventually leaves are masked regular logarithmic terms whose threshold-squared expressions have zero first derivative at zero and derivatives through order three bounded by D_i on the unit ball; $\pm z_i$ are healthy; both cutoffs equal one near the origin in the two deformation parameters. Here upper rate a means an eventual bound e^{rH_i} for every $r > a$. Writing P_i for the paved expansion, for every $\varepsilon > 0$, eventually $\|D_y^2(P_i(z_i; y) + P_i(-z_i; y))|_{y=0}\| \leq \eta_{*,i}^{2-\varepsilon}$. `mirror_price`.
- Fix a representation, degree-d frame and section readable at the origin for the representation's budget. Two prescribed coefficient presentations with equal section densities, remainder bounds a, b and the same positive forced parameter satisfy $\|x_P - x_Q\|_{\mathrm{coeff}} \leq K_{F,H} e^{-a_* m}(a+b)$ for every natural m with $10(m(4B+2)+2B) < n$. Regular terms have supports containing zero, vanish at identity and have zero local degree-d projection on nonwrapping supports; nonzero forced supports contain zero and satisfy eligibility, support, gauge-invariance and measurability conditions. `unique_part`.

Further estimates concern the post-block Hessian (`design`), cubic and quartic coefficients (`tanker`), and trace insertion (`cloche`). *Status:* theorems under the displayed hypotheses; only statements of these supplementary estimates are included in the notes.

3.5.2 2. What it uses

- I.1: $\kappa > 0$; III.1: calendars, histories, branches; III.9: reference quantities and Wilson startup; III.10: one-step construction and divisible periods.
- III.11–III.12, III.15, VII.2: purchased conditions; III.14: demand and external records; III.16: strip and Laurent norms and jets. Free-kernel and stationary-branch inputs: III.3 and III.6 (probable).
- Intermediate value theorem, compactness, logarithmic asymptotics, countable diagonal enumeration.

The coupling shift with error ω is an input to shooting, supplied by III.10 (`Graces.winnow`). Positivity of κ comes from `slopeConstant_pos`.

3.5.3 3. Strategy

Choose stage data before factors, then costs, protection, delays, barrier and wait. Maxima combine floors; persistent increases combine delays and waits. Long plateaux provide comparison windows; exponential side growth pays volume demands. Shooting narrows continuous Wilson families

into successive coupling bands until the level-zero coupling equals $\bar{\beta}$. The shooting windows are $[\bar{\beta} + (\kappa/2)s_m, \bar{\beta} + 2\kappa s_m] \subset \mathfrak{B}_m$.

3.5.4 4. Steps

1. **Exhaustion.** Long plateaux meet entry and window thresholds; one period offset pays the finite first plateau. `Program.exhaust`, `Wait.ready`.
2. **Narrowing.** Endpoint drift inequalities straddle the next window; first and last hitting points give a subinterval mapping entirely into it. Iterate downward. `gates`, `narrow`, `steps`.
3. **Running law.** The coupling error is bounded by $\sum_{k=1}^{m} \omega_{J_k} = o(s_m)$ using $J_k \to \infty$. `trajectory`.
4. **Diagonal payment.** Raise entry couplings above double exponentials of finite partial sums of array entries. Since $J_m \to \infty$, every fixed row is eventually included. `eventuallyPaid`.
5. **Composition.** Combine sheet and native suppliers, joint and trace purchases and persistent payments by maxima of floors and transitivity of delay and wait bounds. Choose the schedule only after these increases, then assemble the run. `sailsW`, `plentifulWithPrediction`, `engageWithCustodyFactor`.
6. **Analytic gains.** Move differences from vanishing kernels onto crosses: their two bounds give L^{-2}, fine-site summation gives L^4, and d differences give L^{-d}. Reflection cancels odd terms; local projection leaves exponentially small wrapping contributions. `pullback_gain`, `mirror_price`, `unique_part`.

3.5.5 5. Where the difficulty lies

Factor-dependent costs require later persistent thresholds. Shooting needs full-family continuity and band membership throughout each interval. The $\kappa/4$ drift bound separates endpoints; j^{-2} controls accumulated error. Each row is paid eventually; the zeroth post-L row needs separate all-level payment.

3.5.6 6. Notation

$m = 0$ is coarsest, N finest; p indexes volume, j stages. $s_m = \sum_{i<m} \log L_i$, $\kappa = 11C_G/(24\pi^2)$, $\hat{\beta}_m = \bar{\beta} + \kappa s_m$, $\mathfrak{B}_m = [\hat{\beta}_m/2, 2\hat{\beta}_m]$, $M_m = e^{(\log\log\hat{\beta}_m)^2}$, $t_m = \hat{\beta}_m^{-1/2} M_m$, $B_m = \lceil (\log \hat{\beta}_m)^2 \rceil$, $R_m = (J_{m+1} + 3)L_m$. In read-out, n, B, a_* are side, cover size, support exponent; $K_{F,H}$ depends on the frame and holonomy tests. span_{ij} is transition support; a prediction is a predicate on the factor recording all choices. Stage records, protection and admitted entrance conditions are `Stage`, `Protection` and `Paid`.

3.5.7 7. What remains unexplained

Purchased predicates need comparison with their owning units. Underlying analytic suppliers and dictionaries lack independent verification here; drift normalization against III.9 remains unresolved. Odd-cancellation scenes, coefficient and readability interfaces and auxiliary parameters need fuller mathematical accounts. Supplementary Hessian, coefficient-extraction and trace-insertion proofs remain unexplained.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.6 III.3. Free lattice kernels and covariances

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/Next/FREE_KERNELS/` (5,450 lines), `YM/MultiNext/SIGNED_SOURCE_CONTINUATION/` (1,674), `YM/Further/FREE_COVARIANCE_KP/` (924), `YM/UVNext/PRICED_KERNELS/` (533), `YM/Further/CORRECTED_LIFT_COEFFICIENTS/` (248), `YM/Push/CUTOFF_PROJECTORS/` (139), `YM/ProofPorts/GRID_RENEWED_RESERVES/` (30). **Size:** 8,998 lines in the main-theorem closure.

3.6.1 1. What this unit proves

For fine/coarse levels, primes denote coarse data and $V = (\sqrt{c}\,1, R)$, where c is the direct clip coefficient. The diagonal identities (D) and roll identities (R) are

$$\begin{aligned} d^*HdW &= Q^*d^*H'd, & QW &= 1-\kappa^{-1}d^*H'd,\\ T^*d &= dQ, & TD'd &= DdW;\\ M_f^*M_f + M_n^*M_n + O^*O &= 1, & \sqrt{\kappa}M_n^* &= V'd,\\ M_fV'd &= VdW, & M_nV'd &= \kappa^{-1/2}d^*H'd,\\ OV'd &= 0, & M_f^*Vd &= \sqrt{\kappa}M_n^*Q. \end{aligned}$$

The defect identity (Δ) is

$$H'(H'^{-1} - T^*H^{-1}T - \kappa^{-1}dd^*)H' = d_2^*Pd_2.$$

Under (D),(R), for all fine/coarse potentials A, B, setting $Z = A - WB$ gives

$$\langle dA, HdA\rangle + \kappa|QA-B|^2 = \langle dB, H'dB\rangle + \langle dZ, HdZ\rangle + \kappa|QZ|^2.$$

Proposition III.3.1 (catalogue). For every kernel choice K, there exist sequences $\alpha_j, \eta_j, \gamma_j > 0$, $\Omega_j \in \mathbb{R}$, $\lambda_j \geq 0$, $b_j \in \mathbb{N}$, $j \in \mathbb{N}$, depending only on K. For every history of index $j \geq 4$ and clip coefficient $0 < c \leq \gamma_j$, its explicit metric data $(D, D^{-1}, H, H^{-1}, R)$ lie in $\mathcal{S}(\alpha_j, \Omega_j)$, with

$$H^{-1} = D^{-1} + \lambda_j(d_2D^{-1})^*(d_2D^{-1}).$$

On every torus, $DD^{-1} = HH^{-1} = 1$, $c1 + R^*R = H$, $\hat{H}(0) = 1$, $Hd = Dd$; H, D are symmetric with bounds $\eta_j 1 \leq H, D \leq \Omega_j 1$. For every gauge group, d^*Hd is hypercubic invariant; empty histories give $D = 1$.

For every flight (j,t), $j,t \geq 4$, $t \in \{j-1, j\}$, and difference word $|u| \leq j-2$,

$$\nabla_u W \in \mathcal{S}(\alpha_j, \Omega_j L^{-1-|u|}), \qquad \nabla_u T \in \mathcal{S}(\alpha_j, \Omega_j L^{-2-|u|}).$$

Its rolled kernel $\mathcal{P}_{\lambda_j,\lambda_t} \in \mathcal{S}(\alpha_j, \Omega_j)$; if $L \geq b_j$, its symbol is Hermitian with $\eta_j 1 \leq \hat{\mathcal{P}} \leq \Omega_j 1$ at every real momentum. *Status:* theorem. *Lean:* `pledges`.

Proposition III.3.2 (families). For every gauge group G and kernel choice K, a catalogue as above exists such that every sheet $\mathfrak{S}$ with kernel K admits a second table $(\alpha'_j, \Omega'_j, \eta_j)$. For every trajectory with that kernel and the sheet's stage profile satisfying, at every level and step respectively,

$$c_i \leq \gamma_{J(i)}, \qquad L_i \geq b_{J(i+1)},$$

there are pre-repair and output families simultaneously for all $N \geq 0$, $0 \leq i \leq N$. They share the explicit history metrics with amendments $\lambda_{J(i)}$ and flight kernels W, T of index $(J(i+1), J(i))$ for $i < N$; bare histories are empty and $D^{N,N} = 1$. Before repair, III.3.1's bounds and symmetry hold, as do (D) and (Δ) on the schedule torus pairs of each step, with $P \in \mathcal{S}(\alpha_j, \Omega_j)$ Hermitian and bounded below by η_j on $\operatorname{ran} \sigma_2$ at real momenta, $j = J(i+1)$. After repair, (D),(R) hold there; metric and difference bounds use the second table, retaining η_j, and

$$M_n, O \in \mathcal{S}(\alpha'_j, \Omega'_j), \qquad M_f \in \mathcal{S}(\alpha'_j, \Omega'_j L_i^{-2}).$$

Status: theorem under the credit condition (C1), (C2) as hypothesis. The variant `before_paid` (assigned to III.4) discharges it from the III.9 sheet and the III.2 payment record. *Lean:* `retained_support`.

Proposition III.3.3 (all tori). Every family satisfying III.3.2's output conditions, for any table, satisfies (D),(R) for every N, $m < N$ and positive sides c, f with $f = cL_m$, using the same kernels and level clips. *Status:* theorem. *Lean:* `complete_roll`.

Proposition III.3.4 (rolled positivity). For fixed kernel choice K and integers $j, t \geq 4$, there exist $C_0, B > 0$ depending only on these data. For every $\lambda \geq C_0$, $\mu \geq 0$, there exist $\eta, \sigma, \Omega > 0$ such that every flight (j,t) has $\mathcal{P}_{\mu,\lambda} \in \mathcal{S}(\sigma, \Omega)$; if $\mu B \leq L^2$, its symbol is Hermitian with $\eta 1 \leq \hat{\mathcal{P}} \leq \Omega 1$ at all real momenta. *Status:* theorem. *Lean:* `Flight.frontier`.

Proposition III.3.5 (filters). Define $\operatorname{rail}_M(z) = M^{-1} \sum_{s=0}^{M-1} e^{isz}$ for positive integers M, $\operatorname{plank}_M(z) = M^{-1} \sum_{|s| \leq (M-1)/2} e^{isz}$ for odd M, and $a_L(p) = \prod_a \operatorname{plank}_L(p_a)$. For all sizes where defined and all $z \in \mathbb{C}$, both line filters satisfy $F_L(Mz) F_M(z) = F_{LM}(z)$ and $|F_M(z)| \leq e^{M|\operatorname{Im} z|}$; plank is even, 2π-periodic and real on $\mathbb{R}$. Absolute $c_*, c_1 > 0$ give $|\operatorname{rail}_M(z)| \geq c_*$ when $M|z| \leq 3\pi/2$, and $|\operatorname{plank}_M(z)| \geq e^{-c_1 M/L}$ for real z, $M \leq L$, $L|z| \leq 3\pi/2$. Absolute B bounds both filters at $(q_\ell)_a$, $q_\ell = (p + 2\pi\ell)/L$, by $B/(1 + |\ell_a|)$ for odd L and $|\operatorname{Re} p_a| < 5\pi/4$, $|\operatorname{Im} p_a| < \pi/4$. *Status:* theorem. *Lean:* `plank_bind`, `principal_plank`, `alias_rail`.

Proposition III.3.6 (curvature covariance). Fix $\alpha, \eta > 0$, $\Omega \in \mathbb{R}$. A real 2-form kernel f is firm if $f \in \mathcal{S}(\alpha, \Omega)$, $\hat{f}(0) = 1$, and $\eta|\omega|^2 \leq \langle \omega, f_n \omega \rangle \leq \Omega|\omega|^2$ on every torus; symmetry is not assumed. On $U = \{p \neq 0 : |p_a| < 3\pi/2\} \subset \mathbb{R}^4$, set

$$\Phi_f = \sigma_1^* \hat{f} \sigma_1 + \sigma_0 \sigma_0^*, \qquad G_f = \sigma_1 \Phi_f^{-1} \sigma_1^*, \qquad G_\infty = s|p|^{-2} s^*,$$

where s wedges with ip. There exist $b > 0$ and $C_n \geq 0$ for every $n \in \mathbb{N}$, depending only on (α, Ω, η) and additionally on n for C_n, such that for every firm f, $p \in U$ and complex 1-form x,

$$\operatorname{Re}\langle x, \Phi_f x \rangle \geq b|p|^2 |x|^2, \quad \|D^n G_f\| \leq C_n |p|^{-n}, \quad \|D^n (G_f - G_\infty)\| \leq C_n |p|^{1-n}.$$

For scalar $f \in \mathcal{S}(\alpha, \Omega)$, $\hat{f} \in C^\infty$ with $\|D^n \hat{f}\| \leq A_n \Omega$ everywhere, $A_0 = 1$, $A_n = (n/\alpha)^n$ for $n \geq 1$. *Status:* theorems. *Lean:* `tank_full`, `gem_quote`, `radial_prices`.

Proposition III.3.7 (nonlinear calculus). *Status:* theorems.

(a) For $b, k \in \mathbb{N}$, $k \leq b$, there exists $C = C(b,k) \geq 1$. If $L \geq 1$, $M \geq 0$, and $p : \mathbb{Z}^4 \to E$, with E a real normed space, satisfies $\|\nabla_u p(x)\| \leq M L^{-1-|u|} (1 + L^{-1}|x|_1)^b$ for every x and $|u| \leq k$, then its degree-at-most-i Newton pieces $P_i(x) = \sum_{|\nu|=i} \binom{x}{\nu} \nabla^\nu p(0)$ obey $\|P_i(x)\| \leq C M L^{-1-i} (1 + |x|_1)^b$ for $i < k$; the remainder $p - \sum_{i<k} P_i$ obeys the same bound with $i = k$. *Lean:* `capital`.

(b) Let $d \in \mathbb{N}$ and $f : E \to F$ be C^d at 0 between real normed spaces, with finitely many continuous linear $p_v : E \to E$, $\sum_v p_v = 1$, integer weights $Y_v \geq 1$. If all derivatives through d of $f(\sum_v t^{Y_v} p_v z)$ vanish at 0 for every z, then $D^m f(0)(x_1, \dots, x_m) = 0$ for all $m \in \mathbb{N}$ whenever integers $k_i \geq 1$ satisfy $\sum_i k_i \leq d$ and $p_v x_i = 0$ for $Y_v > k_i$. *Lean:* `cancel`.

(c) Let $b, d \in \mathbb{N}$, $b \leq d$, $n \geq 2$, $o \in \Lambda_n$, and $X \subset \Lambda_n$ finite with $10 \operatorname{dist}_\infty(x, o) < n$ for every $x \in X$ (torus distance). Suppose $u : \mathrm{Config}(G, \Lambda_n) \to \mathbb{C}$ is gauge invariant, depends only on links with both endpoints in X, and $u \circ \exp$ is C^b at 0. If $u(U_{J,o}(t))$ has zero derivatives through b for every radial polynomial connection jet through weight d (degree $r - 1$ has weight $r \geq 2$), then $D^m(u \circ \exp)(0)(w_1, \dots, w_m) = 0$ for all $m \in \mathbb{N}$, integers $p_i \geq 1$, $\sum_i p_i \leq b$, whenever the link field w_i is polynomial of degree $\leq p_i - 1$ in centred initial-site coordinates on those links; other links are unrestricted. Here $U_{J,o}(t)$ denotes based holonomies with weight-r components multiplied by t^r. *Lean:* `wiped`.

(d) For every $j \in \mathbb{N}$, every family of homogeneous degree-$(j+1)$ polynomials $Q_a : \mathbb{R}^4 \to \mathfrak{g}$ $(a = 1, \dots, 4)$ decomposes as $Q_a = K_a + \partial_a r$, where $r = (j+2)^{-1} \sum_a x_a Q_a$ has degree $j + 2$ and K has degree $j + 1$ with $\sum_a x_a K_a = 0$. *Lean:* `locale`.

(e) There exists $C(G) > 0$: if $0 \leq b \leq 1$, $\tau \geq C(G) b^2$, v is a Lie-algebra link field with $|v_x| \leq b$ on links within distance 5 of Y, and plaquettes touching Y have chord distance $\leq \tau$ from 1 under $\exp(v)$, those bounds persist under $\exp(sv)$ for all $0 \leq s \leq 1$. *Lean:* `strings`.

3.6.2 2. What it uses

- I.1: lattice, gauge group and signed symmetries; III.1: block averages, clips and trajectories; III.2: sheets and payment.
- III.6: positivity, square roots and linear algebra F06–F09; III.10: repair and rigidity; III.16: Fourier kernels and weighted norms.
- Fourier/Bloch decomposition, Schur complements, lattice Koszul exactness, Newton interpolation and radial gauge. The penalty $\kappa > 0$ comes from K.

3.6.3 3. Strategy

Solve the Gaussian block-spin problem by alias sums. Cancel the singular principal-alias poles before inversion; smoothing and geometric dilation along histories give uniform strip bounds. Amend inverse metrics away from exact forms to make the defect positive, then construct the isometry blocks without changing W, T or the quadratic forms on potentials.

3.6.4 4. Steps

1. History filters have uniform alias decay and a principal-alias lower bound; geometric dilation controls products of filters. `Train.smooth_upper`.
2. Lagrange's identity gives the diagonal curvature metric from the auxiliary covariance; cancelling poles gives analytic invertible symbols. `Train.bend_firm`.
3. Alias algebra gives (D), including zero momentum. `Flight.tangent`.
4. Koszul exactness factors the defect through d_2; amendments replace its 3-form factor by $C + \lambda - \mu S^* S$, with $S^* S = O(L^{-2})$. `Flight.rebased`.
5. Fix the catalogue, impose the credit bounds and apply F06–F09 to construct (R); periodisation limits extend identities to all tori. `Bid.pinning`, `Echo.allAcross`.

6. Coercivity and the resolvent identity give the covariance comparison; weighted Taylor expansion and radial decomposition give the nonlinear vanishing results. `gem_quote`, `covers`.

3.6.5 5. Where the difficulty lies

Bounds must hold uniformly over arbitrary history lengths and block sizes. Positivity is established at real momenta before thinning the analytic strip. Harmonic forms require defect identities on all tori. Repair constants depend on direct clip coefficients; cross-kernel decay is measured on the coarse lattice.

3.6.6 6. Notation

$\Lambda_n = (\mathbb{Z}/n\mathbb{Z})^4$; d_r acts on real r-cochains, $d = d_1$, $\sigma_r = \hat{d}_r$, and $*$ is adjoint. $R \in \mathcal{S}(\alpha, \Omega)$ means, for every entry, $\sum_y e^{\alpha|y|_1}|R_{ab}(y)| \le \Omega$. $\hat{R}, R_n$ denote Fourier symbols and periodisations; aliases use centred representatives. $J(i)$ is the stage index and c_i the direct clip coefficient. D^n denotes full Fréchet derivatives with multilinear operator norm; ∇_u denotes fine forward differences. The cochain inner product counts cells. Histories comprise odd blocks $L \ge 3$, stage indices $J \ge 4$ dropping by at most one; a flight (j, t) adds one block to an index-j history, giving index $t \in \{j-1, j\}$. Q averages potentials, W minimises the quadratic action, T transports curvature, and M_f, M_n, O are the isometry blocks; C, S denote the unamended 3-form defect and transport. `Towers`.

3.6.7 7. What remains unexplained

The iterated Feynman-gauge interpretation remains probable. Gauge and harmonic zero modes, Gaussian determinants and integration-variable changes require further treatment in III.4 and III.10. F06–F07 and analytic coefficient extraction are inputs here. III.3.6 asserts no position-space continuum limit or contact-term statement.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.7 III.4. Gaussian integration: change of Gaussian sector, determinants, window law

Part: III, Ultraviolet construction (renormalization group). **Lean:** about 40 work packages. The main ones are `YM/SectorNext/GAUSSIAN_SECTOR_CHANGE/` (exact change of variables, gap, windows), `YM/AnalyticBridge/RELATIVE_LOGDET_TRACE/` and `YM/PhysicalNext/AFTERNOON_INTERACTING_RANK_COMPARISON/` (determinants and the polymer form of a sector), `YM/AnalyticBridge/ANNULUS_RATIO_INSERTIONS/`, `YM/SectorTools/PHYSICAL_GAUSSIAN_INSERTION/` and `YM/Primitives/CONNECTED_GAUSSIAN_CUMULANTS/` (Gaussian moments), `SECTOR_AFFINE_JETS/`, `SECTOR_COVARIANCE_JETS/` and `SELECTED_ORDERED_KERNELS/` (jets), `LOCAL_GAUSSIAN_LAWS/`, `GAUSSIAN_FIELD_ALIGNMENT/`, `YM/Push/FREE_RESOLVENT/` and `YM/RG08/Gaussian/` (locality, infinite-volume kernels), `YM/SectorNext/LARGE_FIELD_REMAINDERS/`, and the constant-producing statements in `Next/STATIONARY`, `MergeProof` and `PhysicalNext/V9_JOINT_ENDPOINT_LOAD`. **Size:** 37,956 lines in 4,737 declarations of the main-theorem closure.

Conventions. $x = \hat{\beta}_m$ in applications; $R(x) = \lceil (\log x)^2 \rceil$, $t(x) = x^{-1/2} e^{(\log\log x)^2}$, $\beta_F(x, L) =$

$x + s_G \log L$, $t_L(x) = t(\beta_F(x, L))$, $g = \beta^{-1/2}$ and $r_x = \lfloor R(x)/60 \rfloor$. Write $\|A\|_s$ for the maximum weighted row and column sums with weight $e^{s\,\mathrm{dist}(i,j)}$, using coarse positions. Matrix inequalities are quadratic-form inequalities. "For large x" means beyond a threshold depending only on previously fixed constants.

3.7.1 1. What this unit proves

Setting. One block-spin step S maps a fine torus Λ_f (side $f = Lc$, block size L odd) to a coarse torus Λ_c. The push-forward of the density at fixed coarse field V is decomposed (III.1, III.9) into sectors. A sector is given by:

- a store A: the large-field data, made of cost cells, aisles and flagged sites, with a partition of unity bill_A given by an allocation D;
- a choice k of gauge-fixing tags;
- a spanning forest F (tree gauge).

The integration variables are the free links of F, with product Haar measure. In each sector the unit splits the free links as follows (`chill`):

- **frozen links** $\mathcal{F}_A$: a parent block of an endpoint lies within distance $10R(x)$ of A. These are kept as Haar variables Y.
- **active links** $\mathcal{A}_A$: all others. These are written $u_e = \exp(a_e)\,\nu_e$, with $a_e \in \mathfrak{g}$ and $\nu = \iota V$ the copy of V on the fine lattice (`replica`).

We write $n_a = |\mathcal{A}_A| \dim \mathfrak{g}$.

On selected stencils, the step energy contains a sum of affine squares $Q_Y(a) = \frac{1}{2}|Ma + s(Y)|^2$ (`guide`):

- M is a linear map from active coordinates to rows of linearised plaquette curvatures and block constraints;
- $s(Y)$ depends on the frozen links.

From these the unit defines three objects:

- the **precision** $W = M^\top M$ (`Rows.gram`);
- the **covariance** $C = W_{\mathcal{W}}$, the symmetrised windowed inverse of W, with windows of coarse radius $R(x)/20$ (`fog`);
- the **mean** $m(Y) = -(W^{\mathrm{per}})^\top M^\top s(Y)$ (`meanLog`). This is a window approximation to the minimiser $-W^{-1}M^\top s(Y)$; its gradient need not vanish.

Proposition III.4.1 (exact Gaussian change of variables in one sector). Fix the following data:

- a step S and a tube T satisfying the funding conditions of a teller: the prescribed kernel and stage, the multiscale metric bounds, the prescribed fine and coarse cutoffs, bounds on both path families, and compatible free metrics and towers;
- a representation R of the fine density whose paths satisfy their bounds and whose activities satisfy the validity bounds, with $\beta = \beta_R$;
- x, V, k, A, F, an allocation D and a cutoff profile χ;
- a chart ratio J, meaning:
  - $J_{\mathrm{sc}} > 0$ and $\rho_J > 0$, and J is measurable;

  - J is real-analytic and positive on a ball $B_{\rho_J} \subset \mathfrak{g}$;
  - $J(0) = 1$;
  - exp is injective on B_{ρ_J};
  - $J_{\mathrm{sc}}\, J(a)\, da$ is mapped by exp onto normalised Haar measure restricted to $\exp B_{\rho_J}$;
  - $\exp B_{\rho_J}$ is open;
- a number $d > 0$.

Let C be a matrix on the active coordinates and m a map from frozen arrays to active coordinates. Assume they are *permitted* (`Permitted`):

- $C \succ 0$, $\beta > 0$, m measurable, and $4d < \rho_J$;
- the *standing condition* (`Standing`): whenever $\mathrm{bill}_A \neq 0$ and the gauge tags are eligible, every active link equals $e^{a_e}\nu_e$ with $|a_e| \le d$.

Let $\Phi_{A,k} = \mathrm{bill}_A \cdot \mathrm{exposed}_R \cdot \mathrm{enters}_{T,\beta,k,V}$ be the sector integrand. Here $\mathrm{exposed}_R$ is the interacting density of R, and enters is the tagged block-spin kernel times $e^{-\beta \cdot \text{gap energy}}$. Then the following hold (`Licenses`):

- (a) $\Phi_{A,k}$ is integrable on the slice;
- (b) for every Y, $\Psi(Y,\cdot) \in L^1(\gamma_C)$, and $\Psi \in L^1(\mathrm{Haar}^{\mathcal{F}_A} \otimes \gamma_C)$;
- (c) the identity

$$\int \Phi_{A,k}\, d\mathrm{Haar}^F = \mathrm{pref}_S(\beta)\, J_{\mathrm{sc}}^{|\mathcal{A}_A|}\, (2\pi g^2)^{n_a/2} (\det C)^{1/2} \int \mathrm{Haar}^{\mathcal{F}_A}(dY) \int \gamma_C(dz)\, \Psi(Y,z).$$

Here $\gamma_C = N(0,C)$. Set $a = m(Y) + gz$, and let U be the configuration with frozen links Y, active links $e^{a_e}\nu_e$ and tree links 1. Then (`dyed`)

$$\Psi(Y,z) = \tilde{J}_d(a)\, \chi_d(a)\, \mathrm{bill}_A\, \mathrm{lik}_k(U)\, \mathrm{exposed}_R(U)\, \exp\Big(-\beta\, \overline{\mathrm{res}}(Y,U,a) - \beta Q_Y(m(Y)) + \eta\cdot z + \tfrac{1}{2} z\cdot(C^{-1} - W)z\Big).$$

In this formula:

- $\eta = -g^{-1}(Wm(Y) + M^\top s(Y))$;
- $\overline{\mathrm{res}}$ is the orientation-averaged step energy minus the affine squares;
- $\tilde{J}_d(a)$ is the product of the extended link Jacobians, each equal to $J(a_e)$ for $|a_e| \le 2d$; χ_d is the product cutoff, equal to 1 when every $|a_e| \le d$.

The identity holds for every permitted (C, m) without quantitative locality assumptions.

Status: theorem.

Lean: `authorized_change`.

Proposition III.4.2 (licence of every committed RG step; the transfer identity). Let $\mathcal{R}$ be a directed RG run (III.9) whose delays satisfy the rate conditions and the per-stage chart conditions (`Starcash`). Let $m < N$, let p be a period, and let R be a representation at level $m+1$ on its selected history with $\beta_R \in [\hat{\beta}_{m+1}/2,\, 2\hat{\beta}_{m+1}]$. Write:

- S for the step from level $m+1$ to level m;
- $x = \hat{\beta}_m$ and $\beta = \beta_R$;
- $j = J(m+1)$ and $i = J(m)$ for the stage indices;
- $d = a_{ji}\, t(\hat{\beta}_{m+1})$, and w for the allocation width at this stage pair.

Then:

- $w > 0$, $d > 0$, $\beta > 0$, $E_{ji} > 0$, $2d < \rho_{\text{first}}$ and $4d < \rho_J$;
- $|\log J(t)| \le E_{ji}|t|^2$ for $|t| \le 4d$;
- there is $\kappa > 0$ such that for all allocations D of width w, forests H and profiles χ:
  - (i) writing I_A for the signed sector column, its contribution to the push-forward, and L_A for the nonnegative column with absolute fine density, after removing the selected packet energy and kernel prefactor, both have bounded Borel envelopes and $|I_A(V)| \le \mathrm{pref}(\beta) e^{-\beta\kappa t_L(x)^2|A|} L_A(V)$ (`PortionCost`);
  - (ii) for all A, k, V, the chart is admitted (`Admitted`); when $\mathrm{flags}(A) = \mathrm{flags}_{t(x)}(V)$, also $C = W_{\mathcal{W}} \succ 0$, $W \succ 0$ and the selected predictor rows have analytic derivatives with a common bound; and the conclusions of III.4.1 hold with $C = W_{\mathcal{W}}$ and the windowed mean for every store (for wrong flags trivially, since $\mathrm{bill}_A \equiv 0$);
  - (iii) for every V:

$$\mathcal{S}_\beta(\rho_R)(V) = \sum_A \mathrm{sieve}(V) \sum_k \mathrm{pref}\, J_{\mathrm{sc}}^{|\mathcal{A}_A|} (2\pi g^2)^{n_a/2} (\det C_{A,k,V})^{1/2} \iint \Psi \, d\gamma_C \, d\mathrm{Haar}.$$

Here $\mathcal{S}_\beta$ is the block-spin push-forward, ρ_R is the fine density, and $\mathrm{sieve}(V) = \exp(-\beta \cdot$ orientation-averaged coarse energy of $V)$.

Status: theorem, conditional on the run, history and band hypotheses. III.4.12 supplies the rate and chart conditions; III.9 supplies the selected history.

Lean: `committed`, `displaced_wrong`.

Proposition III.4.3 (gap and locality of the actual precision and covariance). Fix G, a patch $P = (L, J)$ and a teller whose kernel satisfies the branch condition. There are $v, \eta, b, E > 0$ and $K \ge 0$ (post-L constants) with the following property. For large x, take any step with patch P and coarse side $c \ge 100(1 + R(x))$, any funded tube, and any A, F, k, V with $\mathrm{flags}(A) = \mathrm{flags}_{t(x)}(V)$. Then:

- (i) $\eta|a|^2 \le a \cdot Wa$ for all active a (`Gap.not_flat`);
- (ii) W has exponential off-diagonal decay in the Schur norm: $\sup_i \sum_j e^{v\,\mathrm{dist}(i,j)} |W_{ij}| \le K$, and the same for columns, where dist is the coarse distance between the parent blocks of the coordinates;
- (iii) $\|C - W^{-1}\|_0 \le E\, e^{-br_x}$, and $C \succ 0$;
- (iv) $C \le K'$ for some post-L constant K' (`open_price`).

The constants do not depend on the volume, the store, the forest or the field.

Status: theorem.

Lean: `opaqueCost`.

Proposition III.4.4 (window law). Let $\sigma, \eta > 0$, $K \ge 0$, and fix a nonnegative function $B : \mathbb{R} \to \mathbb{R}$. There are $u, E, g > 0$ such that for every sufficiently large integer r the following holds, uniformly in the torus and the finite coordinate set. Let the coordinate positions p_j satisfy $\sum_j e^{-v\,\mathrm{dist}(i,p_j)} \le B(v)$ for every coarse site i and every $v > 0$ (`Revenue`). Let A be a finite symmetric matrix with:

- Schur decay $\|e^{\sigma\,\mathrm{dist}} A\|_{\mathrm{Schur}} \le K$;
- spectral gap η;
- windows $\mathcal{W}_i$ that contain the r-ball around i.

Let L be the windowed inverse, $L_{\cdot i} = (A|_{\mathcal{W}_i})^{-1} e_i$ extended by zero, and $A_{\mathcal{W}} = \frac{1}{2}(L + L^\top)$ (`clouded`). Then:

- $\|A_{\mathcal{W}} - A^{-1}\|_{\mathrm{Schur}} \le E e^{-ur}$;
- $A_{\mathcal{W}} \ge g = \eta/(2(K+1)^2)$ and $A_{\mathcal{W}} \succ 0$.

A strengthened form (`same_purchase`) gives a rate $s \in (0, \sigma]$ and a constant K_1 such that A, A^{-1}, L and $(A_{\mathcal{W}})^{-1}$ all have s-weighted Schur norm $\le K_1$, and $\|e^{s\,\mathrm{dist}}(L - A^{-1})\|_{\mathrm{Schur}} \le K_1 e^{-sr}$.

Status: theorem. The constants depend only on (σ, η, K, B).

Lean: `Schur.windows`.

Proposition III.4.5 (log-determinants as local path series). Let $A \succ 0$ with windows $\mathcal{W}$ ($i \in \mathcal{W}_i$). Let P_u be the column-wise windowed inverse of $A+uI$, and $\Delta_u = (A+uI)P_u - I$. Assume the following bounds (`Priced`), at a fixed nonnegative Schur rate, with $C \ge 1$ and $0 \le q < 1$, for every $u \ge 0$:

- $\|A\| \le C$;
- $\|(A+u)^{-1}\|,\ \|P_u\| \le C/(1+u)$;
- $\|\Delta_u\| \le q/(1+u)$;
- $\|S\| \le q$, where $S = AA_{\mathcal{W}} - 1$.

Assume also $A_{\mathcal{W}} \succ 0$. Then

$$\log\det A = \sum_i \Big(\ell_i + \sum_{n\ge 0} r_{i,n}\Big), \qquad \log\det(1+S) = \sum_i \sum_{n\ge 0} \frac{(-1)^n (S^{n+1})_{ii}}{n+1}, \qquad \log\det A_{\mathcal{W}} = -\log\det A + \log\det(1$$

with:

- $\ell_i = \int_0^\infty [(1+u)^{-1} - (P_u)_{ii}]\,du = (\log A|_{\mathcal{W}_i})_{ii}$, the single-window letter;
- $r_{i,n} = -\int_0^\infty (P_u(-\Delta_u)^{n+1})_{ii}\,du$ and $\int|\cdot| \le C q^{n+1}$.

All series converge absolutely. Moreover (`labeled`), the matrix products expand into closed index paths. A term $r_{i,n}$ has one P_u factor and $n+1$ defect factors, hence $n+2$ jumps; the corresponding trace-series term has $n+1$ jumps. The absolute values, summed over the intermediate indices, are bounded by Cq^{n+1} and $q^{n+1}/(n+1)$, respectively.

Status: theorem.

Lean: `Priced.chains`.

For the relative expansions, the sector regularity conditions are: a lawful representation; matching fine paths and cutoff; a local chart with $d > 0$, $4d < \rho_J$ and $2d < \rho_{\mathrm{first}}$; positive coupling and sector width; $0 < t_L(x) \le 1$; output threshold $t(x)$; and the tube's geometric, local-readability, predictor-derivative, covariance-positivity and spectral-gap conditions (`Chips`, `Boosted`).

Proposition III.4.6 (relative normalisation of a sector and its localisation). Let K_F be the flat kinetic precision: the Gram matrix of the move at $V = 1$ with empty store, on all free links of F.

(i) If the fine path family satisfies its path bounds and the kernel satisfies the patch's branch condition, $\det K_F$ does not depend on the forest or on the gauge tags (`Native.shared`).

(ii) Assume:

- the tube is funded, and its fine path family and cutoff are those of R;
- the precision bounds of Lemma 4.21 of the detailed notes holds;
- J is a local chart ratio, $\beta > 0$, and the threshold of the output budget point is $t(x)$.

Then on every sector point $p = (A, F, k, V, Y, z)$ with matching flags, the positive density factor, with the sector masks and polymer product kept outside it, factorises as

$$\mathrm{freight}(p) = \mathrm{loading}(V)\, e^{\Lambda(p)}, \qquad \mathrm{loading}(V) = \mathrm{banker}(V)\, \mathrm{pref}(\beta)\, e^{v_R|\Lambda_f|}\, \mathrm{spare}(\beta)^{|F^{\mathrm{free}}|} (\det K_F)^{-1/2} > 0.$$

Here $\mathrm{freight} = J_{\mathrm{sc}}^{|\mathcal{A}|}(2\pi g^2)^{n_a/2}(\det C)^{1/2} e^{\frac{1}{2} z\cdot C^{-1} z}\, \tilde{J} \cdot (\mathrm{noise}) \cdot e^{\mathrm{exponent}_R}$, and $\mathrm{spare}(\beta) = (2\pi g^2)^{\dim \mathfrak{g}/2} J_{\mathrm{sc}}$ is the Gaussian normalisation per link. The exponent is

$$\Lambda(p) = -\beta\, \mathcal{E}(p) + \eta \cdot z + \tfrac{1}{2} z \cdot (C^{-1} - W) z + \kappa(p) + \sum_e \log \tilde{J}(a_e) + (\mathrm{tails}),$$

$$\kappa(p) = -|\mathcal{F}_A| \log \mathrm{spare}(\beta) - \tfrac{1}{2}(\log\det W - \log\det K_F) + \tfrac{1}{2}\log\det(WC).$$

Here $\mathcal{E}$ is the orientation-averaged row energy, including the subtraction of the affine energy at a and its addition at m:

$$\mathcal{E} = E_{\mathrm{noise}}(U, V) + E^{\mathrm{cut}}_{\mathrm{fine}}(U) - E^{\mathrm{cut}}_{\mathrm{coarse}}(V) - Q_Y(a) + Q_Y(m).$$

The tails are the remaining fine exponent after removing its volume constant and restoring its cut energy, together with the coarse cutoff correction.

(iii) Assume in addition the sector regularity conditions stated above, a nonnegative polymer exponent, and both path-counting bounds of Lemma 4.21 of the detailed notes. Then $\kappa(p) = -|\mathcal{F}_A| \log \mathrm{spare} + \sum_e \mathrm{sect}_e + I + \mathrm{Sh}$ (`book_price`), where:

- $\mathrm{sect}_e = -\frac{1}{2}\sum_{\mathrm{colours}} [\ell_e(\mathrm{actual}) - \ell_e(\mathrm{flat})]$, the difference of the single-window letters of III.4.5 for the identity-completed full precision at (A, V) and at $(\emptyset, 1)$;
- I is the corresponding difference of path letters;
- $\mathrm{Sh} = \frac{1}{2}\log\det(1 + S)$.

(iv) Regrouping (`rate_pass`) gives $\kappa = \sum_o \mathrm{rent}_o + \mathrm{far} + I + \mathrm{Sh}$:

- o runs over the cores of A (the components formed by overlap of the $500R(x)$-halos of its seeds);
- rent_o is the sum of $\mathrm{sect}_e - \log \mathrm{spare} \cdot 1_{e\in\mathcal{F}_A}$ over links within $200R(x)$ of o;
- far is the same sum over links farther than $200R(x)$ from A.

Status: theorem.

Lean: `Dyed.cardinal_spin`, `Dyed.keptEarly`, `Dyed.spoolLift`.

Proposition III.4.7 (polymer form of a sector, with large-field suppression). Fix:

- a teller and a patch D satisfying the branch condition;
- a determinant-jet fund and a local chart ratio J;
- a polymer exponent $e \geq 0$, a chart-width multiplier $h \geq 0$, constants $a, \alpha, \tau > 0$ with $2\alpha \leq a/2$, and bound constants $C_0 \geq 0$, $g_0, b_0, d_0 > 0$, $\eta_0 \in \mathbb{R}$.

Then for large x the following holds. Let $y = \mathrm{retime}(x)$, $z = \beta_F(x, D.L)$ and $\mathrm{hot}(s) = e^{(\log\log s)^{3/2}}$. Eventually $x \leq y \leq x^2$ and $t(y) = x^{-1/2}\mathrm{hot}(x)$. Take a step with patch D, a sector datum with chart J, and positive cutoff parameter W_0, subject to the following assumptions:

- the representation is lawful; the tube has the same fine paths and cutoff as the representation and satisfies the funded, geometric and chart-admission conditions; the chart radius $d = h\,t_{D.L}(y)$ satisfies $d > 0$, $4d < \rho_J$ and $2d < \rho_{\text{first}}$; the output threshold is $t(y)$;
- the step satisfies the free-metric size condition and $c \geq 100(1 + R(y))$, and its fine and coarse path families have a common path bound;
- $\beta \in [z/2, 2z]$; the fine threshold $s = z^{-1/2}\text{hot}(z)$ also lies in $[t_{D.L}(y)/2, 2t_{D.L}(y)]$;
- on each eligible packet the row energy $\mathcal{E}$ is at least as^2 times its size, whenever the sector cutoffs are nonzero;
- the support exponents satisfy the reserve inequality with parameters (e, τ); the budget radius B_0 satisfies $25R(y) + 30 \leq B_0 \leq \text{walls}(y)$, and the fine budget radius is at most $2B_0$;
- the lawful-representation bound can be chosen with activity constant $\Gamma \leq \beta_F(y, D.L)^{1/8}$ and smallness parameter $\delta \leq y^{-1/2}$, and the fine paths satisfy the required row-counting bound;
- the determinant-jet bounds at halo 20, exponential order 2 and derivative order 1, and the precision and two path-counting bounds of Lemma 4.21 of the detailed notes hold at y, with constants $(g_0, C_0, \eta_0, b_0, d_0)$;
- uniformly over flag-matched sectors and frozen arrays, $\|C\|_0$ is bounded by a common constant, every component of m and of $Mm + s(Y)$ has absolute value at most 1, and $|\eta_i| \leq y^{-4}$; on the chart-cutoff support, $|z_i| \leq y^2$.

For such data there is a decomposition (`Plate`)

$$\Lambda(p) = \sum_{o \in \text{cores}(A)} \text{base}_o(p) + \sum_X \text{aux}_X(p)$$

whenever the flags match and the product of the fine-field, block, gauge-tag and chart cutoffs is nonzero. The bounds below hold for every flag-matched p, including points outside that support. In it:

- base_o depends only on the data in the $500R(y)$-neighbourhood of o;
- aux_X depends only on the data near the local support X;
- both are gauge-invariant and lattice-covariant;
- $\text{base}_o \leq -\alpha\,\text{hot}(z)^2|o|$, where $|o|$ counts cost cells, aisles and flags;
- $|e^{\text{aux}_X} - 1| \leq f_X$, with $f \geq 0$ satisfying the polymer bound $\sum_{X \ni y'} f_X e^{e\,N_{B_0}(X)} \leq y^{-1/48}$ at every site y', where N_{B_0} is the cover number at the output budget radius (`Hit`).

Status: theorem.

Lean: `foreshadows`, with hypotheses `Helps`.

Proposition III.4.8 (moments under the sector law). Fix G. For every covariance cap $L > 0$ and $k \in \mathbb{N}$ there is $D \geq 1$ with the following property. Take:

- any finite index set I and any probability space (Ω, μ);
- any $C \succeq 0$ on $\mathbb{R}^{I \times \dim \mathfrak{g}}$ with $C \leq L$;
- any finite family $(r_i)_{i \in Y}$ of measurable insertions with $w_i, d_i \geq 0$, $|r_i(\omega, z)| \leq w_i e^{d_i(1 + |z_{a_i}|^2 + |z_{b_i}|^2)}$, and $\sum_{i:\, x \in \{a_i, b_i\}} d_i \leq c = 1/(24L)$ at every coordinate site x;
- any $\rho, H \geq 0$ and sets $P, J \subset I$;
- any measurable v with $|v| \leq H(1 + \sum_{i \in P} |z_i|^2)^k$ that vanishes unless $|z_i| \geq \rho$ for all $i \in J$.

Then $f = v\prod_{i\in Y} r_i \in L^1(\mu\otimes\gamma_C)$ and

$$\int |f|\, d(\mu\otimes\gamma_C) \le H\, D\, (1+|P|\dim\mathfrak{g})^{2k+1}\, e^{-c|J|\rho^2}\, 2^{\dim\mathfrak{g}\,|J|} \prod_{i\in Y} (e^c\, 4^{\dim\mathfrak{g}} w_i).$$

In a sector, take $\mu = \mathrm{Haar}^{\mathcal{F}_A}$ and $C = W_{\mathcal{W}}$; III.4.3 supplies a uniform covariance cap. Moreover:

- (`r4`) For measurable q_i on the same product space, let $|q_i| \le x_i^2(1+|z_{a_i}|^2+|z_{b_i}|^2)$ with $0\le x_i\le 1$ and $\sum_{i\ni \mathrm{site}} 8x_i \le c$. Then $\int(\prod_i(e^{q_i}-1))^4 \le (\prod_i Mx_i)^4$, with $M = M(G,L)$.
- (`oasis`) Fix a patch, a teller satisfying its branch condition and a covariance cap $L>0$. For large x, every funded step with that patch and $c\ge 100(1+R(x))$, matching flags, $0<d\le 1$, $2d<\rho_{\mathrm{first}}$ and $\beta\in[\beta_F(x,L_{\mathrm{block}})/2, 2\beta_F(x,L_{\mathrm{block}})]$ has the fourth-moment insertion bounds above. The single-site exponents come from $\eta\cdot z$ and the pair exponents from $\frac12 z\cdot(C^{-1}-W)z$. Their weights are $De^{-uR(x)/480}$ for sites and exponentially decaying in distance for pairs; the constants are fixed before the threshold in x, which may depend on the covariance cap.

Status: theorem.

Lean: `same_law`.

Proposition III.4.9 (Gaussian insertions supported on selected coordinates). Fix $p\ge 1$, a finite jet order, G, a patch and a teller satisfying the branch condition. There are post-L constants $K, M, s, b > 0$ and $B\ge 0$ such that, for large x, the following holds. Take a funded step with that patch, $c\ge 100(1+R(x))$ and matching flags. Fix any $g, d, \theta > 0$, frozen array and slot set U. Restrict the linear and quadratic correction to U, obtaining $t\cdot z_U + \frac12 z_U\cdot \mathcal{J} z_U$, where $t=\eta$ and $\mathcal{J} = C^{-1}-W$. Its tilted law is $N(\mathcal{A}t_U, \mathcal{A})$, with $\mathcal{A} = (C^{-1}-\mathcal{J}_U)^{-1} \succ 0$. The likelihood ratio of the U-marginals satisfies

$$\mathbb{E}_{\gamma_{C,U}}[(dN_U/d\gamma_{C,U})^p] \le \exp\Big(2p(|U|\ell^2/(4\theta) + 2K|U|(\theta+\varepsilon/2))\Big)$$

whenever $Kp(\theta+\varepsilon/2)\le 1/4$. The parameters are:

- $\varepsilon = Ke^{-bR(x)}$;
- $\ell = g^{-1}M^2 e^{-sr_x} B\,(t(x)+d)$.

The restriction is on z; coefficients may depend on all background and frozen data.

Status: theorem.

Lean: `Physical.blocks`, built on `insert_letter`.

Proposition III.4.10 (jets of the block coefficients and of the covariance). Fix G, a patch P, a teller satisfying the branch condition, and an order n. There are $r, d, v, \eta, \lambda_0, C_* > 0$ such that for large x the following holds. Take every funded step with patch P and $c\ge 100(1+R(x))$, every A, F, V with matching flags, every k, and every logarithmic perturbation B of the background with $\|B\|\le r$. The store, forest and tags remain fixed as B varies. Then:

- the move matrix $M(B)$ is real-analytic;
- M, $W = M^\top M$ and C satisfy weighted Schur bounds up to order n. For every list ℓ of at most n coordinate directions, with coarse positions p_b, and output and input positions p_i, p_j,

$$\prod_{b\in\ell} e^{d\,\mathrm{dist}(p_b,p_i)} \sum_j e^{d\,\mathrm{dist}(p_i,p_j)} |\partial_\ell X_{ij}(B)| \le C_*,$$

and the analogous column bound holds with the external weights anchored at p_j;

- $W(B) \geq \eta I$, $C(B) \geq \lambda_0 I$ and $C(B) \succ 0$;
- for every $u \geq 0$, the full, principal and windowed resolvents have the same marked jet bounds with cap $C_*/(1+u)$ at order zero and $C_*/(1+u)^2$ at positive orders;
- the window defect has marked jet cap $C_* e^{-vR(x)}/(1+u)$, and the difference between the windowed and full resolvents has cap $C_* e^{-vR(x)}/(1+u)^2$.

The selected predictor rows are real-analytic, with k-jets bounded by constants depending only on (G, P, K, k), on a ball of fixed radius (`sounds`, `chosen_branch`). The precision W is also real-analytic; covariance and resolvent conclusions are finite-order jet bounds.

Status: theorem.

Lean: `assembly`, `fromTeller`, `coverall`, `stores`.

Proposition III.4.11 (affine squares in the large-field annulus). Fix a teller, a block size L, $W_0 > 0$, $K \in \mathbb{R}$ and an integer halo $D \geq 0$. There are $C \geq 1$ and an eventually nonnegative function $e(x)$ with $(1+R(x))^n \beta_F(x, L) e(x) \to 0$ for every $n \in \mathbb{N}$, such that for large x the following holds. Take a funded step of block size L with $c \geq 100(1+R(x))$, stores O, A, a forest F, fields U, V and tags k. Assume U is in tree gauge. In the $(D+5)R(x)$-halo of O, assume the following local sector conditions: fine plaquettes whose support meets that halo and which belong to A have toll at least $t_L(x)/4$, those outside A have toll at most $t_L(x)/2$; block bonds whose support meets that halo and which belong to A have discrepancy at least $W_0 t_L(x)$, those outside have discrepancy at most $2W_0 t_L(x)$; the coarse flags match V; and the chosen frames have positive mass. Then there is a direction a, vanishing on the tree links, such that for every pair of windows H, Q with $H_{\mathrm{face}} \subseteq Q_{\mathrm{face}}$ and $H_{\mathrm{noise}} \subseteq Q_{\mathrm{noise}}$, and with the face, noise and coarse rows of Q in $\{p \notin A^{(2R)},\ p \in O^{(DR)}\}$:

- Q is charted by a with constant C;
- the affine series on H converge, the required curvature and predictor derivatives exist with the stated linearisation, and $\mathrm{work}_Q \geq \mathrm{squares}_H(a) - e(x)|O|$;
- for every other a' charting H with constant K, the same convergence, derivative statements and bound hold with a'.

Here $|O|$ is the number of cost cells, aisles and flags. The comparison with a' uses uniqueness of the logarithm on the selected stencils.

Status: theorem.

Lean: `annulus_pay`, with `cleans_at` and `reacts`.

Proposition III.4.12 (production of the step constants). Assume:

- G is simple;
- the kernel choices K satisfy the branch condition for every patch;
- the initial stage is at least 4; the jet orders, position orders and support exponents are nondecreasing, the support exponents are positive, and at stage i the two orders are at least $\max(6, i)$ and $\max(2, i)$, respectively;
- costs $C_i \geq 1$ are given.

Then there are pre-L data with kernel K and the prescribed orders, two families of path prices, free-field and slope bounds, and a cost bound, with the following property. For every tariff and scope there is a sequence of block-size floors. After any increase of those floors and any choice of block sizes above them, there is a stagewise Hessian spectral gap η. For every compatible post-L cost choice (spectral gap η, activity cap at least 1, and slope bound below the chosen cap), protection and initial delays, the delays can be enlarged so that the rate and chart conditions hold for all further enlargements. On every paid run with these data there are stationary path families, including the full stationary estimates in the retained family. For each adjacent stage pair (j, i) above the initial stage, the per-step chart conditions hold with a_{ji}, E_{ji} chosen after L_j (`sponsored_permit`). `issue_cosmos` replaces each entry value by the maximum of its old value and the thresholds for (j, j) and $(j, j-1)$, preserving the rate conditions.

Status: theorem under these kernel and order assumptions.

Lean: `sailsV`, using `geared`.

3.7.2 2. What it uses

- **III.1:** block-spin maps, tree gauge, tags and predictors (`StepData`); the branch condition requires unique selected minimisers and an eligible axial gauge.
- **II.2, III.5, III.8:** representations, activity bounds, large-field allocations and polymer summability (`Rep.Lawful`).
- **III.10:** funded tubes, work rows, cutoffs, path bounds and compatible free metrics (`Funds`).
- **III.3, III.6–III.7:** free-field estimates, stationary paths and fourth-order matching (`FullPaths`).
- **III.2, III.9:** ordered constants, directed runs, selected histories and coupling bands.
- **I.1, III.16:** Haar measure, exponential charts, analytic jets and Gaussian measures.
- **Standard results:** Gaussian affine transport, Combes–Thomas, the resolvent integral for log, Neumann series, Gaussian integration by parts and analytic-strip Fourier decay.

Constants are chosen before L, after L, then before the threshold in x. Packet-energy, support-exponent, chart, noise and representation assumptions in III.4.7 remain separate inputs.

3.7.3 3. Strategy

Keep links near the store as Haar variables and put the remaining links in exponential coordinates. Split off affine squares, shift by the windowed mean and divide by a Gaussian density; the linear and quadratic errors remain in the integrand. A local Poincaré inequality gives the precision's spectral gap, and decay estimates make its windowed inverse positive and exponentially close to the full inverse.

Expand relative log-determinants into window and path terms, then collect terms near large-field cores. Packet energy absorbs their logarithmic cost; the remainder has a small polymer norm. Gaussian moments control exponential insertions, marked jets control background derivatives, and annulus estimates recover affine squares away from the store.

3.7.4 4. Steps

1. **Chart transport and Gaussian scaling.** Under III.4.1's support and chart hypotheses, extended Jacobians and cutoffs preserve the sector integral and its integrability. Translate

and dilate the Lebesgue variables, then expand the affine square to obtain the exact Gaussian formula. Lean: `authorized_change`.

2. **Precision and windows.** Under III.4.3's hypotheses, a linkwise gauge choice compares predictor derivatives with the flat field, and the local Poincaré estimate absorbs the errors. Combes–Thomas bounds the window defect, giving III.4.4 and positivity. Lean: `Gap.not_flat`, `Schur.windows`.

3. **Admission along the run.** Under the funding and size hypotheses, axial gauge bounds active-link chords by a constant times $t_L(x)$, so the logarithmic chart gives the standing condition. With III.4.2's run and band assumptions, enlarge stage thresholds, apply the sector identity to matching flags, and use vanishing for wrong flags to recover the push-forward. Lean: `matinee`, `committed`.

4. **Relative determinants.** Identity completion puts actual and flat precisions on the same coordinates. Under III.4.5–III.4.6's assumptions, the resolvent integral and Neumann series give window letters and closed paths, which are regrouped near cores. Lean: `Priced.chains`, `rate_pass`.

5. **Core suppression.** Under III.4.7's hypotheses, packet energy dominates core normalisation costs; the remaining terms have powers $y^{-1/8}$, $y^{-1/24}$ or exponential decay. Clamping preserves core values on the cutoff support, and symmetrisation gives the $y^{-1/48}$ bound. Lean: `foreshadows`. *(proof read; some supporting path and interaction bounds statement only)*

6. **Gaussian moments and local insertions.** Under III.4.8–III.4.9's assumptions, exponential quadratic moments and polynomial moments control insertions and tails. A correction supported on selected coordinates has a likelihood ratio determined by those marginals. Lean: `same_law`, `insert_letter`. *(moment and insertion proofs read; marginal-support assertion statement only)*

7. **Jets.** Under III.4.10's hypotheses, transplant predictor stencils to a fixed torus and use a local gauge to obtain uniform analytic rows. Perturbation preserves the precision's spectral gap and supplies the weighted inverse and window jet bounds. Lean: `sounds`, `assembly`. *(predictor and precision proofs read; transport and cut-error bounds statement only)*

8. **Annulus squares and constants.** Under III.4.11's hypotheses, clean logarithmic coordinates give cubic row errors; summing the rows and discarding unselected positive squares proves the annulus bound. Under III.4.12's assumptions, choose stationary paths and block sizes, then enlarge delays and chart thresholds. Lean: `annulus_pay`, `sailsV`. *(assembly and constant selection proofs read; clean-field construction and row recovery statement only)*

3.7.5 5. Where the difficulty lies

The spectral gap requires a different gauge comparison for each link; its smallness radius comes from continuity. Inverse bounds spend decay rate while remaining uniform in volume and sector. Relative determinants compare different active dimensions, and the large-field threshold must absorb their core costs while preserving polymer summability.

3.7.6 6. Notation

Symbol	Lean	Meaning
γ_C	`Gaussian.cloud`	Centred Gaussian $N(0, C)$.
$\mathcal{F}_A, \mathcal{A}_A$	`chill`	Frozen and active free links.
$W = M^\top M$	`Rows.gram`	Precision of the affine squares.
$C = W_{\mathcal{W}}$	`fog`	Symmetrised windowed inverse.
$m(Y)$	`meanLog`	Windowed mean.
$\kappa(p), \Lambda(p)$	`keptEarly`	Relative log-normalisation and full log-integrand.

$N_{B_0}(X)$ is the cover number; store size counts cost cells, aisles and flags.

3.7.7 7. What remains unexplained

The simultaneous supply of III.4.7's packet-energy and support-exponent assumptions on the selected run, and the transfer of stationary paths to work rows, remain unexplained. Several supporting kernel, counting, transport and clean-field proofs were read only as statements. The free-kernel comparison is for flat sectors without frozen links under compatibility assumptions; no direct comparison for arbitrary interacting sectors was located. Downstream reset and repeatable-step results and the additional assigned packages remain unchecked; precise bibliographic references remain open.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.8 III.5. Cluster expansion of the polymer gas

Part: III, Ultraviolet construction (renormalization group). **Lean:** work packages `YM/Push/ABSOLUTE_SUMS/` (3,624 lines), `YM/AnalyticNext/CONNECTED_ACTIVITY_BOUNDS/` (3,589), `YM/Push/SECTOR_MEASURE/` (1,153), `YM/SectorNext/VACUUM_NORMALIZATION/` (935), `YM/Progress/STATIONARY_SCENES/` (445). **Size:** 9,746 lines in 1,097 declarations used by the main theorem.

3.8.1 1. What this unit proves

All propositions are **theorems** under the stated hypotheses. Infinite nonnegative bounds include summability.

Proposition III.5.1. For finite labels, reflexive symmetric incompatibility R and real w, put $\ell_m = \sum_{z \in \iota^{m+1}} \phi_R^T(z) \prod_a w_{z_a}/(m+1)!$. If $|\ell_m| \le C q^m$ for every $m \ge 0$, $C \ge 0$, $0 < q < 1$, then $\Xi(w) = \exp(\sum_m \ell_m)$. `Gas.exact_polynomial`

Proposition III.5.2. On every finite graph with a distinguished vertex, the absolute alternating sum over connected spanning subgraphs is bounded by its number of spanning trees. `Graph.pay`

Proposition III.5.3. For arbitrary labels, supports $X_i \subseteq \Lambda_n$, $B \in \mathbb{N}$, $w \ge 0$, $a \ge 0$, $R \subseteq$ overlap

and Ready$(w; a, \gamma, \delta, q)$, every site x and $m \geq 0$ satisfy

$$\sum_{k \geq m} \sum_{z \in \iota^{k+1} : x \in X(z)} \frac{|\phi_R^T(z)| \prod_a w_{z_a}}{(k+1)!} e^{a\mathrm{cov}_B(X(z))} \leq \frac{\gamma q^m}{1-q}.$$

The same bound holds for connected finite sets through x of at least $m+1$ labels, weighted by $\prod_i w_i e^{a\mathrm{cov}_B(X_i)}$. If $Q \subseteq \iota$ has total inflated weight $\sum_{i \in Q} w_i e^{(a+\delta)\mathrm{cov}_B(X_i)} \leq b$, the order-$m$ cluster sum marked by Q, with individual covering weights, is $\leq bq^m$. `Ursell.log_hit_tail` `Ursell.cluster_at`

Proposition III.5.4. For the same supports, real w, reflexive symmetric $R \subseteq$ overlap, $a \geq 0$ and Ready$(|w|; a, \gamma, \delta, q)$, the gas converges absolutely with absolute sum $\leq e^{n^4\gamma}$; its cluster series converges absolutely to $\log \Xi(w)$, with absolute sum $\leq n^4\gamma/(1-q)$ and order-m bound $n^4\gamma q^m$. In particular $\Xi(w) > 0$. `Gas.price_log`

Proposition III.5.5. Under III.5.4, with nonempty supports and overlap incompatibility, deleting all polymers meeting any $Y \subseteq \Lambda_n$ gives $\Xi(w^Y) = \Xi(w)e^{-p}$. p sums clusters meeting Y; their absolute tail from order m is $\leq |Y|\gamma q^m/(1-q)$ for every $m \geq 0$. `Gas.restores_log`

Proposition III.5.6. On one lattice, let unforced (X_i, u_i) and forced (Y_k, d_k) have nonempty supports, $u, d \geq 0$, $R \subseteq$ overlap. Assume $A \geq 0$, $\delta' > 0$, $t \geq 1$, $0 \leq \sigma < 1$, $\mu \geq 0$, $\eta > 0$, $q' < 1$, Ready$(u; A + \delta', \gamma, \delta, q)$, $\mathrm{Hit}_B(d; A + \delta', c)$, and

$$\frac{te^{\mu}c + t\gamma/(1-q)}{1-\sigma} \leq \eta \leq \sigma, \qquad 4\mathrm{ball}_B\eta/\delta' \leq q'.$$

Atoms are forced polymers and unforced cluster tuples with absolute cluster weights. For every site x and $m \in \mathbb{N}$, connected atom multisets through x, containing a forced atom and total multiplicity $\geq m$, have total weight $e^{A\mathrm{cov}_B(\cup)} \prod_\alpha w_\alpha^{n_\alpha}/n_\alpha!$ at most $t^{-m}e^{-\mu}\eta/(1-q')$. `OmitClusters.restore`

Proposition III.5.7. For a step of block side L, $l, B \in \mathbb{N}$, connected packets with $0 \leq p_A \leq Ce^{(s-g)|A|}$, $a, C \geq 0$, and $\rho = K_{\mathrm{step}}^2 e^{s-g+a(2l+3)^4} \leq q < 1$ satisfy, for every coarse site x and $m \in \mathbb{N}$,

$$\sum_{A : x \in D(A),\, |A| \geq m} p_A e^{a\mathrm{cov}_B(D(A))} \leq K_{\mathrm{hit}} C q^{\max(1,m)}/(1-q).$$

This includes variants supported within domes whose total packet weight obeys the same envelope. `Packets.packet_income` `Packets.packet_variants_tail`

For $\varepsilon, \delta, \gamma_{\mathrm{root}} > 0$, $a, C, e_s, \mu_s \geq 0$, assume the packet condition at $(a + 2\varepsilon + \delta, s + 2\varepsilon e_s + \mu_s)$ with ratio $q_p < 1$, optional polymers $w_i, \lambda_i \geq 0$ satisfying $\mathrm{Hit}_B(we^{2\varepsilon\lambda}; a + 2\varepsilon + \delta, \gamma_{\mathrm{opt}})$, and $K_{\mathrm{hit}} C q_p/(1-q_p) + \gamma_{\mathrm{opt}} \leq \gamma_{\mathrm{root}}$, $4\mathrm{ball}_B\gamma_{\mathrm{root}}/\delta \leq q < 1$. Connected mixed sets through x, of size $\geq m+1$, seed count $\geq r$ and charge $\mathrm{ch} \geq T$, satisfy

$$\sum_s \prod_{\alpha \in s} w_\alpha (1 + \mathrm{ch}(s))^k e^{a\mathrm{cov}_B(D(s))} \leq k! \varepsilon^{-k} e^{\varepsilon} e^{-\varepsilon T - \mu_s r} \gamma_{\mathrm{root}} q^m/(1-q)$$

for every $m, k, r \in \mathbb{N}$, $T \in \mathbb{R}$. Charge sums individual covering numbers and lengths; packet length is $e_s|A|$. Disconnected packets have weight zero. `Mixed.mixed_hit`

Proposition III.5.8. Fix the group, L, l, $\mu, u > 0$, $a, D \geq 0$, $d \in \mathbb{N}$, $z < 2$. Eventually in x, whenever $s \leq F$, $g \geq M$, $0 \leq C \leq e^F$, where $F = D\mathrm{bill}^d P(\beta_F)^z$, $M = \mu u P(\beta_F)^2$, the packet ratio

is $\le q = e^{-M/2}$ and $K_{\mathrm{hit}} C q^{\max(1,m)}/(1-q) \le e^{-M\max(1,m)/4}$ for every m. Every $\beta \ge u\beta_F$ gives $\beta\mu t^2 \ge M$. `Delay.fits`

For $\varepsilon, \delta > 0$, $\beta \ge u\beta_F$, $C, e_s \ge 0$, $C \le e^F$, $s + 2\varepsilon e_s \le F$, $0 \le \gamma_{\mathrm{opt}} \le \delta/(16\mathrm{ball}_B)$, the mixed numerical conditions eventually hold with $g = \beta\mu t^2$, $\mu_s = M/2$, $q = 1/2$, $\gamma_{\mathrm{root}} = e^{-M/8} + \gamma_{\mathrm{opt}}$. Actual weight bounds remain hypotheses. For every estimator, choose L sufficiently large, then W sufficiently large: some $\mu > 0$, chosen before l, d, u, a, D, z, gives the point charge for all sufficiently large x and funded steps of coarse side $\ge 100(1+B)$, uniformly in volume. `Buy.quoted Grant.order`

Proposition III.5.9. Let J be finite, $\emptyset \ne X_i \subseteq J$, $v \ge 0$, $w \ge 1$ submultiplicative on overlapping unions, $a \ge 0$, $0 \le \delta \le 1 - e^{-a}$, and $\sum_{i: x \in X_i} w(X_i) v_i e^{a|X_i|} \le \delta$ for every x. For every real $|u| \le v$, the disjoint-support gas satisfies $Z_u(B) > 0$, $Z_u(B) = e^{-\sum_{A \subseteq B} \lambda_u(A)}$, $|\lambda_u| \le \lambda_{-v}$ and $\sum_{A \ni x} w(A)\lambda_{-v}(A) \le a$. Deleting any $Q \subseteq J$ multiplies $Z_u(J)$ by e^l, $|l| \le a|Q|$. `ReLog.catalog`

For countably many local, gauge-invariant Borel E_i bounded by v_i on unflagged supports, mask flagged supports. If $M \in \mathbb{N}$, $A, a \ge 0$, $v \ge 0$, $a(2M+1)^4 \le 1$, $\delta \le 1 - e^{-a}$ and $\sum_{i: x \in X_i} e^{(A+1)\mathrm{cov}_M(X_i)} v_i \le \delta$ at every site, the masked gas has these identities and a field-independent envelope $|\lambda_P(V)| \le g(P)$, $\sum_{P \ni x} e^{A\mathrm{cov}_M(P)} g(P) \le a$. Raw coefficients are local, gauge-invariant Borel; masked coefficients vanish on flagged regions. `ReLog.intake`

Proposition III.5.10. For finite real matrices $A > 0$, $C = C^T$, $S = AC - I$, and $\sum_j |S_{ij}| \le q < 1$, $q \ge 0$, both $\det C, \det(I+S) > 0$, and $\log\det(I+S) = \sum_{k \ge 1} (-1)^{k+1} \mathrm{tr}(S^k)/k$ absolutely. `Round.exact_trace`

Proposition III.5.11. Fix a funded step, tube, estimator, threshold t, $W, \kappa > 0$, adapted lawful representation, coupling $\beta \ge 0$, tree-gauge forest and Borel nonnegative allocation summing to one. Require global cell chords $\ge t/4$ on listed seeds, $\le t/2$ otherwise; wire chords $\ge Wt$ on listed bonds, $\le 2Wt$ otherwise; and equality of packet sites with coarse flags wherever allocated weight is nonzero. Assume the point charge $\mathcal{E}_A \ge \kappa t^2 |A|$ on every local sector on every tree-gauge slice; local tests use $D_{5B}(A)$ and positive frame mass. Then, for every V, $\mathrm{push}_\beta \rho_R(V) = \sum_A c_A(V)$ and $|c_A(V)| \le \pi_\beta e^{-\beta\kappa t^2|A|} \ell_A(V)$, with bounded Borel sector outputs and remainders. The point charge comes from III.10 or the committed-scale hypothesis of III.1. `Presented.portionCost`

Proposition III.5.12. For finite symmetric $M \ge rI > 0$ with lattice positions, weighted row and column sums $\le C$ at rate $\eta \ge 0$, $0 \le \theta \le 1$, $\theta C \le r/2$, every $u \ge 0$ satisfies

$$e^{\theta\eta d(a_i, a_j)} |(M + uI)^{-1}_{ij}| \le [\min(1, r/2)(1+u)]^{-1}.$$

Hypotheses persist under principal restriction. `Green.inverse`

Proposition III.5.13. For finite real symmetric $A \ge rI > 0$, fixed positive weighted row/column bounds at positive rate and counting bound $\mathrm{card}\{j : d(y, d_j) \le \rho\} \le U(1+\rho)^4$, windows containing every index within distance R of their addressed index have, uniformly for large R, inverse residual and inverse difference $\le Ke^{-cR} < 1$ for some $K, c > 0$ at a positive rate. Full/windowed inverses are bounded by K throughout $tA + (1-t)I$, $0 \le t \le 1$. At $t = 1$, the symmetrized windowed inverse has positive uniform lower bounds, as does its inverse. `Win.Promise.receive`

Proposition III.5.14. Fix an estimator and patch satisfying branch conditions. For all large x, every funded step with that patch, coarse side $\ge 100(1+B)$, forest, packet and field with

matching flags has completed precision $G = D^T D + I_{\text{inactive}} \geq rI$, uniform counting constant U, and weighted row/column bound K at half the upper path-decay rate, for some $r, U, K > 0$. Endpoint windows of radius $\lfloor B/4 \rfloor$ satisfy III.5.13 with error $K_{\text{win}} e^{-c_{\text{win}} \lfloor B/4 \rfloor} < 1$. `Aff.bedrock` `Aff.install_promises`

3.8.2 2. What it uses

- I.1, II.2: lattice covering bounds and connected supports.
- III.1, III.2: step data, funding, branches, lawful representations and committed-scale energy assumptions.
- III.10: energy grant; III.6: gauge-fixed Poincaré inequality.
- Mayer expansion, Penrose partition, lattice-animal counting, Möbius inversion, Schur test, Combes–Thomas conjugation.

3.8.3 3. Strategy

Component decomposition identifies the finite-gas logarithm. Tree estimates give geometric cluster tails; absolute convergence extends the identity to arbitrary labels. Seed energy dominates packet entropy. Deletion gives another logarithm; exponential conjugation and the spectral gap control windowed inverses.

3.8.4 4. Steps

1. For gas polynomial P and cluster series L, splitting the distinguished component gives $P' = L'P$, hence $Pe^{-L} = 1$. `Coef.recurrence`
2. Penrose intervals cancel free edges; parent factorials cost at most $4^m m!$, absorbed by the reserve exponent. `Graph.pay` `Trees.trees_afford`
3. Count packets by spanning walks and compare their entropy with $P(\beta_F)^2$. `Packets.stores_count` `Delay.fits`
4. Deletion induction bounds Möbius logarithm pieces by negative-envelope pieces. `ReLog.swept`
5. Sum allocations, then extract the orientation-averaged point charge. `Oriented.turn_charge`
6. Poincaré and background comparison give the spectral gap; conjugation gives decay, and window residuals occur beyond R. `Aff.surplus` `Windows.window_price`

3.8.5 5. Where the difficulty lies

The bounds retain pins, covering weights and repeated labels. Global seed tests survive orientations. The sector remainder has no volume-uniform bound here.

3.8.6 6. Notation

$\Lambda_n = (\mathbb{Z}/n\mathbb{Z})^4$, $n \geq 1$; d is maximum distance; cov_B is connected ball-cover number; $\text{ball}_B = (2B+1)^4$. $\text{Hit}_B(w; a, \gamma)$ means every site sum of $w_i e^{a \text{cov}_B(X_i)}$ is $\leq \gamma$. Ready adds $\gamma, \delta > 0$, $4\text{ball}_B \gamma/\delta \leq q < 1$, using exponent $a + \delta$.

Ξ sums compatible finite sets; ϕ^T is the connected spanning alternating sum on the occurrence graph (edges require mutual R); singleton coefficient 1; $X(z) = \cup_a X_{z_a}$. Order m means $m+1$ occurrences. Seeds are fine cells, coarse bonds or sites; packets are finite seed sets; $D_r(A)$ is the radius-r neighbourhood of seed locations (projected cell vertices, bond endpoints, or site).

Skins are individual radius-lB neighbourhoods; $D(A)$ is their union, connectedness their overlap connectivity. $\tau_L = L^4\mathrm{card}\{\text{plaquette orientations}\}+5$, $K_{\mathrm{hit}} = (2lB+3)^4\tau_L$, $K_{\mathrm{step}} = (4lB+5)^4\tau_L$.

$\beta_F = x + b\log L$, $B = \lceil(\log x)^2\rceil$, $P(s) = e^{(\log\log s)^2}$, $t = \beta_F^{-1/2}P(\beta_F)$, $\mathrm{bill} = 1 + B + (\log\beta_F)^2$. λ_u denotes Möbius pieces of $-\log Z_u$. $\mathcal{E}_A$ assesses energy anchored in $D_{2B}(A)$ with truncated walks. π_β is the inverse product of single-bond normalizations; ℓ_A is the remaining absolute sector integral. D is the transported flat row matrix, using truncated curl/path rows and linearized block constraints, replica transport and centred path gauges in every orientation. Keep row centres outside $D_{8B}(A)$ and nontrunk columns with projected endpoints outside $D_{10B}(A)$; normalize by the square root of the orientation count. Its range is $\lfloor B/100000\rfloor + R_{\mathrm{pred}} + 2$. Weighted sums at rate η use factors $e^{\eta d(a_i,a_j)}$. A windowed inverse uses column j of the principal inverse on $P(j)$, extended by zero.

3.8.7 7. What remains unexplained

Imported energy and Poincaré proofs, supporting counts and local Gaussian comparisons remain inputs. Raw logarithm coefficients are local; masked coefficients additionally inspect flags. The transported flat matrix is not identified with the nonlinear derivative.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.9 III.6. The stationary branch: background field of the effective action

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/Bid/Station.lean`, `YM/Bid/Craft.lean`, `YM/Push/STATIONARY_UNIFORM/`, `YM/Planned/STATIONARY_JET_PORT/`, `YM/Extra/ITERATION_NO_LOSS/`. **Size:** 21,329 lines in the main-theorem closure, including attached packages.

3.9.1 1. What this unit proves

Setting. G is the gauge group and $\mathfrak{g}$ its Lie algebra. In dimension four, $f = Lc$, with odd $L \geq 3$ and patch order $J \geq 4$. Sums use counting measure and physical colour coordinates. A step S has kernel choices K, penalty $\kappa > 0$, tagged predictors Q_k, and radius $R_S = (J+7)L$. "Larger" means unique selected minimizing frames and predictors, nonempty fallback axial frames, $100R_S \leq f$, and $c > 1$. A forest H has one block-centred rooted tree per block, height at most $4L$; H^{fr} denotes free bonds.

Here x is a coarse $\mathfrak{g}$-valued bond field and a a free-bond field. The axis field is $U^H(x,a) = e^a\mathcal{R}(e^x)$ on free bonds and 1 on tree bonds; $\mathcal{R}$ copies coarse links onto crossing fine bonds. With the symmetrized Wilson, row-square, cubic and quartic main term $\mathcal{M}_p$,

$$\mathcal{T}_k^H(x,a) = \mathcal{M}_p(W; U^H(x,a)) + \kappa\sum_b \mathrm{price}(e^{x_b}Q_k(U^H(x,a))_b^{-1}), \qquad \mathrm{price}(U) = \|U-1\|^2/(2c_G^2).$$

A critical branch has $\rho(0) = 0$, is analytic at 0, and satisfies $D_a\mathcal{T}_k^H(x,\rho(x)) = 0$ near 0 for every tag, with every tested action jointly analytic there. For a local exponential-chart Haar density $J_{\mathcal{J}}$, its thermal function is

$$\Theta_k^H(x) = \mathrm{loop}_W(U^H(x,\rho(x))) + \tfrac{1}{2}\log\det\mathrm{Hess}_a\mathcal{T}_k^H(x,\rho(x)) - \sum_{e\in H^{\mathrm{fr}}}\log J_{\mathcal{J}}(\rho(x)_e).$$

Proposition III.6.1 (tested second jet; F03). Let S be larger, H a forest, B a bed with floor (a, A) for clip p on Λ_f, and W have bounds (p, σ, C, Λ), be attached to B, with B sworn. Assume main-term analyticity along analytic families through the identity. For every tag k, $\mathcal{T}_k^H$ is analytic at 0, its value and gradient vanish, and for all $(x, a), (x', a')$,

$$D^2\mathcal{T}_k^H(0)[(x,a),(x',a')] = \langle ds, M_B ds'\rangle + \kappa\langle x - Ps, x' - Ps'\rangle, \qquad s = \iota x + a, \quad s' = \iota x' + a'.$$

Free-bond fields are extended by zero on tree bonds. *Status:* theorem. *Lean:* `Bid.F03`.

Proposition III.6.2 (main-term analyticity; F02). If W has bounds (p, σ, C, Λ), X is a finite-dimensional real normed space, $x_0 \in X$, and $A(y)$ is a configuration family with $A(x_0) = 1$ and link matrices analytic at x_0, then $\mathcal{M}_p(W; A(y))$ is analytic at x_0. *Status:* theorem; supplies III.6.1's analyticity assumption. *Lean:* `Bid.F02`.

Proposition III.6.3 (vertical gap; F01). Assume shell comparison E02. For every (G, L, J, K, a) with $a > 0$ there is $\eta > 0$ such that, for all sides c, f, larger S with these parameters, forests H, clips p, beds B with floor (a, A) for clip p on Λ_f for some A, and path data W whose tested actions have the analyticity and jets of III.6.1,

$$D_a^2\mathcal{T}_k^H(0,0)[v,v] \geq \eta \sum_e |v_e|^2$$

for every tag k and free-bond field v. One may take $\eta = \min(a, \kappa)/(1 + \beta + \gamma)$, where $\beta = 2C_{\mathrm{E02}}^2 L^4 \cdot 4 \cdot 6|X|^2$, $\gamma = 2L^4$, and $X = \{-R_S, \dots, R_S\}^4$. Thus η is independent of volume, forest, B beyond its floor, and W. *Status:* theorem conditional on E02, proved in III.10. *Lean:* `Bid.F01`.

Proposition III.6.4 (branch, tangent and value; F04). Let S be larger, H a forest, B, C beds, p a clip, W a path datum, and T towers for (S, B, C) satisfying diagonal identities. Assume every tested action has the analyticity and jets of III.6.1, a vertical gap $a > 0$, and a floor (v, u) for C on Λ_c for some clip q and real v, u. There is a critical branch with

$$D\rho(0) = \mathrm{rel}_H, \qquad D^2[\mathcal{T}_k^H(\cdot, \rho(\cdot))](https://github.com/scottnarmstrong/yang-mills-informalization/blob/n$$

For every tag the stationary value is analytic with zero value and gradient. *Status:* theorem giving a germ. *Lean:* `Bid.F04`.

Proposition III.6.5 (forest independence; F05). Let S be larger, H, H' forests, W have bounds (p, σ, C, Λ), and ρ, ρ' be critical branches with positive gaps a, b, respectively. Given a local chart ratio $\mathcal{J}$, for every tag and x near 0 the stationary values and thermal functions agree between forests. *Status:* theorem; thermal equality has additive constant zero. *Lean:* `Bid.F05`, `Work.F05.same_thermal`.

Proposition III.6.6 (assembled branch). For every (G, L, J, K, a) with $a > 0$ there is $\eta > 0$: for every larger S with these parameters, forest H, beds B, C, clips p, q, path datum W and towers T, assume diagonal identities for (S, B, C), floors (a, A) for (B, p) on Λ_f and (v, u) for (C, q) on Λ_c for some real A, v, u, bounds $(p, \sigma, C_0, \Lambda)$ for W, attachment to B, and B sworn. Then III.6.3 holds with η and there is a branch satisfying III.6.4. *Status:* theorem with no work-item assumptions. *Lean:* `Bid.Scene`, `Integrated.stationary_scene`.

Proposition III.6.7 (volume-independent ball). (a) Assign finite horizontal and vertical coordinate sets ι, τ to torus sites by ℓ, with multiplicity at most K_0, using sup norms and periodic

max-distance. Fix $C_0, a_0, r, m > 0$, integer $n \geq 3$. Suppose $f : \mathbb{R}^{\iota \sqcup \tau} \to \mathbb{R}$ is analytic on $\|z\|_\infty < r$ and, for every such z, $1 \leq k \leq n$, coordinate tuple and $1 \leq j \leq k$,

$$|\partial_{q_1} \cdots \partial_{q_k} f(z)| \leq C_0 e^{-a_0 \sum_i \operatorname{dist}(\ell(q_j), \ell(q_i))}.$$

If $\nabla_\tau f(0) = 0$ and $D^2_{\tau\tau} f(0) \geq mI$, there are $0 < r_{\mathrm{hor}} \leq r_{\mathrm{zone}} \leq r_{\mathrm{safe}} < r$, depending only on (K_0, C_0, a_0, r, m), and a map $g : \mathbb{R}^\iota \to \mathbb{R}^\tau$ with $g(0) = 0$ such that for $\|x\|_\infty < r_{\mathrm{hor}}$, g is analytic at x, has norm at most $r_{\mathrm{zone}}/2$, and is the unique vertical critical point in the closed r_{zone}-ball.

(b) Fix G, real μ, C, order J, choices K, and integer $n \geq 3$. Auxiliary constants can be chosen before L so that (a) applies to every clip p, real Λ, larger S of order J with choices K, forest H, and W with bounds (p, μ, C, Λ). One branch serves all tags; radii depend on the auxiliary constants, p, L, but not c, f, H, Λ or W beyond its bounds. Coarse coordinates are located at fine block roots. *Status:* theorem. *Lean:* `Receipt.throughout`, `Before.graph`.

Proposition III.6.8 (localized derivatives). For every integer $N \geq 0$, assume III.6.7(b), choose auxiliary constants at order $N + 3$, and take any branch satisfying its analytic, size and uniqueness conclusions. For every tag, on the closed joint r_{safe}-ball the vertical Hessian is analytic, $\geq m/2$, with localized coefficient derivatives through $N+1$. Every principal sub-block shifted by tI, $t \geq 0$, is invertible: its inverse is exponentially localized with constant $O((1+t)^{-1})$, has localized derivatives through $N + 1$ and ℓ^2 norm at most $(m/2 + t)^{-1}$; its log determinant is analytic with localized derivatives of orders $1, \ldots, N + 1$, bounded proportionally to $(1 + t)^{-1}$. On the horizontal ball, g and its graph have localized derivatives of orders $1, \ldots, N + 2$, the classical action of orders $1, \ldots, N + 3$, and the Hessian along the branch has the same block package. Constants depend only on the corresponding bounds and radii. No volume-independent value bound for the action or log determinant is asserted. *Status:* theorem. *Lean:* `Counter.service`, `BlockPack`.

Proposition III.6.9 (one iteration level). Assume III.2's pre-L constants, warm-up and paid schedule; a horizon of beds $M_{N,m}$ and towers with prescribed strip/derivative bounds, floors on every torus, diagonal/stack identities and identity diagonal at cutoff (`Suffices`); sworn beds at the periods used; harmonic towers (zero-momentum two-form coefficient independent of fine block residue); a levelwise gap quote supplied by `Iteration.post`; a local chart ratio; and the `fresh-data hypotheses`. These supply bounded fresh cubic, quartic and loop atoms for every admissible datum and family of branches satisfying (i) below, classical/thermal matchings on chosen forests/tags per period (possibly different choices), and cutoff cubic/quartic atoms.

For every $m < N$ and path datum T funded at $m + 1$, there are R funded at m, branches $\rho_{p,F}$ for every period and forest, and $\alpha \in \mathbb{R}$, such that:

- (i) Each branch satisfies III.6.6 with the level-$(m + 1)$ gap, tower map and bed $M_{N,m}$.
- (ii) $|\alpha + \kappa_G \log L_m| + 1 \leq \mathrm{drift}_{m+1}$.
- (iii) For every period, forest and tag, the stationary tested action for T with the level-$(m+1)$ clip has the same jets through order four as $\mathcal{M}_q(R; e^x)$, with level-m clip q.
- (iv) Every thermal function is analytic with positive Hessian determinant near zero. For all tags and real parts w_1, w_2 of complex plane waves with every axis/colour coefficient of modulus at most one,

$$|D^2 \Theta^F_k(0)[w_1, w_2] - D^2(\alpha \mathcal{M}^{(2)}_q(R) + \mathrm{loop}_R)(0)[w_1, w_2]| \leq |\Lambda_c| \Gamma \lambda_{m+1} e^{-\gamma c}.$$

Here $\Gamma, \gamma > 0$ are chosen after L; R, α serve all periods and forests. $\mathcal{M}^{(2)}$ is the full symmetrized Wilson-plus-row-square term, including higher-order dependence. There is also a funded datum

at cutoff N whose main term has Wilson-energy jets through order four. *Status:* theorem under these hypotheses. *Lean:* `Iteration.extend`, `Iteration.peak`.

3.9.2 2. What it uses

- III.1: block-spin data, forest gauges, root-trivial gauge invariance, tower identities and Haar-preserving forest changes (`Profiles.spending`).
- III.10: E02 shell comparison (`Work.E02.answer`); III.3: free kernels; III.2: constant schedule; III.9: one-loop slope κ_G.
- I.1, II.1–II.2: lattice forms, group geometry, path bounds; III.16: analytic derivatives and jets.
- Analytic implicit/inverse function theorems, Weierstrass normal convergence, change of variables, Banach contraction, Combes–Thomas estimates; Bałaban's background-field method.

3.9.3 3. Strategy

Forest gauge removes blockwise gauge directions; curvature and the predictor penalty control the remaining links. The vertical gap gives a stationary germ, with tangent and quadratic value identified by the towers. Under forest changes, the Hessian's squared Jacobian cancels the Haar-density ratio in the thermal function.

Exponential inverse-Hessian decay supplies a volume-independent sup-norm bound for Newton contraction. Differentiated stationarity and resolvent identities control localized derivatives. The iteration joins transported atoms to supplied fresh classical and one-loop data.

3.9.4 4. Steps

1. **Analyticity and jets.** Exponential summability gives normally convergent holomorphic row/atom series; quadratic limits and the floor identity give III.6.1 after chart composition. `Work.F02.answer`, `Work.F03.answer`. *(proof read; metric and cubic/quartic limits: statement only)*
2. **Gap.** Linearize E02, square and sum link bounds, then use the floor and penalty to control vertical directions. `Work.F01.answer`.
3. **Branch and tangent.** Tags agree near identity; apply the analytic implicit function theorem, then differentiate stationarity and use $D^2\mathcal{T}(0)[(y,b),(x,\mathrm{rel}_H x)] = \langle dy, M_C dx\rangle$ to identify the tangent and stationary quadratic value. `Work.F04.answer`.
4. **Forest comparison.** Transport critical graphs by root-trivial gauge transformations; Haar change of variables cancels the determinant Jacobian. `Work.F05.answer`.
5. **Uniform radius.** Combes–Thomas decay gives summable inverse rows; Newton contraction yields the ball, using a direct shell estimate and Wilson positivity without beds or attachment. `inverse_sealed`, `Receipt.throughout`, `Before.graph`. *(proof read; direct shell/action-derivative estimates: statement only)*
6. **Localized jets.** Differentiate $D(M^{-1}) = -M^{-1}(DM)M^{-1}$ and $D\log\det M = \mathrm{tr}(M^{-1}DM)$, control the index sums by exponential convolution, and compose with the stationary graph. `inverse_lines`, `Counter.service`. *(proof read for inverse induction and assembly; convolution and composition: statement only)*
7. **Iteration.** Join fresh atoms to inherited atoms, whose cubic and quartic charges carry factors L^{-2} and L^{-4}; use the schedule for the drift bound. Match jets on reference forests

and transfer the classical and thermal matchings to every forest and tag. `Iteration.extend`. *(proof read for assembly; inherited-atom bounds and matching: statement only)*

3.9.5 5. Where the difficulty lies

Curvature alone leaves flat and gauge directions. The gap is uniform in volume; no lower bound uniform in L is asserted. A spectral gap alone gives no uniform sup-norm radius: exponential summability is also needed. The conclusions concern local stationarity and ball uniqueness, without global constrained minimization.

3.9.6 6. Notation

- P: predictor derivative; ι: linearized replica; d: lattice exterior derivative.
- A bed B consists of translation-invariant plaquette kernels, with metric M_B, roots R_B and diagonal kernel D_B (`Bed`). A floor (a, A) includes $M_B = c_p I + R_B^\top R_B$, metric/diagonal inverse identities, $\widehat{M}_B(0) = I$, $M_B d = D_B d$, symmetry, and quadratic bounds $aI \le M_B, D_B \le AI$ (`Floor`). Attachment identifies row coefficients, summed by offset and plaquette type, with R_B; swornness means signed-coordinate-permutation invariance of $\langle dA, M_B dA\rangle$ (`Attached`, `Sworn`).
- Towers satisfy `Towers.Diagonal`. With $K_B = d^\top M_B d$ and linearized block average $\bar{Q}$, their map obeys $K_B \mathcal{W} = \bar{Q}^\top K_C$, $\bar{Q}\mathcal{W} = 1 - \kappa^{-1} K_C$; $\mathrm{rel}_H = (\widehat{\mathcal{W}}_H - \iota)|_{H^{\mathrm{fr}}}$, where $\widehat{\mathcal{W}}_H$ is its forest-gauge transform.
- A clip p carries $c_p > 0$. Path data W contain row series of transported plaquette variables and invariant loop, cubic and quartic atoms. Bounds (p, σ, C, Λ) require exponential-length summability, row/cubic/quartic bounds C, loop bound Λ, row normalization $c_p \delta_{ij} + \sum_k r_{ki} r_{kj} = \delta_{ij}$, and summable vanishing loop zeroth and first offset moments in every offset component, for every pair of plaquette types and colours (`Bounds`). Here r_{ij} sums row-i coefficients of plaquette type j. "Funded" further specifies root-kernel rows on fixed coordinate paths, level bounds and weighted cubic/quartic caps (`Work item`).
- A local chart ratio has positive measurable analytic density $J_{\mathcal{J}}$, $J_{\mathcal{J}}(0) = 1$, on a positive-radius ball where exp is injective with open image; a positive scale times this density pushes forward to probability Haar measure restricted to that image (`LocalRatio`). The thermal Hessian uses the orthonormal colour basis.
- Localization means $|\partial_{q_1} \cdots \partial_{q_k} A_{ij}| \le C_k e^{-a_k(\mathrm{dist}(i,j) + \sum_\nu \mathrm{dist}(i, q_\nu))}$ using assigned sites; omit j for maps, and use III.6.7(a) for scalars.

3.9.7 7. What remains unexplained

The fresh classical/quantum matching assumptions remain undischarged here: `IterationRecipe.stocked` retains `Heating`. Comparison of uniform radii with downstream small-field thresholds, the all-level induction (III.12), and geometric realization of period comparison remain unverified. F06–F09 and primary-integral/atom-regularity supporting packages have statements only; rotation/Ward material belongs to VI.1.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.10 III.7. Matching the Wilson energy to fourth order

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/Next/STATIONARY/`, `YM/WilsonBulk/Spectral/`, `YM/UVExtension/Projection/`, and a few lines of `YM/UVExtension/Reward/`. **Size:** 19,945 lines in the main-theorem closure: 19,391 in `Next/STATIONARY`, 286 in `UVExtension/Projection`, 215 in `WilsonBulk/Spectral`, and about 50 elsewhere.

3.10.1 1. What this unit proves

The leading action and one-loop term are represented by exponentially weighted curvature paths. The leading action matches Wilson at the finest level and the stationary tree action at each step through order four; the one-loop coupling shift has slope $\kappa = 11C_G/(24\pi^2)$.

Proposition III.7.1 (path data, beds, Fourier kernels). A leg transports the Lie-algebra component $\mathcal{F}(u)$ of $(u - u^{-1})/2$ from a plaquette along a lattice walk; its length is the walk length plus four. A k-atom contracts k legs with a coefficient tensor a; it is invariant when the resulting polynomial is Ad-invariant, and its weight is $\|a\|_1 e^{\sigma\,\mathrm{length}}$. A path datum R comprises rows $\ell_R(U)_{(x,P)} = \sum_k \lambda_{P,k} \log_{P,k}(U,x)$ for six plaquette orientations and sequences of two-, three- and four-leg atoms. For a clip with $c_{\mathrm{dir}} > 0$ and absolutely convergent series,

$$\mathrm{main}_{\mathbf{c},R} = \frac{1}{384} \sum_s \left[c_{\mathrm{dir}} E_W + \tfrac{1}{2} \sum_p |\ell_R(U)_p|^2 + \sum_{x,k} (\mathcal{A}_k^{(3)} + \mathcal{A}_k^{(4)})(U,x) \right]_{U \to sU};$$

loop_R similarly averages two-leg atoms over signed axis permutations.

$\mathrm{Bounds}(\mathbf{c}; \sigma, C, \Lambda)$ requires $\sigma > 0$, weighted row sums and cubic/quartic sums at most C, loop sum at most Λ, invariant atoms, normalization $c_{\mathrm{dir}}\delta_{PQ} + \sum_{P'} \hat{\ell}_{P'P}(0)\hat{\ell}_{P'Q}(0) = \delta_{PQ}$, and summable loop coefficients with zero zeroth and first displacement moments for each orientation-colour pair.

A bed $B = (D, \tilde{D}, H, \tilde{H}, \rho)$ consists of translation-invariant plaquette kernels; attachment means rows, summed by endpoint plaquette and displacement, give ρ. On Λ_n, $\mathrm{Floor}(B, \mathbf{c}, n; \varepsilon, C)$ means $D\tilde{D} = H\tilde{H} = 1$, $\Phi^{\mathsf{T}}\Phi = H$ for $\Phi = (\sqrt{c_{\mathrm{dir}}}\,1; \rho)$, $\hat{H}(0) = 1$, $Hd_1 = Dd_1$, and symmetric H, D between εI and CI. The strip bound is $\sum_y e^{\sigma|y|_1}|K(y)| \le C$ entrywise.

A kernel is strip-analytic of width σ and size K if it is holomorphic, 2π-periodic in every scalar momentum coordinate, and bounded by K where $|\mathrm{Im}\, p_{j,a}| < \sigma$. It records the complexified coefficient $D^r F(0)/r!$ with error ϵ if the coefficient on every tuple of canonical unit plane waves differs by at most ϵ from $|\Lambda_n|$ times the kernel when momenta sum to zero, and from zero otherwise. For two legs, α_b is twice the real s^2-coefficient of $\hat{K}_{(b,b),(1,1)}(se_0)$; the bed quadratic kernel has $\alpha_b = 1$ when its metric has positive strip width and $\hat{H}(0) = 1$.

Status: definitions; path specifications are from III.1. *Lean:* `RG.PathData`, `PathData.Bounds`, `Floor`, `Recorded`, `alphaPay`.

Proposition III.7.2 (Wilson jet). For every gauge group G, $\sigma_0 > 0$ and $A, Z \ge 0$, there exist $0 < \sigma \le \sigma_0$ and $C \ge 0$ such that: if $n_i \to \infty$, $c_{\mathrm{dir}} \le Z$, $\rho \in \mathrm{strip}(\sigma_0, A)$, Floor holds for every side with some bounds a, b, the bed gauge form is signed-permutation invariant on each Λ_{n_i}, and $D = 1$ there, then a datum Q attached to B has straight-route rows, zero loops, and

$$Q \in \mathrm{Bounds}(\mathbf{c}; \tau, k, \Lambda) \quad (0 < \tau \le \sigma,\ k \ge \max(C, 6Ae^{4\sigma_0}),\ \Lambda \ge 0).$$

On every Λ_{n_i}, $\mathrm{main}_{\mathbf{c},Q}(e^x)$ and $E_W(e^x)$ are real-analytic at zero with equal derivatives of every order $j \le 4$.

Status: theorem. *Lean:* `Stationary.launch`.

Proposition III.7.3 (one-loop slope). For simple G, $\sigma, \varepsilon, \tau > 0$ and $C, Z, A \ge 0$, there is $D \ge 0$ such that, for every odd $L \ge 1$, $n_i \to \infty$, clip $c_{\mathrm{dir}} \le Z$, bed with $H \in \mathrm{strip}(\sigma, C)$, Floor bounds (ε, C) on every side and signed-permutation-invariant gauge form on these tori, and every attached $R \in \mathrm{Bounds}(\mathbf{c}; \tau, A, \Lambda)$ with $\Lambda \in \mathbb{R}$, the function

$$\Pi_\mu(x) = \tfrac{1}{2} \log\det(\mathcal{H}_{\mathbf{c},R}(e^x) + d_U d_U^* + \mathcal{M}_\mu) - \log\det(d_U^* d_U + \mathfrak{m}_\mu), \qquad \mu = 1/L,$$

has a strip-analytic two-leg recording kernel $\hat{Y}$ with errors $|\Lambda_{n_i}| e_i$, $e_i \to 0$ exponentially in n_i at an L-dependent rate, and $|\alpha_b(\hat{Y}) + \kappa \log L| \le D$ for every colour b. Here $\mathcal{H}(U)$ is the Hessian at $y = 0$ of $\mathrm{main}_{\mathbf{c},R}(e^y U)$, $(d_U \phi)_e = \phi(e_-) - \mathrm{Ad}(U_e)\phi(e_+)$, $\mathfrak{m}_\mu = \mu^2[\mu^2(d_U^* d_U + \mu^2)^{-1}]^9$, and $\mathcal{M}_\mu$ applies it componentwise at initial endpoints.

Status: theorem; D is independent of L. *Lean:* `pouring`.

Proposition III.7.4 (one step and iteration). Fix simple G, kernel parameters K with penalty $a > 0$, stage J, $\varepsilon, w, \delta > 0$ and $D, Z, C, Z' \ge 0$. There exist output bounds $A_* \ge 0, s > 0$; for every input $\tau > 0, A \ge 0$ there exist $d \ge 0, \gamma > 0, u$ and functions $\Theta_0(L), \Lambda_0(L)$, under these hypotheses:

- Admissible steps $\Lambda_{Lm_i} \to \Lambda_{m_i}$ use kernel K and one stage-J patch, $L \ge u$, $m_i \to \infty$, $m_i > 1$, and fine sides at least 100 times the patch extent.
- Fine/coarse clips satisfy $p_{\mathrm{dir}} \le Z, q_{\mathrm{dir}} \le Z'$. Fine $H, \rho \in \mathrm{strip}(w, D)$ with Floor (ε, D) on every side; coarse $H_T, \rho_T \in \mathrm{strip}(\delta, C)$ with Floor (ε', C) on every side for some real ε'. Both gauge forms have signed-permutation symmetry on their tori.
- A tower satisfies III.1's quadratic interpolation, curl-transport and diagonal-transport identities. Its interpolation kernel obeys strip bounds $(w, D/L^{1+|\nu|})$ for every fine-coordinate difference list $|\nu| \le 2$.
- Fine $R \in \mathrm{Bounds}(\mathbf{p}; \tau, A, \Lambda)$ is attached to B, $\Lambda \ge 1$. For each i, an axial forest F_i and branch g_i with $g_i(0) = 0$ are given. The branch and fine action are real-analytic at zero, and g_i is stationary for every block-kernel choice t; the free-variable second variation at $(0,0)$ is at least $\eta \sum_e |v_e|^2$ for a common $\eta > 0$ and every t. Their derivatives are the axial free interpolant minus the copied-field derivative, and their tree-level second jets equal the coarse quadratic form. A local Haar chart ratio J is given.

Then there exist Y attached to T with straight-route rows and $\alpha \in \mathbb{R}$ such that

$$Y \in \mathrm{Bounds}(\mathbf{q}; \eta', A', \Lambda') \quad (0 < \eta' \le s,\ A' \ge A_*,\ \Lambda' \ge \max(\Lambda_0(L), \Lambda/2)), \qquad |\alpha + \kappa \log L| + 1 \le d.$$

For every i, t, the tree action $S_t(x, g_i(x))$ and $\mathrm{main}_{\mathbf{q},Y}(e^x)$ have equal jets through order four. Here

$$S_t(x, z) = \mathrm{main}_{\mathbf{p},R}(U_{x,z}) + a \sum_e \left(1 - d_{\mathrm{rep}}^{-1} \mathrm{Re}\,\mathrm{tr}(V_e \Pi_t(U_{x,z})_e^{-1})\right),$$

where $V = e^x$, $U_{x,z}$ is the forest lift with free variables z, and Π_t is the block predictor. The one-loop function

$$\mathrm{th}_{i,t} = \mathrm{loop}_R(U_{x,g_i(x)}) + \tfrac{1}{2} \log\det \mathrm{Hess}_z S_t(x, g_i(x)) - \log \prod_e \mathsf{J}(g_i(x)_e)$$

is analytic at zero, with positive Hessian determinant nearby. For every $\Theta \geq \Theta_0(L)$, $c' \leq \gamma$ and coarse real plane waves $v = \mathrm{Re}(e^{ip\cdot x}\xi)$, $w = \mathrm{Re}(e^{ip'\cdot x}\zeta)$ whose direction-colour amplitudes have modulus at most one,

$$\left|D^2\mathrm{th}_{i,t}(0)[v,w] - D^2(\alpha\,\mathrm{quad}_{\mathbf{q},Y} + \mathrm{loop}_Y)(0)[v,w]\right| \leq |\Lambda_{m_i}|\Theta e^{-c'm_i}.$$

Both one-loop functions are real-analytic at zero; quad omits cubic/quartic atoms. The drift is independent of L; $\Lambda_0 = a_0L^4 + b_0\log L + c_0$ and $\Theta_0 = b_1L^4$ with nonnegative constants.

For iteration, fix positive stage widths/lower bounds, stage amplitudes, K and clip bounds. Choose maximal path widths σ_j and minimal amplitudes A_j; for every $0 < \tau_j \leq \sigma_j, A'_j \geq A_j$ choose drifts $d_j \geq 0$, rates $\gamma_j > 0$, minimal factors u_j; for every compatible choice of stage parameters with direct clip coefficients below their prescribed bounds, drifts at least d_j and block-factor lower bounds at least u_j, and then every admissible choice of block factors, choose volume thresholds and Hessian bounds $\eta_j > 0$; for every admissible protection sequence choose coupling-entry thresholds.

For any subsequent $\Gamma_j \geq 1$, $0 < c_j \leq \gamma_j$, compatible history meeting III.2's thresholds, free-field horizon satisfying III.10's strip, Floor, symmetry, top-identity, interpolation and three-form coercivity conditions, second horizon with identical beds/interpolation/curl kernels, and Haar ratio, there are kinetic data $(\alpha_{N,m}, P_{N,m}, g_{N,m})$.

Every $P_{N,m}$ is attached with level-m bounds and loop amplitude $\mathrm{loopCap}_m$; $P_{N,N}$ is the Wilson top datum on every volume with zero loops. For every $m < N$, volume and axial forest, the branches satisfy stationarity, the derivative condition, second-variation bound $\eta_{J_{m+1}}$, classical match, $|\alpha_{N,m}+\kappa\log L_m|+1 \leq \mathrm{drift}_{m+1}$, and the finite bound with $(\Gamma_{J_{m+1}}, c_{J_{m+1}})$ at loop amplitude both $\mathrm{loopCap}_{m+1}$ and 1.

Status: step and iteration theorems, conditional on the stated inputs. *Lean:* `transition_cap`, `geared`, `Kin.Planned`.

Proposition III.7.5 (completed free inverse). For stage j, kernel parameters K, $\sigma, \varepsilon > 0$, $\Omega \geq 0$, there exist $\eta > 0$ and, for every L, $A_L \geq 0$: every admissible stage-j patch and bed with $H \in \mathrm{strip}(\sigma/L, \Omega)$ and Floor (ε, D) on every side have a completed block-Bloch inverse $\mathcal{G}_B \in \mathrm{strip}(\eta, A_L)$ satisfying $\mathcal{C}_B\mathcal{G}_B = 1$ on every step, for any coarse side and kernel choice. The joint free form is $\langle dv, Hdv\rangle + a\|-Qv + d_ch\|^2$, completed by $F_{1/L}^{\mathsf{T}}F_{1/L}$; $F_\mu = d_1^*P_f + \mathfrak{m}_\mu X$, with X minus the fine rooted-frame logarithm derivative. Matrices use physical counting bases; η is uniform in L.

Status: theorem. *Lean:* `WilsonBulk.uniform_backing`.

Proposition III.7.6 (Faddeev–Popov identity). For an admissible step, clip $\mathbf{p}$, $R \in \mathrm{Bounds}(\mathbf{p}; \sigma, A, \Lambda)$ with $\sigma > 0, A, \Lambda \in \mathbb{R}$, an axial branch $g(0) = 0$ analytic and stationary for every kernel choice with second variation bounded below by $\eta\sum_e |v_e|^2$, $\eta > 0$, every $\mu > 0$, kernel choice t and local Haar chart ratio, for coarse fields x near zero

$$\det \mathcal{P}_\mu > 0, \qquad \mathrm{th} = \mathrm{loop}_R(U_{x,g(x)}) + \tfrac{1}{2}\log\det\mathcal{P}_\mu - \log\det\mathcal{G}_\mu.$$

Here $\mathcal{P}_\mu$ acts on all fine-link and coarse-frame directions: it is the left-coordinate Hessian of

$$\mathcal{S}_t(V;U,h) = \mathrm{main}_{\mathbf{p},R}(U) + a\sum_e \left(1 - d_{\mathrm{rep}}^{-1}\mathrm{Re}\,\mathrm{tr}(V_e[\mathrm{gauge}_h\Pi_t(U)]_e^{-1})\right)$$

at $V = e^x$ and the lifted joint stationary configuration, completed by $F_\mu^{\mathsf{T}} F_\mu$; $F_\mu = d_U^* P_f + \mathfrak{m}_\mu X$, where X is the rooted-frame derivative at that configuration, and $\mathcal{G}_\mu = d_U^* d_U + \mathfrak{m}_\mu$. The underlying identity is $T^2 \det \mathcal{P}_\mu = \det M (\det \mathcal{G}_\mu)^2$, with axial Hessian M and Haar Jacobian T.

Status: theorem. *Lean:* `tribute`.

3.10.2 2. What it uses

- I.1: gauge group, Wilson action and Casimir; `casimir`.
- III.1: path, block-step and kinetic specifications; III.2: compatible histories and constant thresholds, `Supply.Paid`.
- III.6: analytic stationary branches and free-interpolation derivative, `Integrated.stationary_scene`.
- III.10: bed symmetry and free-field horizon conditions, `Craft.Rack`; supplied globally by III.4.
- II.1: Haar chart, `chart_ok`. III.16/III.3: Laurent/Bloch strip calculus.
- Standard results: polarization, lattice Poincaré/Koszul lemma, Schur's lemma, Faddeev–Popov procedure, background-field method, Balaban propagator bounds and rotation-invariant shell integration. No one-loop coefficient is assumed.

3.10.3 3. Strategy

After lower jets are matched, gauge invariance makes residual Fourier Taylor kernels transversal. Quantitative lattice Poincaré division factors them through curvature; exponentially decaying Laurent coefficients give cubic and quartic path atoms. This matches Wilson at the top and the stationary tree action at each coarse level.

The Faddeev–Popov identity replaces axial determinants and Haar factors by covariant vector and ghost determinants. Uniform inverse bounds permit comparison with the infrared-regulated fine determinant. Its continuum bubble determines the logarithmic slope; subtracting that multiple of the coarse quadratic kernel leaves a fourth-order momentum zero, represented by loop atoms.

3.10.4 4. Steps

1. **Fourier kernels and Ward identities.** Local analytic translation sums give strip-analytic kernels; vanishing lower jets and gauge invariance give transversality in the infinite-volume limit. `gathered`, `withdrew`.
2. **Curvature factorization.** Hadamard division of $e^{ip_j} - 1$ solves each leg with controlled strip loss; invariant Laurent coefficients give atoms. Successively remove cubic and quartic residuals. `allleg`, `assembly`.
3. **Determinants and inverses.** Axial determinant factorization and Haar density give III.7.6; real coercivity followed by complex-strip perturbation bounds the completed inverses uniformly in block-scaled norms. `tribute`, `post_cert`.
4. **Compute the slope.** Vector and ghost bubbles have angular average $-(11/3)C_G|k|^{-4}$; normalized integration over $1/L < |k| \le 1$ yields $-\kappa \log L$. The shell error $O(|k|^{-3} + L^{-2}|k|^{-6})$ is integrable with an L-independent bound; the infrared ball and outer region also contribute bounded remainders. `poleSpin`, `absoluteGeode`.
5. **Compare and subtract.** Contact terms do not change α; coarse/fine determinant comparison costs $O(1)$, and aliasing preserves α. The residual two-leg kernel yields loop atoms with

zero zeroth and first moments. `quoting`, `pole_reduced`, `color_pipe`.

6. **Iterate.** Input-dependent classical terms carry L^{-1}, inherited loops L^{-2}; large L fixes output bounds, and coupling thresholds absorb newly generated $O(L^4)$ loops. Downward induction gives the kinetic data. `transition_cap`, `geared`.

3.10.5 5. Where the difficulty lies

The coefficient calculation must reach the blocked determinant with error uniform in L, through contact cancellation and every Fourier alias. Curvature factorization must preserve decay and previously imposed leg constraints. Output widths and classical amplitudes precede input amplitudes. The auxiliary regulator respects gauge transformations and unsigned permutations; full signed-permutation symmetry is asserted for the blocked action.

3.10.6 6. Notation

Λ_n has n^4 sites; $U = e^x$. $E_W(U) = \sum_p (1 - d_{\text{rep}}^{-1} \operatorname{Re} \operatorname{tr} U_p)$ in representation dimension d_{rep}; C_G is the adjoint Casimir. ρ, H, D are bed roots, metric and diagonal; d_1 is the lattice curl, while d_1 is d_U at $U = 1$. P_f projects onto fine links, Q is the block average, d_c the coarse gradient. Levels $m = 0, \dots, N$ run from coarse to fine, and $J_{m+1} \in \{J_m, J_m + 1\}$. loopCap_m is the allowed loop amplitude.

3.10.7 7. What remains unexplained

Subsidiary quotient, strip-perturbation, tadpole/bubble, determinant/Haar, contact and alias estimates remain partly at statement level. The reason for regulator power nine is unsettled. Global hypotheses have not been checked in every downstream application; the complete compactness and limiting-jet argument belongs to III.13.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.11 III.8. Connected activities and backward dependence cones

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/RG08/Steam/`, `YM/RG08/Forms/`, `YM/RG08/Skiff/`, `YM/RG08/Jetty/`, `YM/Push/GRID_APPROXIMATION/`. **Size:** 4,147 lines in the main-theorem closure.

3.11.1 1. What this unit proves

3.11.1.1 Part A Let I be any set, d a pseudometric, $\sigma \geq 0$. Rank-n tensors are real functions on I^n with ordered slots; $\operatorname{diam}_d(x)$ is the largest pairwise distance, zero for an empty tuple. Use

$$\|F\|_\sigma = \max_i \sup_z \sum_{x : x_i = z} |F(x)| e^{\sigma \operatorname{diam}_d(x)}, \qquad [\Gamma]_\sigma = \max\left(\sup_u \sum_v |\Gamma(u,v)| e^{\sigma d(u,v)}, \sup_v \sum_u |\Gamma(u,v)| e^{\sigma d(u,v)} \right).$$

Empty maxima and suprema are zero, including rank-zero tensor norms. Propagators need not be symmetric. A pair is (B, δ)-controlled if both norms are at most B and its difference norm is at most $B\delta$. A polynomial majorant satisfies $0 \leq P(x) \leq C x^q$ for $x \geq 1$, for some $C \geq 0$, $q \in \mathbb{N}$.

A formal potential $V = (V_{e,n})$ is eligible if $V_{0,n} = 0$ and nonzero coefficients satisfy $n \le e+2$, $e+n$ even. Let Δ_Γ and $\star_\Gamma$ contract and join slots, averaging over output permutations:

$$\Delta_\Gamma H(x) = \mathrm{Sym} \sum_{u,v} \Gamma(u,v) H(u,v,x), \qquad (H \star_\Gamma G)(x,y) = \mathrm{Sym} \sum_{u,v} H(u,x)\Gamma(u,v)G(v,y).$$

Define $V^{(0)} = V$ and

$$V^{(j+1)}_{e,n} = \frac{1}{j+1}\left[\binom{n+2}{2}\Delta_\Gamma V^{(j)}_{e,n+2} + \sum_{c=0}^{n}\sum_{i=0}^{j}\sum_{a=0}^{e}\frac{(c+1)(n-c+1)}{2} V^{(i)}_{a,c+1} \star_\Gamma V^{(j-i)}_{e-a,n-c+1}\right].$$

Proposition III.8.1. Fix $j, e, n \in \mathbb{N}$. Put

$$T_j(e,n) = \{(a,r) : 1 \le a < e,\ 1 \le r \le a+2\} \cup \{(e,r) : 1 \le e,\ n < r \le e+2\},$$

adding (e,n) if $j = 0$, $e \ge 1$, $1 \le n \le e+2$. There is a polynomial majorant P, depending only on (j,e,n), such that for all $W \ge 1$, $\delta \ge 0$, eligible V, V' and propagators Γ, Γ', control (W,δ) of the propagators and coefficient pairs in $T_j(e,n)$ implies control $(P(W),\delta)$ of the output pair $V^{(j)}_{e,n}$. The majorant is uniform in I, d, σ, δ. At $n = 0$ this is vacuous. *Status:* theorem. *Lean:* `navigation`.

Proposition III.8.2. For every eligible V and arbitrary Γ, all $V^{(j)}$ are eligible, nonzero coefficients satisfy $n + 2j \le 3e$, and positive-j outputs are symmetric. For every real t,

$$\mathcal{E}_t(V)_{e,n} = \sum_{j=0}^{3e} t^j V^{(j)}_{e,n} = V_{e,n} + \mathcal{R}_t(V)_{e,n}.$$

Here $\mathcal{R}_t$ sums $j \ge 1$, and $\mathcal{E}_t(V)$ is eligible. The series vanishes for $n > 3e$; otherwise its degree is at most $\lfloor (3e-n)/2 \rfloor$. *Status:* theorem. *Lean:* `preserves`, `live`, `coal`.

Proposition III.8.3. Let $\mathcal{S}$ consist of smooth real functions on $\mathbb{R}^4$ with polynomially growing derivatives, with

$$\|f\|_{k,\sigma} = \max_{|\alpha| \le k} \sup_y |\partial^\alpha f(y)| e^{-\sigma|y|}.$$

For marked tensors valued in real-linear functionals on $\mathcal{S}$, use $\|\lambda\|^*_{k,\sigma} = \sup_{\|f\|_{k,\sigma} \le 1} |\lambda(f)|$ and $\|M\|_h = \sum_x \|M(x)\|^*_{k,\sigma} e^{\sigma h(x)}$, where $h : I \to [0,\infty)$ is 1-Lipschitz and $h(x) = \max(0, \max_i h(x_i))$. Set $M^{(0)} = F$; the recursion replaces the contraction input by M and the join by $M^{(i)} \star_\Gamma V^{(j-i)}$, doubling its coefficient.

For each $j, e, n \in \mathbb{N}$ there is a polynomial majorant P, depending only on these indices, such that for all $k \in \mathbb{N}$, $\sigma > 0$, $W \ge 1$, $U, \delta \ge 0$: if V, V' are eligible, propagators and unmarked inputs $1 \le a \le e$, $1 \le r \le a+2$ are (W,δ)-controlled, and marked inputs with $a < e$, or $a = e, r > n$, or $j = 0, (a,r) = (e,n)$ are (U,δ)-controlled, the marked output pair is $(P(W)U,\delta)$-controlled and real-linear.

For every $l \in \mathbb{N}$, eligible V and F supported on $r \le l + a$ give nonzero outputs only if $n + 2j \le l + 3e$ and $n \le l + e$. For fixed l, e, n, there is a polynomial majorant P depending only on these indices such that the paired sums $\sum_{j=1}^{l+3e} t^j M^{(j)}_{e,n}$ are $(P(W)U,\delta)$-controlled for $|t| \le W$, under the preceding bounds omitting the marked input (e,n). They comprise the whole positive-time part if both marked inputs satisfy that support condition. *Status:* theorem. *Lean:* `journey`, `registry`, `full`.

3.11.1.2 Part B Use I.1's gauge group and Haar probability measures. A block step has odd $L \geq 3$, $J \geq 4$, patch radius $\rho = (J+3)L$, embedding ι, and $\mathcal{B}F(U) = \int k_\beta(U,V)F(V)\,dV$. Locality on X means dependence only on links with both endpoints in X; $\mathcal{I}(X)$ adds gauge invariance, measurability and boundedness.

The patch hypothesis requires unique minimising frames and best predictors whenever their branches are used, and nonempty fallback tags for every cover configuration.

Proposition III.8.4. Assume the patch hypothesis, $\beta \in \mathbb{R}$. For bounded measurable complex F local on X, $\mathcal{B}F$ is local on any $Y \supseteq \bigcup_{x\in X} \Lambda_{10\rho}(\iota x)$. If also $k_\beta(U^g, V^{g\circ\iota}) = k_\beta(U,V)$ for every fine gauge transformation g, then $F \in \mathcal{I}(\Lambda_R(o))$ implies $\mathcal{B}F \in \mathcal{I}(\Lambda_{LR+10\rho}(\iota o))$, with the same sup bound. *Status:* theorem. *Lean:* `back_local`, `back_cylinder`.

Histories satisfy the patch hypothesis, gauge covariance and covariance under translations and signed axis permutations at every step. For each N, p, $\mathcal{B}_m$ uses coupling $\beta_{N,p,m+1}$; levels grow finer, $a_{m+1} = a_m/L_m$, $J_{m+1} \leq J_m + 1$, $\rho_m = (J_{m+1}+3)L_m$. Define

$$\mathcal{B}_{k\to m} = \mathcal{B}_{m-1} \circ \cdots \circ \mathcal{B}_k, \quad r_{k\to k}(r) = r, \quad r_{k\to m+1}(r) = L_m r_{k\to m}(r) + 10\rho_m,$$

with L_m odd, $L_m \geq 3$, $J_m \geq 4$, identity at $m = k$, $\tau_m = 30a_m(J_m+5)$, and $\mathrm{room}(h,k,r)$ meaning $2(a_k r + \tau_k) < a_h$.

Proposition III.8.5. Under the patch hypothesis, for every β, bounded measurable F, G local on disjoint coarse sets satisfy $\mathcal{B}(FG) = \mathcal{B}F\mathcal{B}G$; likewise finite products of $F_i \in \mathcal{I}(C_i)$ on distinct tiling cells. For every history, N, p, $h \leq k \leq m$, finite level-h set l, and $F_i \in \mathcal{I}(\Lambda_r(\iota_{h\to k} i))$, $\mathrm{room}(h,k,r)$ implies

$$\mathcal{B}_{k\to m} \prod_{i\in l} F_i = \prod_{i\in l} \mathcal{B}_{k\to m} F_i.$$

This also holds for tile copies of one $F \in \mathcal{I}(\Lambda_r(0))$, with level-$m$ copies on the right. *Status:* theorem. *Lean:* `back_factors`, `backTo_copies`.

Proposition III.8.6. For every history, N, p, $k \leq m$, pull-back preserves bounded measurability, is linear, fixes constants, commutes with conjugation and decreases the sup norm. For bounded measurable complex F at level k and integrable complex g at level m, the composite push-forward from level m to k is integrable and

$$\int (\mathcal{B}^*_{k\to m} g) F = \int g \mathcal{B}_{k\to m} F, \qquad \int \mathcal{B}^*_{k\to m} g = \int g.$$

For $h \leq k$ and every level-h site i, pull-back sends $\mathcal{I}(\Lambda_r(\iota_{h\to k} i))$ into $\mathcal{I}(\Lambda_{r_{k\to m}(r)}(\iota_{h\to m} i))$. On bounded measurable functions it commutes with signed axis permutations followed by translation by $\iota_{h\to\cdot} i$, and centred level-h copy maps. *Status:* theorem. *Lean:* `pushTo_dual`, `backTo_local`, `backTo_copy`.

Proposition III.8.7. For every such history and nonnegative integer levels and radii,

$$\tau_{m+1} \leq \tfrac{2}{3}\tau_m, \qquad \tau_m \leq (2/3)^m \tau_0 \to 0, \qquad a_m r_{k\to m}(r) + \tau_m \leq a_k r + \tau_k \quad (k \leq m).$$

If $h \leq k$ and $\mathrm{room}(h,k,r)$ holds, every cone box at $m \geq k$ lies in its centred level-h cell, of physical side a_h. For fixed h, r, room holds for all sufficiently large $k \geq h$. *Status:* theorem. *Lean:* `scales_budget`, `room_fits`, `plenty`.

3.11.2 2. What it uses

- I.1: periodic lattices, Haar measure, `Schedule`.
- III.1: probability kernels, the patch hypothesis proved in `Next.Foundations.branches`, covariance, `History`.
- IV.1: cells, copies, `Tiling.planted` (chessboard estimate).
- Polchinski's formal heat equation; weighted ℓ^1 estimates; Fubini. III.14 supplies Part A's inputs.

3.11.3 3. Strategy

For finite I, eligible symmetric V and symmetric Γ, the unmarked series represents $\log \mathbb{E}_{t\Gamma} e^V$ formally by Wick's rule. Contractions and joins generate connected coefficients. Diameter and height inequalities make exponential weights multiply; induction and telescoping give polynomial and difference bounds.

Given the fine field, site tags are independent; coarse links are independent conditional on those tags. Unused variables integrate out. Locality iterates through the affine radius recurrence; decay of the physical budget keeps cones within cells.

3.11.4 4. Steps

1. For finite input norms and positive output rank, $\|\Delta_\Gamma H\|_\sigma \le \|H\|_\sigma[\Gamma]_\sigma$ and $\|H \star_\Gamma G\|_\sigma \le \|H\|_\sigma[\Gamma]_\sigma\|G\|_\sigma$. Diameter inequalities absorb internal sums; telescoping gives difference factors $2, 3$. `Held.bridge`.

2. Heat-order induction gives bounds; slot counting gives support. The marked norm also controls scalars. `journey`.

3. Frame routes stay within 9ρ; uniqueness makes selected frames local; connectors fit within 10ρ. `patch_dep`.

4. Disjoint coarse sets use disjoint tags and bonds, giving factorisation. `back_factors`.

5. Physical radius increments are at most $\tau_m/3$; combine with budget decay and iterate. `scales_next`.

6. For $h \le k \le N$, bounded measurable O at level N and $F \in \mathcal{J}(\Lambda_w(0))$ at level k satisfying $\mathrm{room}(h, k, w)$, expand replicated differences over subsets and apply factorisation and duality; the remaining factors stay inside one push-forward. Joint.restored.

7. For central-cell observables under IV.1's chessboard hypotheses, expand differences of products over nonempty subsets and bound each by its chessboard roots. tariff.

3.11.5 5. Where the difficulty lies

Positive heat order uses higher ranks at the same coupling order; the polynomial may depend on the fixed orders. Unmarked scalar bounds are vacuous. Conditional factorisation allows overlapping fine windows without asserting independence under the fine Gibbs law.

3.11.6 6. Notation

e, n, j: coupling order, rank, heat order. $\Delta_\Gamma, \star_\Gamma$: `lap`, `bridge`. Λ_r: periodic ℓ^∞ box (`window`). $\iota_{h\to m}$: embedded site. $\mathcal{B}^*$: iterated push-forward (`pushTo`). Copies translate, reflect and conjugate on odd tiles.

3.11.7 7. What remains unexplained

III.14 must supply covariance bounds; the relation to polymer activities remains tentative. The marked normalised Gaussian interpretation remains provisional (probable). Non-summable infinite-lattice sums are assigned zero; unmarked scalars require separate convergence arguments; no convergence in g is asserted. Support packages need separate outlines.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.12 III.9. The marginal flow: the running coupling and its reset

Part: III, Ultraviolet construction (renormalization group). **Lean:** chapter 21: coupling reset, verified reset, Wilson start-up. **Size:** 7,215 lines.

3.12.1 1. What this unit proves

Let $G \subset U(n)$ be closed, with Lie algebra $\mathfrak{g}$.

Proposition III.9.1 (slope). For simple nonabelian $\mathfrak{g}$ with orthonormal basis (b_i) in $\langle X, Y\rangle = -\mathrm{Retr}(XY)/n$, define $C_G = -(\dim \mathfrak{g})^{-1} \sum_i \mathrm{tr}_{\mathfrak{g}}((\mathrm{ad} b_i)^2)$ and $\kappa = 11C_G/(24\pi^2)$. Then $\kappa > 0$. *Status:* definition and positivity theorem. *Lean:* slopeConstant_pos.

Proposition III.9.2 (reference scales). Fix $\kappa > 0$, $\bar\beta > 1$, an infinite sequence of odd $L_m \geq 3$. Set

$$s_m = \sum_{i<m} \log L_i = \log(1/a_m), \quad \hat\beta_m = \bar\beta + \kappa s_m, \quad M_m = e^{(\log\log \hat\beta_m)^2},$$

$$t_m = \hat\beta_m^{-1/2} M_m, \quad \varepsilon_m = \hat\beta_m^{-11/10}, \quad \tilde\varepsilon_m = e^{-M_m^{1/5}}, \quad B_m = \lceil (\log \hat\beta_m)^2 \rceil, \quad \mathfrak{B}_m = [\hat\beta_m/2, 2\hat\beta_m].$$

Then $\hat\beta_{m+1} = \hat\beta_m + \kappa \log L_m$, $\hat\beta$ is nondecreasing and diverges. Additional scale hypotheses are: $t_m \leq 1$; sides at least $100(B_m + 1)$ and $100(R_m + 1)$ at levels $m, m+1$, for every volume; $J_m \geq 4$ nondecreasing, divergent, increasing by at most one; nondecreasing derivative orders, spatial powers, support exponents and stage factors ℓ_j, with $L_m = \ell_{J_{m+1}}$; antitone $0 < \theta_{J_m} \leq 1/J_m$; $\log L_m/s_m \to 0$; $\omega_{J_{m+1}}/\log L_m \leq J_{m+1}^{-2}$ and $\log(1 + \text{paths} + \text{width}^{-1} + \text{columns} + \text{relaxation} + \text{support exponent})_{J_{m+1}}/\log L_m \leq J_{m+1}^{-2}$. *Status:* definitions and elementary theorems; scale hypotheses supplied by III.2. *Lean:* Calendar.Fits.

Proposition III.9.3 (selected histories). A selected history assigns representations $\rho_{N,p,m}$ for all N, p and $m \leq N$, with volume-independent paths $P_{N,m}$, level norms and clip, stage path bounds and loop cap $2M_m^{\theta_{J_m}}$, and $\beta_{N,p,m} \in \mathfrak{B}_m$. Regular activities are local, gauge invariant, vanish at the flat field, with zero affine Hessian on nonwrapping supports and bound ε_m; forced activities obey II.2 with bound $\tilde\varepsilon_m$; the family respects translations and signed permutations. At the top, $v_N = 0$, $\rho_N = e^{-\beta_N S_W}$; at the bottom, $\beta_0 = \bar\beta$. For every $m < N$,

$$\rho_m = T^{(m)}_{\beta_{m+1}} \rho_{m+1}, \qquad |\beta_{m+1} - \beta_m - \kappa \log L_m| \leq \omega_{J_{m+1}}.$$

For every $\epsilon > 0$, eventually $|\beta_{N,p,m} - \widehat{\beta}_m| \le \epsilon s_m$ uniformly in $N \ge m, p$. Consequently $\beta_{N,p,m}/s_m \to \kappa$ uniformly, including $m = N$, and $\int \rho_{N,p,m}\, d\mathrm{Haar} = Z_W(\Lambda_{N,p}; \beta_{N,p,N})$ for every $m \le N$. *Status:* specification; realized in III.13 using III.9.5 and III.10. *Lean:* History.Selected.

Proposition III.9.4 (reset along a curve). Fix scales meeting entry thresholds and kernel conditions, compatible horizon, Haar chart, positive post-block constants, planned kinetic data and recipe. For $N, p, m < N$, let ρ'_x, $x \in I = [l, u]$, be continuous admissible fine representations, with $l \le u$; write $\beta'_x = \beta(\rho'_x)$. Fix stage-pair reset constants $c > 0$, $0 < r \le 1$, prescribed fine-stage $g > 0$, and coarse-stage sensor $\eta > 0$. Assume:

- $2(c+1)\varepsilon_m \le \eta t_m^2 r$; a small-field probe satisfies $\mathcal{K}(V) \ge |\Lambda_m| \eta t_m^2$.
- Trial activities $F(x,d)$, $|d| \le r$, have fixed addresses, valid allowances, coefficient continuity, and common summable norm majorants along subsequences of every convergent parameter sequence. For $H_x = T^{(m)}_{\beta'_x} \rho'_x$, $H_x(1), H_x(V) > 0$ and

$$|(\beta'_x + \alpha)\mathcal{K}(V) + \mathcal{L}(V) - \log(H_x(1)/H_x(V))| \le |\Lambda_m| c\varepsilon_m,$$

$$|\mathcal{E}_{F(x,b)}(V) - \mathcal{E}_{F(x,a)}(V)| \le |\Lambda_m| c\varepsilon_m |b - a|.$$

- The trial shape $e^{-(\beta'_x+\alpha+d)\mathcal{K}-\mathcal{L}-\mathcal{E}_{F(x,d)}}\mathcal{G}_{F(x,d)}$ differs from $H_x(U)/H_x(1)$ by at most $C_{x,U}|f(x,d)|$, with finite $C_{x,U} \ge 0$ independent of d; forced activities have bound $e^{-gM^2_{m+1}}$; every exact trial passes the recipe.

There is continuous $d : I \to [-r, r]$ giving continuous admissible $\rho_x = H_x$, coupling $\beta'_x + \alpha + d(x)$ and $v_x = |\Lambda_m|^{-1} \log H_x(1)$, retaining the stronger forced bound. Without the last bullet, the unique residual root still matches $1, V$. *Status:* theorem. *Lean:* advance.

Proposition III.9.5 (explicit reset; program). Under III.9.4's datum, curve and constants, assume simplicity, $\beta'_x \in \mathfrak{B}_{m+1}$, and the early presentation

$$H_x = e^{\tau_x|\Lambda_m| - \beta'_x\mathcal{K} - \mathcal{L} - \sum_i \chi_{X_i} E_{x,i}}\, \mathcal{G}_x e^{-\alpha\mathcal{S}}.$$

Assume local gauge-invariant smooth regular functions with nonnegative sup-jet bounds $h_{x,i}$ satisfying, at every anchor, $\sum_{i:o_i=o} e^{(a+3)\mathrm{cover}_{B_m}(X_i)} h_{x,i} \le H^E_x$, forced-family conditions and site-pinned bound w_x at exponent $a+1$, translation and point-group symmetry, fixed addresses and III.9.4's continuity/tails. Require priced reset, compatible clip, width and cubic conditions, $H^E_x \le c\varepsilon_m/4$, regular/forced reorganization estimates fitting numerical bounds e_x, k_x that imply both allowances and $e^{-gM^2_{m+1}}$, and recipe acceptance and continuation. Then

$$d(x) = \delta_x, \quad t_m^2|\delta_x| \le H^E_x, \quad v_x = \tau_x - |\Lambda_m|^{-1} \sum_i E_{x,i}(1).$$

All III.9.4 conclusions hold. A Gaussian-sector formulation additionally requires integration licenses and equality of that integral with the push. If these inputs hold at every level along every continuous admissible curve with couplings in $\mathfrak{B}_{m+1}$, and start-up curves pass initial and retention tests, they give, for every N, p, an admissible continuous Wilson curve on $\mathfrak{B}_N$ with coupling parameter β and $v = 0$, and reset outputs along every continuous admissible input curve with couplings in $\mathfrak{B}_{m+1}$ on every closed interval, for all $m < N$. Output couplings need not remain in the coarse band before shooting. *Status:* conditional theorems; physical inputs supplied by III.10. *Lean:* Return.reset, program.

Proposition III.9.6 (Wilson start-up). For compatible clip/norm pairs ($0 < t \le 1$, $c_{\mathrm{dir}} \le 1$, $B \ge 2$, $te^{(B-2)/c_{\mathrm{rel}}} \ge 2$), Wilson top paths, point-group-averaged regular activities $\beta(1 - c_{\mathrm{dir}})R_p$ on small-field plaquettes and forced activities $e^{-\beta\,\mathrm{charge}(D,U)}$ on padded large-field components give $\rho_\beta^{\mathrm{start}}(U) = e^{-\beta S_W(U)}$ for every real β and configuration U, with $v = 0$. For every G, k, q, some $C > 0$, $0 < \delta \le 1$ give residual sup-jet bound Ct^5 for every norm choice of order k and power q, every torus, plaquette and site of that plaquette when $t \le \delta$. At each fixed stage with $c_{\mathrm{dir}} \le 1$, some β^* ensures compatibility, small threshold and both norm allowances for every realized choice of levels and every N at that stage with $\hat{\beta}_N \ge \beta^*$. The band curve is continuous; admissibility additionally requires Wilson top paths, their stage bounds and recipe tests. Given old planned data and a revised horizon with Wilson top quadratic datum, unchanged lower data and initial guesses, bounded Wilson top paths can replace the old top while preserving α and planned conditions. *Status:* theorem. *Lean:* Start.evaluates.

Proposition III.9.7 (thresholds). For a fixed pre-block profile and simple G, sensors $\eta_j > 0$ precede block sizes. Fix block factors, $0 < \theta_j \le 1$, reset constants, decay forecasts and post-block costs. Assume, for each $i \le j \le i + 1$, $8f_{ji} \le q_i \ell_j^{\min}$ and $u_{ji}(x) \to 0$, where f_{ji} is the inherited coefficient, $\ell_j^{\min}$ the preassigned block-size floor, and $q_i = 1/[64(1 + A_i)]$ for the regular reorganization coefficient $A_i \ge 0$. Every threshold record V admits $U \ge V$ such that every $W \ge U$ ensures pricing, cubic, width, reorganization and inherited inequalities beyond thresholds; clip compatibility requires $c_{\mathrm{dir}} \le 1$. Realized levels must share these data and satisfy $W_{J_{m+1}}^{\mathrm{entry}} \le \hat{\beta}_m$. Physical bounds also require $4q_i \le c_{ji}$ and $4g_j \le \mathrm{rate}_{ji}$. Start-up thresholds are likewise monotone when every used clip has $c_{\mathrm{dir}} \le 1$. *Status:* conditional numerical theorem. *Lean:* Time.beforeWait.

Proposition III.9.8 (marker sums). Fix $M, \mu \in \mathbb{N}$, $a \in \mathbb{R}$, $c \ge 0$. On a four-torus let arbitrarily indexed polymers X_i have $f_i \ge 0$ and subsets of finitely many marker names, at most μ markers per site. Every positive-weight polymer contains one of its markers' sites. If every marker-rooted sum $\sum_{i:v\in\mathrm{markers}(i)} e^{(a+1)\mathrm{cover}_M(X_i)} f_i \le c$, then every site-pinned sum $\sum_{i:x\in X_i} e^{a\mathrm{cover}_M(X_i)} f_i \le \mu e(2(16M+9))^4 c$, uniformly in volume. All nonnegative sums must be summable. *Status:* theorem. *Lean:* root_hit.

3.12.2 2. What it uses

- II.2: polymer norms, masks and gas estimates; III.1: normalized block kernels and representation interfaces.
- III.2: constant order, scales and shooting; III.7: $|\alpha + \kappa \log L| + 1 \le \omega$.
- III.4, III.10: Gaussian-sector identity, early presentation, local bounds, continuity and recipe acceptance; III.13: compatible kinetic data and selected history.
- Intermediate value theorem, Taylor's formula, Schur's lemma, lattice reflections and permutations, Fourier analysis.

3.12.3 3. Strategy

Extract the isotropic quadratic coefficient of the early regular activities into β. Subtract flat values and quadratic projections, compensate path terms, and absorb failed-mask discrepancies into the forced gas. A scalar equation identifies this exact reorganization with the continuous reset; shooting chooses the bare coupling with endpoint $\bar{\beta}$.

3.12.4 4. Steps

Under the corresponding hypotheses in §1:

1. Taylor expansion through order four gives the Wilson residual; connected large-field components give the exact gas. Start.gas_raw.
2. A small-field stripe gives $\mathcal{K}(V) \geq |\Lambda|\eta t^2$. Pricing and the activity-motion bound make $f(x,d) = (\beta'_x + \alpha + d)\mathcal{K}(V) + \mathcal{L}(V) + \mathcal{E}_{F(x,d)}(V) - \log(H_x(1)/H_x(V))$ increasing with slope at least $\mathcal{K}(V)/2$, yielding a unique continuous root. paidResponse.
3. Simplicity and lattice symmetry make the projected Hessian isotropic; exact masked splitting preserves every density value and gives $d = \delta$. Local.exponent.
4. Fixed-stage asymptotics make the reset inequalities eventual; prefix maxima combine their thresholds monotonically. collect_upper.
5. Shooting restricts intervals to keep couplings in their bands and reach $\beta_0 = \bar{\beta}$; summing drift errors gives uniform running. Shoot.trajectory.

3.12.5 5. Where the difficulty lies

Two-point matching requires full masked reorganization to imply exact transfer. Activity-norm continuity must survive hard configuration masks. The running estimate gives an $o(s_m)$ remainder, without a bounded remainder or two-loop expansion.

3.12.6 6. Notation

N, p, m: cutoff, volume, level; $|\Lambda_m|$: site count; $\alpha = \alpha_{N,m}$: one-loop shift; a: support exponent; cover_M: polymer cover size; J_m: stage; ω_j: drift allowance; R_m: patch radius. $\mathcal{K}, \mathcal{L}, \mathcal{E}, \mathcal{G}$: kinetic, loop, regular and forced-gas terms; $\mathcal{S}$: masked quadratic receipt; δ: isotropic Hessian coefficient; χ_X: small-field indicator. Admissible means prescribed paths/norms/clip, lawfulness and recipe acceptance. Curve continuity means continuous coupling and vacuum constant, fixed supports, inventories and eligibility tests, and continuity in weighted regular sup-jet/forced uniform norms; regular jets are measured on small-field supports at the prescribed order and power.

3.12.7 7. What remains unexplained

Expand reorganization constants and connected-polymer estimates. Nonlinear inputs belong to III.4/III.10; analytic existence and uniqueness within the specified continuation class belong to III.13.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.13 III.10. One full nonlinear renormalization step: bounds on irrelevant terms

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/Work/` (items E01–E09, H01–H04, I01–I04), `YM/Bid/`, `YM/Next/FLUCTUATIONS/`, `YM/AnalyticNext/CONNECTED_ACTIVITY_JETS_E6/`, `YM/PhysicalNext/RG_NEXT_COMPLETE_REPEATABLE_NONLINEAR_STEP/`, `YM/PhysicalNext/V9_JOINT_ENDPOINT_LOAD/`, `YM/Pressure/` (definitions). **Size:** 49,448 lines in the main-theorem closure.

Conventions. Levels are indexed from coarse to fine, as in III.9.

- *Step and couplings.* One step S goes from a fine torus of side $f = cL$ to a coarse torus of side c, with odd block factor $L \geq 3$. Write x for the coarse reference coupling $\hat{\beta}_m$. The fine coupling is $\beta_F = x + b_G \log L = \hat{\beta}_{m+1}$ (`clock_next`), where $b_G \geq 0$ is the one-loop slope of III.9; $b_G > 0$ when the Lie algebra of G is simple.
- *Thresholds, radius and cutoff.*

  $t_c(x) = x^{-1/2} e^{(\log\log x)^2}$ (coarse field), $\quad t = t_c(\beta_F)$ (fine field), $\quad T = \beta_F^{-49/100}$ (large field proper),

  with cluster radius $B = \lceil (\log x)^2 \rceil$ and path-length cutoff $R = \lfloor B/10^5 \rfloor$.
- *Sizes.* Since $e^{(\log\log\beta)^2} = \beta^{o(1)}$, we have $t = \beta_F^{-1/2+o(1)} \ll T$. On the other hand $\beta_F t^2 = e^{2(\log\log\beta_F)^2}$ grows faster than every power of $\log\beta_F$.
- *Field sizes.* $\chi(g)$ is the chordal distance from $g \in G$ to 1. For a plaquette p, $|U_p| = \chi(U_{\partial p})$. For a coarse bond b, $w_b = \chi(V_b Q_b^{-1})$ measures how far the coarse field V is from the block-spin predictor $Q = Q(U, k)$ of the fine field U with gauge-fixing tags k (III.1).
- "For all large x" means $x \geq x_0$, with x_0 independent of the period and volume. Other dependencies are specified in each statement.

3.13.1 1. What this unit proves

One-step energy, activity, derivative and Gaussian bounds support the effective-action construction of III.9; III.10.6 supplies start-up fusion and marker estimates.

Data of a step.

- *Request.* A request P (`Teller`) consists of:
  - kernel choices K, with penalty $a > 0$ and cuts;
  - a fine stage and a coarse stage, each with clip constants $\alpha, \lambda, \rho > 0$ satisfying $\lambda < \alpha/16$ (`Clip`), a path width and a path budget;
  - a step index j and constants $\eta > 0$, C.
- *Tube.* A tube $\mathcal{T}$ (`Tube`) carries the towers Φ, Φ', N and the fine and coarse path data. Φ maps coarse plaquette coordinates to fine plaquette coordinates; Φ' maps them to coarse-bond ("interior") coordinates; N maps them to gauge coordinates.
- *Admissibility.* A step with tube $(S, \mathcal{T})$ is admissible for P (`Teller.Funds`) if:
  - S uses K and index j;
  - the patch is well posed (`Branches`: unique minimizers whenever the frame or predictor selector uses minimization, and a nonempty set of eligible axial tags for every configuration);
  - the scalar tower coefficients have exponentially weighted absolute sums at rate η bounded by C/L^2 for the fine kernel and C for the interior and gauge kernels. Finite differences of order $\ell \leq j - 2$ of the one-form and two-form cross kernels have bounds $C/L^{1+\ell}$ and $C/L^{2+\ell}$, respectively (`YM.Precision.Towers.Deep`);
  - the clips and path budgets are those of P;
  - the flat, uncut tower matrices satisfy the exact identities `Towers.Roll`, among them the resolution of the identity

$$\Phi_0^* \Phi_0 + \Phi_0'^* \Phi_0' + N_0^* N_0 = 1.$$

The transported kernels cut at length R need not satisfy this identity; their defects are estimated in Steps 5–6.

Seeds. A store $A = (A_F, A_I, A_\flat)$ (`Stores`) consists of finite sets of fine plaquettes ("costs"), coarse bonds ("aisles") and coarse sites ("flags"). $\sharp A$ is the sum of their cardinalities, including coincident seeds of different types. $D_r(A)$ is the coarse r-neighbourhood of their positions: all parent sites touched by a fine plaquette, both endpoints of a coarse bond, and each flag site. A is *consistent* with (U, k, V) at (x, W) (`Plot`) if the following hold:

- whenever a parent site touched by p meets $D_{5B}(A)$, $p \in A_F$ implies $|U_p| \geq t/4$, and $p \notin A_F$ implies $|U_p| \leq t/2$;
- whenever an endpoint of b meets $D_{5B}(A)$, $b \in A_I$ implies $w_b \geq Wt$, and $b \notin A_I$ implies $w_b \leq 2Wt$;
- $A_\flat$ agrees with the flags of V at threshold $t_c(x)$ on $D_{5B}(A)$;
- each site tag based in $D_{5B}(A)$ has positive frame weight.

Nothing is assumed outside $D_{5B}(A)$. The *spent* seeds A^{sp} are the costs with $|U_p| > T$ and the aisles with $w_b > T$.

Action gap.

- *Clipped density.* For a stage with clip (α, λ, ρ) and threshold s, the clipped density of a plaquette is $\phi_s(|U_p|) + \mathbf{1}_{\mathrm{kept}} \frac{1}{2} |\ell_p(U)|^2$ (`crown`). Here:
  - $\phi_s(r) = \alpha r^2/2$ for $r \leq s$ and $\phi_s(r) = \lambda s r$ for $r > s$ (`clipped`);
  - $\ell_p(U)$ is the path series of the stage, cut at length R;
  - "kept" means $|U_q| \leq s\, e^{\mathrm{dist}(p,q)/\rho}$ for all q (`kept`).
- *Actions.* $\mathcal{A}_F(U)$ is the fine clipped action (fine stage, threshold t) and $\mathcal{A}_c(V)$ the coarse one (coarse stage, threshold $t_c(x)$).
- *Step functional.* $\mathcal{E}_A(U, V)$ (`assessed`) is the sum of the row contributions of Step 1 over the rows located in $D_{2B}(A)$. Summed over all rows of the torus, the same contributions give exactly
$$\mathcal{A}_F(U) + \frac{a}{2} \sum_b w_b^2 - \mathcal{A}_c(V).$$
So $\mathcal{E}_A$ is the action gap of the step (fine action, plus block-spin penalty, minus coarse action), localized near the seeds.

Proposition III.10.1 (energy per seed). Let P be a request. There is L_0 such that for every block factor $L \geq L_0$ there is $W_0(L)$ such that for every $W \geq W_0$ there is x_0 with the following property. Take $x \geq x_0$ and:

- a step S of block factor L with coarse side $c \geq 100(1 + B)$, and a tube $\mathcal{T}$ admissible for P;
- a spanning forest F of the block structure, and a fine field U equal to 1 on the trunks of F;
- a store A, tags k and a coarse field V such that A is consistent with (U, k, V) at (x, W).

Then
$$\kappa\, t^2\, \sharp A \;\leq\; \mathcal{E}_A(U, V), \qquad \kappa = \tfrac{1}{2} \min\{\lambda_F/8,\ a/32,\ \alpha_c/64\},$$
where λ_F is the linear clip constant of the fine stage and α_c the quadratic clip constant of the coarse stage.

Status: theorem under the stated admissibility, tree-gauge and consistency hypotheses.

- The Lean statement (`Teller.cash`, `Teller.Grants`) asserts only some $\kappa > 0$ chosen after L and W. The value above is the one the proof produces (Step 7), and it is independent of L and W.
- This deterministic estimate precedes the Gaussian tail and activity bounds of III.10.2.

Lean: proved by `Fluctuations.grants`. This is `Integrated.pressure_grants` applied to `Fluctuations.bare`, through `Bid.issues`.

Proposition III.10.2 (activities of one step). Let P be a request and $\mathcal{T}$ target data (`Target`): constants $\varepsilon, \delta > 0$, $e \geq 0$, a rate $r \in (-\frac{1}{2}, 0)$, a halo $h \geq 500$ and a local exponential chart of G, with 4ε below the support exponent of the fine stage and 8ε below both stage widths. Put $a_* = e + 2\varepsilon + \delta$.

There is L_0 such that for every $L \geq L_0$ and every large W there is $\mu > 0$ with the following property. For every patch D of block factor L satisfying the branch condition there are window constants o (`Target.Order`: a decay rate, a spectral gap, a shell width K, a Lipschitz constant γ, a drift cap, a plateau a_o and a trigger level ξ) and a constant $C \geq 0$ such that the following holds for all large x, every step S on D with $c \geq 100(1 + B)$ and every tube admissible for P (`Works`).

- (a) The conclusions `GrantBodies` hold:

  1. the bound of III.10.1 holds at this x with constant μ (`PointCharge`);
  2. if U is in tree gauge and, within coarse distance $B/4$ of the bond's parent site, site tags have positive frame weight, fine plaquettes have size at most $t/2$, and coarse mismatches are at most $2Wt$, that link lies within Kt of the replicated coarse link; the Lie logarithm is reliable at that radius (`ShellAccess`);
  3. in every sector whose flags are the actual flags of V (`manifest`), the Gaussian precision matrix is symmetric, has exponentially decaying entries and a positive spectral lower bound, and the covariance formed from its local inverses is positive definite (`Rotator`);
  4. the Gaussian centring drift has norm at most $\mathrm{cap} \cdot t$ at each fine bond. If the tube also satisfies the compatibility condition in (b), then, for every $\beta > 0$, a fluctuation satisfying $|z_j| \leq \xi t\sqrt{\beta}$ on all coordinates within distance B of a site farther than $30B$ from the seeds has site and bond tag masses one, $|U_p| < t/4$ for plaquettes based there, and $w_b < Wt$ for bonds based there;
  5. numerical margins hold: $4a_o t$ is below the chart radius, and, for the window's positive inverse-kernel constants τ_0, C_0 and window bound $Q_0 \geq 1$, the quantity $\epsilon_B = e^{-\tau_0 B}(Q_0 C_0 + 1)$ satisfies $\epsilon_B < 1$ and $Q_0 C_0 \epsilon_B < 1$.

- (b) The conclusions `Invoices` hold for every tube satisfying the free-kernel compatibility condition `Stamp`, and for every input representation K of the fine-level density as in `Intake`. The input has the fine-stage support exponent, threshold t, coupling $\beta \in [\beta_F/2, 2\beta_F]$, the path data and clips of $\mathcal{T}$, cluster radius $B(\beta_F)$, and activities obeying the incoming bounds of III.9: E-norm $\beta_F^{-11/10}$, K-norm $e^{-M^{1/5}}$ and loop budget $2M$, where M is the fine monitor. The conclusions are:

  1. the cost record with gain $\beta\mu t^2$ per seed and spend $C\,\mathrm{bill}(x)^5$, where $\mathrm{bill} = 1 + B + (\log \beta_F)^2$, passes the tree-graph summability test `Buy`;
  2. the letters of the new gas carry weights k_i, y_i, E satisfying a rooted Kotecký–Preiss condition with exponent ε and total β_F^r (`Sales`). For a letter with support X_i and extra

length d_i, this means nonnegative weights u_i with

$$\sum_{i:o\in X_i} u_i \exp\left(a_* N_B(X_i) + 2\varepsilon d_i\right) \le \beta_F^r \qquad \text{for every coarse site } o.$$

Here $N_B(H)$ is the minimum number of radius-B balls whose connected union contains H. The letters are the forced activities of K, the logarithmic letters, and the warning letters. Their weights are respectively k_i, y_i and E times a fixed patch-dependent warning-amplitude bound; warning letters have $d_i = 0$;

3. in every manifest sector, the following bounds hold:
   - the old activities are bounded by k_i;
   - the logarithmic letters ℓ_i and the differences $e^{\ell_i} - 1$ are bounded by y_i;
   - every packet weight satisfies $|\mathrm{Pack}| \le \exp\left((C\,\mathrm{bill}^5 - \beta\mu t^2)\,\sharp A\right)$ (`Pack`);
   - for every set v of warning addresses, the Gaussian measure of the catastrophe set is at most $E^{\sharp v}$. The catastrophe set consists of the fluctuations z for which, near the anchor of each address in v, some coordinate within distance $2B$ satisfies $|z_j| \ge \xi t\sqrt{\beta}$. Every fluctuation that triggers the warnings on v lies in that set.

Status: theorem. The target data, fine input bounds and free-kernel compatibility are hypotheses. The input bounds are supplied along the RG trajectory (III.9); the compatibility construction is discussed in §7.

- *Quantifier order:* L_0, then W, then μ (uniform in D), then (o, C) for each patch, then x.
- *Size of the per-seed factor.* Since $\beta\mu t^2 \ge \frac{\mu}{2} e^{2(\log\log\beta_F)^2}$ and $\mathrm{bill}^5 = O((\log\beta_F)^{10})$, the factor $e^{C\mathrm{bill}^5 - \beta\mu t^2}$ is eventually smaller than every power of β_F.

Lean: `Target.issue`. `Target.quotesTogether` repackages the output as the bundle `Cash` (the activity estimates together with witnesses for the letter weights).

Proposition III.10.3 (derivatives of the new activities). Let P and $\mathcal{T}$ be as in III.10.2. There is L_0 with the following property. Fix $L \ge L_0$, a patch D satisfying the branch condition, an order $i \in \mathbb{N}$, a sector cutoff χ, $W, \mu > 0$, window constants o, and chart constants (A, f) satisfying `YM.ConnectedActivityE6.Blaze.Eider.Canary`. The latter require $A, f > 0$, $\log(\exp u) = u$ for $|u| \le 2A$, $\exp(\log(\exp u\, g)) = \exp u\, g$ for $|u| \le A$ and $\|g - 1\| \le f$, and analyticity of $M \mapsto \log(\exp u\, M)$ for $|u| \le 2A$, $\|M - 1\| \le f$. Fix a function $\mathcal{V}$ of rate $-1/2$ that is eventually nonnegative (rate as defined in §6).

There are $C \ge 0$, functions q, ρ, p of rates $-1/2, 0, 0$, respectively, and $l \in \mathbb{N}$ such that, for all large x, the following holds for every step on D satisfying the compatibility condition, every fine representation R, and every bound bundle H of III.10.2 at (W, x, μ). One has $8a_o t \le A/2$ and a derivative-bound structure Ξ (`Utopia`) with the following properties in every sector at x, of shell width K from o, cutoff χ, and with the actual coarse flags.

- Its nonpacket weights satisfy the rooted bound of III.10.2(b) with total β_F^r. Its catastrophe probabilities, with trigger $\xi/2$, are at most $(\mathrm{rad}^2)^{\sharp v}$.
- Let $V(y)$ be a family of coarse fields on a finite-dimensional real normed parameter space, based at the sector field $V(y_0) = V_0$. Suppose each link matrix is C^i at y_0 and

$$\|D^m V_e(y_0)\| \le c_G \mathcal{V}(x)^m \qquad (0 \le m \le i),$$

where c_G is the fixed matrix-norm bound in `Mode`. Transport the fluctuation coordinates compatibly with this family: the physical and Gaussian coordinates represent

the same field, their physical displacement has all derivatives through order i bounded by $q(x)$, and, near y_0, the resulting fine field agrees with the held fine field on every cold bond and on every hot bond whose physical coordinate has norm at most $8a_o t$ (`YM.ConnectedActivityE6.Blaze.Silica.Clutch`). In a region disjoint from the coarse flags, each letter a_j is C^i at y_0 and satisfies

$$\|D^m a_j(y_0)\| \le \mathrm{cost}_j(\rho(x)\mathrm{span}_j)^m \qquad (0 \le m \le i)$$

(`Brush`). Packet spans are one; other spans include their support and path lengths. The costs use the packet estimate and the nonpacket weights before extracting the catastrophe-probability factor.

The same Ξ satisfies an integrated estimate (`Herd`). Let $\mathfrak{p}$ be a finite molecule with connected support-overlap graph and total support disjoint from the flags. Require that its packet stores are connected with pairwise disjoint radius-hB neighbourhoods, that its old forced supports are pairwise disjoint, and that every site in an old forced inventory touches a fine-plaquette seed in the union of the packet stores. Along a family $V(y)$ as above, the prescribed transported integrand is C^i at y_0 and has every derivative of order at most i bounded by an integrable function g. With $d\nu = d\lambda_F\, d\gamma$, the product of tree-slice Haar measure and Gaussian fluctuation measure,

$$\int g\, d\nu \le |p(x)| \prod_{j \in \mathfrak{p}} \mathrm{fare}_j\, (1 + \mathrm{shape}(\mathfrak{p}))^l.$$

The fares include a factor rad for each distinct warning address. For one measurable, gauge-invariant source letter of amplitude at most w_M, local with its eligibility tests included, the same estimate replaces its fare by w_M. A forced source must vanish outside the original eligibility set; a regular source occupies an unused regular-letter address. The remaining warning and shape factors are unchanged.

Status: theorem, conditional on the input bound bundle and compatibility. The constants and functions are uniform in the volume. `Captain` and `rocketsC` also control iterated left derivatives in spatially weighted directions.

Lean: `Utopia.whittle` (letter jets), `timing` (integrable majorants).

Proposition III.10.4 (Gaussian sector of the step). Fix a patch D, a request Q for which D satisfies the branch condition, an aperture $B_0 > 0$, a sector cutoff χ, and $k, p \in \mathbb{N}$. There are $\sigma, d > 0$ and $\pi \ge 1$ (`Charges`) such that the following holds for all large x.

Take $0 < \beta \le 2\beta_F$, a budget point of power p and threshold $t_c(x)$, a step on D with $c \ge 100(1+B)$ admissible for Q, a forest, tags, and a store A whose flags equal the flags of V on the whole coarse torus. Freeze A, the forest and tags while differentiating. Let Γ be the sector precision matrix, $\tilde{\Gamma}$ its local approximate inverse, m the smooth sector mean at aperture $B_0 t$, and $h = -g^{-1}$ times its stationarity residual, with $g = \beta^{-1/2}$. These are the objects in `Value`. Then Γ and $\tilde{\Gamma}$ are positive definite, and m equals the stationary logarithmic mean for every frozen-link input.

For a coarse site o and $0 \le m \le k$, let $\mathcal{D}^m$ be an iterated left derivative in m directions satisfying $|A_e| \le t_c(x)(1 + \mathrm{dist}(e, o))^p$. If x_i is the coarse location of coordinate i, put

$$G_i = (1 + \mathrm{dist}(x_i, o))^{pk}.$$

The bounds are

$$|\mathcal{D}^m \Gamma_{ij}| \le \pi e^{-\sigma \mathrm{dist}(x_i,x_j)} G_i^m, \qquad |\mathcal{D}^m (\Gamma\tilde{\Gamma} - 1)_{ij}| \le e^{-dB} \pi e^{-\sigma \mathrm{dist}(x_i,x_j)} G_i^m,$$
$$|\mathcal{D}^m m_i| \le \pi t G_i^m, \qquad |\mathcal{D}^m h_i| \le e^{-dB} \pi G_i^m.$$

Thus the spatial exponent at derivative order m is pkm.

Status: theorem. `JointLoad.stage` also supplies a covariance cap and a trigger-tail estimate; its scales are given in §7.

Lean: `JointLoad.capital`.

Proposition III.10.5 (schedule).

- (a) Let P be a sheet of pre-L data (III.2) and K a price map. There are floors k_i with the following property. Take block factors $L_i \ge k_i$, any protection, any delays V and any $m \ge 0$. Then there are delays W, differing from V only by larger entry couplings β_i^*, and numbers $F_{ij} \ge m$, such that for every pair of stages $j \le i \le j+1$ with j at least the first stage of P the bound of III.10.1 holds at every coarse coupling $x \ge \beta_i^*$, for the request with fine stage i, coarse stage j and step index i, with block factor L_i, aisle multiplier F_{ij} and some $\mu_{ij} > 0$ uniform in x, period and volume (`Ringing`).
- (b) If G is simple, then for every window datum of III.2 (entry and plateau requirements) there is a level schedule that meets all requirements of III.2 (`Paid`) and whose periods are divisible: for every side c and every z there is a period index $p \ge z$ with $c \mid \mathrm{period}(p)$.
- (c) Along a directed trajectory satisfying (a), fix an adjacent stage pair $i = J(m)$, $j = J(m+1)$ with i at least the first stage of P. There is $\mu > 0$, uniform in the cutoff N, level m and period, such that for $m < N$ at this pair and $x = \hat{\beta}_m$, one has $\mu\, t_c(\hat{\beta}_{m+1})^2 \sharp A \le \mathcal{E}_A$ whenever the fine field is in the chosen tree gauge and the store is consistent at multiplier F_{ji}.

Status: theorems. Part (a) is unconditional (`pressure_pricing` applies `Bid.pricing_order` to III.10.1). Divisibility is what lets torus tilings fit in III.12–III.13.

Lean: `Wait.engine`, `notation_ticket`.

Proposition III.10.6 (start-up fusion H01–H04; markers I01–I04).

(H) *Setting.* Fix a budget point and:

- a *pipe* X: regular and forced addresses, and supports of logarithmic letters;
- a *tap* d: regular activities, forced activities K_i and logarithmic letters ℓ_i.

The *fused* gas (`fused`) groups the forced activities and the factors $e^{\ell_i} - 1$ into connected clusters (pellets). Each pellet contains at least one forced activity, and the forced supports in a pellet are pairwise disjoint.

Hypotheses. Fix a bijective enumeration of the pellets. Let $\kappa, \gamma \ge 0$.

- d is *digested* (`Digests`): the gas is valid with specified regular and forced bounds; the logarithmic family has a summable uniform majorant, and each letter is measurable, gauge invariant, local with the flag tests included, vanishes on healthy configurations, and is translation and lattice-symmetry covariant.
- d is *bought* with constants κ, γ (`Buys`): $|K_i| \le e^{-\kappa} f_i$ and $|e^{\ell_i} - 1| \le f_i$, where $f \ge 0$ satisfies the rooted Kotecký–Preiss condition at support exponent one larger than the budget exponent, with total γ, and $(2B+1)^4 \gamma \le 1$.

Conclusions.

- (H01) For every configuration V, the fused gas equals the original gas times $\exp(\sum_{i=0}^{\infty} \ell_i(V))$.
- (H02) The fused activities satisfy the K-bound with any price $v \geq e^{-\kappa}(e^{\gamma} - 1)$.
- (H03) The fused system is a forced family and admits the lattice-symmetry moves.
- (H04) Fix the paths, clip, budget point, addresses and pellet enumeration. Suppose the original representations form a continuous curve on a subset of $\mathbb{R}$, each forced or exponentiated logarithmic letter varies continuously in the supremum norm over configurations, and a single nonnegative rooted majorant as above works throughout the parameter set. Then the fused representations form a continuous curve in the same representation norm. This is a fixed-volume assertion.

(I)

- (I01) Let $B \geq 8$, let F be a set of at least two markers with pairwise separation at least B^2, and let $F \subset H$. Then $B\sharp F \leq 64\, N_B(H)$, where $N_B(H)$ is the minimum number of radius-B balls with connected union covering H. The markers lie on a finite periodic lattice with its max-distance; no nonwrapping hypothesis is required.

- (I02) Fix a representation with budget order n. Suppose marker bands satisfy `YM.Mix.Stubs.Holds` at $r > 0$: valid local addresses, r times every unit amplitude at most one, $B \geq 8$ and separation at least B^2 for each pair of distinct markers, factorially weighted derivative majorants, and separate rooted budgets at most one for single-marker and multiple-marker terms. For the multiple-marker budget one may instead use positive weights t_v and an extra exponent $\delta \geq 0$, with $r \leq t_v e^{\delta B/64}$ for every marker v and a rooted budget at most one using support exponent increased by δ. Then, at weight $r/n!$, the bands satisfy `YM.Overlay.Bands.Radial`: positive weights, unit-amplitude bound one, and rooted ordinary derivative and forced-entry bounds at most two each.

- (I03) Let a family satisfy the marker schedule condition: $B_{r+1}^2 \leq 2B_r^2$ for every r, and $(L_r B_r^2)^4 \mathcal{F}_{r+1} = o(\mathrm{scale}_{r+1})$. Here a_r is the lattice spacing, $L_r = a_r/a_{r+1}$, and the vacuum bound is
$$\mathcal{F}_r = 2(2 + 4P_r) + 1 + 6(\alpha_r + P_r^2) + 4d_G(\log 16 + \tfrac{1}{2}\log \hat{\beta}_r),$$
with P_r the stage's path budget, α_r its direct clip coefficient and $d_G = \dim_{\mathbb{R}} M_m(\mathbb{C})$ for the defining matrix representation. This is the hypothesis of `YM.Stock.Tabs`. Fix $k \geq 1$ and levels $m \leq i$ such that $kB_i^2 \leq a_m/a_i$ (an open stop). For every $N \geq i$ and every period, the centred copies at level N are the embedded sites from level m; their parents agree at consecutive levels from m to N; and distinct embedded sites are separated by at least kB_r^2 for every $i \leq r \leq N$.

- (I04) Assume G is simple. For every family, every $p > 0$ and $r \in \mathbb{N}$, if $\mathrm{order}_i! \leq \mathrm{columns}_i$ for all i, then the widths
$$w_i = \frac{\hat{\beta}_i^{-r/2}}{p\,\mathrm{columns}_i \prod_{m<i} \mathrm{columns}_{m+1}}$$
are positive, and for every $\varepsilon > 0$ eventually $5 - \log(w_i/\mathrm{order}_i!) \leq \varepsilon\, \mathrm{scale}_i$.

Status: theorems. In the main-theorem closure:

- H01–H04 are used by the start-up representation at the finest level;
- I01, I03 and I04 are used by the replicated-marker constructions;

- the assemblies `Oust.begins` and `Bid.installs` are not in the closure.

Lean:

- Statements: `H01`, `H02`, `H03`, `H04`, `I01`–`I04`.
- Proofs: `H01.gathered`, `H02.answer`, `H03.fused_forcedFamily`, `H04.answer`, `I01.answer`, `I02.answer`, `I03.answer`, `I04.answer`.

3.13.2 2. What it uses

- **III.1:** block-spin patches, tree gauge, predictor and tags; predictor-curvature reduction by L^{-2} (`qdc`). The branch condition holds for the construction's kernel choices (`Foundations.branches`).
- **III.2:** sheets, block factors, protection, delays, entry and plateau requirements.
- **III.3–III.4:** flat tower identities and exponential decay (III.3 attribution probable), Gaussian integration and rectangle tails.
- **III.5–III.8:** polymer representations and Kotecký–Preiss bounds, stationary means and jets, quartic matching, connected-activity locality.
- **III.9:** b_G, thresholds, monitor and the incoming allowances $\beta_F^{-11/10}$, $e^{-M^{1/5}}$, $2M$.
- **I.1, II.2, III.16:** lattice geometry, Haar measure, chordal distance, activities and Taylor jets.

Standard results: Balaban's lattice-gauge renormalization method, rooted Kotecký–Preiss and tree-graph bounds, Schur's test, Gaussian exponential moments and Taylor estimates on compact charts. Request and target data precede L; window constants follow L, W and the patch. Simplicity is required in III.10.5(b) and I04.

3.13.3 3. Strategy

Seeds cost κt^2 each. At $\beta \geq \beta_F/2$, their Boltzmann suppression pays every fixed polylogarithmic entropy loss. Completed squares isolate positive row contributions and two kernel defects: affected cells are paid by spent seeds, while local gauge fixing and flat identities reduce unaffected defects to chart errors and exponentially small tails.

Gaussian integration produces packet, old forced, logarithmic and warning letters. Energy and annulus estimates bound packets; simultaneous Gaussian excursions bound warnings. Rooted bounds sum connected molecules. Compatible fluctuation transport controls derivatives, and Cauchy–Schwarz preserves one square root of the simultaneous-warning probability in integrated bounds.

3.13.4 4. Steps

Steps 1–7 use III.10.1's admissibility, tree gauge and consistency hypotheses; all thresholds are independent of period. Steps 8–10 use III.10.2–III.10.3's activity, compatibility and derivative hypotheses; Steps 11–12 use III.10.6 and III.10.5, respectively.

1. **Complete the squares.** Fine, interior, coarse and gauge rows, together with the linear and completeness defects, sum exactly to $\mathcal{A}_F + \frac{a}{2}\sum_b w_b^2 - \mathcal{A}_c$ by adjoint cancellation; $\mathcal{E}_A$ retains rows in $D_{2B}(A)$. `Tube.locates`.

2. **Pack seeds and compare links.** Fine plaquettes have bounded macro-box multiplicity; coarse plaquettes above a fixed $\delta > 0$ lie within $4B/5$ of spent seeds, and the relevant neighbourhood contains $O(B^4 \sharp A)$ coarse plaquettes. Closing fine links by tree paths bounds their distance from replicated coarse links by macro-box curvature plus crossing mismatches. `Work.E01.answer`; `Work.E02.answer`.

3. **Bound kernels and predictions.** Absolute row and column sums control transported kernels and path series; in particular $|v| \le At$, $|\Phi v| \le At/L^2$, $|\Phi' v| \le At$, with A independent of L, W. Norm-one adjoint transport and exponentially weighted coefficient sums give these estimates. `Work.E03.answer`; `Work.E04.answer`.

4. **Obtain positive row charges.** Large L separates fine seeds from their predictions, and large W separates mismatch seeds; dropped path terms are paid after summation by column bounds. Coarse rows above t_c and below a fixed δ are positive, while small coarse rows lose $O(t_c^4)$; affected-cell losses are charged to large fine inputs, mismatches and $O(t^2 B^4 \sharp A^{\mathrm{sp}})$. `Work.E05.answer`; `Work.E06.answer`; `Work.E07.answer`.

5. **Cancel flat defects.** Outside the affected set, local axial gauge fixing makes fine and coarse links $h = CB^{10}T$-close to identity; the flat uncut tower identities cancel both linearized defects. Comparing transported, truncated quantities with their flat versions leaves errors governed by $\omega = (1+R)h + e^{-\tau R}$. `Work.E08.answer`; `roll_l`; `roll_d`; `global_chart_cost`.

6. **Sum unaffected defects.** Gauge covariance, locality and extension to global log-fields apply the chart estimates locally; the total defect loss is at most $\varepsilon t^2 \sharp A$ for every $\varepsilon > 0$ and all large x. The factor B^4 is absorbed because $B^n \omega \to 0$ and $B^n h\omega/t \to 0$ for every n. `interior_cost`; `Fluctuations.bare`.

7. **Assemble the energy bound.** Choose L, then W, then x: the first absorbs fine cross terms, the second mismatch losses, and the last all remaining losses, using spent-seed charges of size tT. The row sum gives exactly III.10.1's κ. `Work.E09.answer`.

8. **Bound packets and other letters.** The annulus remainder has rate $-3/2$ and is absorbed into $C\,\mathrm{bill}^5$; retained energy gives the packet estimate. Incoming bounds and rate calculus control old and logarithmic letters, including both ℓ_i and $e^{\ell_i} - 1$. `Coat.earns`; `steam_prices`; `recommends`. *(annulus statement only; other proofs read)*

9. **Control Gaussian warnings.** Coercivity and exponentially decaying rows give a positive precision and covariance, an $O(t)$ drift and a small residual; compatibility gives the no-new-seed margins. Gaussian rectangle tails, summed over local witnesses, make catastrophe weights smaller than every inverse power of β_F; these bounds combine with Step 8 into III.10.2. `shoots`; `Coat.ready`; `Coat.alarm_bills`; `Target.issue`.

10. **Differentiate letters and integrate molecules.** Compatible coordinate transport gives letter jets with rate-zero derivative scale and catastrophe probability $\mathrm{rad}^{2\sharp v}$. Leibniz bounds and Gaussian moments, followed by Cauchy–Schwarz, retain $\mathrm{rad}^{\sharp v}$ in the integrable molecule majorant; a source amplitude replaces its fare. `Utopia.whittle`; `timing`.

11. **Fuse and propagate markers.** Absolute convergence justifies Mayer regrouping; a retained forced-letter factor and rooted tree sum give $e^{-\kappa}(e^{\gamma} - 1)$, while common domination gives continuity. Connected ball-cover traversal proves I01; extra support decay and factorial rescaling give I02, composing centred embeddings gives I03, and subscale logarithmic col-

umn growth gives I04. `H01.gathered`; `H02.answer`; `H04.answer`; `I01.answer`; `I02.answer`; `I03.answer`; `I04.answer`. *(proof read; H03 symmetry moves statement only)*

12. **Choose levels and periods.** Adjacent-stage floors and larger entry couplings enforce the energy estimates; finite-request scheduling supplies the remaining requirements. Periods $4(M+p+1)$ admit arbitrarily late multiples of every side. `Bid.pricing_order`; `Wait.engine`.

3.13.5 5. Where the difficulty lies

Fine-input coefficients must remain independent of L, and mismatch coefficients independent of W, until those parameters absorb the cross terms. Transported, truncated kernels lack exact completeness: only their flat versions cancel, so local gauge fixing and tail estimates must control the defects after neighbourhood summation. Derivative integration must also retain a gain for every distinct warning address under compatible fluctuation transport.

3.13.6 6. Notation

$\sharp A$ counts all three seed types, including coincident types; $D_r(A)$ uses coarse max-distance. Φ, Φ', N are transported towers, with subscript zero denoting flat uncut operators; v denotes thinned coarse coordinates. $N_B(H)$ is the minimum connected radius-B ball-cover size. A function has rate a when, for every $v > a$, its absolute value is eventually at most β_F^v (`Rate`). In III.10.3, ρ is the rate-zero jet scale and rad is the square root of a warning-probability bound; in III.10.4, $\Gamma, \tilde{\Gamma}, m, h$ denote precision, approximate inverse, mean and scaled stationarity residual.

3.13.7 7. What remains unexplained

- The full conversion of rooted letter and packet/tree bounds to III.9's outgoing E- and K-norms, including reset, remains unwritten.
- Compatibility is constructed on the auxiliary torus by `YM.PhysicalNext.Repeat.Pane.ovation` assuming lattice symmetry of the beds at every period; that symmetry for the selected trajectory and the exact III.3 attribution remain unchecked.
- Underlying stationary coercivity, local-inverse, packet-jet and individual letter-jet proofs have not all been checked through; H03's full symmetry lift also remains unchecked.
- Additional reset estimates and the full use of `JointLoad.stage`'s covariance and trigger bounds remain outside the numbered outputs. The trigger uses $1 + B$ and $e^{(\log\log\beta_F)^2}$, absorbing fixed powers of the former into $e^{-ce^{2(\log\log\beta_F)^2}}$ per distinct endpoint.
- Exact correspondences with Balaban's lemmas, thresholds and normalizations remain open.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.14 III.11. Comparison with the free Gaussian measure: interacting rank and determinant budgets

Part: III, Ultraviolet construction (renormalization group). **Lean:** chapter 24: `YM/Next/RANK/`, `YM/PhysicalNext/AFTERNOON_INTERACTING_RANK_COMPARISON/`, `YM/PhysicalNext/RANK_COVARIANCE_ATLAS/`, `YM/MergeProof/` (with `West/`, `North/`, `East/`, `South/`). **Size:** 42,556 lines (about 5,300 declarations).

3.14.1 1. What this unit proves

The unit has two outputs. The first is the **rank comparison**. In the continuum limit, the space F_d of local fields of degree d is isomorphic to the corresponding space of the free Maxwell field, compatibly with complex conjugation and with the lattice symmetry group. Here F_d is the space of labels of degree d modulo labels whose correlations vanish. The second output is the **stress tensor of the limit**: a nonzero conserved stress tensor that satisfies the translation and rotation Ward identities and is normalized against the free Maxwell stress tensor along a scaling sequence. Both outputs rest on quantitative estimates proved here: a per-site determinant budget, chessboard bounds for connected parts of corrected observables, and comparisons of covariances with the free curvature kernel. The trial systems test a finite separating family of free one- and two-point correlations.

Notation. Set $\mu_k = \exp((\log\log\hat{\beta}_k)^2)$. The centered estimate and the trial systems use this fixed schedule quantity. G is a compact matrix group, assumed simple only where stated; $\dim G$ is the dimension of its Lie algebra; h is normalized Haar measure; $\mathrm{chord}(u) \geq 0$ is the chordal distance from u to 1; and $\mathrm{shell}(r) = \{u \in G : \mathrm{chord}(u) \leq r\}$. The torus of side $n \geq 2$ is $\Lambda_n = (\mathbb{Z}/n)^4$, with $c = n^4$ sites, $4c$ bonds and $6c$ plaquettes. The prior μ_{prior} is product Haar measure on bonds. An RG representation R on Λ_n has an inverse coupling $\beta = \beta(R)$, an exponent $\mathrm{exponent}_R$ with constant part $\mathrm{scalar}(R)$, and an unnormalized density $e^R = e^{\mathrm{exponent}_R} \cdot (\text{polymer gas})$.

Proposition III.11.1 (determinant budget on the torus, and its per-site price). (a) Let R be a lawful representation on Λ_n, $n \geq 2$, with lawful-class parameters $(\sigma, A, \Lambda, \delta, w)$, and assume $e^R \geq 0$ pointwise. Let $0 < s \leq 1 \leq b$ be a Haar budget (Step 1). Let $g > 0$ with $g \leq \mathrm{threshold}(R)$ and $\beta g^2 = 1$. Let F be a μ_{prior}-integrable complex function and $z \geq 0$, $X \in \mathbb{R}$ such that $z|F(U)| \leq \exp(\mathrm{exponent}_R(U) + cX)$ for every configuration U. Then

$$\frac{z\left|\int F\, d\mu_{\mathrm{prior}}\right|}{\int e^R\, d\mu_{\mathrm{prior}}} \;\leq\; \exp\Big(c\,\pi(R, X, g) - 4\dim G\,\log g\Big),$$

where the per-site price is

$$\pi = 2\,\mathrm{exCost}(A, \Lambda, \delta) + 6 + \beta\,\mathrm{baseBudget}(A, g) + X + 3\,\mathrm{ent} - 4\log s, \qquad \mathrm{ent} = \log b - \dim G\,\log\sigma_R,$$

with $\sigma_R = \min(1, \ell_R)$, where q_R and ℓ_R are the quadratic and linear clip coefficients. Here $\mathrm{exCost} = \delta + \mathrm{loopsPrice}$ bounds the loop and additive activity terms per site, and $\mathrm{baseBudget} = 3(q_R + A^2)g^2$ bounds the remaining action on configurations whose bonds all lie in $\mathrm{shell}(g/4)$. (b) Let $(\mathcal{A}_m)$ be the levels of an RG schedule for simple G, and let $O = (s, b)$ be a Haar budget. For all sufficiently large levels m, every period p, every representation P on the level-m torus (with $c_{m,p}$ sites), every $k \in \mathbb{N}$, every $X \leq 1 + k$, and every (g, Λ, δ) admissible at level m, the following holds:

$$\pi(P, X, g) - \frac{4\dim G\,\log g}{c_{m,p}} \;\leq\; \mathrm{workitem}(O, m, k).$$

Admissible at level m means: same clip as the stage of m, $\beta(P) \leq 4\hat{\beta}_m$, $\beta(P)g^2 = 1$, $0 < g \leq \mathrm{threshold}(P) \leq \mathrm{threshold}_m$, $\delta \leq 1$, $\Lambda \leq \mathrm{loopCap}_m$. The work item is

$$\mathrm{workitem} = 2(8\,\mathrm{paths}_m + 2) + 6 + 3(\mathrm{direct}_m + \mathrm{paths}_m^2) + (1 + k) + 3\,\mathrm{ent} - 4\log s + 1.$$

In (b), π uses $A = \mathrm{paths}_m$ and the clip of stage m. The work item depends on that stage, O and k, uniformly over the admissible couplings and periods. *Status:* theorems. *Lean:* `determinant`, `protection`, `HaarBudget`; (b) `affordable`, `Meets`, `work item`.

Proposition III.11.2 (free pair terms). Fix G, $d, n \in \mathbb{N}$, and two colored plaquette derivative stencils L, S based at the origin, with scaling degrees $\Delta_L = 2+\mathrm{ord}(L) \le d$ and $\Delta_S = 2+\mathrm{ord}(S) \le d$. There is $C \ge 0$, depending only on these data, such that the following holds. Let $x \in (\mathbb{R}^4)^n$ have distinct entries and let $i \ne j$. Suppose $D \ge 0$ satisfies $|\mathrm{Re}\,\Gamma_{(i,a),(j,b)}| \le D$ for every pair of free-current keys a, b of scaling degree at most d. Then, for every $r \ge 1$,

$$r\left|r^{\Delta_L+\Delta_S}\,\langle \mathrm{ev}_L \otimes \mathrm{ev}_S, B_r(i,j)\rangle - \mathrm{light}(L,S)(x_i,x_j)\right| \le CD.$$

Here

$$B_r(i,j) = \sum_{a,b} \mathrm{Re}\,\Gamma_{(i,a),(j,b)}\; r^{-\mathrm{gen}(a)} t_a \otimes r^{-\mathrm{gen}(b)} t_b,$$

where t_a is the polynomial jet sample of key a, $\mathrm{gen}(a)$ is its scaling degree, and ev_L applies the stencil L. The kernel $\mathrm{light}(L,S)$ is the corresponding derivative of the free Maxwell curvature two-point function. *Status:* theorem. *Lean:* `Prayer.fence`, `deck`.

Proposition III.11.3 (window estimate for the free covariance, uniform in the regulator). Let $\alpha, \eta > 0$, $\Omega \in \mathbb{R}$ and $n \in \mathbb{N}$. Then there is $C \ge 0$ with the following property. For every regulator f in the family $\mathrm{Firm}(\alpha, \Omega, \eta)$ of III.3, every G, and every pair of colored plaquette stencils a, b, the frequency chart $K = K_f(a,b)$ is measurable, vanishes for $|\xi| > 2$, is C^n away from 0, and satisfies:

- $|K(0)| \le C$;
- $|\partial^i K(\xi)|\,|\xi|^i \le C$ for $0 < |\xi| \le 2$ and $i \le n$;
- $|\partial^i (K - H_{a,b})(\xi)|\,|\xi|^i \le C|\xi|$ for $0 < |\xi| \le 1$ and $i \le n$.

Here ∂^i denotes the i-th Fréchet derivative, measured in operator norm, and $H_{a,b}$ is the degree-zero Maxwell curvature symbol for the plaquettes and colors of a, b, with heat parameter $(2\pi)^2$. If a, b have different colors, then $K = H_{a,b} = 0$. *Status:* theorem. *Lean:* `window_prices`, `Window`, `Cast`.

Proposition III.11.4 (frequent good scales produce corrected trial systems). Let G be simple and C a Wilson family satisfying the construction record. Let d be a degree and $\mathcal{F}$ a stage package for degree d (fixed band $b_0 \ge \max(6,\, s_d + 3)$, with s_d the label's scaling saving, and Taylor order $\ge d$). Let X consist of a real basis of F_d^{free}, starting with the unit, and finitely many generic one- and two-point tests that separate this space. Choose representatives of that basis in the label space. Let O be a Haar budget, J a chart ratio, and B a binding of the RG history of C to its price lists. Fix $w \in \mathbb{R}$. Assume that at infinitely many levels k the following conditions hold simultaneously:

- the four price conditions: tender, metering with (O, w), and, for some common auxiliary parameters l, r, hours with (J, B, l, r, w) and vents with (J, B, l, r);
- the lantern-mark condition for X.

Then there is a trial system Z for X that travels with the station and satisfies the purchase conditions:

- depths $k_1 < k_2 < \ldots$ and test scales $s_i = a_{k_i} \mu_{k_i}^{b'}$ (fixed $b' > 0$) strictly decreasing to 0, with strictly increasing width levels $\mathrm{width}_i \le k_i$ and the stage index constant on $[\mathrm{width}_i, k_i]$;
- for each trial i, coefficient vectors $v_i(N,p)$ of corrected observables that are real, bounded uniformly in (N,p), and have first entry the reference unit;
- generic test configurations lying strictly inside the centred width cell, whose rescalings converge to the prescribed configuration of each row;

- for each trial i, eventually in N and for every period p, on the centred tiling of the cutoff torus: for every row, every basis column and every nonempty set s of insertion modes, the chessboard root, at coupling $\beta_{N,p}$, of the s-connected part of the corrected observable (taken relative to its free Maxwell one- and two-point profile) is at most $\gamma_i = \mu_{k_i}^{-1/32} \to 0$.

Status: theorem, with the frequent-scale conditions carried as hypotheses. In the main chain they are discharged as follows. In `Accent.rated`, the lantern marks hold eventually in k by `bursts`. The four price conditions are supplied to the rank assembly by `franchise` (this unit) inside the assembly `jointOffers`. That assembly is closed by `uv_log` (III.13), whose only hypotheses are that G is simple and $b > 0$. *Lean:* `cast_off`, `wheeled`, `v_culminates`, `v_obtain_purchases`, `Purchases`.

Proposition III.11.5 (the field spaces of the limit are those of the free Maxwell field). Let G be simple. Write $\mathcal{L}_d$ for abstract labels and $\mathcal{V}_d$ for their fixed coordinate space. Let C be a Wilson family satisfying the construction record and the centred estimate: for each d and $R, \varepsilon > 0$ there is $A \geq 0$ such that, at every level $m \geq \mathrm{ref}(d)$, eventually in N and for all p, every label of degree d and norm $\leq R$ has centred budget $A\, a_m^{-d-\varepsilon}$. Assume:

- (i) a projector presentation of the labels: for each d and cutoff (N, p), a projector $\Pi_{d,N,p}$ on the coordinate label space $\mathcal{V}_d$ whose kernel is the image of the Maxwell equation ideal $\mathcal{E}_d \subset \mathcal{L}_d$ under an invertible corrected label map $\mathcal{L}_d \to \mathcal{V}_d$. For each d, both are real and equivariant for all sufficiently large N, uniformly in p. Assume also affordability and relay-cost bounds along every extraction;
- (ii) the one-step cost package along the gauge-door maps between successive RG levels;
- (iii) panels Q_d, the mark exponents $c_d = 2(w_d + 1)$, a chart ratio and a binding such that, for each d, the four price conditions of III.11.4 hold simultaneously at infinitely many levels, with $w = -3$. The panels are specialized to the family using c_d.

Then for every extraction order f there are a refinement g, a smooth limit table T with a differentiation plan E, and the extraction record of IV.1 along $f \circ g$ (raw limits, bins, spans, cleaning to T, Euclidean axioms), together with a seed record for $(C, f \circ g, T)$. The seed record consists of: cutoff, ordered and generic point limits, Euclidean axioms, cell hypothesis, and corrected rank widths. Moreover the following holds: there is a real structure on the free Maxwell table and, for every d, a linear isomorphism

$$\Phi_d : \; F_d(T) \;\xrightarrow{\sim}\; F_d^{\mathrm{free}} = \mathcal{L}_d/\mathcal{E}_d$$

that intertwines complex conjugation and the action of the signed permutations of the four axes. More precisely, there is a real, equivariant linear isomorphism $L_d : \mathcal{L}_d \to \mathcal{V}_d$ carrying $\mathcal{E}_d$ onto $\mathrm{null}_d(T)$. Thus $\dim F_d(T) = \dim F_d^{\mathrm{free}}$; the identification of null spaces uses this corrected map. *Status:* theorem with hypotheses (i)–(iii). All three are discharged in the main chain: (i) by `reception` through `chaletrail` (this unit), (iii) by `franchise`, and the binding by `castle`. The one-step package (ii) and the centred estimate are also included in `Chateau` and consumed by `uv_log` (III.13). *Lean:* `Accent.rated`, `Maxwell.Ranks`, `Spec.Seed`, `Scotia`, `Wcmain`.

Proposition III.11.6 (the stress tensor of the limit, normalized against free Maxwell). (a) Let C satisfy the construction record and f be an extraction order with ordered and generic point limits, giving a table S. Assume:

- S satisfies the Euclidean axioms (with spin structure H) and the cell hypothesis;
- W is a differentiation plan for S with its inclusion and derivative laws;

- $\dim F_d(S) \le \dim F_d^{\mathrm{free}}$ for every d;
- $\mathcal{T}$ is a free Maxwell comparison model, with Euclidean axioms a, inclusion, derivative and orientation laws, and a divergence-free stress;
- the station hypothesis linking C, f and a holds;
- the shell estimate of C holds.

Then W satisfies the orientation law, and there are a stress record I (stress tensor $\mathcal{J}$, translation parts, spin rows) and a scaling X of S such that:

- $\mathcal{J} \neq 0$ and $\mathcal{J}$ lies in the kernel space of H;
- the scaled tables approach the isotropic panel, and the scaling is stable and compatible between the spin structures of S and $\mathcal{T}$;
- the scaled $\mathcal{J}$ is metered against the free stress tensor $\mathcal{T}.\mathcal{J}$;
- $\mathcal{J}$ is conserved and symmetric and satisfies the translation and rotation Ward identities (III.11.7);
- the action law relating $\mathcal{J}$ to the free model holds.

(b) Let G be simple, let C carry the shell estimate, and assume the cofinal grid approximation condition and weak completeness of the corrected comparison tests for f, together with the seed record of III.11.5 for (C, f, S). Then for every differentiation plan W with inclusion and derivative laws there are Maxwell data M and a stress record I with $\mathcal{J} \neq 0$ satisfying the UV interface record $\mathrm{UV}(C, f, S, W, M, I)$. *Status:* theorems. In (b), the free model comes from `maxwell_supplies` (VI.2). The station hypothesis is built from the seed and weak completeness, and the hypothesis `DimAbove` from the seed's corrected rank widths; this is how III.11.5 feeds III.11.6. The shell budget is proved by `laurelE` and `fugacious` (this unit) inside `uv_log`. *Lean:* `Acme.readyA`, `Acme.emanate`, `Aroma`.

Proposition III.11.7 (Ward identity: the flux of the stress against a Killing field is a local field). Let C satisfy the construction record, let f be an extraction order with the cell hypothesis and generic point limits S, and suppose S satisfies the Euclidean axioms with real structure H. Let W satisfy the derivative laws, and assume the shell estimate of III.11.6. Let:

- $\mathcal{J}$ be a stress tensor of S that is conserved with respect to W, and $v \in F_d(S)$;
- U be a smooth Killing field on $\mathbb{R}^4$, homogeneous of degree p ($U(ry) = r^p U(y)$ for $r > 0$);
- $d, D, p \in \mathbb{N}$ and $D + 3 + p = d + 4$.

Then there is $B \in F_D(S)$ with the following property. For every n, all fields $q_1, \dots, q_n$, every configuration of distinct points $x_0, \dots, x_n$, and every window χ adapted to x (smooth, compactly supported, equal to 1 near x_0, with support avoiding $x_1, \dots, x_n$), the flux of $\mathcal{J}$ against $U(\cdot - x_0)$ and χ, inserted with v at x_0 and q_k at x_k, equals the correlation of B at x_0 with the q_k at x_k. The sign convention is

$$\text{flux} = -\sum_{i,j} \int_{\mathbb{R}^4} \partial_i \chi(y)\, U_j(y - x_0)\, \langle \mathcal{J}_{ij}(y)\, v(x_0)\, q_1(x_1) \cdots q_n(x_n) \rangle \, dy.$$

Nonzero homogeneous Killing fields have $p = 0$ (translations, $D = d + 1$) or $p = 1$ (rotations, $D = d$). *Status:* theorem. *Lean:* `annular_fieldA`, `tenet`, `shells_law`.

3.14.2 2. What it uses

- **I.1, II.2:** the torus, Wilson measure, product Haar prior, lawful RG representations, construction record, cells and chessboard roots; `prior`, `Rep.Lawful`.
- **II.1:** label spaces, their real structures, the Maxwell equation ideal and the free Wick realization; `equationIdeal`.
- **III.2:** stages, thresholds, clip coefficients, loop caps, delays and waiting windows; `ruese`.
- **III.3–III.4:** free covariances, regulator symbol bounds, Gaussian integration and window-to-kernel transfer; `intended`.
- **III.9–III.10:** coupling growth, exact RG histories and vacuum identities, one-step bounds and the waiting-window engine; `Wait.engine`.
- **IV.1:** extraction, refinements, raw limits, spans and cleaning to smooth tables; `Base`.
- **VI.1–VI.2:** windows and flux, finite faithfulness of testers, free Maxwell data and stress, and production of the current; `maxwell_supplies`, `produce_current`.
- **III.13:** the common Wilson family closes the conditional chain; `uv_log` has only the hypotheses that G is simple and $b > 0$.

Standard results: the chessboard estimate and reflection positivity of the Wilson action; axial tree gauge and Haar invariance; small-ball Haar volume estimates; Wick expansion; Schur complements; frequency-symbol estimates; finite-dimensional duality; and Ward identities for conserved symmetric stress tensors and Killing fields.

Constants and assumptions: the Haar budget is fixed independently of stages and volumes by `integral_budget`. The projector, one-step, centred, frequent-scale, binding, shell, grid-approximation and weak-completeness hypotheses remain explicit in §1; their main-chain producers are identified there. The older one-step interfaces in `LedgerGate.lean` are conditional definitions, not consequences of lawful representations.

3.14.3 3. Strategy

The Maxwell equations give the upper bound on the interacting field-space dimensions. Corrected observables tested at short distances give the reverse bound: their one- and two-point correlations approach a finite separating family of free correlations, so their classes are eventually independent. The two bounds identify the limit null space with a limit projector kernel. Conjugating a cutoff projector to the limit projector then gives the isomorphism compatible with conjugation and signed permutations.

The quantitative comparison uses the exact RG history to carry reflected products to a coarse torus, then bounds a ratio of Haar integrals. Plaquette elimination gives $3c - 3$ numerator factors, while tree gauge leaves $3c + 1$ vacuum factors; the remaining coupling cost is only $g^{-4\dim G} = \beta^{2\dim G}$ in total. Its chessboard root is affordable, allowing connected errors to tend to zero along infinitely many good scales. For the stress tensor, comparison with free Maxwell fixes its normalization along a scaling sequence; conservation, shell estimates and finite-dimensional duality represent each Killing-field flux by a local field.

3.14.4 4. Steps

1. **Haar budget.** There are $0 < s \le 1 \le b$ with $h(\mathrm{shell}(r/4)) \ge s r^{\dim G}$ for $0 < r \le 1$ and $\int e^{-\mathrm{chord}(u)/t}\, dh(u) \le b t^{\dim G}$ for $t > 0$. Exponential coordinates give the local volume estimates; compactness and a shell sum give the integral estimate. `integral_budget`

2. **Plaquette elimination.** For $n \geq 2$ and continuous $\varphi : G \to [0,1]$, $\int \prod_p \varphi(U_p)\, d\mu_{\text{prior}} \leq (\int \varphi\, dh)^{3c-3}$. A triangular sweep assigns $3c-3$ plaquettes to distinct bonds, whose successive Haar integrations factor; the unselected factors are at most 1. `elimination`, `charged_integral` *(proof read; sweep count and triangularity statement only)*

3. **Vacuum and numerator.** Under III.11.1(a), tree gauge and restriction of the $3c+1$ free bonds to shell$(g/4)$ give the vacuum lower bound, while the clipped action retains exponential chord decay for the numerator. Applying Steps 1–2 and dividing gives $\exp(c\pi)g^{-4\dim G}$; the four-factor difference between the counts accounts for the remaining power of g. `vacuum_bound`, `exponent_retains`, `determinant`

4. **Per-site price and chessboard root.** Under III.11.1(b), the stage bounds give $\pi \leq$ workitem-1 and $-4\dim G \log g/c_{m,p} \leq 1$ eventually, uniformly in admissible parameters and periods. In the trial comparison, the exact vacuum identity and local activity decomposition transfer the determinant estimate to the chessboard root; deleting additive activities costs the 2δ already included in the price. Allocation and affordability, with allowance e^{2I} and $I \geq$ workitem, then give root error e^{-I} for every nonempty connected part. `affordable`, `s_postage`, `coupon` *(proof read; flow identity and cell-stamp bookkeeping statement only)*

5. **Free pair and window estimates.** In III.11.2, stencil evaluation kills keys below the stencil degree, the equal-degree terms give the free curvature kernel, and every remaining term gains r^{-1}, giving the bound CD/r on the scaled remainder. For III.11.3, the regulator chart is a symbol of order zero and differs from the heat symbol by a symbol of order one; the change of variables $\zeta = 2\pi\xi$ and the finite number of plaquette pairs give a uniform constant. `Prayer.fence`, `window_prices`

6. **Covariance splitting.** For an atlas compatible with the schedule and retained unmasked quadratic forms, satisfying symmetry, coercivity, inverse and crossing laws, and a step $i < N$ with a tree forest and the required torus size, the complete curvature covariances satisfy

$$\mathcal{C}_{i+1} - W_i \mathcal{C}_i W_i^{\mathsf{T}} = d_f \Gamma_i d_f^{\mathsf{T}}, \qquad \Gamma_i \succeq 0.$$

   This is the Schur-complement identity on the curl image; uniform resolvent estimates give the propagation constant for fixed kernel choices and stage indices $j, t \geq 4$ in the range $d-1 \leq j-2$ (natural-number subtraction). `one_fold`, `train_issue`

7. **Local polynomial comparison.** For regular local observables with the prescribed derivative bounds, Taylor expansion in link coordinates gives bounded polynomial coefficients and a remainder of the required order. For zero-mean formal Gaussian data with symmetric covariance vanishing within each location, Wick expansion runs over cross-location pairings without requiring positive covariance; under the comparison's common-mask, support, coefficient, mean and covariance hypotheses, these moments give the corrected connected-component bounds. The allocation, integrability, locality and compatibility hypotheses give exact sector decomposition and factorization over disjoint supports. `minted`, `headed`, `console`, `rendered`, `peer_beams` *(proof read; local comparison and allocation inputs statement only)*

8. **Corrected trials.** Under III.11.4, one correction map and polynomial approximations serve all test rows; the free-profile correction cancels the leading terms, and the one-step estimates propagate the remaining source bounds from the middle level to the width level. Steps 3–4 give the purchase bounds, and an inductive choice among the good levels gives increasing

depths and widths, decreasing test scales, and bounded real coefficient vectors with the reference unit first. `v_paid_factory`, `v_culminates`, `v_conditioned`, `v_obtain_purchases` *(proof read; analytic inputs and helper estimates statement only)*

9. **Supply of good levels.** For simple G, the construction record and one-step package with the prescribed weights and multiplicities at least 1, a fixed degree-d column family and stage package with mark exponent $c \geq 2(w_d + 1)$ satisfy the lantern-mark condition at every sufficiently large metered level. Compatible pre-L prices and one-step bounds, stage thresholds below block lengths, eventual presence of each degree index, nonnegative mark exponents and a Haar budget give panels and an enlarged waiting window on which every Wilson family paid on that window and every binding with those prices have infinitely many simultaneous good levels for each d. `bursts`, `franchise` *(proof structure read; analytic inputs and helpers statement only)*

10. **Equations and independent columns.** Under III.11.5(i), free probes separate the on-shell sector and give $\dim \mathcal{L}_d \leq \dim \ker \Pi_{d,N,p} + \dim F_d^{\mathrm{free}}$ at each cutoff. For a Wilson family satisfying the construction record, with generic point limits and continuous kernels integrable against every separated test function, purchases and iterated coefficient limits give moment errors $P_k \gamma_k \to 0$, so the limiting corrected columns are independent modulo the integrated-test null space for all sufficiently large trials. Relay estimates put the limit projector kernel inside the null space; with the centred port, raw limits, convergent relay projectors, equation counts and trials in every degree, extraction and the dimension comparison give a smooth table T, its seed, and $\mathrm{null}_d(T) = \ker \Pi_\infty$. `savings`, `measured`, `arriving_seed` *(proof read; free positivity, chessboard and analytic inputs statement only)*

11. **Equivariant identification.** Under the projector presentation and its extraction record, if $A_i \to \Pi_\infty$, then $Y_i = \Pi_\infty A_i + (1 - \Pi_\infty)(1 - A_i) \to 1$ and $Y_i A_i = \Pi_\infty Y_i$. For large i, Y_i is invertible, real and equivariant; composing it with the corrected label map carries $\mathcal{E}_d$ onto $\mathrm{null}_d(T)$ and induces III.11.5's quotient isomorphism. `drumroll`, `thrush`, `Accent.rated`

12. **Flux becomes a local field.** Under III.11.7, conservation permits windows on shrinking annuli; the shell estimate shows that every finite combination of grid testers annihilating F_D also annihilates the flux. Finite-dimensional duality supplies a matching field for each finite tester family, finite faithfulness gives a single field for all testers, and translation covariance, continuity and grid density extend the identity to all distinct configurations. Constant and rotational Killing fields give the translation and rotation maps. `tenet`, `annular_fieldA`, `shells_law` *(proof read; annulus comparison inputs statement only)*

13. **Stress and ultraviolet records.** Under III.11.6(a), the orientation law and station equivalences allow the free comparison to produce a nonzero current and scaling sequence; zero current divergence gives conservation, the weak scaling comparison gives the action law, and Step 12 gives the stress laws. Under III.11.6(b), canonical free Maxwell data, the seed's corrected rank widths and weak completeness supply the comparison model, dimension bound and station hypothesis needed for (a). `Acme.readyA`, `Acme.emanate` *(proof read; helper estimates statement only)*

3.14.5 5. Where the difficulty lies

The determinant comparison must avoid a per-site loss of $\log \beta$: that would overwhelm the vanishing chessboard error. Tree-gauge counting leaves only the volume-independent factor

$\beta^{2\dim G}$; this is a Haar integral estimate rather than a Hessian determinant calculation. The lower rank bound requires short-distance comparison at infinitely many scales, whereas the upper bound is algebraic at every cutoff. The isomorphism comes from conjugating projectors, and its symmetry statement concerns complex representations with compatible conjugations under signed permutations. Constants must be chosen before block lengths and uniformly in the period; eventual estimates require the specified delays before use on comparison windows.

3.14.6 6. Notation

Symbol	Meaning and Lean link
μ_k, w_d	Fixed schedule factor $\exp((\log\log\widehat{\beta}_k)^2)$; coefficient exponent. `monitor`, `winter`
(s,b), σ_R, π, work item	Haar budget, clip strength, per-site determinant price and stage allowance of III.11.1. `HaarBudget`, `strength`, `protection`, `work item`
$\mathcal{L}_d$, $\mathcal{V}_d$, $\mathcal{E}_d$	Abstract labels, coordinate labels and Maxwell equation ideal; $F_d^{\mathrm{free}} = \mathcal{L}_d/\mathcal{E}_d$. `equationIdeal`
$F_d(T)$, $\mathrm{null}_d(T)$	Local fields of degree d and labels with vanishing correlations. `Maxwell.Ranks`
$\Pi_{d,N,p}$, Π_∞	Cutoff and limit projectors; their kernels identify the equations after correction. `Scotia`
Stage package and price conditions	Stage data; tender, metering, hours, vents and lantern marks in III.11.4. `Cohesion`, `Tender`, `Metering`, `ShowHours`, `Vents`, `LanternMarks`
$\mathcal{C}_i$, W_i, d_f, Γ_i	Complete curvature covariance, passage map, fine exterior derivative and uncut tree covariance. `one_fold`
$\mathcal{J}$, shell budget	Stress tensor; annular estimates with radii tending to zero. `Aroma`
Seed, station, grid approximation, weak completeness	Records used in III.11.5–III.11.6. `Spec.Seed`, `Stations`, `Grill`, `CompleteWeak`

3.14.7 7. What remains unexplained

- The discrete stencil, interpolation and shell estimates underlying the annulus bound have not been independently checked through their analytic dependencies; `zone`.
- The nonlinear integration and remainder estimates beneath the one-step comparison were not checked in full. The correspondence between the older conditional interfaces in `LedgerGate.lean` and all restarted replacements remains to be documented.
- Complete mathematical estimates for the stage, launch, hours, vents, relay, station and shell records remain to be written. Their structural hypotheses cannot be replaced by the displayed inequalities alone.
- Sweep triangularity and the auxiliary estimates were checked at statement level. The lower-level proofs supplying the centred estimate, one-step package and schedule existence were not checked in full.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.15 III.12. Bounds uniform in the volume and convergence as the spacing tends to zero

Part: III, Ultraviolet construction (renormalization group). **Lean:** chapter 30, `YM/FullVolume/`. **Size:** 18,973 lines in the main-theorem closure.

3.15.1 1. What this unit proves

Proposition III.12.1 (kernel bounds uniform in the volume). Let R be a run for a simple gauge group G, with Wilson family C. Fix n, labels $v_i \in \mathrm{Lab}_{d_i}$, a finite list λ of pairs (slot, axis), and $r > 0$. There are $B \geq 0, N_0$ such that, for every $N \geq N_0$, every p and every integer configuration x with $a_N \mathrm{dist}(x_i, x_j) \geq r$ for $i \neq j$,

$$\left| \Delta_\lambda \mathbb{E}_{N,p} \prod_{i=1}^{n} O_{N,p}(v_i; x_i) \right| \leq B.$$

Each difference acts in its indicated slot; distance is periodic ℓ^∞ distance. Constants may depend on $R, n, (v_i), \lambda, r$, but not p. *Status:* theorem for every run. *Lean:* `genuine`.

Proposition III.12.2 (derivative closure). For a run with simple G, every sequence $w_{N,p} \in \mathrm{Lab}_d$ bounded by some $K \geq 0$ eventually in the mesh, and every finite axis list l, the trial $\Delta_l O(w)$ is funded. *Status:* theorem. *Lean:* `Run.financed`.

Proposition III.12.3 (closure lemma). Fix d. Assume a consistent Wilson family C and:

1. the centred budget;
2. the scale-selection data;
3. sound sections V_e and linear automorphisms W_e of abstract labels for every degree e, satisfying the section, symmetry and exact local identities described below, with `Portfolio`;
4. a presentation π and the full `section-comparison` data, including quantitative comparison with V_e, identification of $\ker \pi$ with the dressed equation ideal, and $\sup_{N,p} \|\pi_{N,p}\| < \infty$ in each degree;
5. degree-d detector rows with the placement and bounded-lift conditions and short-distance comparison of `VitalRank`.

If F is funded, every $F_{N,p}(o)$ is continuous in the gauge field, l is an axis list, and x is a degree-d coin with $O(x) - \Delta_l F$ owed at cost $u_{N,p} \to 0$ in the mesh, then $\Delta_l F$ is funded. *Status:* conditional theorem; a run's ties and short-distance estimates supply these hypotheses. *Lean:* `improved`.

Proposition III.12.4 (uniform rank detection). Under hypotheses 1–5 of III.12.3, fix l. For all large k and all sufficiently small $b > 0$, there is $Z \geq 0$ such that, eventually in the mesh,

$$\|\pi_{N,p} x\| \leq Z \|\mathcal{P}_{k,b,l;N,p}(\pi_{N,p} x)\| \quad (x \in \mathrm{Lab}_d).$$

The detector takes correlations with fixed partners and successively averages the first slot in the directions of l over lengths between b and $2b$; repeated axes give repeated averages. *Status:* conditional theorem. *Lean:* `trained`.

Proposition III.12.5 (negligible equation labels). Assume consistency of C and hypotheses 2–4 of III.12.3. Every degree-d coin x with $\pi_{N,p}x_{N,p} = 0$ for all (N,p) has $O(x)$ owed at cost $a_N^{1/32}$. *Status:* theorem. *Lean:* `banked`.

Proposition III.12.6 (coupling power-series bounds). Let R be a run with a `Campaign`, `Pillars` and bounds `Sky`. Let $\bar\beta$ be the target level-0 coupling and $w > 0$ the fan width. Fix p, an n-tuple of labels V_0 and $r > 0$. There exist $b, m \in \mathbb{N}$, $m \geq 1$, $B \geq 0$ and $c \in (0,1]$ such that the following holds for every $m \leq j < N$ and $a \in [-1,1]$.

Take any family $F(s)$, $s \in [-1,1]$, of admissible campaign members with common block sequence, sections and labels, whose level-0 inverse coupling at (N,p) is $\bar\beta + was$. For every r-separated configuration o in $\Lambda_{j+1,p}$ embedded into $\Lambda_{N,p}$,

$$\mathbb{E}_{N,p}^{F(cs/2)} \prod_i O(V_{0,i}; o_i) = \sum_{i \geq 0} P_i s^i, \qquad \sum_i |P_i| \leq B a_{j+1}^{-b} \quad (s \in [-1,1]).$$

Expectation and representatives belong to the attached auxiliary transfer family; its correlations agree with the successful trajectory at positive cutoffs. Constants depend only on campaign data, p, V_0, r. *Status:* conditional theorem; VIII.1 supplies the campaign records. *Lean:* `Face.financed`.

Proposition III.12.7 (fixed-volume Cauchy property). Fix simple G, one RG construction with stationary-path bounds, free amplification, observable-transfer bounds and common reference sections, and assume its uniform perturbation estimates. Fix p. Let C_n come from histories at cutoffs $f(n) \to \infty$. Assume:

- Each history is a selected branch from level k, agreeing with a fixed kinetic path; stationary kinetic paths converge with strict reserve in their admissible bounds. At every fixed $i \geq k$, inverse couplings are Cauchy and activities have Cauchy representatives in the common weighted space.
- Each degree used has reference level $t(d) \geq k$ with $d + 3 \leq J(t(d))$; transfer data are based at its reference section and compose branch transfers.
- For every sufficiently large base level r, multiscale remainder bounds hold for all sufficiently large n. Label allowances $\zeta(d,r)$ are eventually positive for each degree used; occupancy allowances are at least 1 at every level.

For every finite label list and every configuration x with $x_i - x_j \notin L_p\mathbb{Z}^4$ for $i \neq j$, lying exactly on some common lattice $x = a_r z$, z integer, the cutoff-$f(n)$ correlations at coordinatewise ceiling addresses of x are Cauchy. *Status:* conditional theorem. A run with campaign, pillars, fixed-level convergence and uniform perturbation estimates gives $C_n = C$, $f(n) = n$. *Lean:* `Rising.cinema`.

Proposition III.12.8 (positional ultraviolet limit). Let C be a Wilson family and e a strictly increasing mesh sequence. Assume the kernel-bound conclusion of III.12.1 and that ceiling-address correlations are Cauchy along e whenever $x_{i\mu} - x_{j\mu} \notin L_p\mathbb{Z}$ for every $i \neq j$ and axis μ. For every p, n and label list there is a function f on $(\mathbb{R}^4)^n$, smooth on $\mathrm{NoHit}_p = \{x : x_i - x_j \notin L_p\mathbb{Z}^4 \ (i \neq j)\}$, with these properties:

- For every smooth compactly supported t supported off collisions and strictly inside the centred period cell in every coordinate, smeared lattice sums converge along e to $\int tf$.
- Fix any finite list of coordinate directions (slot, axis), with integers $q_-, q_+ \geq 0$, $q_- + q_+ > 0$ in each direction, using the stencil $[F(x + q_+e_\mu) - F(x - q_-e_\mu)]/[(q_- + q_+)a_N]$. Write g for the corresponding iterated derivative of f. Along $e(s_i)$ for every $s_i \to \infty$, these differences

at lattice configurations whose physical positions converge to $x \in \mathrm{NoHit}_p$ converge to $g(x)$. Their multilinear interpolations converge locally uniformly there. For every test t above, tg is integrable and smeared differences converge to $\int tg$.

Status: conditional theorem; the run hypotheses following III.12.7 give the full sequence $e(s) = s$. *Lean:* `positioned`; full-sequence instance `Campaign.realUV`.

3.15.2 2. What it uses

- **II.1–II.2, III.14:** labels, representatives, consistency and the `Run` record, whose ties supply the inputs of III.12.3.
- **III.2, III.11:** numerical parameters, scale selection, section comparisons, regularity, local matching and short-distance estimates; the centred budget `Wcmain` states: for every degree d, $R > 0$ and $\varepsilon > 0$, there is $A \geq 0$ such that for every $m \geq t(d)$, eventually in the mesh, every $\|v\| \leq R$ and site o give an observable localized in the cell centred at o, with root at most $Aa_m^{-d-\varepsilon}$.
- **IV.1:** `Seam` gives chessboard and root triangle inequalities; positional Lipschitz bounds and dense-set convergence. (The offset comparison comes from II.1.)
- **III.6, III.8–III.10:** histories, selected branches, stationary paths and transfer estimates. **VIII.1:** `Winter` supplies fixed-level convergence, kinetic reserve and free amplification; assembly supplies uniform perturbation inputs.
- **Standard results:** Wilson reflection positivity (Osterwalder–Seiler); chessboard estimates and seminorm (Fröhlich–Israel–Lieb–Simon, Fröhlich–Lieb, Shlosman, Biskup); summation by parts, finite-dimensional injectivity stability, Arzelà–Ascoli, Taylor's theorem, absolutely convergent power series and dominated convergence. Volume-uniform control of effective actions: Balaban (1984–1989).

The section hypotheses require the prescribed polynomial realization and finite regularity bounds; dressings commute with signed lattice symmetries and conjugation. Dressed equation labels have zero Wilson expectation against every bounded measurable spectator depending only on links beyond a fixed lattice distance, for every nonzero inverse coupling and sufficiently large torus; spectators need not be gauge invariant.

Full section-comparison data include terminal-coordinate automorphisms $A_{d;N,p}$, auxiliary columns and sub-power $E_d(N)$. Eventually, uniformly in p, $A_{d;N,p}$ is the symmetry-averaged column map and its inverse increases coefficient size by at most 2. For each column and its signed or conjugated versions, the centred root of the auxiliary-minus-chosen section evaluation is at most $C_d(m,k)E_d(N)a_N^{1/8}$ for every admissible $k < N$ and $m \leq N$; scale selection chooses k for each large m. The kernel is the image of the Maxwell equation ideal under dressing, threshold rescaling, terminal map and $A_{d;N,p}$. Projection symmetries and finite-cutoff equation and difference comparisons are also required.

Detector rows use test configurations with distinct projections on every axis in a box of side $a_{\mathrm{width}(k)}$, and lifted partner labels bounded uniformly in N, p and finite indices for fixed k. Their short-distance cumulant comparisons approach the prescribed free Maxwell tests.

3.15.3 3. Strategy

Normalized differences introduce a_N^{-1}, beyond the centred budget for bounded labels. Local matching replaces each differenced observable by a primary with sub-power inverse terminal

coordinates and vanishing root error. Averaged partner correlations bound its projected physical label through an injective free Maxwell detector; equation labels contribute a vanishing root. This funds differences, and chessboard estimates bound their separated products.

At fixed volume, compare full correlations with scalar-normalized correlations at an intermediate level. Fixed-level convergence makes the latter Cauchy, while an exponentially small tail in the budget radius tends to zero. Positional bounds extend convergence from common-lattice configurations; Taylor estimates identify derivatives. For coupling dependence, polynomial coefficient bounds and smaller parameter intervals turn level differences into summable coefficient errors.

3.15.4 4. Steps

1. **Match differences.** For a run with simple G and an eventually bounded degree-d label sequence w, normalized differences preserve regularity with sub-power bounds after increasing degree. Local matching gives a degree-$(d + |l|)$ coin x with $O(x) - \Delta_l O(w)$ owed at $a_N^{1/16}$. `prepared_rows`.
2. **Remove equation labels.** Under III.12.5's hypotheses, auxiliary equation sections have zero root: their exact distant-spectator identity annihilates the reflected product. Section comparison leaves cost $a_N^{1/32}$. `banked`.
3. **Detect projected labels.** Under III.12.3's hypotheses, averaging differences telescopes and bounds tests by $(2/b)^{|l|}$ times undifferenced correlations. Short-distance comparison approaches the injective free Maxwell matrix; finite-dimensional stability and the rank bound give III.12.4. `trained`.
4. **Close the budget.** Under III.12.3's hypotheses, set $X = \pi x$. Apply negligible-equation control to $(x - X)/(1 + \|X\|)$, obtaining $\|X\| \le Zc + \frac{1}{2}(1 + \|X\|)$. Thus X is bounded; the centred budget funds $O(X)$, and vanishing errors fund the desired difference. `cap`, `improved`.
5. **Bound kernels.** Once all fixed-label differences are funded, place separated insertions in distinct cells with interior margins; offset comparison and the chessboard inequality bound products. Slot differences act on their insertions, giving III.12.1. `funding`.
6. **Control coupling series.** Under III.12.6's campaign hypotheses, composed transfer bounds give coefficient sums polynomial in a_{j+1}^{-1}. With fixed-level convergence and perturbation inputs, interval reduction converts real $O(a_j)$ level differences to summable $C\sqrt{a_j}$ coefficient bounds. For fixed p, labels, compact $K \subset \mathrm{NoHit}_p$ and $q \in \mathbb{N}$, this gives a positive parameter width independent of cutoff and a neighbourhood of K with limits C^q in position and analytic in the complex parameter; kernels with spatial stencils of length at most q, their interpolations and admissible smeared sums have analytic continuations whose parameter derivatives of every fixed order converging uniformly on compact subdiscs and compact spatial subsets. `Fan.standing`.
7. **Take the mesh limit.** Under III.12.7's hypotheses, fixed-level effective insertions and normalized densities are uniformly Cauchy; Haar integration and positive partition-function bounds give Cauchy correlations. For large fixed j, the full correlation differs by at most $2F_0 e^{-B_j/64}$, uniformly in sufficiently large cutoffs, with F_0 independent of j and cutoff and $B_j \to \infty$. `Sever.echoes`, `Rising.arrives`.
8. **Recover spatial derivatives.** Under III.12.8's hypotheses, equicontinuity extends dense-set convergence. Taylor comparison with differences over a fixed physical interval identifies successive derivative limits; stencil averaging, multilinear interpolation and dominated convergence give the remaining conclusions. `positioned`.

3.15.5 5. Where the difficulty lies

Matching controls inverse terminal coordinates; rank detection must bound physical labels while section comparison controls the projection kernel. Bounds are volume-uniform, but convergence fixes p first. Every derivative order is fixed before choosing the cutoff threshold; analytic widths may depend on volume, compact spatial set and derivative order. The additive action scalar need not converge. Collision extension belongs to VII.4.

3.15.6 6. Notation

- $a_N = (\ell_0 \cdots \ell_{N-1})^{-1}$; $\Lambda_{N,p}$ has physical period L_p. $O(v; o)$ represents $v \in \mathrm{Lab}_d$; $\mathbb{E}_{N,p}$ is Wilson expectation.
- Eventually in the mesh: all $N \geq N_0$, all p; N_0 may depend on a fixed coarse level. Sub-power growth means $\|w_{N,p}\| \leq a_N^{-\varepsilon}$ eventually for every $\varepsilon > 0$. A `Coin` is a label sequence x with $\|\tau_{d;N,p}^{-1} x_{N,p}\|$ of sub-power growth, where τ is the terminal coordinate isomorphism.
- $\Delta_\mu F(o) = [F(o + e_\mu) - F(o)]/a_N$; lists indicate iteration. A trial is a site-indexed family of gauge-field functions; it is local if eventually translation covariant, gauge invariant, measurable, bounded and supported on one fixed finite link stencil.
- $R_m(H)$ is the absolute reflected-product expectation to power $1/(L_p/a_m)^4$ on centred cells of physical side a_m, conjugating odd reflections. A local trial is **funded** if, for all large m, eventually in the mesh, its value at 0 is localized in the central cell and has root at most $u(m)$ independent of N, p; **owed** replaces $u(m)$ by a cost $v(N, p)$ independent of m.
- $\pi_{N,p}$ is an idempotent label map; its dressed-equation kernel is an additional hypothesis. Ceiling addresses are $\alpha_N(x)_{i\mu} = \lceil x_{i\mu}/a_N \rceil$. $B_j = \lceil (\log \hat{\beta}_j)^2 \rceil$ is the budget radius.

3.15.7 7. What remains unexplained

The local matching estimate, construction of auxiliary equation sections and relation to lattice Schwinger–Dyson identities remain imported or unverified here. One-step transfer capacity, intermediate weighted analytic estimates, spatial derivative identification for analytic limits, and multiscale remainder and omission bounds need their source analyses.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.16 III.13. Extension to all labels: one common Wilson family

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/SharpGram/Common.lean`, `YM/SharpGram/Measured.lean`, `YM/SharpGram/Assembly/`, `YM/UVExtension/`. **Size:** about 22,100 lines in the unit inventory.

3.16.1 1. What this unit proves

Throughout, G is a compact matrix group with simple nonabelian real Lie algebra and $b > 0$ is the Gaussian block-spin penalty. Levels run from the coarsest, 0, to the cutoff, N; p indexes periods. Notation is collected in §6.

Proposition III.13.1 (one ultraviolet and logarithmic family). There exists a Wilson family C, with penalty b, satisfying all the following.

1. It is calibrated: one full choice of constants and a directed history generate its history, with the per-step estimates of III.9–III.12.
2. It is consistent: the trajectory is selected, local sections are sound, and, for every degree d, $d+3 \le J(\mathrm{ref}(d))$, $d+1 \le \mathrm{order}_m$ and $d \le \mathrm{power}_m$ for $m \ge \mathrm{ref}(d)$. Every degree-d stencil point satisfies $20(|x_i|+1) \le \mathrm{side}(N,p)$ in each coordinate whenever $N \ge \mathrm{ref}(d)$. Placed basis sections are regular on their footprints, with the prescribed derivative bounds of size $\mathrm{columns}_N \theta_N^q$ for factor degree q; transfer is calibrated from $\mathrm{ref}(d)$ onwards.
3. For every d and $R, \varepsilon > 0$, there is $A \ge 0$ such that, for every $m \ge \mathrm{ref}(d)$ and all sufficiently large N, every p and every $v \in \mathcal{L}_d$ with $\|v\| \le R$, the observable at every site is supported in its centred level-m reflection cell and its chessboard root is at most $A a_m^{-d-\varepsilon}$. Here $N \ge \mathrm{ref}(d)$ is required.
4. It is the centre of the analytic shooting family of III.13.7.
5. The OPE entrance conditions hold: for every d, D, h with $D + h = d + 4$, there exist $t > 1$ and $j(i) \to \infty$ with $\sigma_{j(i)}/\sigma_i \to t$ for which the III.11 shell bounds hold for every choice of labels and extraction order; cover sizes satisfy $\mathrm{cs}_{i+1}^2 \le 2\mathrm{cs}_i^2$ and for every $\varepsilon > 0$, $(L_i \mathrm{cs}_i^2)^4 (\text{vacuum cost}) \le \varepsilon \sigma_{i+1}$ eventually; positive radii satisfy $\log \rho_i/\sigma_i \to -1$.
6. There exist coefficient-map bounds W, allowances $r : \mathbb{N} \to \mathbb{N}$ and transfer data giving the logarithmic OPE package: III.11 transfer/remainder estimates, triangular label steps, auxiliary comparisons, separation bounds, product sup bounds and VII.4 arity bounds. Also $L_m, \mathrm{columns}_m, \mathrm{columns}_{m+1} \le \hat{\beta}_m$, $r \to \infty$, and the support exponent is 100.
7. For these data and every d, N, p, m with $\mathrm{ref}(d) \le m < N$, there is a linear isomorphism T with
$$\|x_m(v) - T(V_{L_m} x_{m+1}(v))\|_1 \le \mathrm{barunit}(L_m) \|\mathrm{core}(x_{m+1}(v))\|_1 \quad (v \in \mathcal{L}_d).$$
On a basis vector of engineering grade e and factor degree q, the only output of grade $\le e$ is that vector times $(\theta_m/\theta_{m+1})^q$; no lower factor degree occurs. The map preserves the unit, commutes with core projection and lattice symmetries, and $\|T\|_1, \|T^{-1}\|_1 \le K(J(m+1))$, where K comes from W.
8. Coverage holds: some $g(k) \to \infty$, $g(k) \le k$, satisfies $a_{g(k)} \le \hat{\beta}_k a_k$ eventually, with three depth-d probe panels poised at level $g(k)$ for every d and all large k.
9. Along every extraction order satisfying IV.1's cell hypotheses, grid approximation holds at arbitrarily fine fixed levels for bounded degrees and III.11's allowed reflection pairs. Along each such order, complete weak comparison systems exist for every finite demand and degree horizon, positive radius bound and real-spin structure on the smooth free comparison table.
10. Every extraction order f has a refinement g, extraction base tables $A, (L_k), T$, a bin and derivative plan along $f \circ g$, and a seed for $(C, f \circ g, T)$ with Maxwell ranks for T.

Status: theorem. *Lean:* `uv_log`.

Proposition III.13.2 (coordinated family). There exists one C of penalty b, consistent, calibrated and floral, with the chessboard bound above, III.13.5's fixed-volume limits, the logarithmic package, calibrated label steps, coverage and complete weak comparison systems along every extraction order satisfying the cell hypotheses. For every extraction order F, there exist a refinement g and table T giving the ordered limit along the full mesh sequence and the periods of $F \circ g$, VII.4's OPE services, and a II.2 construction record for $(C, F \circ g, T)$.

Status: theorem; C precedes F. *Lean:* `coordination`.

Proposition III.13.3 (all phases). For every extraction order f, there exist C, a refinement g, a table T, a construction record F for $(C, f \circ g, T)$, and a V.3 reconstruction record $\mathcal{A}$ for every phase

of F, satisfying III.13.2 and completed reconstruction: Wightman realization, surjective comparison maps onto the Osterwalder–Schrader Hilbert spaces, unique vacuum, phase-uniform light-cone and heat-kernel bounds, trivial commutant, continued stress tensor, continuous phase dependence of boundary correlations, and barycentre identities. The derivative sector has a prescription P satisfying VII.4's estimates for every fixed finite cluster tree and separated spectators, with arbitrary positive power loss.

Status: theorem; C follows f in the statement, but the proof uses III.13.2's common choice. *Lean:* `common`.

Proposition III.13.4 (moving-frame Gram limits). For every extraction order f, there exist $C, g, T, F, \mathcal{A}, P$ satisfying III.13.3, with coverage at the cutoff of F. For every ultrafilter $\mathcal{U}$ finer than the filter of $f \circ g$ (period large, then mesh large) and every depth D:

- Depth-D portfolios exist: levels $k'(k) \to \infty$, $k' \le k$, eventually satisfy $a_k \le a_{k'} \le \widehat{\beta}_k a_k$, with three poised panels for all sufficiently large k, and label-matrix limits along $\mathcal{U}$ for every sufficiently large k.
- For every portfolio X, there are isomorphisms $Y_k(d)$ from the free Maxwell degree-d field space onto that of T, for all $d \le D$ and k. Eventually they respect grading through D; for each $d \le D$, their Gram forms are bounded and coercive uniformly for all sufficiently large k.
- For each $d \le D$, some $q_d \in \mathbb{N}$ gives, for every sufficiently large k and every field x, representatives $v_{N,p} \to v$ along $\mathcal{U}$, projecting to x, with eventual source cost at most $\widehat{\beta}_k^{q_d} a_k^d \|Y_k(d)^{-1} x\|$. Conversely, some $K_d \ge 0$ gives $\|Y_k(d)^{-1} x\| \le K_d a_k^{-d} z$ for every convergent representative family of cost at most z along any nontrivial filter finer than $\mathcal{U}$, for all sufficiently large k.
- For each $d \le D$, the Gram form of X in basis $Y_k(d)$ converges, as $k \to \infty$, to the free Maxwell Gram form.

Status: theorem. *Lean:* `habitat`.

Proposition III.13.5 (fixed-volume derivative limits). For C as in III.13.1 and every $d, k \in \mathbb{N}$, all sufficiently large p (depending on d, k), every n and labels v_i of degrees $\le d$, there exists $g : (\mathbb{R}^4)^n \to \mathbb{C}$, continuous on Ω_p^n, such that the following hold along the whole sequence $N \to \infty$.

- Periodic correlations, extended as step functions by coordinatewise ceiling projection, converge to g uniformly on separated compact sets.
- Normalized forward differences of every coordinate word u of length $\le k$ converge there to $\partial_u g$. Multilinear interpolation gives the same limits for fixed words of length $\le k$ of normalized differences with positive integer steps and fixed integer stencil offsets, including backward and centred differences.
- $\partial_a \partial_u g = \partial_{(a,u)} g$ for $|u| < k$ on Ω_p^n.
- For every admissible test t, tg is integrable and

$$a_N^{4n} \sum_r t(r) \Big\langle \prod_i O_{v_i}^{N,p}(r_i) \Big\rangle_{N,p} \longrightarrow \int tg,$$

  summing over centred torus representatives.

Status: theorem. *Lean:* `Poppy.blossoms`, `Poppy.Margins`.

Proposition III.13.6 (separated lattice points). Fix p and a member $C^{\mathbf{s}}$ of III.13.7 whose shifts satisfy $s_{N,p} \to s$, $|s| < r_*$. Assume its chessboard bound, slow growth, label-transfer and remainder bounds. For every J, every finite label list u_i with $\mathrm{ref}(\deg u_i) \le J$, and pairwise distinct level-J torus sites o_i, the correlations $\langle \prod_i O^{N,p}_{u_i}(o_i) \rangle_{N,p}$ converge as $N \to \infty$. III.13.7 supplies these bounds for every shift assignment; the centre has zero shifts.

Status: theorem. *Lean:* `Hearth.food`.

Proposition III.13.7 (analytic shooting family). For C of III.13.1 there exist a full choice of per-step constants, a shooting structure with an arc, a label-transfer bundle and reference levels such that:

1. Some $r_* > 0$, depending only on those constants, works for every N, p: each real $|s| < r_*$ determines exactly one Wilson coupling whose fixed-recursion trajectory stays in the prescribed scalar bands and reaches $\bar{\beta} + s$ at level 0. Assignments $|s_{N,p}| < r_*$ define $C^{\mathbf{s}}$ with the bundle's labels, sections and reference levels, and $C = C^{\mathbf{0}}$.
2. For $m \le N$, $\beta_m(N, p; s)$ and normalized activity readings extend holomorphically to $|s| < r_*$, bounded by $3\widehat{\beta}_m$ and soft $=$ rim/8, respectively. For fixed m, p, they converge locally uniformly with every shift derivative as $N \to \infty$.
3. The centre has the calibration package with constant allowance, priced sup bounds and extraction base data. Every $C^{\mathbf{s}}$ has label-transfer, remainder and chessboard bounds.
4. Some common disc $|z| < a$ lies in the shooting domain and carries holomorphic coefficient and transfer data, including inverse terminal label maps, for all d, N, p with $\mathrm{ref}(d) \le N$.
5. Fix d, m. For all large p, every n and labels of degree $\le d$, there exist $b' > 0$, holomorphic lattice correlations $W_N(z, x)$ on $|z| < b'$ and limits $f(z, x)$. At real s, W_N equals the correlation whenever $s_{N,p} = s$ and all reference levels fit. For every complex z, $f(z, \cdot)$ has continuous iterated coordinate derivatives through order m on Ω_p^n; step differences and interpolated differences converge to them with every z derivative, uniformly on products of separated spatial compacts and disc compacts. Real-shift limits agree with constant-shift fixed-volume limits. For admissible t, $\int t f(z, \cdot)$ is holomorphic and smeared lattice correlations converge to it locally uniformly with every z derivative.

Status: theorem, included in III.13.1. *Lean:* `Oasis.Floral`.

3.16.2 2. What it uses

- I.1; II.1–II.2: Wilson measures, labels, observables, extraction orders and construction records.
- III.1–III.2; III.4–III.12: kernels, order of constants, chessboard and per-step estimates, selected trajectories and histories; III.7 supplies the one-loop annulus coefficient.
- III.15–III.16: label costs and analytic tools; IV.1: extraction, cell hypotheses and Arzelà–Ascoli.
- V.3; VII.1, VII.3–VII.4; VIII.1: phase reconstruction, coverage, Gram comparison, OPE services and construction records.
- Balaban block-spin renormalization; Banach contraction; Vitali–Porter and the identity principle; Arzelà–Ascoli; dominated convergence; reflection-positive chessboard estimates; ultrafilters.

3.16.3 3. Strategy

Choose constants in their required order, raising thresholds while preserving earlier estimates. Increasing Taylor orders and powers accommodate each label degree from its reference level onwards. Shooting gives trajectories with prescribed coarse coupling; uniform contraction and decay of dependence on remote ultraviolet levels give full-sequence limits at each fixed level. Uniform holomorphic bounds then permit Vitali's theorem.

Approximate each renormalized observable by a conditional average of a coarse observable with convergent coefficients. Exponential improvement beats subexponential coefficient costs, and chessboard estimates control products. Dense-grid convergence and large-period slicing yield derivative limits. Extraction, OPE and reconstruction results then apply to the same family.

3.16.4 4. Steps

1. **Choose constants.** For simple G, combine the per-step estimates, then raise thresholds for shooting, label transfer, logarithmic OPE and weak comparison. Instantiate first stage 4, Taylor order $6 + i$, power $2 + i$, support exponent 100, allowances 1 and $j + 1$, and target rate $-7/16$; the final wait also supplies coverage. `uv_log`.
2. **Take fixed-level limits.** For the constructed arc, contraction has rate $1/2$ uniformly in cutoff; continuity and kinematic convergence give convergence of each iterate. Dependence on levels above $m+v$ decays as soft^v, giving real-shift limits, then Vitali gives every complex-shift derivative. `Camp.artistry`, `Camp.viva` *(proofs read; contraction and decay estimates used as stated).*
3. **Pass to correlations.** For convergent real shifts and the bundle bounds, effective densities converge uniformly and partition integrals stay positively bounded below. The RG identity and coarse approximation reduce separated products to convergent effective expectations; chessboard errors tend to zero, proving the Cauchy property. `Hearth.food`.
4. **Obtain derivatives and analyticity.** Large-period slicing bounds lattice differences through one extra order; dense-grid convergence and Arzelà–Ascoli give spatial derivatives and smearing. Holomorphic transfer bounds and Vitali give the analytic correlation limits. `Poppy.blossoms`, `Beach.sparksP` *(proofs read through stated transfer estimates).*
5. **Assemble common outputs.** Uniqueness identifies full-mesh smeared limits with extracted limits. Apply OPE services, every-phase reconstruction and Gram comparison; coverage supplies portfolios along every finer ultrafilter. `coordination`, `common`, `habitat` *(report proofs read).*

3.16.5 5. Where the difficulty lies

Uniform control over arbitrarily many ultraviolet levels and the balance between coarse-approximation gains and coefficient costs are the main analytic difficulties. Derivative limits require large periods; plain smeared limits hold at every fixed period. The coupling/activity disc is uniform in cutoff and period, whereas the correlation disc may depend on labels and period.

3.16.6 6. Notation

$\mathcal{L}_d$ is the degree-d label space; $\mathrm{ref}(d)$ its reference level. $a_0 = 1$, $a_{m+1} = a_m/L_m$, $\sigma_m = \log(1/a_m)$; ℓ_p is the torus side. $\Omega_p^n \subset (\mathbb{R}^4)^n$ consists of configurations distinct modulo $\ell_p\mathbb{Z}^4$; admissible tests are complex-valued, smooth, compactly supported on distinct configurations, with closed support

inside $(-\ell_p/2, \ell_p/2)^{4n}$. $O_v^{N,p}$ denotes the renormalized observable and brackets Wilson expectation. $\bar{\beta}$ is the central coarse coupling, $\widehat{\beta}_m$ the reference coupling, θ_m the threshold, $J(m)$ the stage. x_m are coefficient maps; V_L multiplies grade e by L^{d-e}. $f \circ g$ denotes refinement of an extraction order. A chessboard root is the absolute reflected-product expectation to the reciprocal number of cells.

3.16.7 7. What remains unexplained

Quantitative transfer, remainder, shell and comparison contracts remain those of III.7, III.11 and VII.4. Internal proofs of several constant choices, contraction/remote-level decay, one-step label estimates and auxiliary estimates were used as stated; moving-frame proofs belong to VII.1/VII.3. The penalty's normalization relative to Balaban's parameter remains unverified. The main theorem uses III.14's separate construction: equality of its family with this one, or transfer of III.13.5 to it, is not established.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.17 III.14. The construction run: flat representations and the ultraviolet rate

Part: III, Ultraviolet construction (renormalization group). **Lean:** ConstructionRun, NativeComponentRecipe, Flat. **Size:** 14,290 lines (2,509 declarations) in the main-theorem closure.

3.17.1 1. What this unit proves

Throughout, G is a compact matrix gauge group; "simple" means its real Lie algebra is simple and nonabelian. Level systems and full horizon sufficiency are those of III.1. A sheet fixes the kernel, stage profile and floors before block factors are chosen. Notation and norms are specified in §6.

Proposition III.14.1 (construction run). For simple G and $b > 0$, there exist a run R, a campaign T and pillars W for (R, T), with penalty b, satisfying the full thirteen-clause funded condition of III.15 relative to Δ_{10}, and carrying an attached native component recipe. The selection uses $Q(i) = i + 2$ and support exponent 100 at every stage.

Attachment means that there exist a sheet s, price map q, recipe C of III.14.2 and floors $n \geq \max(n_s, E_C)$ such that R uses s with floors raised to n and primary price map $P_{s,q}$; the recipe is realized on its own levels and horizon. Here n_s is the sheet's floor, E_C combines the three component floors and two numerical floors, and B_C is the maximum of the two post-length bills. Set $U_C(r, j) = \rho_C(r, j)^{-1}$ for the positive recipe widths ρ_C, and $V_C(\ell, r, j) = B_C(\ell, j)$. For both request families $F(r, j) = U_C(r, j)$ and $F(r, j) = V_C(\ell, r, j)$, every fixed r admits m_0 such that $|F(r, j)| \leq 1 + \log\log \widehat{\beta}_m$ for $m \geq m_0$, $j \leq J_{m+1}$; also $B_C(\ell, J_{m+1}) \leq \widehat{\beta}_m$ for every m. Realization includes III.14.3 with $e = 0$.

Status: theorem. *Lean:* constructionRun, Funded, Attached.

Proposition III.14.2 (recipe). For simple G and every sheet s, there exist a secondary price map X and floors $E^{\flat}, E^{\mathrm{sw}}, E^{\mathrm{du}} : \mathbb{N} \to \mathbb{N}$. For every level system with the kernel and profile of s:

- If $L_i \geq E^{\flat}(J_{i+1})$ for all i, every entry stage e satisfying the delay condition in III.14.3 admits its conclusions with this X, simultaneously for every sufficient horizon.
- If $L_i \geq E^{\mathrm{sw}}(J_{i+1})$ for all i, every horizon sufficient for some price map has, for every $i \leq N$, every torus and all colour-valued link fields A, B, colour-summed bilinear form

$\sum_a \langle A^a, \mathcal{K}^{\mathbb{T}}_{N,i} B^a\rangle$ invariant under simultaneous pullback by signed axis permutations fixing the origin. Reversing link orientation reverses the pullback sign.

- If $L_i \geq E^{\mathrm{du}}(J_{i+1})$ for all i, then for every such horizon, every $m < N$ and every level-system torus, the canonical free background map of III.3 satisfies III.14.5 for the fine/coarse metrics $(M^{\mathbb{T}_f}_{N,m+1}, M^{\mathbb{T}_c}_{N,m})$, hence minimizes the actual Gaussian variational problem.

Status: theorem. *Lean:* Choice, available.

Proposition III.14.3 (Gaussian ultraviolet rate). For simple G and every sheet s, there exist a price map X and $E : \mathbb{N} \to \mathbb{N}$ with this property: for every level system with the kernel and profile of s satisfying $L_i \geq E(J_{i+1})$ for all i, and every $e \in \mathbb{N}$ satisfying $\Omega^X_{J_i} \leq \log \hat{\beta}_i$ for $i \geq e$, there are link kernels F_i $(i \geq e)$ and constants $A, B \geq 0$ with the following properties. For every horizon H of that system sufficient for some price map, every $k \in \mathbb{N}$, $i \geq e$, $N \geq i$ and links u, v,

$$\|F_i\|_{w_{J_i}} \leq B\Lambda_{k,i}, \qquad \mathcal{O}^H_{n,i}(u,v) \longrightarrow F_i(u,v), \qquad \|\mathcal{O}^H_{N,i} - F_i\|_{w_{J_i}} \leq A\Lambda_{k,i}(a_N/a_i)^{1/4}.$$

Here $n \to \infty$, $w_j = \min(1, \alpha^X_j, \alpha^X_{j+1})$, and $A, B \geq 0$ depend only on G and the lattice, independently of s, the levels, e and H.

Status: theorem, order zero. *Lean:* Flat.entrance_gate.

Proposition III.14.4 (abstract flat representations). Fix a price map P, $\theta > 0$, positive widths w_j, $A_0 \geq 0$, $d_0 \in \mathbb{N}$, and a family $(\mathcal{L}_x, H_x, r_x, N_x, n_x, k_x, Z_x)_{x \in X}$ with $r_x \leq N_x \leq n_x$, $k_x \in \mathbb{N}$, $Z_x \geq 0$. Put $\Lambda_x(i) = 2 + Z_x + |s_i - s_{k_x}|$, using each system's scales. Assume each H_x is sufficient for P, its plaquette transfers at cutoff N_x and $r_x \leq i < N_x$ are canonical free kernels, and $w_j \leq \alpha_j$ for all j. On those stage ranges assume

$$w_{J_i} \leq \min(1, \alpha_{J_i}, \alpha_{J_{i+1}}), \quad 2w_{J_i} \leq L_i w_{J_{i+1}}, \quad q_i(w_{J_{i+1}}) \leq L_i^{-\theta},$$

and assume $2|\mathcal{F}|\Omega_{J_{N_x}} \leq A_0\Lambda_x(N_x)^{d_0}$ for every x. Then, simultaneously for all x and $r_x \leq i \leq N_x$, there are flat representatives of $\mathcal{K}_{n_x,i} - \mathcal{K}_{N_x,i}$ of width w_{J_i} and cost $A\Lambda_x(i)^{d_0}(a_{N_x}/a_i)^{\min(1,\theta/2)}$, where $A \geq 0$ depends only on A_0, d_0, θ.

Status: theorem. *Lean:* Flat.World.overlap.

Proposition III.14.5 (Schur identity and flat step). Fix fine/coarse tori with side ratio L, block average Q and $b > 0$. The background-field relations are

$$\mathcal{K}(M)W = Q^\top \mathcal{K}(M'), \qquad QW = 1 - b^{-1}\mathcal{K}(M').$$

For symmetric positive semidefinite M and every coarse field A', they make WA' a minimizer of $\langle A, \mathcal{K}(M)A\rangle + b|A' - QA|^2$, with minimum $\langle A', \mathcal{K}(M')A'\rangle$. If two triples $(W_j; M_j, M'_j)$ satisfy these relations and M_2, M'_2 are symmetric, then

$$\mathcal{K}(M'_2) - \mathcal{K}(M'_1) = W_2^\top(\mathcal{K}(M_2) - \mathcal{K}(M_1))W_1.$$

Let B be a neutral translation-invariant plaquette representative of the fine difference, with $\|B\|^{\mathrm{pl}}_\tau \leq r$, $\tau > 0$, $r \geq 0$. Suppose T, U are order-two strip-bounded curvature transports in width ω, with constants $q, p \geq 0$, scaled covariance $T(x + Lv, y + v) = T(x, y)$ and likewise for U. Require their periodizations C_2, C_1 to satisfy $dW_2 = C_2 d$, $dW_1 = C_1 d$, and $\sum_z U((x,\varphi),(z,\psi))$ to be independent of x for each orientation pair. If $0 \leq \sigma \leq \omega$ and $2\sigma \leq L\tau$, then

$$B'(y,z) = \sum_{x,w} T(x,y)B(x,w)U(w,z), \qquad \|B'\|^{\mathrm{pl}}_\sigma \leq \frac{2|\mathcal{F}|^2 qp}{\tau L} r.$$

It is neutral, translation invariant and represents the coarse difference.

Status: theorem. *Lean:* Flat.poised, Flat.toward.

Proposition III.14.6 (recursion rate). For each $x \in X$, fix scales with $L_i \geq 3$, stages $r_x \leq i \leq t_x$, reporting scale $k_x \in \mathbb{N}$, and $\Lambda_x(i) = 2 + Z_x + |s_i - s_{k_x}|$, $Z_x \geq 0$. Fix uniform $\theta, c_0 > 0$, $A_0 \geq 0$, $d_0 \in \mathbb{N}$. If $f_x, g_x \geq 0$ satisfy $f_x(i) \leq L_i^{-\theta} f_x(i+1) + g_x(i)$ below t_x, $f_x(t_x) \leq g_x(t_x)$, and $g_x(i) \leq A_0 \Lambda_x(i)^{d_0} (a_{t_x}/a_i)^{c_0}$ throughout, then

$$f_x(i) \leq A \Lambda_x(i)^{d_0} (a_{t_x}/a_i)^{\mu}, \qquad \mu = \min(c_0, \theta/2).$$

One may take $A = A_0(K+1)^{d_0}/(1 - 3^{-\theta'/2})$, $\theta' = \theta - \mu$, $K = 2(d_0+1)/\theta'$.

Status: theorem. *Lean:* Rate.Stair.fade_solve.

3.17.2 2. What it uses

- I.1: group data and one-loop slope s_G; III.1: levels, sufficiency and background-field relations.
- III.2: selection with nonempty demands and countable numerical requests; III.3: sufficient free horizons, canonical transfers and point-group invariance.
- III.8: weighted kernel bounds; III.12: runs, pillars and primary price maps; III.15: full funded condition.
- Gaussian block-spin minimization (Balaban; Gawędzki–Kupiainen), exponentially weighted ℓ^1 composition, discrete Grönwall, dominated and monotone convergence.

3.17.3 3. Strategy

Compare cutoffs at their common finest stage, where kinetic normalization makes the metric difference neutral. Transport this difference by the mixed Schur identity; neutrality gains L^{-1}, floors give contraction $L^{-1/2}$, and the recursion estimate gives exponent $1/4$. A shadow with altered clip parameters supplies free data; uniqueness transfers its Gaussian forms to every sufficient horizon.

Choose recipe and floors before block factors, then raise coupling thresholds to meet the post-length bill and countable requests. III.2 retains these conditions on the selected funded run. VIII.1 uses its penalty, pillars and funded property, discarding the attached-recipe conjunct.

3.17.4 4. Steps

1. **Initialize and propagate.** The metric difference at stage N has cost $2|\mathcal{F}|\Omega_{J_N}$ and zero orientation-wise row sums. The Schur identity transports it without inverting the curvature form; subtracting the transport kernel at the first base point gains L^{-1}, using half the input exponential weight. Flat.toward.
2. **Obtain the rate.** Floors ensure compatible widths and $q_i(w_{J_{i+1}}) \leq L_i^{-1/2}$; downward induction and III.14.6 absorb age growth. spacing_choice.
3. **Take the limit.** Diverging-torus periodizations identify link kernels; composing with $d^\top, d$ gives the pairwise cutoff estimate. Its Cauchy limit preserves the weighted bounds. supplies.
4. **Transfer from the shadow.** Downward Schur induction from the bare terminal form proves equality of Gaussian forms for sufficient horizons on systems differing only in clip parameters. This transfers the rate, symmetries and free background identities. same_path.

5. **Realize and select.** The combined floor and bill enforce the recipe with $e = 0$ on the run's own horizon. Apply III.2 with reciprocal-width requests and $V_C(\ell, r, j) = B_C(\ell, j)$. on_run.

3.17.5 5. Where the difficulty lies

Gauge zero modes obstruct curvature-form inversion. Orientation-wise neutrality supplies the gain, but preserving it requires constant row sums, beyond scaled covariance. Floors precede block factors; bills follow them. The rate concerns only the Gaussian link kernel: it gives no rate for the undetermined part of the plaquette metric, positive-order coefficients or source kernels. The formal recipe does not identify physical marked observables.

3.17.6 6. Notation

$L_i = \ell_{J_{i+1}} \geq 3$ are odd block factors; $a_0 = 1$, $a_{i+1} = a_i/L_i$, $s_i = \log(1/a_i)$; $J_i \leq J_{i+1} \leq J_i + 1$ are smoothing indices. Cutoff N has spacing a_N. Put $\hat{\beta}_i = \bar{\beta} + s_G s_i > 1$, $\Lambda_{k,i} = 5 + \log \hat{\beta}_k + |s_i - s_k|$. A price map is $(\alpha_j, \Omega_j, \eta_j)$: positive widths, upper bounds, positive coercivity bounds. A horizon supplies metric kernels $M^{\infty}_{N,i}$ and transfer data; $M^{\mathbb{T}}_{N,i}$ is the periodization. The lattice exterior derivative is d; $\mathcal{K}(M) = d^{\top} M d$, $\mathcal{O} = d^{\top} M^{\infty} d$, without a factor $1/2$. The local-estimate package Δ_{10} is garment10.

Write $x(u)$ for a base point. For link kernels, $\|F\|_{\sigma} = \sup_u \sup_{\phi} \sum_{y \in \mathbb{Z}^4} |F(u,(y,\phi))| e^{\sigma |x(u)-y|_1}$, with ϕ a link orientation; for plaquettes, $\|B\|^{\mathrm{pl}}_{\sigma} = \sup_u \sum_{v \in \mathbb{Z}^4 \times \mathcal{F}} |B(u,v)| e^{\sigma |x(u)-x(v)|_1}$, summing all target orientations $\mathcal{F}$. A flat representative is translation invariant, has zero row sums in each target orientation, and on every level-system torus periodizes to the quadratic-form difference after applying $d^{\top}(\cdot)d$; cost bounds this norm. In III.14.4, $q_i(u) = 2|\mathcal{F}|^2 h_i^2/(uL_i)$, $h_i = e^4 |\mathcal{F}|^2 \Omega^2_{J_{i+1}} \Omega_{J_i}$.

3.17.7 7. What remains unexplained

The scale and extraction choices are constructed in III.2; the meanings and proofs of the full funded clauses belong to III.15 and later units. The free-kernel existence proof and lattice kernel estimates are used without explanation here. The auxiliary curvature-lift, marked-kernel and formal Ward estimates still require their full admissibility conditions and proofs. Removing the recipe demand would not identify the resulting run with this selected run.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.18 III.15. Cost accounting for observables and the remainder comparison

Part: III, Ultraviolet construction (renormalization group). **Lean:** chapter 32: `YM/RemainderComparison/`, `YM/Progress/DIRECTED_RECIPE/`, `YM/Next/SCHEDULE/`, with readout and integration fragments. **Size:** 7,382 lines, 1,053 declarations; about 4,200 lines concern launch data.

3.18.1 1. What this unit proves

Proposition III.15.1 (pair-step rates). Fix $\rho > 0$, $C \geq 0$, $r > \rho$, $d \in \mathbb{N}$, $e \in \mathbb{R}$, $q > 0$. Put

$$s = p(y)\sqrt{\beta/y}, \quad a = Ce^{-rs^2}, \quad b = C(1 + \mathcal{B}(y))^d p(y)^e g(y)^q, \quad \kappa = \rho s^2, \quad \gamma = e^{\kappa} a + 2b.$$

For all sufficiently large $y > 1$, uniformly for $\beta \in [y/4, 4y]$ and integers $0 \le B \le \mathcal{B}(y)$,

$$\kappa, a \ge 0, \qquad 0 \le b \le 1, \qquad (2B+1)^4\gamma \le 1, \qquad \mathrm{quote}(\kappa, \gamma) \le e^{-\kappa}.$$

The threshold depends only on the fixed parameters. *Status:* theorem. *Lean:* `YM.Progress.DirectedRecipe.Payments.Pairing.currency`.

Proposition III.15.2 (hard–soft merging). Fix a parameter point with $K \ge 6$, direction power $P_0 \ge 2$, $B_P \ge 1$, $\sigma > 0$, $0 < t_P \le 1$. Assume countably indexed regular terms E_i, forced activities F_i on X_i, and real corrections ℓ_i on nonempty X_i'. Require support regularity, flat-field vanishing and vanishing affine Hessian on nonwrapping supports for E_i; gauge invariance, measurability and dependence on supported links and final flags for F_i, their eligibility tests and ℓ_i; eligibility whenever $F_i \ne 0$; translation/reflection covariance for all families, with prescribed permutations, support and inventory transformations for regular and forced families; and $\ell_i = 0$ on its small-field region. Suppose the regular family's nonnegative derivative bounds through K on small-field supports, in prescribed directions, have weighted anchored sum $\le \delta$; $|F_i(U)| \le p_i$, $|\ell_i(U)| \le q_i$ for all i, U, with $p_i, q_i \ge 0$; and

$$\mathrm{Hit}_{B_P}(p; \sigma + 1) \le a, \qquad \mathrm{Hit}_{B_P}(q; \sigma + 1) \le b.$$

Assume III.15.1's inequalities with $B = B_P$, replacing its last bound by $\mathrm{quote}(\kappa, \gamma) \le w$. Then the fusion input condition holds with (δ, w): original bounds δ, a persist, the soft family has a summable uniform majorant, and $(F_i) \sqcup (e^{\ell_i} - 1)$ has weights $(e^{\kappa} p_i) \sqcup (2q_i)$ and hit bound γ, with $|F_i| \le e^{-\kappa}(e^{\kappa} p_i)$ and $|e^{\ell_i} - 1| \le 2q_i$. III.10's fusion gives norm $\le w$. *Status:* theorem. *Lean:* `YM.Progress.DirectedRecipe.Payments.tender`.

For III.15.3–4 fix $K \ge 6$, $P \ge 2$, $0 < t \le 1$. Directions satisfy $\|A_e\| \le t(1 + \mathrm{dist}(e, o))^P$.

Proposition III.15.3 (slots). For every $\ell \in \mathbb{N}$, slot s with $|s| \le \ell$, finite $S \subset \mathbb{Z}^4$ containing its stencil, and site o, the slot is K times left-differentiable. If U is small-field on wrapped $o + S$, every derivative of order $m \le K$ in admissible directions has norm $\le C_s(P, K, \ell)t$, where

$$C_s(P, K, \ell) = (1 + \kappa_G^{-1}\|1\|)[2(\kappa_G + 1)2^P(\ell + 1)^{P+1}]^K.$$

At order zero the bound is t; constants are independent of volume and t. *Status:* theorem. *Lean:* `YM.Next.Scheduling.Wages.slotPays`.

Proposition III.15.4 (atoms). For every $k \in \mathbb{N}$, k-leg atom a, site o, and U small-field on its footprint, $a(\cdot, o)$ is K times left-differentiable, with every admissible derivative through K bounded by

$$|a| 2^K [2^K C_s(P, K, \mathrm{len}(a))t]^k.$$

Status: theorem. *Lean:* `YM.Next.Scheduling.Wages.fin`.

Proposition III.15.5 (launch data). Let G be simple. Fix a sheet of profiles P_j, kernel, compatible price table, targets R_{ji} with $\rho_{R_{ji}} < -3/8$ for all j, i, tolerances $h_{ji} > 0$, libraries C_{ji} of order P_i, and marks r_{ji}. There exist $f : \mathbb{N} \to \mathbb{N}$ and $C'_{ji} \ge 1$ giving quotation bounds with fine/coarse exponents $(\sigma_j, \sigma_i + 2)$ and factor $C'_{ji} L^{-5}$. These require support enlargement by $r_{ji}\mathcal{B}(x)$, coarse power equal to that of P_i, fine exponent σ_j, $B_P = \mathcal{B}(x)$, $B_B \le 2B_P$, $t_P \le 2t_B$, and acceptance. For every $u \ge 0$ satisfying local analytic smallness and every $\tau \ge 0$, the bound

holds. The quotation inequality is $e^{(\sigma_i+2)N_{B_P}(Y)}\tau\mathrm{returnFee} \le C'_{ji}L^{-5}e^{\sigma_j N_{B_B}(X)}\tau$. For each stage pair (j,i) and every block side $L \ge f_j$, III.15.6 holds and

$$(\mathrm{rel}_{R_{ji}}+1)10c_{h_{R_{ji}}+5}/L + 2\varepsilon_{R_{ji}} \le \sigma_j, \quad 20(\sigma_i+2)\mathrm{toll}(r_{ji}) \le L\sigma_j, \quad C'_{ji}/L \le h_{ji}.$$

Status: theorem. *Lean:* `YM.RemainderComparison.Chron.sails`.

Proposition III.15.6 (large-side comparison). For simple G, kernel K and target Q with $\rho_Q < -3/8$, there exists n such that for every $L \ge n$ and branching patch, every $W, \mu > 0$, order datum at W, songbook and scrollwork at side L, χ, stages E, E', rays satisfying the heart condition for (E', e_Q), library of order E', record with parameter $\le 1/4$, $j \in \mathbb{N}$, reserve f, and $\eta, \delta > 0$, there exist a financed quarry D at rate ρ_Q, with $d(D) > 0$ and paint supply, and P_1. For all $x \ge P_1$, poster comparison, chart conditions and instructions hold; active, resonance and tolerance conditions hold for every $\beta \in [\beta_F/4, 4\beta_F]$ and every point tempered to E'. The threshold n is independent of auxiliary data; D, P_1 may depend on them. *Status:* theorem. *Lean:* `YM.RemainderComparison.Cadenza.buoys`.

Proposition III.15.7 (delay enlargement). Fix a sheet, compatible target/expense tables R, X, sides L_j, protection, register, firmament, tolerances, libraries and marks as in III.15.5, and records u_{ji}. The register supplies positive W_{ji}, μ_{ji} and orders on stage pairs; each finite request family supplies scrollwork, songbook, grade/orders, reserve, positive constants, and an extra condition eventually true for every financed quarry. Assume launch thresholds $k_j \le L_j$, $\varepsilon(u_{ji}) \le 1/4$ on stage pairs with i at least the first stage, and $\sigma_i + 3 < \mathrm{e}_{R_{ji}}$ for all j, i. For every initial delay T_0, there exist $W \ge T_0$ and quarries V_{jig} for every request slot at every (j,i), chosen before any $U \ge W$. For every such U, launch data persist; on branching stage pairs with $j \ge 4$ and i at least the first stage, every request has the heart property, positive d, paint supply, III.15.6's allotment at threshold $\beta^*_j(U)$, and its extra condition above that threshold. On all stage pairs with i at least the first stage, requests are tiled whenever $x \ge \beta^*_j(U)$ and

$$c \ge \mathrm{shells}_j(U)(1+\mathcal{B}(x))^{\mathrm{growth}_j(U)}.$$

Tiling means $2z_g(x) < c$ and $4\mathrm{halo}_g(x) + 16 \le c$. Delay ordering increases entry couplings, scale-entry functions, shell counts and growth exponents componentwise. *Status:* theorem. *Lean:* `YM.RemainderComparison.Chron.grows`.

3.18.2 2. What it uses

- III.1: scales, slots, atoms; II.1: left-derivative calculus; II.2: hit bounds; III.5: logarithmic cluster tails.
- III.10: fusion, quotation, connected-activity and tree estimates; III.2: selected targets and delay ordering.
- Leibniz and chain rules, Mayer expansion, Kotecký–Preiss bounds, elementary asymptotics, $|e^z - 1| \le 2|z|$ for $|z| \le 1$.

3.18.3 3. Strategy

Retain $e^{-\rho s^2}$ from the hard bound and use the remaining Gaussian decay to absorb support counting; soft bounds decay by their negative coupling power. Express slots as single group words: unitarity controls derivatives, and Leibniz gives t^k for atoms.

Choose the block-side threshold before auxiliary data, then comparison functions and entry couplings. Negative activity rate survives the tree weight $\beta_F^{3/8}$; finitely many requests per stage permit common delay enlargement.

3.18.4 4. Steps

1. **Fusion.** Rescaled hard weights and doubled soft weights add to γ; the exponential inequality bounds the Mayer terms. `YM.Progress.DirectedRecipe.Payments.tender`.
2. **Observable bounds.** Differentiate the detour word, use unitarity, then multiply slot estimates. `YM.Next.Scheduling.Wages.slotPays`; `YM.Next.Scheduling.Wages.fin`.
3. **Comparison functions.** Choose reserve functions conditional on local derivative estimates, charts and coefficient matching. `YM.RemainderComparison.Herbal.compile` *(proof read; analytic inputs not expanded).*
4. **Marked outputs.** For the conditional marked-output estimate, separate single remaining occurrences from clusters with further occurrences and from large supports. The L^{-5} quotation summed over L^4 sites supplies the C_0/L term; the other contributions have negative rate. Occurrences are counted outside inserted leaves, with at least one marked. `YM.RemainderComparison.Solo.atlas` *(proof read; singleton input not expanded).*
5. **Launch and enlargement.** Combine comparison and tail bounds with quotation thresholds, then take finite stagewise maxima. `YM.RemainderComparison.Chron.sails`; `YM.RemainderComparison.Chron.grows`.

3.18.5 5. Where the difficulty lies

The bounds must be uniform in coupling bands and volume; the block-side threshold precedes all auxiliary choices, while comparison functions precede the entry coupling. Part C's conditional comparison conclusions reach the selected run but are discarded by the final construction-run theorem.

3.18.6 6. Notation

$g(\beta) = \beta^{-1/2}$, $p(x) = e^{(\log\log x)^2}$, $t(x) = g(x)p(x)$, $\mathcal{B}(x) = \lceil (\log x)^2 \rceil$, $\beta_F = x + b_G \log L$; x is coarse inverse coupling. $\mathrm{quote}(\kappa, \gamma) = e^{-\kappa}(e^{\gamma} - 1)$. Hit means $\sum_{i : z \in X_i} f_i e^{a N_M(X_i)} \le \gamma$ at every site z, for weights $f_i \ge 0$; N_M counts a minimal connected covering by M-balls. Stage pairs have $i \in \{j, j-1\}$; σ_j is the profile's support exponent. A quarry is a record of bounding functions; financing specifies reserve/rate bounds. Rate $\le a$ means $|w| \le \beta_F^v$ eventually for every $v > a$; a reserve satisfies $|w| \le p(\beta_F)^k$ eventually for some $k \in \mathbb{N}$.

A slot is curvature transported along a walk ω, with length $|\omega| + 4$; atoms are coefficient-weighted products of slots, with coefficient norm $|a| = \sum_b |c_b|$ and total length $\mathrm{len}(a)$. Small-field means every touching plaquette has chord distance $\le t$ from identity; κ_G is the colour scale and $\|1\|$ the Frobenius norm of identity.

For a target Q, $\mathrm{rel}_Q = \mathrm{e}_Q + 2\varepsilon_Q + \delta_Q$; h_Q is its halo and c_{h_Q+5} its pre-cost. Targets carry their chart/local-ratio conditions, $\rho_Q \in (-1/2, 0)$, $\mathrm{e}_Q \ge 0$, $\varepsilon_Q, \delta_Q > 0$, $h_Q \ge 500$, $4\varepsilon_Q$ below the top support exponent and $8\varepsilon_Q$ below both chart widths. `YM.ConnectedActivityE6.Target`.

The heart condition is

$$0 \le \sigma_*, \quad \sigma_{E'} \le \sigma_*, \quad \sigma_* + 1 \le a_*, \quad \varepsilon_* > 0, \quad a_* + 3\varepsilon_* \le \mathrm{e}_Q,$$

$$\operatorname{ord}(E') \le J_*, \quad \operatorname{ord}(E') + (p_* + 1) + p_* + 2 \le n_*, \quad p_* + 1 \le s_*, \quad k_* + 1 \le s_*,$$

with starred ray parameters. Tempered points have radius $\mathcal{B}(x)$, threshold $t(x) \le 2t(\beta_F)$, and the stage's order, power and support exponent.

3.18.7 7. What remains unexplained

Local analytic, compatibility and readiness conditions remain to be written out. Comparison-function existence does not supply coefficient matching. The singleton, weighted-cluster, full-bill and analytic-expansion inputs are not all checked here; readout, RG12 and R2I fragments also remain outside the detailed verification.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.19 III.16. Analytic toolkit

Part: III, Ultraviolet construction (renormalization group). **Lean:** eight folders, none a numbered chapter of its own: `YM/Complex/`, `YM/TimeKernel/`, `YM/Analysis/`, `YM/Laurent/`, `YM/CellFourier/`, `YM/Work/F07/`, `YM/Maxwell/`, `YM/Labels/`. **Size:** 5,318 physical source lines in 41 files, plus elementary helper lemmas in other folders.

3.19.1 1. What this unit proves

Proposition III.16.1 (integrated positivity). Let X be compact metric with finite Borel measure μ, $\mathcal{A}, \mathcal{B}$ sets, $K : \mathcal{A}^2 \to \mathbb{C}$ positive semidefinite, and $g : \mathcal{B} \times X \to \mathcal{A}$. If $(x, y) \mapsto K(g(b, x), g(c, y))$ is continuous for every b, c, then $\tilde{K}(b, c) = \iint K(g(b, x), g(c, y))\, d\mu(y) d\mu(x)$ is positive semidefinite. *Status:* theorem. *Lean:* `YM.Analysis.NNKernel.accumulate`.

Proposition III.16.2 (trace Hölder). For finite complex square matrices of the same size $A \succeq 0, b_1, \dots, b_n$, integers $q, M \ge 1$, $Z_A = \operatorname{tr} A^M > 0$, and $v \in \mathbb{C}^n$ with $\operatorname{Re} v_k \ge 0$ and $n/(2q) + \sum_k \operatorname{Re} v_k / M = 1$,

$$\left| \operatorname{tr} \prod_k b_k A^{v_k} \right| \le C_A, \qquad C_A = \prod_k \|b_k\|_{2q} Z_A^{1 - n/(2q)}.$$

Status: theorem. *Lean:* `YM.Holder.estimation`.

Proposition III.16.3 (moving matrix gaps). For such matrices, integers $q, w \ge 1$, $M = 2qw$, $Z_A > 0$, and $Y, Y' \in \mathbb{N}^n$, suppose $nw + \sum_k Y_k = M$ and $Y' = Y + \ell(e_i - e_j)$ for $i \ne j$, $\ell \in \mathbb{N}$, with e_i the coordinate vectors. If $r > 0$, $Y_k > 2r$ for every k, and $\ell \le r$, then

$$\left| \operatorname{tr} \prod_k b_k A^{Y'_k} - \operatorname{tr} \prod_k b_k A^{Y_k} \right| \le C_A \ell / r.$$

Status: theorem. *Lean:* `YM.Holder.finite_lip`.

Proposition III.16.4 (moving kernel gaps). On X, μ as in III.16.1, let K be continuous positive semidefinite and $B_1, \dots, B_n$ continuous complex kernels. For integers $q, w \ge 1$, $M = 2qw$, real $Z = \operatorname{Tr}(K^M) > 0$, and Y, Y', ℓ, r satisfying all gap conditions of III.16.3, assume real c_k satisfy $|\operatorname{Tr}((B_k^* B_k)^q)|^{1/(2q)} \le c_k Z^{1/(2q)}$. Then

$$|\operatorname{Tr}(B_1 K^{Y'_1} \cdots B_n K^{Y'_n}) - \operatorname{Tr}(B_1 K^{Y_1} \cdots B_n K^{Y_n})| \le Z \prod_k c_k \, \ell / r.$$

Status: theorem. *Lean:* `YM.TimeKernel.passage`.

Proposition III.16.5 (Laurent estimates). Let α be finite and $\eta > 0$. For holomorphic, 2π-periodic $u : S_\eta \to \mathbb{C}$, define $\hat{u}_p(x) = \int_{[0,1]^\alpha} u(p + 2\pi\theta)e^{-2\pi i\theta\cdot x}d\theta$, $\hat{u} = \hat{u}_0$. The torus mean is independent of $p \in S_\eta$, and $\hat{u}_p(x) = e^{ip\cdot x}\hat{u}(x)$. If $|u| \le K$, then $|\hat{u}_p(x)| \le K$ and, for $H \ge 0$, $b > 0$, $H + b < \eta$,

$$\sum_x e^{H|x|_1}|\hat{u}(x)| \le K(2(1+b)/b)^{\operatorname{card}\alpha}.$$

If $|u| \le K$, the Laurent series reconstructs u absolutely throughout S_η: $u(p) = \sum_x \hat{u}(x)e^{ip\cdot x}$.

Conversely, for $f : \mathbb{Z}^\alpha \to \mathbb{C}$, if $\sum_x e^{\eta|x|_1}|f(x)| \le W$, then $|\hat{f}| \le W$ on $|\operatorname{Im} p_a| \le \eta$. For every $0 < \sigma < \eta$, $\hat{f}$ is holomorphic on S_σ, with derivative norm and Lipschitz constant at most $W/(\eta-\sigma)$ for the sup norm. For real p, $\widehat{\operatorname{Re} f}(p) = (\hat{f}(p) + \overline{\hat{f}(-p)})/2$. *Status:* theorems. *Lean:* `full_turn`, `Vane.issues`, `Vane.paints`, `Width.estimate`, `readDeriv`, `readDiff`, `readRealPart`.

Proposition III.16.6 (Fourier and Bloch identities). For positive integers n, c, L, finite internal index sets α, β, and real entrywise summable $R : \mathbb{Z}^4 \to \mathbb{R}^{\alpha\times\beta}$, put $R_n((y,a),(y',b)) = \sum_{x\in\mathbb{Z}^4 : y+x\equiv y' \pmod n} R_{ab}(x)$. Then

$$F_nF_n^* = F_n^*F_n = n^4 I, \qquad R_n(F_n \otimes I_\beta) = (F_n \otimes I_\alpha)\operatorname{diag}_p \hat{R}(2\pi p/n).$$

This includes $p = 0$. With $f = cL$, reindexing fine momenta as $k_{p,l} = 2\pi(p + cl)/f$, $p \in \Lambda_c, l \in \Lambda_L$, gives $\Phi\Phi^* = f^4 I$ and the same intertwining identity with Φ, R_f, and blocks $\operatorname{diag}_{p,l}\hat{R}(k_{p,l})$. For every $p \in \mathbb{C}^4$, $E_L(\bar{p})^*E_L(p) = L^4 I$. *Status:* theorems. *Lean:* `circulant`, `gram`, `coGram`, `readingsGram`, `aligns`, `straightens`.

Proposition III.16.7 (matrix root). For $m > 0$, nonempty finite matrix indices and complex square X,

$$\Phi_m(X) = \sqrt{m}\sum_{k\ge 0}\binom{1/2}{k}(X/m - I)^k$$

is holomorphic on $\|X/m - I\| < 1$, bounded on every smaller closed ball, and commutes with entrywise conjugation. If $v > 0$ and X is Hermitian with $vI \le X \le mI$ as quadratic forms, its value is Hermitian, squares to X, and lies between $\sqrt{v}I$ and $\sqrt{m}I$. *Status:* theorem. *Lean:* `binRoot_on_band`, `binRoot_deriv`, `binSum_bound`, `binRoot_conj`.

Proposition III.16.8 (Maxwell jets). Fix the compact matrix gauge group and orthonormal real colour coordinates. For every integer $j \ge 0$, curl is an isomorphism $\mathcal{A}_j \to \mathcal{C}_j$, with inverse $A_i = (j+2)^{-1}\sum_k x_k F_{ki}$. For every smooth antisymmetric colour-valued field satisfying Bianchi and Maxwell on an open $U \subset \mathbb{R}^4$, its j-th Taylor coefficient lies in $\mathcal{S}_j$ at every $x \in U$.

Off zero, $G_0 = (4\pi^2|x|^2)^{-1}$ is harmonic; set $\psi_{b,km} = \delta_{kb}\partial_m G_0 - \delta_{mb}\partial_k G_0$, $C_{ab,km} = \partial_a\psi_{b,km} - \partial_b\psi_{a,km}$. In either index pair, C is smooth, antisymmetric and solves both equations; $C_{ab,km}(z) = C_{km,ab}(z) = C_{ab,km}(-z)$.

For all $j, l \ge 0$, real linear probes p, q on $\mathcal{S}_j, \mathcal{S}_l$ have a bilinear contraction $\kappa(p,q)(z)$, independent of extensions to unrestricted tensors, smooth for $z \ne 0$, with $\kappa(p,q)(z) = \kappa(q,p)(-z)$. For coordinate probes $p(F) = F_{ab}[u]_c$, $q(F) = F_{km}[v]_d$, $u \in (\mathbb{R}^4)^j, v \in (\mathbb{R}^4)^l$,

$$\kappa(p,q)(z) = \delta_{cd}(-1)^l c_l(t \mapsto c_j C_{ab,km}(t)[u])(z)[v].$$

Status: theorems; contraction definition. *Lean:* `potential`, `WaveOn.jets`, `Bare.equation`, `legKernel`, `reads_mixed`.

Proposition III.16.9 (labels). Fix real bases of $\mathcal{A}_j$; coordinate weight is $j+2$. For each integer $d \geq 0$, Label_d consists of complex polynomials of weight $\leq d$ invariant under simultaneous adjoint action. Real jet evaluation determines polynomials. There are finitely many weight-$\leq d$ monomials, so Label_d is finite-dimensional. Every label decomposes into invariant bihomogeneous components (weight and degree both $\leq d$), commuting with conjugation and $O(4)$. For $s \in O(4)$, the action is $(s \cdot P)(A) = P(T_{s^{-1}}A)$, where $(T_s A_j)_i(x) = \sum_k s_{ik}(A_j)_k(s^{-1}x)$, and satisfies the group law. Simultaneously for all d, there exist complex bases of real bihomogeneous labels containing $e_0 = 1$, nested as sets of polynomials with no further prescribed ordering. *Status:* theorems. *Lean:* `eval_ext`, `labels_finite`, `decompose`, `rotation_law`, `frames`.

Proposition III.16.10 (Wick hierarchy). For every $n \geq 0$, integer cutoffs $d_i \geq 0$, distinct $x_i \in \mathbb{R}^4$, and $P_i \in \mathrm{Label}_{d_i}$, restrict labels by $A_j = \mathrm{curl}^{-1}F_j$, $F_j \in \mathcal{S}_j$. Sum products of $\kappa(\cdot,\cdot)(x_i - x_j)$ over pairings of monomial occurrences at different points, extending multilinearly. This correlation is smooth, kills any insertion vanishing on all shell jets, respects simultaneous permutations and insertion of unit labels at additional distinct points, and equals $\prod_i a_i$ for constant insertions $a_i \in \mathbb{C}$, including the empty product 1. *Status:* definitions and theorems. *Lean:* `Wick.correlation`, `Wick.smooth`, `Wick.kills_equation`, `Wick.exchangeSites`, `Wick.adjoin`, `Wick.constants`.

3.19.2 2. What it uses

- I.1: lattices, subdivision, gauge colours, adjoint action and distinct-point configurations: `Lattice.Site`, `Gauge.Color`.
- III.1: summability budgets: `RG.SumBudget`.
- Three-lines theorem, Cauchy estimates and contour formulas, mean value inequality, spectral theorem, Schur products, dominated convergence, Fourier inversion, Banach-algebra binomial series, symmetry of derivatives and polynomial identity theorem. All analytic assumptions are stated above; parameters are chosen in the applications.

3.19.3 3. Strategy

Moving an insertion transfers units between two gaps. Continue these gaps holomorphically while preserving the Hölder balance; Cauchy's derivative estimate controls the change by ℓ/r. Sampling all kernels and adjoints with one partition passes this dimension-free estimate to cyclic integrals.

Shift Fourier contours to exchange strip holomorphy for exponential decay; diagonalize finite tori and construct matrix roots by a binomial series on a thinner strip. Radial gauge identifies potentials with closed jets; the free kernel descends to shell probes and supplies Wick correlations.

3.19.4 4. Steps

1. Factor matrices into normalized entire paths with operator norm ≤ 1 on $0 \leq \mathrm{Re}\, z \leq 1$ and trace norm ≤ 1 on its right boundary. At exponents with nonnegative real parts summing to 1, three-lines induction bounds the product trace by 1, giving dimension-free Hölder. Lean.
2. Vary gaps along $Y + z(e_i - e_j)$; Cauchy's estimate on radius-r circles gives III.16.3. Lean.
3. Positive quadratic forms pass through quadrature, proving integrated positivity. Sample kernels with square-root cell masses; positivity, adjoints and cyclic traces survive, and quadrature passes the trace estimate to kernels. `accumulate`, Lean.

4. Shift integration contours and sum geometric tails; Fourier inversion reconstructs symbols; termwise differentiation gives the reverse derivative estimate. `Vane.paints`, `readDeriv`.
5. Character orthogonality diagonalizes periodized kernels and their alias blocks, retaining zero momentum. `circulant`, `straightens`.
6. Diagonalization reduces the binomial matrix series to scalar square roots; Banach-algebra convergence gives holomorphy. `binRoot_on_band`.
7. Euler's identity and Bianchi invert curl; shell restriction makes contractions independent of extensions. Lean, Lean.
8. Finite Wick sums give smoothness; real bihomogeneous decomposition gives nested frames. Lean, Lean.

3.19.5 5. Where the difficulty lies

The trace bound must remain independent of matrix dimension as gaps vary and quadrature converges at fixed exponents and period. Strict gap inequalities avoid the exceptional zeroth-power convention. Laurent estimates lose strip width; complex alias adjoints require $\bar{p}$. Maxwell contractions retain factorials, the colour normalization and the second-endpoint sign.

3.19.6 6. Notation

$\|\cdot\|$: operator norm; $\|b\|_{2q} = (\mathrm{tr}((b^*b)^q))^{1/(2q)}$. Products are ordered. $A^z = \sum_{\lambda>0} e^{z\log\lambda} P_\lambda$ for spectral projections P_λ; A^0 projects onto the range. For $L \geq 1$, $\mathrm{Tr}(k_1 \cdots k_L) = \int_{X^L} \prod_{l=1}^{L} k_l(x_l, x_{l+1})\, d\mu^{\otimes L}$, $x_{L+1} = x_1$; powers repeat kernels; $B^*(x,y) = \overline{B(y,x)}$. Positive semidefinite means every finite Gram matrix, allowing repeated points, is Hermitian positive semidefinite.

$|x|_1 = \sum_a |x_a|$, $S_\eta = \{p \in \mathbb{C}^\alpha : |\mathrm{Im}\, p_a| < \eta \ \forall a\}$, $\hat{f}(p) = \sum_x f(x) e^{ip\cdot x}$. $\Lambda_n = (\mathbb{Z}/n\mathbb{Z})^4$; $F_n = (e^{2\pi i p\cdot x/n})$, $\Phi = (e^{ik_{p,l}\cdot y})_{y\in\Lambda_f,(p,l)\in\Lambda_c\times\Lambda_L}$, $E_L(p) = (e^{i(p+2\pi l)\cdot x/L})_{x,l\in\Lambda_L}$. Residues use representatives $0, \ldots, n-1$.

Colour coordinates are orthonormal for $-(\dim\rho)^{-1}\mathrm{Re}\,\mathrm{tr}(XY)$ in the fixed defining representation. $\mathcal{A}_j$: colour-valued 4-tuples of homogeneous degree-$(j+1)$ potentials with $x\cdot A = 0$; $\mathcal{C}_j$: antisymmetric homogeneous degree-j curvature jets satisfying $\partial_a F_{bc} + \partial_b F_{ca} + \partial_c F_{ab} = 0$; $\mathcal{S}_j$: additionally divergence-free. $c_j = D^j/j!$.

3.19.7 7. What remains unexplained

III.6 supplies norm conversion and strip choices for square roots; IV.1 supplies gap parameters needed for uniformity. External helper lemmas and longer dependency lists remain unchecked individually.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.20 IV.1. Infinite volume: extraction of a subsequence and the Euclidean axioms of the limit

Part: IV, infinite volume (chapter 10). **Lean:** `YM/Extraction`, `YM/Cells`, `YM/Slice`, `YM/Euclid`, `YM/Dash`, `YM/Next/EXTRACTION`, `YM/Lane`; D01–D06. **Size:** 7,444 lines in 1,049 declarations used by the main theorem.

3.20.1 1. What this unit proves

Let C be a Wilson family for a compact matrix gauge group. **(H1)** is its specification (II.1–II.2, III.13; `Family.Makes`). **(H2)** comprises fixed-volume centered budgets, ordered-tail power budgets, reflection positivity and chessboard inequalities, including root subadditivity (`CenterPort`). Extractions select strictly increasing spacing and volume indices; ordered tails mean k large, then j large depending on k. For every fixed p and label v, at every sufficiently fine scale m, (H2) gives a constant bounding all centered roots and the interior condition for all sites and large j, with $N_j \geq \max(m, \mathrm{ref}(d))$. For every degree d, $R > 0$ and $\varepsilon > 0$, it also gives $m_0 \geq \mathrm{ref}(d)$ and $A \geq 0$ such that, for every $m \geq m_0$ and $\eta > 0$, eventually along ordered tails with $N \geq m$, all labels $\|v\| \leq R$ and sites satisfy the centered interior condition and root bound $Aa_m^{-d-\varepsilon} + \eta$. Positivity and chessboard laws hold at every cutoff and volume.

Proposition IV.1.1 (raw limits). Under (H1)–(H2) along any extraction f, there exist a refinement $((N_j), (p_k))$ and multilinear tables S, P_p such that, for every arity n and label list v:

(a) For every p, $P_p(v;\cdot)$ is continuous on $\mathrm{Gen}_n^{(p)}$. For every compact K therein and $\varepsilon > 0$, eventually in j, $|\mathcal{C}_{N_j,p}(v;r) - P_p(v;a_{N_j}r)| < \varepsilon$ for all lifted lattice tuples r with $a_{N_j}r \in K$.

(b) $S(v;\cdot)$ is continuous on Gen_n, and $P_{p_k} \to S$ locally uniformly there.

(c) For every $t \in \mathcal{D}_n$, t is eventually admissible at p_k; at every admissible p, $V_p = \lim_j \mathcal{C}_{N_j,p}(v;t)$ exists, and $V_{p_k} \to \int tS(v;\cdot)$.

(d) $tS(v;\cdot)$ is integrable for every $t \in \mathcal{D}_n$.

Status: theorem. *Lean:* `Extraction.crops`.

Proposition IV.1.2 (label derivatives). Suppose S satisfies IV.1.1(c)–(d), with (H1)–(H2) along f. Assume the relay properties of §2 for every degree, projections $\pi_d(N,p)$ on Label_d converging in operator norm along ordered tails to projections π_d, and $\dim \mathrm{Null}_d(S) \leq \dim \ker \pi_d$. Then $\ker \pi_d = \mathrm{Null}_d(S)$, and there exist linear maps $\kappa_{d,e} : \mathrm{Label}_d \to \mathrm{Label}_e$ for every $d \leq e$, and $\tau_{d,a} : \mathrm{Label}_d \to \mathrm{Label}_{d+1}$ for every axis, such that for all arities, slots, labels and $t \in \mathcal{D}_n$,

$$\int tS(v_{i\to\kappa v_i}) = \int tS(v), \qquad \int tS(v_{i\to\tau_a v_i}) = -\int (\partial_{x_i^a} t)S(v).$$

Status: conditional theorem. *Lean:* `Work.D04.answer`, `lifts_model`.

Proposition IV.1.3 (smooth limit). If S satisfies IV.1.1 along an extraction where IV.1.2 applies, there exists T, smooth on every Conf_n, equal to S on Gen_n, satisfying IV.1.1(a)–(d) with the same extraction and P_p, and the basic identities below. The maps descend to $\mathrm{rise}_{d,e} : F_d \to F_e$ and $\partial_{d,a} : F_d \to F_{d+1}$. For every arity, label list, slot and $x \in \mathrm{Conf}_n$,

$$T(\dots, \mathrm{rise}\, u, \dots) = T(\dots, u, \dots), \qquad T(v_{i\to\partial_a v_i}; x) = \partial_{x_i^a} T(v;x).$$

The smooth ordered limit on Conf_n along a given extraction is unique. *Status:* theorem. *Lean:* `Org.Arrives.serves`.

Proposition IV.1.4 (Euclidean hierarchy). Assume (H1), IV.1.1(a)–(c) for T along f, and smoothness on every Conf_n. Then, for every arity, degree list and labels, on Conf_n:

- **(E1)** Simultaneous permutations preserve T; the empty correlation is 1; inserting 1_d deletes that slot.
- **(E2)** $T(\rho(s)v; sx) = T(v; x)$ for every $s \in H_4$, and $T(\bar{v}; x) = \overline{T(v; x)}$.
- **(E3)** $T(v; x + b) = T(v; x)$ for every $b \in \mathbb{R}^4$.
- **(E4)** For every axis a, finite collection of words w_i with distinct points within each word and $x^a > 0$, and $c \in \mathbb{C}^m$, $\sum_{i,j} \bar{c}_i c_j T(\Theta_a w_i w_j)$ is real and nonnegative.
- **(E5)** The same holds for smeared words $(v^{(i)}, t_i)$, $t_i \in \mathcal{D}_{n_i}$ supported where all $x_k^a > 0$, with entries $\iint \overline{t_i(y)} t_j(z) T(\Theta_a(v^{(i)}, y)(v^{(j)}, z))$; these integrals converge absolutely.

With **(E7)**, this gives (E1)–(E7): **(E6)** is smoothness; **(E7)** requires, for every arity, degree list, labels and $j \geq 0$, some $A \in \mathbb{N}, C \geq 0$ with $\|D^j T(v; x)\| \leq C r^{-A}$ for all $0 < r \leq 1$ and all r-separated configurations. The constants are independent of r, x; D^j has its real Fréchet operator norm. *Status:* theorem, conditional on (E7) for assembly. *Lean:* `Order.assembles`.

Proposition IV.1.5 (short-distance bound). Assume (H1)–(H2) along f, smoothness of T for every arity and labels, and IV.1.1(a)–(b). For every arity, degree list (d_i) and labels v, some $C \geq 0$, independent of r, x, satisfies $|T(v; x)| \leq C r^{-4n(1+\max_i d_i)}$ for all $0 < r \leq 1$ and r-separated x. The empty maximum is 0. With IV.1.3, B10 (V.1) supplies (E7). *Status:* theorem. *Lean:* `Work.D05.answer`.

Proposition IV.1.6 (symmetries of derivatives). For smooth, permutation-symmetric T satisfying (E2), let $(\mathrm{rise}, \partial)$ satisfy IV.1.3's identities. For every $d \leq e$, axis and $s \in H_4$, these maps commute with conjugation, rise commutes with $\rho(s)$, and

$$\rho_{d+1}(s)\partial_b = \sum_a s_{ab} \partial_a \rho_d(s).$$

Status: theorem. *Lean:* `Work.D06.compat`.

3.20.2 2. What it uses

- II.1–II.2, III.13: Wilson family specification (H1).
- III.11: centered budgets (H2), supplied by `MergeProof.Accent.gardena`.
- III.10–III.11: relay approximations to differences and inclusions, null-label detection and the separate dimension comparison (`MergeProof.Scotia`, `savings`).
- III.16: positive-kernel trace Lipschitz estimates. I.1, III.2, V.1: gentle scales and subpolynomial column products. V.1: B10.
- Osterwalder–Seiler positivity; chessboard estimates; Schatten Hölder, three-lines and Cauchy estimates; Arzelà–Ascoli, dominated convergence and Fourier regularity.

The relay hypotheses require bounded projections at each fixed volume and along ordered tails; degree-$(d+1)$ approximants to degree-d forward differences and degree-e approximants to degree-d labels for $d \leq e$, with subpolynomial cutoff coordinates; and a null-label detector. For each approximation error there must be one rate $y(N, p) \to 0$: for every integer $q \geq 0$ and lower scale m, some $k \geq m$ gives, eventually along ordered tails with $N \geq k$, interior errors with roots $\leq y(N, p)$ in every translated bond cell of side a_k, at relative lattice coordinates satisfying $M \leq z^a$ and $z^a + M + 1 \leq a_k/a_N$, where $M = \lfloor (a_k/a_N)/(8(q+1)) \rfloor$. Projection-kernel labels with subpolynomial growth must be chessboard-close to bounded measurable observables whose separated spectator correlations vanish exactly, eventually along ordered tails. Subpolynomial

means $\|B_{d,N,p}^{-1} w(N,p)\| \le a_N^{-\varepsilon}$ eventually for every $\varepsilon > 0$, where B is the terminal transfer to reference coordinates. The dimension comparison is an additional hypothesis (`RelayBin`).

3.20.3 3. Strategy

Chessboard bounds control separated correlations. Transfer kernels turn movement of an insertion into movement between gaps; complex interpolation gives equicontinuity. Two compactness arguments select spacing and volume subsequences. Relay approximations give distributional derivatives as raised-label kernels, hence smoothness. Lattice symmetries and positivity pass to the limit; scale estimates give polynomial bounds.

3.20.4 4. Steps

1. Under (H1), reflection positivity and repeated Cauchy–Schwarz give chessboard estimates; translation log-convexity transfers centered bounds to interior positions. `Around.interior_from`.
2. Under (H1)–(H2), translated cells separate insertions and bound correlations on distinct-point compact sets. `Around.size_done`.
3. Under the same hypotheses, slab traces and chessboard column moments give equicontinuity on generic compact sets through interpolation and Cauchy's estimate. `Difference.completed`.
4. Two Arzelà–Ascoli extractions give IV.1.1(a)–(b); domination gives (c)–(d). `first_surface`, `Order.ordinary`.
5. Under IV.1.2's hypotheses, summation by parts identifies weak derivatives; small chessboard errors vanish. Normalized projected approximants and the dimension count give fixed derivative labels. `Work.D03.answer`, `Dash.brace`.
6. Iterating these identities gives rapid localized Fourier decay and the smooth representative. `Dash.awake`.
7. Under IV.1.4's hypotheses, density passes symmetries; limiting Gram matrices pass positivity. Under IV.1.5's hypotheses, two scale choices give its bound. The derivative plan then promotes it to (E7). `Order.realPositive`, `Work.D05.answer`, `Work.B10.capped_tags`.

3.20.5 5. Where the difficulty lies

Uncentered roots and transfer-kernel equicontinuity require the main estimates. Fixed derivative labels require normalized remainders and the separate rank comparison. Equicontinuity holds on generic configurations; smoothness extends to all distinct-point configurations. The label maps are determined only modulo null labels, while their quotient maps are intrinsic. Continuous rotation symmetry comes later.

3.20.6 6. Notation

Label_d is the finite-dimensional complex label space; $F_d = \mathrm{Label}_d/\mathrm{Null}_d$, where null labels have zero smeared kernels in every arity and slot against all spectator labels and tests. Conf_n means distinct points; Gen_n means separation in every coordinate, modulo L_p for $\mathrm{Gen}_n^{(p)}$. $\mathcal{D}_n = C_c^\infty(\mathrm{Conf}_n)$; admissibility means support inside $(-L_p/2, L_p/2)^{4n}$. $a_N \downarrow 0$ and $L_p \uparrow \infty$ are spacings and periods. $\mathcal{C}_{N,p}(v;r) = \langle \prod_i I_{N,p}(v_i, r_i) \rangle$; smearing sums against $t(\hat{r})$ with weight a_N^{4n}, where $\hat{r}$ is the physical site position. Lifted tuples are reduced modulo L_p/a_N for evaluation. $v_{i\to w}$ replaces slot i by w. Separation uses the supremum norm. ρ_d acts on labels and their quotients. H_4 consists of signed

coordinate permutations, with matrices (s_{ab}); s_a reverses axis a; Θ_a reverses words, reflects points and replaces v by $\rho(s_a)\overline{v}$.

3.20.7 7. What remains unexplained

The upstream relay, rank and full fixed-volume kernel arguments remain unexplained, as does the budget's relation to III.12. Extraction is subsequential in both indices; III.13 supplies the full-spacing upgrade (`UVExtension.Reports.useFull`). Independence from the volume subsequence is not asserted.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.21 V.1. Decomposition of the limit into phases

Part: V, phases and reconstruction. **Lean:** `YM/OS`, `YM/Phase`, `YM/Ground`, `YM/Heat`, `YM/SNAG`, `YM/Paley`, `YM/Spectral`, the work items B01–B18 in `YM/Work`, and the assembly files `Bid/Phase.lean`, `Bid/Fetch.lean`, `Bid/Market.lean`, `Org/Phase.lean`. **Size:** about 5,700 lines in declarations used by the main theorem, including about 2,700 in B01–B18, plus the assembly files.

3.21.1 1. What this unit proves

S is the limit table; words are separated unless stated otherwise. "Hierarchy" means permutation symmetry, normalization and unit labels, signed-axis symmetries and reality, translation invariance, point and smeared reflection positivity for every coordinate axis, smoothness on separated configurations, and caps on all derivatives (IV.1). A cap bounds each fixed correlation derivative by Cr^{-A} for $0 < r \le 1$ and mutual sup-distance $\ge r$, with $C \ge 0$, $A \in \mathbb{N}$ depending on arity, degrees, labels and derivative order. The OS inner product is $\langle b, b' \rangle = S(\Theta b \cdot b')$, and $\Omega = |\emptyset\rangle$.

Proposition V.1.1 (chambers and ground space). For every S satisfying the hierarchy and coordinate axis a, its OS space carries a strongly continuous self-adjoint contraction semigroup and a commuting strongly continuous spatial unitary group, $E_t|b\rangle = |b + te_a\rangle$, $U_y|b\rangle = |b + y\rangle$ $(t \ge 0$, $y \perp e_a)$. These identities hold for every a-positive b. This is a chamber. The fixed space $\mathcal{H}_0$ of all E_t contains Ω, and $E_t\psi \to P_0\psi$ for every ψ, where P_0 projects onto $\mathcal{H}_0$. *Status:* theorem. *Lean:* `OS.implement`, `OS.Semigroup.emerges`.

Proposition V.1.2 (spectral calculus). In every chamber, writing $E_t = e^{-tH}$, there is a strongly continuous unitary group determined by $e^{isH}E_1\psi = E_1^{1-is}\psi$ through continuous functional calculus. The strongly continuous unitary representation $V(x) = e^{ix^aH}U_{x-x^ae_a}$ has a projection-valued measure P; for every ψ, $\mu_\psi(B) = \langle \psi, P(B)\psi \rangle$ is supported in $p^a \ge 0$, and, for every x and $\operatorname{Im} z \ge 0$,

$$\langle \psi, V(x)e^{i(\operatorname{Re} z)H}E_{\operatorname{Im} z}\psi \rangle = \int e^{ip\cdot x}e^{ip^a z}\, d\mu_\psi(p).$$

Status: theorem. *Lean:* `Semigroup.unit`, `Spectral.Orbit.total`, `Semigroup.heat`.

Proposition V.1.3 (slab operators). Under the hierarchy and bench V.1.7, every time-axis chamber has unique bounded $T_d(w)$ for $d > 0$, w in $(0, d)$, satisfying $T_d(w)|b\rangle = |w \cdot \tau_d b\rangle$ for every positive b. For every $D \in \mathbb{N}$, $\rho > 0$, some $C > 0$, $\kappa \in \mathbb{N}$ give

$$\|T_d(w)\| \le C^{|w|}(|w|/r)^{\kappa|w|} \tag{B}$$

for all $d > 0$, $0 < r \le 1$, and r-buffered w with degrees $\le D$, label norms $\le \rho$; the empty-word bound is 1. With the plan's derivative identities, for fixed d and labels, $T_d(w(x))$ is norm-C^∞ on the open slab configuration set, and coordinate differentiation replaces the corresponding label by $\partial_j v$. More generally this smoothness holds for any slab family satisfying (B) on a time-axis chamber of S', provided S, S' satisfy the hierarchy, S' respects S, and S' satisfies the derivative identities of the plan on S. *Status:* theorem conditional on the bench. *Lean:* `Work.B01.answer`, `Work.B02.answer`.

Proposition V.1.4 (compressed algebra). Assume the hierarchy, $O(4)$-covariance, a time-axis chamber and a slab family satisfying V.1.3's insertion identity. Then spatial translations fix $\mathcal{H}_0$ pointwise. For every separated w, $M(w) = P_0 T_d(w + s)|_{\mathcal{H}_0}$ is independent of admissible s, d; $M(b)\Omega = P_0|b\rangle$ for positive b, and $\langle\Omega, M(w)\Omega\rangle = S(w)$. For any finite collection of separated words q_i and shifts $a_i(n)$ with $t(a_j(n) - a_i(n)) \to +\infty$ for $i < j$, the concatenations are eventually separated and their correlations tend to $\langle\Omega, \prod_i M(q_i)\Omega\rangle$, regardless of spatial shifts. The generated norm-closed $*$-algebra $\mathcal{A}$ is commutative; Ω is cyclic and separating. With smooth slab laws, M is permutation/translation invariant, satisfies $M(\emptyset) = 1$, permits dropping unit labels together with their points, satisfies $M(w)^* = M(\Theta w)$, is multilinear through products of $F_d(S)$ and norm-smooth in positions; $\mathcal{A}$ is separable. *Status:* theorem. *Lean:* `Ground.adjunction`, `Work.B03.answer`.

Proposition V.1.5 (decomposition). Under V.1.4 including smoothness, $X = \mathrm{Spec}(\mathcal{A})$ is compact metrizable, with full-support Borel probability ν. Its points bijectively parametrize characters χ_ξ; every evaluation is continuous and $\int_X \chi_\xi(p)d\nu = \langle\Omega, p\Omega\rangle$ for all $p \in \mathcal{A}$. Each $S^\xi(w) = \chi_\xi(M(w))$, extended by zero off separated configurations, is a table; evaluations are continuous in ξ, and $S(w) = \int_X S^\xi(w)d\nu$ for every separated w. *Status:* theorem. *Lean:* `Ground.resolution`, `Resolution.barycenter`.

Proposition V.1.6 (phase laws). Assume V.1.3–V.1.5 with smooth laws for the field-algebra plan, the Seed, grid hypothesis and field-algebra laws on S. For every separated word w, $M(w) = M(\vartheta w)$ with arbitrary complex labels, and $M(w) = M(Kw)$ for every $K \in O(4)$. For every $\xi \in X$, S^ξ satisfies the hierarchy, respects S, is $O(4)$-covariant for the same action, inclusion/derivative identities and stress-tensor Ward identities, and $N_d(S^\xi) = N_d(S)$ for every degree d. For every $D \in \mathbb{N}$, $\rho > 0$, some $K > 0$, $\kappa \in \mathbb{N}$ give $|S^\xi(w)| \le K^{|w|}(|w|/r)^{\kappa|w|}$ uniformly in ξ, $0 < r \le 1$, and words with degrees $\le D$, label norms $\le \rho$, mutual sup-distance $\ge r$. *Status:* theorem. *Lean:* `Work.B05.answer`, `Retain.lift_detect`.

Proposition V.1.7 (bench). For every $D \in \mathbb{N}$, $\rho > 0$, some $C_0 > 0$, $\kappa \in \mathbb{N}$ work for every $d > 0$, $0 < r \le 1$, nonempty r-buffered w with degrees $\le D$, norms $\le \rho$: there are a lattice level b and $s \le d$, $s \in a_b\mathbb{N}$, with w in $(0, s)$, such that every positive v has B_v satisfying

$$|S(\Theta v_n \cdot v_n)| \le B_v A^{4n} \quad (n \ge 0), \qquad A = C_0^{|w|}(|w|/r)^{\kappa|w|},$$

where $v_0 = v$, $v_{n+1} = \tau_s(\Theta w) \cdot \tau_s(w \cdot \tau_s v_n)$. *Status:* hypothesis here, proved in VII.4 for every Seed table. *Lean:* `Next.Phase.seats`.

Proposition V.1.8 (vacuum uniqueness). Given V.1.5 and the hierarchy for every phase, fix a time-axis chamber for each ξ. Then $\mathcal{H}_0^\xi = \mathbb{C}\Omega_\xi$ for ν-almost every ξ, so such phases exist. Every time-axis phase chamber has slab operators with $\|T_d^\xi(w)\| \le \|T_d(w)\|$, inheriting (B) whenever the original family has it, and, under V.1.6, smoothness. *Status:* theorem. *Lean:* `Cluster.lifts`, `Work.B13.answer`.

Proposition V.1.9 (selected-phase certificate). For every Wilson family with extraction order and limit S, assuming its Seed, grid hypothesis and field algebra with laws, there exist a time-axis chamber of S with spatially fixed ground space, a smooth slab family satisfying (B), decomposition as V.1.5 with every phase satisfying V.1.6's hierarchy, field laws and null-space equality, and a selected phase with a time-axis chamber having ground space $\mathbb{C}\Omega$ and smooth slab family. *Status:* theorem. *Lean:* `Phase.Loom.cleanse`.

3.21.2 2. What it uses

- IV.1: hierarchy and `Spec.Seed`, including cell bounds and corrected-field data.
- VII.4: bench; III.11: `Org.Grill`, the grid hypothesis with cutoff-independent bounded approximants.
- VI.1: `Spec.Algebra` and field laws; V.2: `Kernel.Gram` completion; I.1, II.1: lattice and Wilson family.
- OS doubling; chessboard estimates; continuous functional calculus; SNAG; Paley–Wiener; Gelfand–Naimark; Riesz–Markov; dominated convergence; Cartan–Dieudonné.

3.21.3 3. Strategy

Compress bounded slab insertions to the ground space. Their commutative algebra describes infinite-time-separated observables; characters yield phase tables and the vacuum state supplies their mixing measure. Operator identities give laws in every phase; an integrated variance identity selects a unique-vacuum phase almost everywhere.

3.21.4 4. Steps

1. Construct contractions by OS doubling; functional calculus and SNAG give spectral measures, with positive energy from half-plane analyticity. `OS.contract`, `Semigroup.nonneg`.
2. Turn bench train bounds into slab norms by the power argument; absorb leftover width with E_{d-s}. `Drive.against`.
3. Rotate spatial displacements into time to fix ground vectors; exchange middle clusters in a four-cluster limit to prove commutativity. `Ground.inert`, `Ground.exchanges`.
4. Gelfand–Riesz gives a compact metrizable spectrum and the vacuum measure; separation of Ω gives full support. `Ground.resolution`.
5. For real labels, $D = M(w) - M(\vartheta w)$ is skew-adjoint. Positivity of variances of many bounded lattice approximants forces $-\|D\Omega\|^2 \geq 0$; separation gives $D = 0$. `Work.B05.answer`.
6. Transfer positivity and Ward identities before applying characters. Corrected-field detector moments, full support and finite-dimensional injectivity preserve null spaces. `Work.B08.answer`, `Work.B09.answer`, `Retain.lift_detect`.
7. Dominated convergence makes $\|P_0^\xi|b\rangle\|^2 - |S^\xi(b)|^2$ have integral zero; one common countable total family gives uniqueness almost everywhere. `Cluster.exp_real`, `Work.B14.universal`.

3.21.5 5. Where the difficulty lies

Phase reflection needs lattice variance positivity; null-space retention needs quantitative corrected-field estimates. Neither follows merely from the abstract character decomposition. No clustering, unique vacuum or spectral gap is assumed.

3.21.6 6. Notation

$t(x)$: last coordinate; e_a: coordinate unit vector; $v \in V_d$: degree-d label. Separated means distinct points. $|w|$: word length; $w \cdot w'$: concatenation; τ_s: time translation; Θ: antilinear OS reflection with reversal; ϑ: linear time reflection. Positive words have positive times; r-buffered means mutual sup-distance $\geq r$ and times in $[r, d-r]$. $N_d(S)$ comprises labels vanishing against every separated spectator family; $F_d(S) = V_d/N_d(S)$. A table S' respects S when $N_d(S) \subset N_d(S')$ for every d.

3.21.7 7. What remains unexplained

Ultraviolet packing, bounded approximants and corrected-field estimates remain external inputs. With copy number and approximation index fixed, the variance argument takes cutoff limits before large translations, then sends copy number to infinity and approximation error to zero. Vacuum uniqueness for every phase belongs to V.3; selection here is noncanonical.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.22 V.2. Osterwalder–Schrader reconstruction of a Wightman theory

Part: V, phases and reconstruction. **Lean:** A16, C01–C29, Boundary, Fields, Shift. **Size:** 6,456 lines; 705 declarations.

3.22.1 1. What this unit proves

All arities include zero; statements hold for every label tuple and Schwartz test unless restricted. Tables S, T share a register; T respects S means $N_d(S) \subseteq N_d(T)$ for every degree, where N_d consists of labels vanishing with arbitrary insertions. Write (H) for: both Euclidean hierarchies; respect; differential-plan inclusions compatible with both tables and derivatives with T; a spread satisfying the slab bound, smoothness in insertion positions on every valid slab, and insertion derivative laws. The slab bound is $\|A_d(w)\| \leq C^{|w|}(|w|/r)^{a|w|}$ for buffers and pairwise maximum-norm distances at least $r \in (0, 1]$, with $C > 0$, $a \in \mathbb{N}$ uniform in width and length for fixed degree and representative-norm bounds.

(SC) means continuity in strict complex times and spatial positions, holomorphy in times, and agreement with T at ordered imaginary times. (PG) means $|F_n^y(x)(v)| \leq Ck^{-m}$ whenever all height gaps are at least $k \in (0, 1]$, with $C \geq 0$, $m \in \mathbb{N}$ depending on n, v. Put $\psi_v(y; f) = \int F_n^y(x)(v) f(x)\, dx$; (I), (HC), (CR) mean integrability, continuity on E_n and $\partial_a \psi(y; f) = \psi(y; -i\partial_{\underline{a}} f)$ for every real height direction a within E_n. (BV) means $\psi_v(y; f) \to W_n(v)(f)$ as $y \to 0$ through ordered heights, without a fixed direction.

Proposition V.2.1. A16 and C01–C29 hold simultaneously, each with its own hypotheses. *Status:* theorem. *Lean:* reconstructionArcs.

Proposition V.2.2. Under (H), linear bounded-operator insertions B_p, $p \geq 0$, exist, given for $p > 0$ by the central one-point slab operator of width $2p$, with $\|B_p(u)\| \leq Cp^{-a}$ for $0 < p < 1$, for each u, with $C > 0$, $a \in \mathbb{N}$. The empty product is $\|\Omega\|^2$. Their vacuum products, separated by $e^{-(\operatorname{Im} w - 2p)H} e^{i \operatorname{Re} w H} U(\vec{a})$, are independent of widths $0 < p < 1$ with gaps $> 4p$, agree with T at imaginary times, and define F satisfying (SC), (PG). *Status:* theorem. *Lean:* C01, C02, C03.

Proposition V.2.3. (SC), (PG) imply (I), (HC), (CR). Separately, (I), (HC), (CR), (PG) produce a multilinear tempered boundary family W satisfying (BV), hence every ordered radial limit. *Status:* theorem. *Lean:* C04, jointly.

Proposition V.2.4. Under (SC), (PG), (CR), (HC), (BV), for ordered y and compactly supported f, $W_n(v)(\hat{f}) = \psi_v(y; \widehat{e^{\langle y,\cdot\rangle} f})$ (L). Under (SC), (PG), (I), (L), Fourier support lies in $\mathcal{P}_n = \{\sum_k p_k^0 = 0,\ \sum_{k\geq j} p_k^0 \geq 0\ \forall j\}$. With Lorentz covariance it lies in $\mathcal{V}_n = \bigcap_\Lambda \tilde{\Lambda}^{-1}\mathcal{P}_n$; thus total momentum vanishes and suffix momenta belong to $\bar{V}_+$. *Status:* theorem. *Lean:* C05, C06, tents.

Proposition V.2.5. Unconditionally, spatial rotations and coordinate boosts generate $\mathcal{L}_+^\uparrow$; $g \mapsto g_* f$ is Schwartz-continuous and $t^{-1}(\Lambda_a(t)_* f - f) \to -D_a f$, where $D_a = \sum_k (x_k^a \partial_{x_k^0} + x_k^0 \partial_{x_k^a})$. Assume both hierarchies, inclusions compatible with S, respect, Euclidean covariance, rotation laws and spin compatibility. Under (SC), (PG), (I), (CR), (HC), infinitesimal rotations give $\sum_k \psi_{v^{(k)}}(y; f) = \psi_v(y; D_{a,y} f)$, for $y \in E_n$ and each spatial axis a, where $D_{a,y} = D_a + i\sum_k y_k \partial_{x_k^a}$ and $v^{(k)}$ replaces v_k by $-i\rho(e_0 \wedge e_a)v_k$. Under (SC), (PG), that identity and (BV), $W_n(K(\Lambda)v)(g_* f) = W_n(v)(f)$ for every Poincaré transformation $g = (a, \Lambda)$. With S symmetric, inclusions compatible with S, spin compatibility and induced star J, this covariance makes the word Gram form invariant and word matrix elements continuous. *Status:* theorem. *Lean:* C07, C09, C28.

Proposition V.2.6. For $p \in \mathcal{V}_n$, $\langle p, \eta\rangle \geq 0$ on the closed imaginary cone, and locally $\langle p, \eta\rangle \geq c|p|$ inside, for some $c > 0$. Smooth cutoffs χ_n, equal to one on the 1-neighbourhood of $\mathcal{V}_n$, supported in its open 3-neighbourhood and with polynomially bounded derivatives, exist. For η in the closed imaginary cone, $e^{-\langle \eta,\cdot\rangle}\chi_n$ has temperate growth and its product with each test varies continuously in Schwartz space. The function $\mathcal{W}_n(v)(z) = (2\pi)^{-4n}(\mathcal{F}W_n(v))(e^{i\langle\cdot,z\rangle}\chi_n)$ is holomorphic on $\mathcal{T}_n$. For interior η, its integrals represent $W_n(v)[\eta](https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/f) = (\mathcal{F}W_n(v))(e^{-\langle\eta,\cdot\rangle}\chi_n \check{f})$. Assuming damping, cone support, holomorphy, this integral representation, (SC), (PG), (I), (L), it equals F_n^y at $x + iy$ (TC). Cone support and covariance imply $\mathcal{W}_n(K(\Lambda)v)(\Lambda z) = \mathcal{W}_n(v)(z)$. *Status:* theorem. *Lean:* C10, C15, C16.

Proposition V.2.7. Assume (SC), (TC), cone support, damping, spin compatibility, tube covariance, symmetric S, T, T smooth off coincidences, respect and inclusions compatible with S. There are spectator profiles approaching zero, equal at the adjacent pair and otherwise ordered, each admitting imaginary-boost paths into both tubes. Spin boosts extend entirely; the two resulting holomorphic germs agree on the zero-real-time slice and locally represent partial boundary values. Then for every adjacent swap σ, $W_n(v)(f) = W_n(v \circ \sigma)(f \circ \sigma)$ whenever $\mathrm{supp} f \subset \{x : x_{i+1} - x_i \text{ spacelike}\}$. This joint locality implies exchange of adjacent letters in any word whose two test supports are spacelike separated. *Status:* theorem. *Lean:* actor, C27.

Proposition V.2.8. Heated vectors are the insertion products of V.2.2 acting on Ω, multiplied on the left by $e^{-(y_1-p)H}e^{ir_1^0 H}U(\vec{r}_1)$ at heights y and real points r; the empty vector is Ω. Under (H), V.2.2 and oriented P, heated OS vectors have inner product $F^{(-x_n,\dots,-x_1,y_1,\dots,y_m)}(r_n, \dots, r_1, q_1, \dots, q_m)(Jv_n, \dots, Jv_1, w_1, \dots, w_m)$ for positive ordered heights x, y and widths in $(0, 1)$, with heights and gaps exceeding four times their respective widths. Under (H), slab insertions and spin compatibility, their smeared vectors $\Phi_t(b)$, using heights $(t, 2t, \dots, nt)$ and width $\min(t, 1)/8$, have Bochner-integrable defining integrands and satisfy $\Phi_t(a \cdot b) = U(a)\Phi_t(b)$ for every $t > 0$, word b and translation a. Under (H), (SC), (BV), these properties and the pointwise Gram identity, $\langle \Phi_s(b), \Phi_{s'}(c)\rangle \to W(b^* c)$ independently as $s, s' \downarrow 0$. Tensor products

are dense in Schwartz space. With this density, equal null spaces, both hierarchies, inclusions compatible with both tables, (SC), (TC), vanishing of $W(a(u,f)c)$ for all a, c, f implies $u = 0$. Every boundary family W and antilinear J give bilinearity of $(u,f) \mapsto W(a(u,f)c)$ and continuity of $(f,g) \mapsto \langle (u,f)a, (u',g)c\rangle_W$. *Status:* theorem. *Lean:* C24, C26.

Proposition V.2.9. A realization comprises spin-compatible K, induced J, fields bilinear in labels and tests on a dense invariant domain equal to the span of vacuum word vectors, a unit vacuum fixed by a strongly continuous unitary Poincaré representation, covariance $U(g)\varphi(u,f) = \varphi(K(\Lambda)u, g_*f)U(g)$, the identity $\langle \varphi(u,f)\psi, \chi\rangle = \langle \psi, \varphi(Ju, \bar{f})\chi\rangle$ on the domain, continuous smeared-field vectors, local commutativity, and faithfulness: $u \neq 0$ implies $\varphi(u,f) \neq 0$ for some f. It has vacuum expectations W, translation spectral measure supported in $\bar{V}_+$, and only scalar multiples of the vacuum translation invariant. *Status:* definition. *Lean:* Realizes.

Proposition V.2.10. Fix rotation and real structures, induced involutive J, and spin compatibility. Assume word bilinearity, two-test continuity, word locality, label faithfulness, continuous word matrix elements and invariant Gram form. If vectors satisfy V.2.8's Gram limit, $\Phi_s(\emptyset) \to \Omega$, and eventual translation covariance for each word and translation, and T is normalized, unital and permutation symmetric, V.2.9 holds except possibly vacuum uniqueness. If the heat-fixed space is $\mathbb{C}\Omega$, uniqueness also holds. *Status:* theorem. *Lean:* harvest.

3.22.2 2. What it uses

- I.1, II.1: quotient labels and common finite-dimensional levels.
- IV.1, V.1: Euclidean hierarchy, OS space, heat semigroup, translations and slab bound.
- VI.1: rotation, spin and star data. VIII.1 supplies phase data and selected-phase heat-fixed uniqueness.
- Standard results: OS reconstruction, tempered holomorphic boundary values, Fourier–Laplace theory, Kolmogorov construction, Riesz–Markov and SNAG.

3.22.3 3. Strategy

Bounded slab insertions construct strict-time correlations. Boundary values and boosts give covariant distributions with forward spectral support. Fourier–Laplace continuation and permutation symmetry give locality. Heated vectors transfer positivity and translations to the word Hilbert space, where left multiplication defines fields.

3.22.4 4. Steps

1. Heat factors telescope, removing the slab width; contractions give polynomial growth. C02.matrix_pad.
2. Taylor's integral remainder controls boundary limits uniformly over nondecreasing heights. Waves.maintains.
3. Continue rotation identities, then integrate boost derivatives to obtain covariance. C09.boost_invariant.
4. The Laplace identity forces nonnegative suffix energies; Lorentz covariance upgrades this to cone support. tents.
5. Fourier–Laplace damping extends F to the tube. Exponential decay gives complex differentiability in Schwartz space. C13.analytic_interior.

6. Imaginary boosts construct two germs; equal Wick-slice jets identify them, yielding boundary exchange. C20.complex_germ.
7. Gram limits give an isometric embedding into the OS space; left multiplication defines fields, and the embedding transfers positive energy. universal_embed.

3.22.5 5. Where the difficulty lies

Boost derivatives require finite-dimensional label levels and Schwartz estimates. Locality requires complete jets at equal pair times. Independent heating scales require unrestricted boundary limits.

3.22.6 6. Notation

$\mathcal{H}, \Omega, H \geq 0, U$: OS space, vacuum, Hamiltonian and translations. $\mathcal{F} = \bigcup_d F_d$: quotient labels; J: reflection adjoint; b^*: reversed starred word. $E_n = \{y_1 < \cdots < y_n\}$; $F_n^y(x) = F_n(x^0 + iy; \vec{x})$; $\underline{y} = ((\vec{0}, y_k))_k$. $\mathcal{T}_n = \{z : \mathrm{Im}(z_{k+1} - z_k) \in V_+\}$, with unrestricted first imaginary point. $V_+ = \{x : |\vec{x}| < x^0\}$. Pairings are Euclidean; $\tilde{\Lambda} = \mathsf{r}\Lambda\mathsf{r}$, with spatial reversal r. $\mathcal{L}_+^\uparrow$ is the proper orthochronous Lorentz group; K is its label action. The closed imaginary cone replaces V_+ by $\bar{V}_+$. $\mathcal{F}W(f) = W(\hat{f})$; Fourier phases are $e^{-i\langle x,p\rangle}$, inverse normalization $(2\pi)^{-4n}$. $g_* f = f \circ g^{-1}$ slotwise; ρ is the rotation Lie action.

3.22.7 7. What remains unexplained

The upstream slab laws, compact Borel functional calculus and some Fourier–Laplace differentiation estimates remain outside the detailed checks. Vacuum uniqueness applies to the selected phase; it is not supplied for the mixed table.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.23 V.3. Reconstruction in every phase

Part: V, phases and reconstruction. **Lean:** `YM/EveryPhase` (34 files). Its hypotheses are recorded in `YM/Spec/Closed.lean` and `YM/Phase/Mixture.lean`. **Size:** 5,268 lines.

3.23.1 1. What this unit proves

Hypotheses throughout. Fix a Wilson family, extraction and limit table S^{mix} satisfying `Final`, constructed in VIII.1. This includes the Euclidean axioms; a compact metric phase space X with full-support Borel probability ν; continuous characters χ_ξ, indexed bijectively by X, of the commutative C* ground-space slab algebra $\mathfrak{A}$, with $\xi \mapsto \chi_\xi(M)$ continuous and $\int \chi_\xi(M) d\nu = \langle \omega, M\omega \rangle$ for every $M \in \mathfrak{A}$; spatial translations trivial on the mixed ground space; phase tables satisfying the Euclidean axioms and $S^\xi(w) = \chi_\xi(M_w)$ for distinct-point words, where M_w is their time-translated compressed slab operator; the selected reconstruction at ξ_* and continuations of $S^{\mathrm{mix}}, S^{\xi_*}$; and the VI.1 field algebra, derivatives, rotations, Euclidean stress tensor and phase Ward identities.

Proposition V.3.1 (every phase). There exist OS spaces of S^ξ with slab operators, continuations F^ξ of S^ξ with boundary distributions W^ξ, and theories $\mathfrak{W}_\xi = (\mathcal{H}_\xi, \mathfrak{D}_\xi, \Omega_\xi, \phi_\xi, U_\xi, P_\xi)$ for every $\xi \in X$, agreeing with V.1–V.2 at ξ_*. The common label space $\mathbb{F}$, generated by the F_d, has the

selected star operation and Poincaré action; $\mathfrak{D}_\xi$ is the dense invariant word domain, Ω_ξ a unit vacuum, and U_ξ a strongly continuous unitary Poincaré representation on the Hilbert space $\mathcal{H}_\xi$. Fields $\phi_\xi(a,f)$ use $a \in \mathbb{F}$, $f \in \mathcal{S}(\mathbb{R}^4)$. For every phase:

1. Vacuum field products equal W^ξ; $\phi_\xi(a,f)^*$ extends $\phi_\xi(a^*,\bar{f})$; spacelike separated fields commute on $\mathfrak{D}_\xi$; U_ξ implements Poincaré covariance; the translation spectrum lies in $\overline{V}_+$. Every nonzero $a \in \mathbb{F}$ has some $f \in \mathcal{S}(\mathbb{R}^4)$ with $\phi_\xi(a,f) \neq 0$ on $\mathfrak{D}_\xi$; translation-invariant vectors are multiples of Ω_ξ.
2. A unitary $J_\xi : \mathcal{H}_\xi \to \mathcal{H}_\xi^{\mathrm{OS}}$ sends Ω_ξ to ω_ξ, intertwines translations and spectral projections, and sends word vectors to boundary values of the corresponding Euclidean vectors at imaginary times.
3. The OS ground space is $\mathcal{D}_\xi = \mathbb{C}\omega_\xi$.
4. For every $x \in \mathcal{H}_\xi$, $P_\xi^0 x = x$ iff $x \in \mathbb{C}\Omega_\xi$.
5. For every compactly supported finite sum of words A, some $M \geq 0$ satisfies $\mu_A^\xi(\{0 < p^4 \leq u\}) \leq Mu^2$ for all $\xi \in X$, $0 < u \leq 1$.
6. For every such A, some $M \geq 0$ satisfies, for all $\xi \in X$, $s \geq 1$,

$$0 \leq \langle A\Omega_\xi, (e^{-sH_\xi} - P_\xi^0)A\Omega_\xi\rangle \leq Ms^{-2}.$$

7. A bounded complex-linear B is scalar iff, for every a, f and $u, v \in \mathfrak{D}_\xi$, $u \in \operatorname{dom}\phi_\xi(a,f)^*$ and $\langle \phi_\xi(a,f)^* u, Bv\rangle = \langle u, B\phi_\xi(a,f)v\rangle$.
8. Proposition V.3.y holds.
9. For every n, $a \in \mathbb{F}^n$, $g \in \mathcal{S}((\mathbb{R}^4)^n)$, $\xi \mapsto W_n^\xi(a;g)$ is continuous and $\int_X W_n^\xi(a;g)d\nu = W_n^{\mathrm{mix}}(a;g)$. For every distinct-point word w, $\int_X S^\xi(w)d\nu = S^{\mathrm{mix}}(w)$.

Status: theorem under the stated hypotheses. *Lean:* `all_characters`. Items 5–7 are also Propositions V.3.n (`light_bound`), V.3.o (`inverse_square`) and V.3.t (`commutant`).

Proposition V.3.2 (no exceptional phase). For every $\xi \in X$ and $x \in \mathcal{H}_\xi$, $P_\xi^0 x = \langle \Omega_\xi, x\rangle \Omega_\xi$. *Status:* theorem. *Lean:* `all_plant`.

Remark V.3.3 (additional consequences). Under V.3.1, for every ξ and $x \in \mathcal{H}_\xi$, being fixed by e^{-sH_ξ} for all $s \geq 0$, being translation invariant, and belonging to $\mathbb{C}\Omega_\xi$ are equivalent. For every degree d and degree-d token coefficient vector v, nullness for S^ξ is equivalent to vanishing of the field of its F_d class on every test. For every $O \in O(4)$, $b \in \mathbb{R}^4$, coordinate a, finite family of distinct-point words w_i in $\{x : (O^{-1}(x-b))^a > 0\}$ and coefficients $c_i \in \mathbb{C}$,

$$\sum_{i,j} \overline{c_i} c_j S^\xi(\theta_{O,b,a} w_i\, w_j) \geq 0.$$

Here $\theta_{O,b,a}$ reverses the word, reflects its points across the hyperplane, and transports the a-axis label reflection through O. *Status:* theorem consequences, unused by the main theorem. *Lean:* `fixedHeat`, `exactly_fixed`, `recognizes`, `allPlanes`.

Proposition V.3.y (stress tensor). For every $\xi \in X$:

- **(T0)** F^{ξ} extends holomorphically to the full forward tube (consecutive imaginary differences in the open forward cone), with continuous tested partial boundaries on the closed cone, equal to W^{ξ} at zero. Each adjacent pair has label-independent tied heights tending to zero with equal exchange germs at spacelike separation.
- **(T1)–(T4)** For every proper orthochronous Poincaré transformation g, Schwartz h, indices $a, b \in \{1, 2, 3, 4\}$ and $u, v, x \in \mathfrak{D}_{\xi}$, $U_{\xi}(g)\mathfrak{D}_{\xi} \subset \mathfrak{D}_{\xi}$ and

$$U_{\xi}(g)T_{ab}(h)x = \sum_{i,j} \Lambda(g)_{ia}\Lambda(g)_{jb}T_{ij}(h \circ g^{-1})U_{\xi}(g)x,$$

$$T_{ab} = T_{ba}, \qquad \langle T_{ab}(h)u, v\rangle = \langle u, T_{ab}(\bar{h})v\rangle, \qquad \sum_{a} \eta_{aa}T_{ab}(\partial_a h) = 0.$$

- **(T5)** Fix $B \in F_d$, a Killing field K, compact $\mathcal{O} \subset \mathbb{R}^4$, $m \in C_c^{\infty}(\mathbb{R}^4)$ supported in $\mathcal{O}$, real $f \in C_c^{\infty}(\mathbb{R})$ with $\int f = 1$, and complex $c \in C_c^{\infty}(\mathbb{R}^3)$ equal to 1 on an open ball. Suppose every $(\vec{y}, s)$ over its complement, for $s \in \operatorname{supp} f$, is strictly spacelike to every point of $\mathcal{O}$. On $\mathfrak{D}_{\xi}$,

$$i[Q_K(f, c), \phi_{\xi}(B, m)] = \phi_{\xi}(\Sigma_K B, m) - \sum_{a} \phi_{\xi}(B, \partial_a(K^a m)).$$

- **(T6)** Under these conditions, $i[Q_K(f, c), A]$, for every finite sum of words A supported in $\mathcal{O}$, is the sum replacing one factor at a time by (T5), preserving order, for any representatives in the F_d.

Status: theorem. *Lean:* `Stress.force`.

3.23.2 2. What it uses

- **IV.1:** Euclidean axioms (`Seed`).
- **V.1:** character decomposition, copy bounds, OS realization and almost-everywhere vacuum uniqueness (`Cluster.lifts`).
- **V.2:** reconstruction, polynomial one-point slab bounds and C01–C29 analytic estimates (`Relativistic`).
- **VI.1:** field algebra and Euclidean stress Ward identities (`Locals`).
- OS reconstruction; spectral theorem; Vitali–Montel, Morera and Liouville theorems; dominated convergence.

3.23.3 3. Strategy

Uniform slab domination transfers reconstruction to every phase. Character continuity passes through holomorphic continuation to boundary distributions. Locality supplies uniform low-energy and heat decay; full support and continuity extend vacuum-subtracted decay from almost every phase to every phase. Its large-time limit identifies the vacuum projection. Separately, Euclidean vector recovery proves surjectivity; continued Ward identities give stress charges.

3.23.4 4. Steps

1. **Uniform reconstruction.** Write $\Pi_\xi(d,w)$ and $\Pi(d,w)$ for phase and mixed slab operators. Repeated reflection and taking roots give $\|\Pi_\xi(d,w)\| \le \|\Pi(d,w)\|$ for every positive thickness and distinct-point slab word. Apply V.2 phasewise. Lean: `sameBound`, `almanac`.
2. **Boundary continuity.** Uniform polynomial bounds, Vitali's theorem and a Taylor formula with integrable remainder prove continuity and barycenters of W^ξ. Lean: `word_continuity`, `barycenter`.
3. **Surjectivity.** Tests with compact frequency support and exponential damping recover positive-height Euclidean vectors in $\operatorname{ran} J_\xi$; totality and closedness give surjectivity. Lean: `onto`.
4. **Low energy.** For compactly supported A, locality makes the time-averaged odd vacuum commutator compactly supported in space. The nonnegative weight $(1-\cos E)/E$ (zero at $E=0$) and a momentum bump give shell mass $O(u^2)$: spatial volume contributes u^3, the weight contributes u. Dyadic summation yields item 5; Laplace integration yields item 6. Lean: `light_bound`, `inverse_square`.
5. **Every vacuum.** Heat forms are continuous in ξ. Almost-everywhere uniqueness and full support extend vacuum-subtracted decay everywhere; strong heat limits and density give V.3.2. Lean: `extends_decay`, `all_plant`.
6. **Irreducibility.** Glue bounded half-plane matrix elements by weak centrality; Morera and Liouville make them constant. Vacuum uniqueness forces the commutant to be scalar. Lean: `commutant`.
7. **Stress charges.** Continue Euclidean contact identities through spatially integrated current jumps, then remove spectator cutoffs and take word-vector matrix elements to obtain (T5). Lean: `Stress.through_jump`, `Stress.local_action`.

3.23.5 5. Where the difficulty lies

Uniform analytic boundary bounds, Euclidean vector recovery and joint current limits require care. Exchange germs agree only at spacelike separation; current limits are spatially integrated. The u^2 estimate does not imply a mass gap. Uniqueness of the phase itself remains undecided.

3.23.6 6. Notation

$\mathcal{D}_\xi$ is the fixed space of the OS heat semigroup; $\mathfrak{D}_\xi$ is the Wightman word domain. Time is coordinate 4; $P^0_\xi = P_\xi(\{p^4 = 0\})$, $\mu^\xi_A(B) = \langle A\Omega_\xi, P_\xi(B)A\Omega_\xi\rangle$, and $e^{-sH_\xi} = J^*_\xi e^{-sH^{\mathrm{OS}}_\xi} J_\xi$. A compactly supported word sum has all tests in C^∞_c. Write $\eta = \operatorname{diag}(-1,-1,-1,1)$, $\zeta_4 = -i$, $\zeta_j = 1$, $T_{ab}(h) = \phi_\xi(-\zeta_a\zeta_b T^{\mathrm{E}}_{ab}, h)$, and $Q_K(f,c) = \sum_b T_{4b}(K^b(c\otimes f))$. Here $K(x) = k + Vx$, $V^{\mathsf{T}}\eta + \eta V = 0$, and Σ_K is its infinitesimal label action.

The reconstruction family is `Almanac` (`L.chambers`, `L.contours`, `L.actors`, `L.J`); word sums are `SumOfWords`; the charge is `Stress.Charge`.

3.23.7 7. What remains unexplained

The C01–C29 estimates remain analytic inputs from V.2. The current-jump argument needs explicit tube faces and spacelike neighborhoods; integrated limits do not supply pointwise stress limits.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.24 VI.1. Ward identities, the rotation action and the conserved stress tensor

Part: VI, symmetry: stress tensor and Ward identities. **Lean:** `YM/Bid/Town.lean`, `YM/Next/ROTATIONS`, `YM/Ward`, `YM/Spin`. **Size:** approximately 7,704 assigned lines, plus 6,100 of collinear flux estimates.

3.24.1 1. What this unit proves

Except for group statements, assume S satisfies the Euclidean axioms: permutation symmetry, empty normalisation, unit deletion, translation invariance, B_4- and conjugation-covariance, coordinate reflection positivity for point words and smeared tests, smoothness, and for each arity, labels and derivative order j, constants $C \geq 0, A \in \mathbb{N}$ with $\|D^j S(x)\| \leq C r^{-A}$ whenever $0 < r \leq 1$ and maximum-norm separation is at least r, uniformly in positions. The plan consists of complex-linear maps $\partial_a : F_d \to F_{d+1}$. Its three laws are: derivative law $S(\dots, \partial_a v_k, \dots; x) = \partial_{x_k^a} S(v; x)$, inclusion law $S(\dots, \iota v_k, \dots) = S(v; x)$, and orientation: inclusions commute with M, Γ, derivatives commute with Γ, and $M_\sigma \partial_b = \sum_a \sigma_{ab} \partial_a M_\sigma$.

The raw maps are complex-linear $p_a : F_d \to F_{d+1}$ and real-linear $Y : \mathfrak{so}(4) \to \mathrm{End}_{\mathbb{C}} F_d$; spin rows T have the latter type. Write (R) for: $J_{ab} \in F_4$ is symmetric, Γ-real, $\sum_a \partial_a J_{ab} = 0$ in F_5, $M_\sigma J_{ab} = \sum_{ij} \sigma_{ia} \sigma_{jb} J_{ij}$, and $(J; p, Y)$ obeys the stress laws. For every separated $(x_0, \dots, x_n)$, labels B, q, and real $\chi \in C_c^\infty$ equal to 1 near x_0 with closed support excluding the other points, these laws for $(J; P, A)$ are

$$-\sum_{a,b} \int \partial_a \chi(y) V_b(y) S(J_{ab}(y), B(x_0), q)\, dy = \begin{cases} S(P_j B(x_0), q), & V = e_j, \\ S(A_K B(x_0), q), & V = K(y - x_0). \end{cases}$$

All identities concern separated insertions; no contact term is evaluated. Define

$$\mathcal{W}_K[T, P](https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/q; x$$

Proposition VI.1.1. Assume the derivative and orientation laws and (R). For every arity n, degree d, coordinate axis $\mathbb{R}e_l$, every $x \in \mathrm{Conf}_n$ on that axis, all tags q and $K \in \mathfrak{so}(4)$, $\mathcal{W}_K[Y, p](https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/q; x) = 0$. For every $j \neq l$ and $v_1, \dots, v_n \in F_d$, $\sum_i S(\dots, p_j v_i, \dots; x) = 0$. *Status:* theorem. *Lean:* `Outer.outer`, `Bid.LineSum`.

Proposition VI.1.2. Assume all three plan laws, the ultraviolet record and free-Maxwell data of §2. There exists real $c \neq 0$, independent of degree and axis, with $p_a = c\partial_a$ on every F_d. For all degrees, axes and K, the rows $D = c^{-1} Y$ satisfy (C) $\partial_b D_K + \sum_i K_{ib} \partial_i = D_K \partial_b$ and (L) $\mathcal{W}_K[D] = 0$ on every coordinate axis, for all labels and K. *Status:* theorem. *Lean:* `Bid.pressured`, `Bid.formed`.

Proposition VI.1.3. Assume the derivative law and spin rows T satisfying (C),(L). Then $\mathcal{W}_K[T](https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/q; x) = 0$ for every arity, separated configuration, tags and K. *Status:* theorem. *Lean:* `Ward.global`.

Proposition VI.1.4. Assume the inclusion and derivative laws, VI.1.3's conclusion, and compatibility of T: it commutes with Γ, ι, kills $\mathbf{1}$, and $M_\sigma T_K = T_{\sigma K \sigma^{-1}} M_\sigma$. Assume internal row erasure: for some N, every $d \geq N$, and every alternating real-bilinear family $\Delta(u, v) \in \mathrm{End}_{\mathbb{C}} F_d$ that is Γ-real, covariant under determinant-one signed permutations, kills $\mathbf{1}$ and preserves every $\iota_{e \to d}(F_e)$, $e \leq d$, vanishing insertion sums for all words, separated configurations and u, v imply $\Delta = 0$. Then $[T_u, T_v] = T_{[u,v]}$ for all d, u, v. *Status:* theorem; erasure is proved here. *Lean:* `Work.A08.answer`.

Proposition VI.1.5. Under VI.1.2's hypotheses, for every d there is a complex-linear isomorphism $V_d : F_d^{\mathrm{free}} \to F_d$ intertwining M, Γ and satisfying $D_K V_d = V_d \mathcal{L}_K^{\mathrm{free}}$ for every K. Thus $R_d = V_d R_d^{\mathrm{free}} V_d^{-1}$ is a continuous $O(4)$ representation, equals M on B_4, commutes with Γ, has generators D satisfying the global Ward identity of VI.1.3. The V_d are chosen independently across degrees. Such an action with specified generators is a rotation port. *Status:* theorem. *Lean:* `Bid.services`.

Proposition VI.1.6. Assume the inclusion and derivative laws and a rotation port R with generators T satisfying VI.1.3's global Ward identity. For all $k \in O(4)$, tags and separated x, $S(R(k)v; kx) = S(v; x)$. Moreover R fixes $\mathbf{1}$, commutes with inclusions, retains the port properties, and $R_{d+1}(k)\partial_a v = \sum_i k_{ia} \partial_i R_d(k) v$ for every d, a, v. *Status:* theorem. *Lean:* `Work.A14.answer`.

Proposition VI.1.7. Assume the derivative law, a covariant rotation action satisfying VI.1.6's laws, (R), $p = c\partial$ with $c \neq 0$, and generators $c^{-1}Y$. There exists symmetric, real, divergence-free J' with $R_4(k) J'_{ab} = \sum_{ij} k_{ia} k_{jb} J'_{ij}$ for every k, satisfying the stress laws with $(P, A) = (\partial, c^{-1}Y)$. The translation shell, Ward identity (13), therefore equals $\partial_{x_0^b} S(B(x_0), q)$ for every window and labels. *Status:* theorem. *Lean:* `Work.A15.answer`.

Proposition VI.1.8. Spatial rotations and coordinate boosts generate $SO^+(1,3)$. Under VI.1.2's hypotheses, choose V, R, J' as above: $B_d(\Lambda) = V_d B_d^{\mathrm{free}}(\Lambda) V_d^{-1}$ commutes with every inclusion $F_d \hookrightarrow F_e$, $d \leq e$, for every Λ. These actions give a representation on the union, continuous on each F_d, equal to R_d on spatial rotations and commuting with $M_{\theta_t}\Gamma$. For every d, spatial i and $v \in F_d$, its boost derivative at zero is $-iD_{K^{(3i)}} v$. *Status:* theorems. *Lean:* `Work.A16.answer`, `Work.A17.answer`, `Spin.reach`.

Proposition VI.1.9. Given §2's free data, infrared triple and ultraviolet record, all six assembly clauses hold: raw compatibility, normalisation, rotation descent, covariance, averaged stress laws, and Lorentz compatibility. With the inclusion and derivative laws, there exist $c, (V_d), J'$ satisfying VI.1.2 and VI.1.5–8, all three plan laws and the resulting local field laws. *Status:* conditional theorem. *Lean:* `Outer.commission`, `Commission.picks`.

3.24.2 2. What it uses

- IV.1: limit table `Table.Hierarchy`, differential plan, commuting derivatives and injective inclusions.
- V.1: `OS.Chamber`, OS Hilbert spaces, contraction semigroups and joint spectral measures, constructed from the axioms.
- VI.2: `Org.MaxwellData`; free plan, covariant rotation and Lorentz actions, stress laws for $(J^{\mathrm{free}}; \kappa\partial^{\mathrm{free}}, \kappa\mathcal{L}^{\mathrm{free}})$ with $\kappa \neq 0$, derivatives nonzero on F_4^{free}, generator relations, Casimir polynomials and row erasure. At every sufficiently large degree, finitely many collinear readouts have common kernel $\mathbb{R}\partial^{\mathrm{free}}$ within the free vector-map space. Their producers assume the gauge group simple.
- VI.2 and III.11: `Org.IR` and `Org.UV`, including (R), orientation, lattice/extraction data, corrected-field bounds, grid approximations and weak scaling comparison of correlation jets

and stress with the free table.

- Weak comparison means: for every arity bound N, degree bound d, derivative bound m and $R \geq 1$, there exist $r_k \downarrow 0$, $\xi_k > 0$ and complex-linear isomorphisms $U_{k,e} : F_e^{\mathrm{free}} \to F_e$. Through degree $\max(d, 4)$ these intertwine M, Γ, units and inclusions; for $n \leq N$, degrees $\leq d$ and $j \leq m$, $D_y^j S(U_k v; r_k y) \to D^j S^{\mathrm{free}}(v; y)$ uniformly where $\|y\| \leq R$ and separation is at least R^{-1}. Also $\xi_k^{-1} r_k^4 U_{k,4}^{-1} J_{ab} \to J_{ab}^{\mathrm{free}}$ for every a, b. The system may depend on all four bounds.
- Standard results: spectral theorem, holomorphic semigroups, analytic vectors, analytic identity theorem, Haar averaging, Cartan–Dieudonné, Givens rotations, McWeeny purification and linear ODE uniqueness.

3.24.3 3. Strategy

Conservation turns shell laws into collinear Ward identities; free comparison fixes their normalisation. Positivity converts collinear two-point jets into analytic OS vectors, allowing propagation to every separated configuration. Exact intertwiners transport the free rotations; covariance then gives compatibility across degrees. Rescaling and averaging give the tensorial stress, and boost ODE uniqueness gives Lorentz compatibility.

3.24.4 4. Steps

1. Moving the centre inside a fixed window gives $\partial_b Y_K + \sum_c K_{cb} p_c = Y_K \partial_b$ and $\partial_b p_i = p_i \partial_b$ by differentiating its shell laws. `Work.A02.answer`.
2. Cutoffs of widths $\lambda^{100}, \lambda^{150}, \lambda^{300}$ cancel two terms by stress symmetry; reflection positivity and spectral estimates bound the exterior flux by $O(\lambda^{-1})$, giving the collinear identity without a mass gap or clustering. `Outer.blows`. *(supporting analysis partly read)*
3. Finite free collinear tests have a one-dimensional kernel; persistence of rank under scaling forces $p = c\partial$. `Climb.extends_form`.
4. Analytic comparison determines two-point jets; positivity gives factorial vector-derivative bounds. Extreme-point moves and density propagate vanishing angular sums. `Analysis.blossoms`, `Ward.global`.
5. Free row erasure removes the bracket defect. Relative-Casimir projection and iteration $x \mapsto 3x^2 - 2x^3$ produce exact intertwiners: the compressed defect satisfies perfect commutator relations, which force sufficiently small operators to vanish. `Work.A08.answer`, `Spin.meet`.
6. Group generation integrates covariance; Haar averaging of $c^{-1} J$ preserves the stress laws. Boost ODE uniqueness proves compatibility across degrees. `Work.A14.answer`, `Work.A15.answer`, `run_boost`.

3.24.5 5. Where the difficulty lies

Diagonal stress components require a conservation estimate beyond reflection-parity arguments. Vanishing smooth jets alone do not justify continuation; positivity supplies analyticity. Exact intertwiners identify the required $O(4)$ action and signed permutations.

3.24.6 6. Notation

S: correlation table; F_d: finite-dimensional complex label spaces of degree $\leq d$ modulo null labels; ι: inclusions; $\mathbf{1}$: unit; B_4, M: signed permutations and their action; Γ: conjugation; Conf_n: distinct

points in $\mathbb{R}^4$. $K \in \mathfrak{so}(4)$ is real skew, with $[u, v] = uv - vu$ and $K^{(ba)} = E_{ba} - E_{ab}$; time is axis 3, and θ_t reverses it. $\lambda = e^k$ is the cutoff scale.

3.24.7 7. What remains unexplained

Supporting flux analysis, including contact-extension normalisation, was not checked fully (probable). The ultraviolet hypotheses of the input producers are not discharged here. Centring of J' is not proved here; neither sign of c, κ is specified. Lean elaboration and transitive axiom dependencies were not checked.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.25 VI.2. The free Maxwell stress tensor and the local Ward charge

Part: VI, symmetry: stress tensor and Ward identities. **Lean:** `YM/Next/MAXWELL`, `YM/UVNext/FREE_STRESS_CERTIFICATE`, `YM/Next/UV_STRESS`, `YM/Pert`. **Size:** 14,179 lines used by the main theorem (Maxwell 8.2k, free-stress classification 3.1k, UV stress 2.6k, Pert 0.2k).

3.25.1 1. What this unit proves

Throughout, G satisfies I.1, $\mathfrak{g}$ is its Lie algebra, spacetime indices run from zero to three, and c indexes an orthonormal colour basis. A Maxwell wave is a smooth $\mathfrak{g}$-valued two-form satisfying $df = 0$ and $\sum_a \partial_a f_{ab} = 0$. The algebra $\mathcal{W}$ consists of complex polynomials in homogeneous on-shell Taylor coefficients of order j, weighted $j + 2$; $\mathcal{P}_d$ is its $\mathrm{Ad}(G)$-invariant sector of weight at most d. Wick correlations contract only between sites, using the curl–curl of $G_0(x) = (4\pi^2|x|^2)^{-1}$. Set

$$T^{\mathrm{free}}_{ab} = \sum_{c,i} F^c_{ai} F^c_{bi} - \tfrac{1}{4}\delta_{ab} \sum_{c,i,k} F^c_{ik} F^c_{ik}.$$

Let Θ be its bilinear polarization, normalized by $\Theta(f, f) = 2T^{\mathrm{free}}(f)$; $j^v_a = \sum_b v_b \Theta_{ab}$ and $\Phi_\chi(j) = -\sum_a \int \partial_a \chi \, j_a$.

Proposition VI.2.1 (free table). For every register L and $d \geq 0$, substitution gives a complex linear isomorphism $\iota_d : \mathcal{F}^{\mathrm{free}}_d \to \mathcal{P}_d$; a label is null exactly when its polynomial vanishes on every on-shell Taylor tower. Correlations are Wick moments. The table satisfies IV.1's Euclidean hierarchy, including point and smeared reflection positivity, and is real-analytic at distinct points. Each fixed correlation derivative is bounded by Cr^{-A} for mutual sup-distances at least r, $0 < r \leq 1$, with $A, C \geq 0$. Taylor differentiation gives $\partial^{\mathrm{free}} : \mathcal{F}^{\mathrm{free}}_d \to \mathcal{F}^{\mathrm{free}}_{d+1}$ with inclusion, derivative and orientation laws. The $O(4)$ action and generators ρ satisfy VI.1's rotation laws and preserve correlations. The stress is symmetric, real, traceless, centred and conserved in degree five. Simplicity is unnecessary. *Status:* theorem. *Lean:* `hierarchy`, `fields`, `realStress`.

Proposition VI.2.2 (local charge). Let U be open, $x \in U$, f a Maxwell wave on U, v a smooth conformal Killing field on $\mathbb{R}^4$, and real $\chi \in C^\infty_c(U)$ equal one near x. Put $f^{p,x}(y) = \langle p(F)(x)F(y)\rangle_0$. For each component probe $p = F^c_{il}$,

$$\Phi_\chi(j^v(f^{p,x}, f)) = -p(\mathcal{L}_v f)(x), \qquad \mathcal{L}_v f = d\iota_v f.$$

If $\mathcal{L}_v f$ is a Maxwell wave, this holds for every probe of every order $j \geq 0$. Translations, centred rotations and dilation give respectively $-\partial_a[p(f)]$, $-(\omega \cdot p)(f)$ and $-(j + 2)p(f)$ at x. Flux

vanishes for two Maxwell waves near $\operatorname{supp}\chi$, and for two generated fields with source x when v is homogeneous about x of degree zero or one. *Status:* theorem. *Lean:* `Maxwell.Free.local_force`, `jet_force`, `two_flux_zero`.

Proposition VI.2.3 (polynomial Ward identities). For $n \geq 0$, distinct $x_0, \dots, x_n$, $p_i \in \mathcal{W}$, and real $\chi \in C_c^\infty$ equal one near x_0 with spectators outside its support,

$$-\sum_{a,b} \int \partial_a \chi(y) v_b(y) \langle T^{\mathrm{free}}_{ab}(y) \prod_i p_i(x_i) \rangle_0 \, dy = -\langle (Dp_0)(x_0) \prod_{i>0} p_i(x_i) \rangle_0 .$$

Here (v, D) is (e_a, ∂_a), $(\omega(y - x_0), \rho(\omega))$ for $\omega \in \mathfrak{so}(4)$, or $(y - x_0, \varepsilon)$, where ε multiplies monomials by weight. Thus τT^{free} has translation/rotation rows $-\tau\partial^{\mathrm{free}}, -\tau\rho$ for every real τ; normalization to $+\partial^{\mathrm{free}}$ gives $-T^{\mathrm{free}}$. *Status:* theorem. *Lean:* `Maxwell.Free.moment_force`.

Proposition VI.2.4 (free stress line). Suppose G is simple and W satisfies the free derivative and orientation laws. Its derivative maps are ∂^{free}. Let $\mathcal{Z}$ be the real space of centred symmetric real signed-covariant degree-four tensors, and $\mathcal{V}$ the real signed-covariant degree-five vectors. Then

$$\operatorname{div} : \mathcal{Z} \twoheadrightarrow \mathcal{V}, \quad \dim_{\mathbb{R}} \mathcal{Z} = 3, \quad \dim_{\mathbb{R}} \mathcal{V} = 2, \quad \ker \operatorname{div} = \mathbb{R} T^{\mathrm{free}}.$$

Every lighthouse with this plan and stress τT^{free}, $\tau \neq 0$, satisfies (FD). *Status:* theorem. *Lean:* `certificate`.

Proposition VI.2.5 (Maxwell data). For simple G, every Wilson family C and real $\tau \neq 0$, $(\partial^{\mathrm{free}}, R, \tau T^{\mathrm{free}}, -\tau)$ satisfies all lighthouse laws and extends to Maxwell data for C. The derivative is nonzero in degree four. For all sufficiently large d, finitely many derivatives of Ward sums at distinct collinear configurations perpendicular to the component axis detect exactly $\mathbb{R}\partial^{\mathrm{free}}$ among complex linear vector maps $\mathcal{F}_d^{\mathrm{free}} \to \mathcal{F}_{d+1}^{\mathrm{free}}$ that kill the unit, are real, signed-covariant and preserve every lower-degree image with degree raised by one.

The datum includes the hierarchy, VI.1's Casimir data, and compatible degreewise continuous proper orthochronous Lorentz representations, agreeing with spatial rotations, commuting with time-reflected conjugation and having boost generator $-i\rho$. For every $d \geq 4$, any alternating real-bilinear family of complex linear endomorphisms of $\mathcal{F}_d^{\mathrm{free}}$ $T(u, v)$, $u, v \in \mathfrak{so}(4)$, that is real, equivariant under orientation-preserving signed permutations, kills the unit and preserves lower-degree images, vanishes if all its Ward sums vanish for every u, v, every arity $m \geq 0$, every degree-d field tuple and every distinct-point configuration. Every Maxwell datum can retain its plan, rotations, hierarchy and Lorentz representation while acquiring stress τT^{free}, full stress rotation covariance and (FD). Its Ward constant is necessarily $-\tau$. *Status:* theorem. *Lean:* `maxwell_supplies`, `FreeStress.fire`, `required`.

Proposition VI.2.6 (strict positivity). For every axis a, $x_a > 0$ and nonzero $p \in \mathcal{W}$, $\langle (\vartheta_a p)(\theta_a x) p(x) \rangle_0 > 0$. Every finite set of distinct coordinate probes has positive-definite reflected propagator Gram matrix there. Finite families of positive-half-space words, with distinct sites within each word, satisfy reflection positivity. For every $d \geq 0$, a polynomial in $\mathcal{P}_d$ annihilated by all distinct-point spectator tests from $\mathcal{P}_d$ is zero. *Status:* theorem. *Lean:* `detects`, `ghosts_out`.

Proposition VI.2.7 (limit stress). Assume II.2's Wilson construction conditions, ordered smeared convergence and generic point limits along F to a table S with Euclidean hierarchy H, an inclusion/derivative plan W on S, a free hierarchy H_0, and a lighthouse with inclusion, derivative, orientation and (FD) laws. Assume VI.2.8's station property and $\dim_{\mathbb{C}} \mathcal{F}_d^S \leq \dim_{\mathbb{C}} \mathcal{F}_d^{\mathrm{free}}$ for all

$d \geq 0$. There exists an approaching stable scaling $X = (r_k, \xi_k, U_{k,d})$, initially with $\xi_k = 1$. For every such scaling, interacting divergence $\mathcal{Z}^S \to \mathcal{V}^S$, defined as in VI.2.4, is onto with one-dimensional real kernel. There exist $0 \neq J \in \mathcal{Z}^S$ with zero divergence and an approaching stable refinement Y by subsequence and change of ξ_k such that

$$\xi_k^{-1} U_{k,4}^{-1}(r_k^4 J_{ab}) \to T_{0,ab}.$$

For every finite arity/degree/derivative bound and $R \geq 1$, tails give tracks intertwining symmetries, units and inclusions through degree $\max(d, 4)$, satisfying the stress limit above, with correlation derivatives converging uniformly on $K_{R,n}$ for each fixed tuple; tails may depend on tuple and tolerance. Thus J satisfies VI.1's comparison law. *Status:* conditional theorem. *Lean:* `Stations.equivalences`, `produce_current`.

Proposition VI.2.8 (stations). Fix the extraction data of VI.2.7. A depth-D station consists of $r > 0$ and complex linear lattice lifts $A : \mathcal{F}_d^{\mathrm{free}} \to \mathrm{Ref}_d$, $d \leq D$. Admissibility requires eventual uniform boundedness, reality, signed covariance and preservation of the degree-zero unit, and, for every tuple of arity $n \leq n_{\max}$ and degrees at most D, smooth complex test supported in $K_{R,n}$, $M > 0$ with $\|t\|_\infty \leq M$, and $\eta > 0$, eventual lattice/free smeared discrepancy along the ordered cutoff tails at most $\varepsilon M \prod_i \|v_i\| + \eta$, using test $r^{-4n} t(\cdot/r)$. An admissible station yields complex linear continuum maps with the same symmetries and estimate without η:

$$\left| \int t(y) \left[\langle \prod_i U v_i(r y_i) \rangle^S - \langle \prod_i v_i(y_i) \rangle_0 \right] dy \right| \leq \varepsilon M \prod_i \|v_i\|.$$

The cutoff tail may depend on the tuple, test, M and η. The station property requires these for every $D, n_{\max} \geq 0$, $R \geq 1$ and $\varepsilon, \delta > 0$, with $r \leq \delta$. Balanced grid, strip and finite-difference estimates of detailed Lemma VI.2.s, for all finite bounds, imply it. *Status:* definitions and conditional theorems. *Lean:* `Station.Admissible`, `harvested`, `realizes_stations`.

Proposition VI.2.9 (normalized limit). Let C be a Wilson family, F an extraction order, S a table with plan W, M a Maxwell datum with lighthouse Ward constant $\kappa \neq 0$, and $I = (J_0, p^I, Y^I)$. Assume VIII.1's ultraviolet-data conditions for these objects. Assume W's inclusion/derivative laws, $0 \neq J_0 \in \mathcal{Z}^S$, (FD), and an approaching stable scaling with $\xi_k^{-1} U_{k,4}^{-1}(r_k^4 J_{0,ab}) \to T_{0,ab}$. Every symmetric real conserved signed-covariant $J \in (\mathcal{F}_4^S)^{4\times 4}$ satisfying translation Ward rows W and some rotation Ward rows obeys

$$U_{k,4}^{-1}(r_k^4 J_{ab}) \to \kappa^{-1} T_{0,ab}.$$

For every finite free-field tuple, any mixture of transported fields and rescaled stress slots converges with every derivative uniformly on compact distinct-point sets to the free correlation with every stress slot replaced by $\kappa^{-1} T_0$; the interacting insertions are evaluated at $r_k y_i$. Canonically $\kappa = -1$, so each stress slot tends to $-T^{\mathrm{free}}$. *Status:* conditional theorem. *Lean:* `Pert.meter_physical`, `Pert.mixed_leading`.

3.25.2 2. What it uses

- III.16: Taylor spaces and Wick moments; IV.1: quotient fields, extraction and dimension bound.
- V.1: smeared positivity and derivative bounds; VI.1: Ward, rotation, track and Casimir laws.

- III.10: Maxwell assembly; III.11: balanced comparison and shell estimates.
- VII.5: normalization ratios; VIII.1: data conditions.
- Standard results: Wick/Fock factorization, Gaussian reflection positivity, Cartan's formula, approximate identities, rank–nullity, compactness and Fourier inversion.

3.25.3 3. Strategy

Strict reflected positivity identifies quotient fields with on-shell polynomials. Local flux gives the Ward identities; finite-dimensional classification gives the conserved stress line. Stations produce comparison isomorphisms. Transporting derivatives and recentering tensors transfers divergence rank to the interacting table. Conservation determines its stress line, and translation Ward identities fix its normalization.

3.25.4 4. Steps

1. Factor reflected Wick pairings through a positive-definite Gram matrix and the Fock inner product; invertibility excludes residual polynomials. `core`.
2. Smooth the Green kernel, integrate by parts twice and remove the approximate identity to obtain $-\mathcal{L}_v f$; differentiate the source for higher probes. `local_force`, `jet_force`.
3. Split Wick contractions by the destinations of the stress legs. Homogeneity and source-free conservation cancel the unmixed terms. `fees`.
4. Classify three tensors and at most two vectors; two independent propagator tests give divergence rank two. `rank_on`.
5. Bounded-degree spectators detect arbitrary weights; dilation and inversion eliminate derivative candidates after subtracting a scalar derivative. Rotational Ward identities and quarter turns eliminate alternating internal rows, completing Maxwell data with the Lorentz and Casimir inputs. `scanner`, `erasing`.
6. Balanced strip-volume and grid-interpolation errors give stations. Compactness of bounded lifts and ordered smeared convergence produce continuum panels. `cutoff_balance`, `harvested`.
7. Correct station panels to preserve inclusions; separating tests give injectivity, and the dimension bound gives isomorphisms. `Stations.equivalences`.
8. Integration by parts gives $r_k U_{k,d+1}^{-1} W_a U_{k,d} \to \partial_a^{\mathrm{free}}$; gradient closure and Fourier inversion give smooth convergence. Recentring then permits rank transfer. `travels_jet`, `differential_rank`.
9. Compactness on a normalized stress sphere selects a conserved limit; (FD) identifies it with T_0 after one sign choice. Writing $c(J - \langle J \rangle \mathbf{1}) = J_0$, comparing Ward rows gives $c/\xi_k \to \kappa$ and the physical limit. `produce_current`, `meter_physical`.

3.25.5 5. Where the difficulty lies

Exact isomorphisms and recentering are needed for rank transfer. Bounded-degree detection and alternating Ward-row rigidity require more than positivity. Conservation fixes a line; Ward normalization fixes its scale and sign.

3.25.6 6. Notation

θ_a, ϑ_a: reflection in $y_a = 0$ and reflected conjugation. A probe of order j is a real linear functional on the homogeneous order-j on-shell Taylor space. $\mathcal{F}_d^S$: degree-d fields. T_0, κ: lighthouse stress and

Ward constant. R, ρ: free rotations and their generators. $\mathrm{div} J = \sum_a W_a J_{ab}$. (FD): onto divergence, one-dimensional kernel, nonzero centred conserved stress. $K_{R,n}$: configurations of total norm at most R and mutual sup-distance at least R^{-1}. Scaling: isomorphisms $U_{k,d}$, strictly decreasing $r_k \to 0$, $\xi_k > 0$; approaching means, for each fixed field tuple, the rescaled interacting/free correlation difference tends to zero uniformly against smooth complex tests supported in any fixed compact subset of Conf_n with sup norm at most one; Conf_n denotes distinct-point configurations; stable means eventual compatibility with reality, signed symmetries, units and inclusions.

3.25.7 7. What remains unexplained

Subsidiary Gram/Fock, invariant-spanning, propagator-table, bounded-order detection, averaging, Lorentz and Casimir proofs remain inputs. III.11 must supply balanced estimates and shell bounds; underlying track Ward estimates also remain inputs. Lean's mixed-correlation theorem specializes to VI.1's assembled stress.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.26 VII.1. Short distances: convergence to free Maxwell fields

Part: VII, identification as Yang–Mills. **Lean:** the chain of `YM/Pert` (folders `Fourier`, `Align`, `Warp`, `Rig`, the jet stencils of `Lattice`, the statement `Position/Crest.finite_leading`, and `Dense`), `YM/Push/FINITE_DETECTION`, `YM/SharpGram/Weak`, `YM/SharpGram/Dense`, `YM/RG08` (about thirty small sub-folders), `YM/PertExtension`, `YM/OPE/Levy`. **Size:** 8,117 lines.

3.26.1 1. What this unit proves

Fix a compact matrix gauge group G with real simple nonabelian Lie algebra $\mathfrak{g}$, invariant inner product and orthonormal colour basis. Let V_d be complex $\mathrm{Ad}(G)$-invariant radial-jet polynomials of engineering weight at most d, F_d their free-null quotient, ρ restriction to Maxwell jets, $\mathcal{J}_d = V_d \cap \ker \rho$, and $\omega_d = \rho \circ \mathrm{sec}_d$ for the VII.3 representatives.

Proposition VII.1.1 (regulator removal). For every integer $L \geq 2$, $d, r \in \mathbb{N}$, label frames, $p \in F_d^r$ and continuous f with compact closed support in Conf_r, set $h_m = L^{-m}$, $N_m = 4L^{m^2+m+1}$. Then

$$\Pi_m(f; p) \longrightarrow \int f(x) \mathcal{W}_x(\omega_d p_1, \dots, \omega_d p_r)\, dx.$$

For any position board with smooth compactly supported profiles in Conf, order-zero readings converge slotwise for every basis-label tuple to free readings. *Status:* theorem. *Lean:* `Rig.zerothProfile`, `Position.finite_leading`.

Proposition VII.1.2 (pointwise limit). For every itinerary (N_i, h_i) with $4 \mid N_i$, $d, r \in \mathbb{N}$, $z \in \mathrm{Conf}_r$, sites $o_{i,j}$ homing to z_j, and $p \in F_d^r$,

$$\Big\langle \prod_j R_{h_i}(\mathrm{sec}_d p_j, o_{i,j}) \Big\rangle_{N_i}^{(0)} \longrightarrow \mathcal{W}_z(\omega_d p_1, \dots, \omega_d p_r).$$

Status: theorem. *Lean:* `Warp.row_limit`.

Proposition VII.1.3 (corrected covariance). For every such itinerary, any index set ι, injective $z : \iota \to \mathbb{R}^4$, homing sites $o_{i,j}$, indices j, l and keys a, b,

$$\mathrm{Cov}_{\mu_{N_i}}(\ell^{h_i}_{o_{i,j},a}, \ell^{h_i}_{o_{i,l},b}) - \delta_{jl} S_{h_i}(a,b) \longrightarrow \begin{cases} K_{ab}(z_j - z_l), & j \neq l, \\ 0, & j = l. \end{cases}$$

Status: theorem. *Lean:* `Align.corrected_limit`.

Proposition VII.1.4 (detection). For every $v \in V_d \setminus \mathcal{J}_d$,

$$\mathrm{Re}\, \mathcal{W}_{(e_t,0)}(\rho v, \rho(\sigma \bar{v})) > 0.$$

More generally, for the diagonalizing and reflected probe substitutions Φ, Ψ, every nonzero shell polynomial q satisfies $\mathrm{Re}\, \mathcal{W}_{(e_t,0)}(\Phi q, \Psi \bar{q}) > 0$. Here Φ is onto and $\rho v = \Phi q$ implies $\rho(\sigma \bar{v}) = \Psi \bar{q}$. *Status:* theorem. *Lean:* `Free.two_detect`, `Free.positive`.

Proposition VII.1.5 (Kelvin formulas). For every real symmetric j-tensor A with harmonic diagonal, and $x \neq 0$,

$$\frac{A(\partial, \dots, \partial)\Gamma(x)}{j!} = \frac{(-2)^j}{4\pi^2} |x|^{-2(j+1)} A(x, \dots, x).$$

For every $\mathfrak{g}$-valued Maxwell shell tensor $Y \in \mathrm{Sh}_j$, contracting the curvature kernel's first rank-j Taylor leg with Y, summing colours, ordered form indices and jet indices without $1/2$, gives $\gamma_j \iota^* Y^\natural(x)$ for $x \neq 0$, where $\iota(x) = -x/|x|^2$, $Y^\natural(x) = Y(x, \dots, x)$ and $\gamma_j = 2^j(j+1)(j+2)/\pi^2$. *Status:* theorem. *Lean:* `Kelvin.harmonic_Green`, `Kelvin.response_eq`.

Proposition VII.1.6 (corrected charts). Fix a realized Wilson family, construction record with limit table S and extraction order f, VII.4 OPE data, $D \in \mathbb{N}$, and a nontrivial filter u on $\mathbb{N}^2$ finer than f's extraction filter. Assume a weak measuring `Portfolio` through D: observable/Haar bounds, Haar chart ratio, three price panels and a dominating label-cost constant; $g(k) \to \infty$ with eventually $g(k) \leq k$ and

$$\mathrm{scale}(k) \leq \mathrm{scale}(g(k)) + \log \hat{\beta}_k;$$

finite-regulator complex linear label maps eventually satisfying `Poised` simultaneously through D at step $g(k)$ (degree availability, monitor bound, tested equivariant maps and all panel estimates); and rescaled read-outs converging along u to $\Phi_{k,d}$ for every $d \leq D$ and reference label. Set $D_{k,d} = \pi^S_d \Phi_{k,d} \lambda_d$, where π^S_d projects to S_d and λ_d is the fixed equivariant representative of the free quotient in reference label coordinates, and let $\chi_k(d) = D_{k,d}$ when bijective, a fixed isomorphism otherwise. There exist complex linear isomorphisms $p_k(d) : F_d \to S_d$ for all k, d, eventually in k, simultaneously for $d \leq D$, unit-preserving, commuting with signed permutations and conjugation, and satisfying

$$p_k(e) \iota^F_{d,e} = \iota^S_{d,e} p_k(d) \quad (d \leq e \leq D);$$

and $\chi_k(d)^{-1} p_k(d) v \to v$ for $d \leq D$, $v \in F_d$. *Status:* conditional theorem. *Lean:* `Weak.Portfolio.tune_grades`.

Proposition VII.1.7 (windows and expansions).

(a) Fix III.2 RG constants, Haar bounds, a Haar chart ratio satisfying the local ratio bounds, and admissible price-panel relay with activation bounds $u_j \leq L_j$, eventual activation of every degree, and all comparison/analytic rank bounds. For any wait U with $w_j(t) = \lfloor \log(\bar{\beta} + c_G t)/\log L_j \rfloor$, $c_G > 0$, there is a wait W with no smaller stage-length/entry bounds

and the same window, and panels and dominating costs at every degree. Every realized Wilson family satisfying W, prescribed observable bounds and polynomial-representative conditions then, for every D, has simultaneous finite-regulator comparison data through D at all sufficiently late steps k whose scale exceeds the stage entry threshold and whose backward window has length at most k and lies in one stage; returned panels satisfy all three comparison inequalities. *Lean:* `Dense.slim_wait_for`.

(b) If U has this window and V has no smaller stage-length/entry/window bounds, each V entry at stage j exceeds or equals U's entry at $j+1$, and $\bar{\beta} + c_G t \geq c_G$ above that V entry, every trajectory satisfying V admits $g(k) \to \infty$ eventually satisfying the scale inequality in VII.1.6, $g(k) \leq k$, and $g(k)$ above its U entry threshold with backward U window of length at most $g(k)$ contained in one stage. *Lean:* `Dense.nearby`.

(c) Fix finite I, $T \subset I$, $p \in T$, $D, k, E, q \in \mathbb{N}$, $R \in \mathbb{R}$, $v > 0$, $q \leq k + E$. There exist $0 < w \leq v$, $C_0 \geq 0$, independent of layout and bound B, such that every layout satisfying the prescribed tensor symmetry, rank and analytic realization conditions (`Sense`), depending only on occupied markers, zero on the empty set, with weights $a_i \in \mathbb{N}$ summing to k on T and anchored v-weighted absolute-sum coefficient bounds B through order E at every leg rank, has this expansion: re-root at x_p, shift order by q, zero terms $q + n < k$; through leg rank $k + E - q$ the coefficients are Taylor coefficients of a finite sum of curvature expressions of ranks $\max(k - q, 0), \ldots, k + E - q$. These depend only on T, retain symmetry and analytic realization, and have anchored absolute-sum bound BC_0 with weight $\exp(w\,\mathrm{girth})$, including relative marker positions. *Lean:* `Dart.coin_pre`. *Status (a–c):* theorems, modulo imported station and curvature-lifting estimates whose full proofs are outside this verification (probable; see §7).

3.26.2 2. What it uses

- I.1/II.1: lattice, symmetries, labels and jets; III.16: Maxwell shells and kernel; VI.1/VI.2: hierarchies and free quotient; VII.3: lattice prescription.
- III.2/II.2: RG scales and price panels; III.11: rank-comparison estimates; VII.4/VII.5/VIII.1: OPE data, chart correction, construction record.
- Wick's theorem, Fourier analysis, Riemann sums, dominated convergence, Hobson's formula and Kelvin inversion; finite-dimensional fitting (VI.1).

3.26.3 3. Strategy

At order zero the normalized expectation is Gaussian. Independent imaginary Gaussians subtract the infinite-lattice self-contractions at each insertion. Fourier analysis gives Maxwell pairings between distinct insertions and zero limits for corrected self-pairings. The common Wick deletion recursion then gives polynomial moments; compactness and dominated convergence give tested limits.

Translated inversion produces a triangular change of probe variables with diagonal positive reflected pairings. Wick matchings give strict positivity on nonzero shell polynomials. Interacting rank estimates transfer detection to raw charts; fixed equivariant retractions remove their filtration defects recursively.

3.26.4 4. Steps

1. Invert the gauge-fixed form: covariance is direct torus Green function minus its reflected image. `Orb.covar_link`.
2. Fourier annuli give separated limits; fixed-offset quadrature controls self-contractions. `Align.chart_limit`, `Align.chart_return`.
3. Induct on distinguishable legs using the shared deletion recursion; compactness and dominated convergence give profiles. `Warp.earth`, `Rig.grid_limit`.
4. Kelvin translation makes probes triangular; diagonal positive pairings give positive Wick monomial norms and finite detection. `Free.pairing_var`, `free_injective`.
5. Rank estimates give moving-detector convergence; free detection and recursive retractions yield compatible charts. `Portfolio.twoC`, `tune_grades`.
6. Raise entry thresholds so all active estimates fit the logarithmic window; near a stage boundary choose the preceding stage's last usable step. `slim_wait_for`, `nearby`.
7. Re-root and reduce exponential weights, then lift and subtract successive lowest coefficients to obtain local curvature expressions with controlled bounds. `coin_pre`.

3.26.5 5. Where the difficulty lies

Oscillatory Fourier cancellation must survive differentiation. The chosen volume growth controls the amplified same-insertion errors; quarter-period centring removes reflected images. Detection needs strict positivity, including constants. No optimal volume growth rate is asserted.

3.26.6 6. Notation

Conf_r: distinct r-tuples in $\mathbb{R}^4$; $\Gamma(x) = (4\pi^2|x|^2)^{-1}$; K: Maxwell jet kernel; $\mathcal{W}$: Wick moments without same-insertion pairings. μ_N: Gaussian on reflection-odd link fields; ℓ^h: linear radial jet; S_h: infinite-lattice self-covariance; $R_h(p,o) = \mathbb{E}_\xi p(\ell^h_o + i\xi)$ with independent $\xi \sim \mathcal{N}(0, S_h)$ (`column`); $\langle\cdot\rangle^{(0)}$: coupling-series constant coefficient. Π_m: centred h_m^{4r}-weighted sum of f times the order-zero product correlation. An itinerary has even N_i, $h_i > 0$, positive integer $N_i h_i \to \infty$, $h_i \to 0$, $N_i h_i^n \to \infty$ for every $n \in \mathbb{N}$; homing means $h_i(o_i - c_{N_i}) \to z$, $c_N = (N/4, \dots, N/4)$, using centred lifts. σ: time reflection; e_t: time unit vector; Sh_j: symmetric Taylor/antisymmetric form tensors with homogeneous fields solving $dF = d{*}F = 0$; Taylor coefficients include $1/j!$. A key $a = (j, \alpha)$ is a coordinate on $\mathfrak{g}$-valued radial-gauge homogeneous potentials of degree $j+1$, with engineering weight $j+2$; K_{ab} uses these coordinates after inverse curl from the shell. A bar denotes complex conjugation; ι^*: 2-form pullback; $\iota^{F,S}$: filtration inclusions.

3.26.7 7. What remains unexplained

Lower Fourier uniformity estimates remain unchecked (probable). Station and curvature-lifting estimates are imported (probable); remaining RG/perturbative-extension packages are unexplained (tentative). The interacting short-distance theorem belongs to VIII.1; this unit gives no all-orders comparison.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.27 VII.2. The trace of the stress tensor and the trace anomaly

Part: VII, identification as Yang–Mills. **Lean:** `YM/TraceExtension`. **Size:** 20,681 lines.

3.27.1 1. What this unit proves

Unless stated otherwise, fix a final record over a Wilson family C, extraction order f and limit table S (VIII.1), with selected phase ξ_* and Ward-normalized stress tensor T (VI.1). The compact matrix group G has simple nonabelian Lie algebra $\mathfrak{g}$. Set

$$\kappa_G = \frac{11C_A}{24\pi^2}, \qquad d_2 = -\frac{\kappa_G}{4}, \qquad A_G = \frac{3\dim \mathfrak{g}}{\pi^4}.$$

Here $\mathcal{O} = \frac{1}{4}\sum_{a,\mu,\nu} : F^a_{\mu\nu}F^a_{\mu\nu} :$ belongs to the free Maxwell table F with $\dim \mathfrak{g}$ species and propagator $1/(4\pi^2|x|^2)$.

Proposition VII.2.1 (centered trace). Define $\Theta = \sum_\mu T_{\mu\mu} - \langle \sum_\mu T_{\mu\mu}\rangle_{\xi_*}\mathbf{1} \in F_4$. It has zero expectation, is real, $O(4)$-invariant and fixed by the time-reflection adjoint; these conclusions also hold when centering in any phase table respecting S. Its reconstructed label $\iota(\Theta)$ is fixed by conjugation J and every Lorentz transformation. For every Schwartz test h, its vacuum expectation vanishes and, on the common invariant field domain,

$$\Phi_\Theta(h) = \Phi_{\sum_\mu T_{\mu\mu}}(h) - \Big\langle \sum_\mu T_{\mu\mu}\Big\rangle_{\xi_*}\Big(\int h\Big)\mathrm{id}.$$

If $r^8\log^2(1/r)G_\Theta(r) \to A > 0$, then $\iota(\Theta) \neq 0$. *Status:* theorem. *Lean:* `pinned_scalar`, `center_field`, `nonzero_strength`.

Proposition VII.2.2 (spectral representation). There is a unique positive Lorentz-invariant measure μ_Θ supported in $\bar{V}_+$ such that:

- Some $C_\mu \geq 0$, $k \in \mathbb{N}_0$ satisfy $\mu_\Theta\{p_0 \leq R\} \leq C_\mu R^k$ for every $R \geq 1$.
- For every $s > 0$, $e^{-sp_0}\mu_\Theta$ is finite and $G_\Theta(s) = \int e^{-sp_0}d\mu_\Theta$.
- For every Schwartz φ, the translation spectral measure of $\Phi_\Theta(\varphi)\Omega$ is $|\hat{\varphi}|^2\mu_\Theta$; for every Schwartz ϕ on $(\mathbb{R}^4)^2$, $W_2(\Theta,\Theta)(\phi) = \int \hat{\phi}^{\mathrm{wave}}d\mu_\Theta$.

Thus G_Θ is real, nonnegative and nonincreasing on $(0,\infty)$. *Status:* theorem. *Lean:* `traceSpectrum_exists`.

Proposition VII.2.3 (one-loop certificates). Let C issue from run data satisfying the RG construction hypotheses in §2, including a chosen bounded block-spin run. For every filter $\mathcal{L}$ on $\mathbb{N}^2$ along which both spacing and volume indices $N, p \to \infty$, there are level-5 lattice work items $t_{k,N,p}$ satisfying the lattice certificate for all large k, $\mathcal{L}$-eventually in (N,p). In particular, their Ward darts satisfy, for every color q,

$$|\mathrm{midPay}_q(J) + 12\kappa_G| \leq \hat{\beta}_k^{-100}.$$

Status: theorem. *Lean:* `agenda`, certificate `WorthB`.

Proposition VII.2.4 (field anomaly; gallery property). Assume II.1 Wilson consistency, the quantitative RG and VII.6 scaling hypotheses in §2, auxiliary depths $n_k \geq 2$ eventually, and degree bounds z_d covering every generator budget of degree at most d and satisfying the degree-d scaling estimates. Let $\mathcal{L}$ be an ultrafilter finer than the fuse filter of f, with work items satisfying VII.2.3's certificate.

For every limit table S along f with ordered Wilson convergence, Euclidean hierarchy, point limits, cell hypothesis and corrected rank estimates, and every field algebra on S satisfying derivative,

rotation, conservation and Ward laws, there exist coupling and label-transfer limits Z along $\mathcal{L}$ and a free Maxwell hierarchy a, with $Z.\beta(k) = \lim_{\mathcal{L}} \beta_{N,p}(k)$ and $\hat{\beta}_k/2 \le Z.\beta(k) \le 2\hat{\beta}_k$ for every level k.

(i) For every scaling sequence (r_i, U_i) approaching a with VII.6 RG lifts and pointwise lattice limits, base levels $k_i \to \infty$ and selected readout levels ℓ_i, put $w_i = Z.\beta(\ell_i)$. Here $r_i \downarrow 0$; the complex-linear identifications $U_i : \mathsf{F}_d \to F_d$ eventually preserve the unit, reality and signed coordinate actions degreewise, with rescaled correlations converging to a. Then

$$w_i \to \infty, \qquad \frac{\log(1/r_i)}{w_i} \to \kappa_G^{-1}, \qquad r_i^4 U_i^{-1} \mathring{T}_{\mu\nu} \to T_{\mu\nu}^{\mathrm{free}}.$$

For every phase hierarchy T' respecting S, real sequences e_i, d_i satisfy, eventually,

$$r_i^4 U_i^{-1} \Theta_{T'} = e_i \mathbf{1} + 4 d_i \mathcal{O}, \qquad w_i e_i \to 0, \quad w_i d_i \to d_2.$$

Thus $w_i r_i^4 U_i^{-1} \Theta_{T'} \to -\kappa_G \mathcal{O}$.

(ii) Every sequence $s_j > 0$ tending to zero has a subsequence that is the scale sequence of a scaling sequence satisfying (i). *Status:* theorem; hypotheses discharged in III.2 for the constructed family, recorded in VIII.1. *Lean:* `gang`, `Store_l`.

Proposition VII.2.5 (short-distance and spectral laws). If the gallery property holds for (C, f), then

$$\lim_{r \downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = A_G > 0$$

in $\mathbb{C}$. For every continuous compactly supported or Schwartz φ, as $\rho \to \infty$,

$$\rho^{-8} \log^2 \rho \int \varphi(p/\rho) d\mu_\Theta(p) \to \frac{A_G}{1536\pi} \int_{V_+} (p^2)^2 \varphi(p) d^4p.$$

The limiting measure has Laplace transform $A_G s^{-8}$. For every Schwartz ϕ on $(\mathbb{R}^4)^2$, setting $\phi_\rho(x, y) = \rho^8 \phi(\rho x, \rho y)$, the same cone integral with $\varphi = \hat{\phi}^{\mathrm{wave}}$ is the limit of $\rho^{-8} \log^2 \rho \, W_2(\Theta, \Theta)(\phi_\rho)$. *Status:* theorem under the gallery property. *Lean:* `outside`, `joint_limit`.

Proposition VII.2.6 (uniqueness of stress). Let $\mathcal{K}(S)$ consist of real, symmetric, S-centered degree-4 tensors covariant under signed coordinate permutations; let $\mathcal{V}(S)$ consist of real degree-5 vectors with that covariance. Fix a derivative plan satisfying derivative laws and the orientation condition. The free Maxwell divergence $\mathcal{K}(\mathsf{F}) \to \mathcal{V}(\mathsf{F})$ is onto with one-dimensional kernel. In VII.2.4(i)'s setting, $\dim \mathcal{K}(S) = 3$, $\dim \mathcal{V}(S) = 2$, and $\dim \ker \mathrm{div} = 1$. *Status:* theorem. *Lean:* `free_rank`, `ranking`.

Proposition VII.2.7 (no finite Wick profile). Suppose $r^8 \log^2(1/r) G_\Theta(r) \to A > 0$. For every finite $W(r) = \sum_\alpha c_\alpha \prod_\beta K_{m_{\alpha\beta}, q_{\alpha\beta}, h_{\alpha\beta}}(r)$, with complex coefficients and

$$K_{m,q,h}(r) = \int_{\mathbb{R}^3} \frac{E_m(p)^q h(p)}{2E_m(p)} e^{-rE_m(p)} dp, \qquad E_m(p) = \sqrt{m^2 + |p|^2},$$

where $m \ge 0$, $q \in \mathbb{N}_0$ and continuous $h : \mathbb{R}^3 \to \mathbb{R}$ is positively homogeneous of nonnegative integer degree, every $\delta > 0$ admits $r \in (0, \delta)$ with $G_\Theta(r) \ne W(r)$. No such collection reproduces the Wightman function by its mass-shell expression on all Schwartz product tests $\varphi \otimes \psi$ with compactly

supported wave transform. This also excludes Wick contractions of two finite polynomial fields from any free-mode palette. *Status:* theorem. *Lean:* `polynomial_excludes`.

Proposition VII.2.8 (free-photon Fock exclusion). Under VII.2.7's positive law, for every $m \in \mathbb{N}_0$, let $\mathcal{F}_m$ be the Fock space of m free-photon species. For every complex-linear isometry $U : \mathcal{H} \to \mathcal{F}_m$ preserving the vacuum, taking values in physical transverse waves and intertwining translations with $e^{ib\cdot p_{\mathrm{tot}}}$, there exist a compactly supported smooth scalar test φ and compactly supported smooth real three-vector tests h_c, $1 \le c \le m$, each causally separated from φ, such that

$$\langle W(-h)\Omega_{\mathcal{F}}, U\Phi_\Theta(\varphi)\Omega\rangle \neq \langle U\Phi_\Theta(\bar\varphi)\Omega, W(h)\Omega_{\mathcal{F}}\rangle.$$

Equality for all such tests would follow from relative locality with the free Weyl operators. The conclusion persists after applying any bounded self-adjoint operator fixing $U\mathcal{H}$ to both coherent vectors, or averaging over any probability family of inner-product-preserving maps on physical waves fixing $U\mathcal{H}$. In the averaged version, for every $r > 0$, tests can be supported in open Q, W with $\bar Q \subset B(0, r)$ and $\bar Q - \bar W$ spacelike, with φ supported in Q and every component of h in W. *Status:* theorem. *Lean:* `excludes`, `regions_excludes`.

Proposition VII.2.9 (position-space coefficient). For every simple G, kernel parameter $b > 0$ and extraction order p, there exist a consistent Wilson family C with parameter b and refinement f of p, admitting final records. For every such final record, there exist a free hierarchy, coupling limits Z and an ultrafilter admitting degree-4 position-space frames. For every admissible pair $P(s), P'(s) : \mathsf{F}_4 \to F_4$ with window cutoff χ, real κ, d, e satisfy, for sufficiently small $s > 0$,

$$P'(s)^{-1}(s^4 T_{\mu\nu}) = \kappa(s) T^{\mathrm{free}}_{\mu\nu} + d(s)\delta_{\mu\nu}\mathcal{O} + e(s)\delta_{\mu\nu}\mathbf{1}, \qquad \kappa(s) \to 1.$$

Admissibility requires position-space Gram normalization, branch/angular alignment and convergence of prescribed finite off-diagonal jets for P; P' positively rescales its centered scalar sector and complement, fixes the unit, and normalizes two shell tests to free values. For every eventually positive g with $\log(1/s)g(s)^2 \to \kappa_G^{-1}$, $d(s)/g(s)^2 \to d_2$. The test-defined coupling satisfies $g_X/g_W \to 1$ and $d - d_2 g_X^2 = O(g_X^4)$, where g_W is the Wilson coupling from Z. Eventually $s^4\Theta = 4d(s)\mathrm{center}_{\xi_*}(P'(s)\mathcal{O})$ exactly. *Status:* theorem for the supplementary construction's family and refinement. *Lean:* `meterQ`, `FieldComparison`.

3.27.2 2. What it uses

- I.1: positive adjoint Casimir; II.2, III.9: schedule $\hat\beta_k = \bar\beta + \kappa_G \log(1/a_k)$.
- III.4–III.15: bounded block-spin run, eventual-level estimates, stationary criticality, fourth-order classical matching, determinant budgets, cluster/polymer, support and small/large-field bounds; III.7: logarithmic one-loop momentum integral. III.2 supplies these run hypotheses.
- IV.1: extraction, point limits, cell hypothesis and chessboard tests; V.1–V.3: phases and reconstruction; VI.1: field, rotation, derivative, orientation and Ward laws.
- VI.2, VII.1, VII.3: free Maxwell calculus; VII.6: scaling sequences, RG lifts and pointwise limits along subsequences of every positive null sequence.
- Spectral theorem, Källén–Lehmann representation, holomorphic identity theorem, Laplace continuity/Tauberian results, coherent states and point-supported distributions (Reed–Simon II; Glimm–Jaffe, ch. 19; Feller II, §XIII.1–5; Widder).

3.27.3 3. Strategy

The classical stress tensor is traceless; the anomaly comes from the running coupling. In Balaban's RG framework, transport a lattice Ward identity through block-spin steps and extract its coefficient from the background-field one-loop logarithmic pole. The canonical tensor has a divergence diagnostic tending to $-\kappa_G$. Conservation forces the pure-trace improvement coefficient to d_2; rank-one uniqueness and Ward normalization identify the conserved candidate with T.

Free-field convergence then gives the two-point law. The factor $(\log(1/r_i)/w_i)^2 \to \kappa_G^{-2}$ cancels the squared anomaly coefficient, leaving A_G. Monotonicity removes lattice rounding; subsequences give the full limit. Laplace convergence gives the spectral law. Finite Wick expansions and relatively local photon Fock realizations each force incompatible short-distance growth.

3.27.4 4. Steps

1. A sound Ward dart represents the action variation along a displacement field through fourth order. Under the criticality, branch, bound, positive price-gap and classical-matching hypotheses, block-spin transport preserves soundness; repeated transport contracts precision errors by $1/2$ per step. Soundness.

2. Under uniform covariance, resolvent and action bounds, the one-loop action regulated at infrared mass $\mu > 0$ has logarithmic coefficient $-12\kappa_G$. A mean-value argument in $\log \mu$, followed by regulated Ward comparison, gives the payoff limit. `awake`.

3. Under VII.2.4's hypotheses, lattice-to-continuum transfer preserves the divergence diagnostic. Solving the two-dimensional improvement equation forces its scalar coefficient to d_2. `quill`.

4. Rank-one conservation identifies the candidate up to a multiple; comparison of Ward fluxes makes that multiple tend to one. `year_spec`.

5. Free Wick normalization gives A_G along scaling sequences; monotonicity removes lattice rounding and the subsequence principle gives the full limit. `outside`.

6. Lorentz invariance converts the law to cone Laplace convergence; uniform energy bounds extend compact-test convergence to Schwartz tests. `joint_limit`.

7. Under the positive trace law, finite Wick power–log expansions cannot produce an inverse logarithm squared. Relative locality in photon Fock space forces point-supported Cauchy data and then momentum-doubling eigencomponents, yielding energy growth at most R^7, incompatible with the spectral law. `no_powerlog`, `saturate`.

8. For the supplementary position-space frames, shell normalization gives $d = d_2 g_X^2 \kappa$ and $\kappa - 1 = -dQ/B_0$, where $B_0 \neq 0$ and Q has a finite limit. Hence $\kappa - 1 = O(g_X^2)$ and $d - d_2 g_X^2 = O(g_X^4)$. `meterQ`.

3.27.5 5. Where the difficulty lies

The Ward transport, exact one-loop coefficient and remainders $o(\hat{\beta}^{-p})$ for every $p \in \mathbb{N}_0$ carry the anomaly proof. Slower coefficient errors must also defeat growing improvement coefficients. The amplitude uses equality of the independently entering schedule and one-loop slopes. The Wick exclusion concerns finite profiles; continuous-mass generalized free fields are outside its scope. Fock exclusion additionally needs point-supported Cauchy data to force dilation identities.

3.27.6 6. Notation

F_d: degree-d labels; $\mathring{T}$: S-centered stress; $G_\Theta(r) = S^{\xi_*}(\Theta(-re_0/2)\Theta(re_0/2))$. $p^2 = p_0^2 - |\vec{p}|^2$; V_+ is the open forward cone; $\hat{\phi}^{\mathrm{wave}}(p) = \int e^{ip\cdot(y-x)}\phi(x,y)d^4x\,d^4y$. T^{free} is the traceless Maxwell stress tensor; g_W, g_X are the Wilson and position-space couplings. midPay$_q$ is the real coefficient selector on the Haar divergence of the origin-rooted displacement field (definition).

For $h = (h_c^a)$, the coherent vector $W(h)\Omega_{\mathcal{F}}$ has n-particle component $e^{-\|\eta\|^2/2}\mathrm{i}^n\eta^{\otimes n}/\sqrt{n!}$, where

$$\eta_c^a(k) = |k|\sum_b\left(\delta_{ab} - \frac{k_a k_b}{|k|^2}\right)\hat{h}_c^b(|k|,k), \qquad d\nu(k) = \frac{d^3k}{2|k|}.$$

The exact run predicates for VII.2.3 are `Bolt`, `Bandage`, `Nest`, `Vigil`, `Acclaim`, `supportQ`; for VII.2.4 they are `PostT`, `Stock.Tabs`, `Scrim`, `Sceptre`, `supportQ`, `Moonlit`, `auraQ`, together with `Binding`, `HaarBudget`, `ChartRatio`, generator bounds and `FiresSky`.

3.27.7 7. What remains unexplained

Full proofs of RG contraction, covariance/resolvent bounds and fast lattice-to-continuum remainders remain inputs. So do the single massive-mode expansion, point-support and analytic uniqueness theorems for Fock exclusion, and existence and shell-observable formulas for position-space frames.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.28 VII.3. The theory is not free: Gram positivity and exclusion of free-field profiles

Part: VII, identification as Yang–Mills. **Lean:** `YM/Extra/FREE_DETECTOR_ALGEBRA`, `YM/Pert/Position`, `YM/Pert/Rigid`, `YM/Pert/Lattice`, and a few declarations of `YM/Push/LOOP_COEFFICIENT`. Proposition VII.3.7 is in `YM/TraceExtension/Wick` and `TraceBridge.lean`. **Size:** about 3,800 declaration lines on the main dependency path in the listed folders, excluding the trace-exclusion files.

3.28.1 1. What this unit proves

All points lie in $\mathbb{R}^4$. Let $\mathfrak{g}$ have its invariant inner product; $\mathfrak{g} = 0$ is allowed in VII.3.1–5. Free moments are Wick-ordered Maxwell correlations with covariance built from $\Gamma(z) = (4\pi^2|z|^2)^{-1}$. Conjugation bars conjugate polynomial coefficients, and orthogonal transformations act on every tensor index. For $x \neq y$, set $\langle P, Q\rangle_{x,y} = \langle P(x)(\mathcal{R}_{x-y}\bar{Q})(y)\rangle_0$.

Proposition VII.3.1 (strict positivity). For every $x \neq y$, this pairing on $\mathcal{W}$ is Hermitian, with real self-pairings positive exactly for $P \neq 0$. For every finite complex-linearly independent family (V_i), $(\langle V_j, V_i\rangle_{x,y})_{ij}$ is positive definite, without degree restriction. *Status:* theorem; only strict positivity is used downstream. *Lean:* `reflected_positive`, `reflected_symm`, `reflected_pd`.

Proposition VII.3.2 (tower Gram form). For $n \geq 0$, $z = x - y \neq 0$, $B_z(p,q) = \langle \ell_p, \ell_q\rangle_{x,y}$ is symmetric positive definite on Tow_n. A block-unitriangular automorphism σ_z satisfies, for all p, q,

$$B_z(\sigma_z p, \sigma_z q) = \sum_{j<n}\frac{(j+1)(j+2)2^j}{\pi^2|z|^{2j+4}}\langle p_j, q_j\rangle.$$

For any $b_1, \dots, b_N \in \mathrm{Tow}_n$ with $B_z(b_i, b_k) = d_i \delta_{ik}$, $d_i \in \mathbb{R}$, let $\Phi_b : \mathbb{C}[Y_1, \dots, Y_N] \to \mathcal{W}$ be the algebra homomorphism with $\Phi_b(Y_i) = \ell_{b_i}$. For all complex polynomials p, q,

$$\langle \Phi_b p, \Phi_b q \rangle_{x,y} = \sum_{\alpha} p_\alpha \overline{q_\alpha} \prod_i \alpha_i! d_i^{\alpha_i}.$$

Status: theorem. *Lean:* `shiftAll_gram`, `towerB_pos`, `duplex_script`.

Proposition VII.3.3 (Wick recursion). For every $n \geq 1$, $x \in \mathrm{Conf}_n$, $W \in \mathcal{W}^n$, insertion i and slot s,

$$\langle \cdots (W_i X_s)(x_i) \cdots \rangle_0 = \sum_{k \neq i} \langle \cdots W_i(x_i) \cdots (\partial_s^{x_i, x_k} W_k)(x_k) \cdots \rangle_0,$$

where $\partial_s^{y,y'}$ is the complex-linear derivation with $\partial_s^{y,y'} X_t = C_{st}(y - y')$. For every $n \geq 0$ and $x \in \mathrm{Conf}_n$, this recursion and $T(1, \dots, 1) = 1$ uniquely determine the complex-multilinear free moment T. Moments are $O(4)$-covariant; conjugating all polynomials conjugates their moment. *Status:* theorem. *Lean:* `Pair.remove`, `byAnnihilators`.

Proposition VII.3.4 (faithfulness). For every $d \geq 0$, $P \in \mathrm{sh}_d$ vanishes if all its correlations with arbitrary finite families $Q_i \in \mathrm{sh}_{d_i}$, $d_i \geq 0$, at distinct points vanish. Thus null labels are exactly $\ker \omega_d$, and ω_d induces $E_d \cong \mathrm{sh}_d$, with injection $\varpi_d : E_d \to \mathcal{W}$. For every basis (b_i) of E_d and $x \neq y$, $v \mapsto (\langle \varpi_d v(x) \varpi_d b_i(y) \rangle_0)_i$ is injective. For any Wilson family, any table on its register, and any pair with $x_\mu \neq y_\mu$ in every coordinate, these readings give a column system: a conjugation-fixed complex basis with unit first, from frame labels with recorded weight and length. *Status:* theorem. *Lean:* `no_residual`, `exactField`, `scanner_inj`, `vane`.

Proposition VII.3.5 (covariance). For every gauge group above, free-table correlations respect signed coordinate permutations and conjugation. For every d, $O(4)$ acts complex-linearly on E_d, with $\varpi_d(Rv) = R\varpi_d(v)$. *Status:* theorem. *Lean:* `table_spin`, `rideO`.

Proposition VII.3.6 (rigidity and position equations). A buffer $B \in \mathrm{Buf}_d$ is complex-linear on E_d, kills 1, preserves every lower-degree subspace, and commutes with conjugation and signed permutations. Buffers form a finite-dimensional real space. A buffer is stationary when inserting B successively in every slot gives total free correlation zero for every field tuple and distinct-point configuration.

(a) If $\mathfrak{g}$ is simple nonabelian, every stationary degree-four buffer vanishes. Indeed $B = 0$ follows from $\langle Bv \rangle_0 = 0$ and $\langle (Bv)(e) w(0) \rangle_0 + \langle v(e)(Bw)(0) \rangle_0 = 0$ for all $v, w \in E_4$, where e is axis 0's unit vector. Also $\mathrm{Buf}_d = 0$ for $d < 4$.

(b) For any gauge group, if stationary degree-d buffers vanish, finitely many sections (r_i, p_i, f_i), $r_i \geq 0$, $p_i \in E_d^{r_i}$, $f_i \in C_c^\infty(\mathrm{Conf}_{r_i}; \mathbb{C})$, and real-linear $g : \mathbb{R}^k \to \mathbb{R}^{\dim_{\mathbb{R}} \mathrm{Buf}_d}$ give the position chart $\mathrm{chart}_Q(w, B) = g(\mathrm{Re}\, w_i((1 + B)p_i))_i$, where w_i are complex-multilinear forms. At the free tested data w_Q and $B = 0$, its buffer derivative is the isomorphism

$$B \longmapsto g\Big(\mathrm{Re} \int f_i(x) \sum_j \langle p_{i1}(x_1) \cdots (Bp_{ij})(x_j) \cdots \rangle_0 \, dx \Big)_i.$$

Such systems are called clean and square.

(c) Assume simple nonabelian $\mathfrak{g}$, a final Wilson record with table S, and a scaling sequence $X = (r_k, U_{k,d}, \xi_k)$: $r_k \downarrow 0$ strictly, $\xi_k > 0$, and complex-linear isomorphisms $U_{k,d} : E_d \to F_d(S)$. Assume VI.2's leading comparison and eventual frame compatibility with unit, conjugation, signed permutations and inclusions. Fix a clean square degree-four system Q and φ satisfying: a local real-analytic solution of $\mathrm{chart}_Q(w, \varphi(w)) = \mathrm{chart}_Q(w_Q, 0)$, with $\varphi(w_Q) = 0$ and local uniqueness; $\pi_Q^{(0)}(L,k) \to w_Q$ for every integer $L \geq 2$; for every $L \geq 2$ and $p, j \geq 0$, eventually all order-j coupling derivatives at zero of the position equations on the order-p lattice truncation and its image under φ equal those of the constant free value; and buffers $A_k \to 0$ with invertible $1 + A_k$, eventually solving the equations and satisfying the displayed comparison below with B_k replaced by A_k, for every clean degree-four R and integer $L \geq 2$.

Then $B_k = \varphi(m_Q(k)) \to 0$; eventually $1 + B_k$ is invertible and $U_{k,4}(1 + B_k)$ solves Q exactly, respects unit, conjugation, signed permutations and degree-≤ 3 inclusions, and

$$m_R^{B_k}(k) - \mathrm{chart}_R(\pi_R^{(0)}(L,k), \varphi(\pi_Q^{(0)}(L,k))) \to 0.$$

Status: (a,b) theorems; (c) conditional theorem, hypotheses discharged in VI.2 and VIII.1. *Lean:* `Rigid.frozen_four`, `square_of_frozen`, `residency`.

Proposition VII.3.7 (finite Wick profiles). A mode has mass $\mathsf{m} \geq 0$, integers $j, q \geq 0$, continuous $h : \mathbb{R}^3 \to \mathbb{R}$ with $h(tp) = t^j h(p)$ for every $t > 0$ and p. Its kernel, for $r > 0$, is

$$K_{\mathsf{m},q,h}(r) = \int_{\mathbb{R}^3} \frac{E^q h(p) e^{-rE}}{2E}\, dp, \qquad E = \sqrt{\mathsf{m}^2 + |p|^2}.$$

A Wick profile is a finite complex linear combination of finite products of these kernels, including empty products. If $r^8 \log^2(1/r) G_\Theta(r) \to A > 0$ as $r \to 0^+$, every profile W differs from G_Θ somewhere in every $(0, \delta)$, $\delta > 0$. In the selected reconstructed theory, every corresponding finite sum of mass-shell contractions differs from the Wightman two-point function on some Schwartz product $\bar{f} \otimes g$ with compactly supported $\widehat{\bar{f}}\hat{g}$, using positive-sign Fourier transforms. Contractions test this product at sums of positive-energy on-shell momenta, with densities $E^q h/(2E)$. *Status:* theorem, assigned to VII.2 and VIII.1; VII.2 supplies $A = 3 \dim \mathfrak{g}/\pi^4$. *Lean:* `Wick.euclidean_excludes`, `Wick.polynomial_excludes`.

3.28.2 2. What it uses

- III.16: Maxwell kernels, jets and response identity; IV.1: tables, quotient fields and columns.
- VI.1, VI.2: scaling sequences, leading Maxwell comparison and degree-four basis; VII.1: order-zero lattice convergence; VIII.1: implicit-position conditions.
- VII.2: trace asymptotic and mode expansions; V.3: positive-energy spectral representation; I.1: Wilson lattice action.
- Wick's theorem, Hobson's formula, Maxwell inversion covariance, finite-dimensional duality and the analytic implicit function theorem.

3.28.3 3. Strategy

Kelvin inversion and translation diagonalize the reflected jet covariance. Wick recursion extends positivity to polynomials, detecting null labels. Degree-four rigidity then supplies finite position equations. Independently, finite Wick profiles have power-log expansions incompatible with the trace's inverse-logarithmic leading term.

3.28.4 4. Steps

1. Hobson's formula identifies multipole fields with Kelvin transforms. Translating at the inverted separation cancels lower Taylor coefficients and gives positive Gram pivots. `reply_kelvin`, `shiftAll_gram`.
2. Wick recursion gives factorial monomial norms; every polynomial lies in a finite tower. Reflected self-tests then prove faithfulness and injectivity. `duplex_script`, `no_residual`.
3. In the 22-dimensional degree-four basis, stationarity forces weighted antisymmetry; signed permutations force symmetry or zero. Finite test functionals detect all buffers, giving invertible linearization. `coeff_barren`, `square_of_frozen`.
4. Local implicit uniqueness identifies B_k with A_k eventually, transferring exact position equations and leading comparison. `residency`.
5. At finite regulator, normalized Gaussian perturbative expectations define lattice profiles; separate imaginary Gaussian averages within insertions implement Wick subtraction. Their tested Riemann sums give the lattice readings. `Full.profile`, `Full.column`.
6. Mode expansions have integer powers and nonnegative logarithmic powers. Dominance along $r = 2^{-n}$ excludes the trace law; spectral damping transfers tested Wightman equality to Euclidean equality. `Wick.no_powerlog`, `Wick.polynomial_excludes`.

3.28.5 5. Where the difficulty lies

Simultaneous diagonalization depends on Taylor normalization and ordered-index contractions. The position comparison is leading-order only. Non-freeness excludes finite profiles, leaving continuous-mass generalized free fields and infinite Wick expansions outside its scope.

3.28.6 6. Notation

Conf_n: distinct-point configurations; Sol_j: real $\mathfrak{g}$-valued homogeneous degree-j 2-forms satisfying $dS = \delta S = 0$; $\mathrm{Tow}_n = \bigoplus_{j<n} \mathrm{Sol}_j$; $\mathcal{W} = \mathbb{C}[X_s]$: polynomials in basis coordinates $s = (j, a)$ of on-shell Taylor coefficients. $\ell_p(A) = \sum_j \langle p_j, A_j \rangle$, using Taylor coefficients $(j!)^{-1} D^j$ and summing over all ordered form and derivative indices, without a factor $1/2$; $\mathcal{R}_z$: reflection in $z^\perp$; C_{st}: jet covariance. ω_d: on-shell evaluation of adjoint-invariant radial-potential labels of weight $\leq d$, with order-j slots weighted $j + 2$; $\mathrm{sh}_d = \mathrm{im}\omega_d$; E_d: free quotient. m_Q, w_Q, $\pi_Q^{(0)}$: tested rescaled correlations in $U_{k,4}$, free readings, and order-zero lattice readings at spacing L^{-k}, respectively; m_R^B: measured readings in frames $U_{k,4}(1 + B)$; chart_Q: tested readings after $1 + B$. $G_\Theta(r)$: centered stress-tensor trace two-point function in the selected phase at $\mp(r/2)e_{\mathrm{t}}$.

3.28.7 7. What remains unexplained

An intrinsic characterization of sh_d needs surjectivity. Interacting rank estimates belong to III.11 and III.12; continuum Wick subtraction to VII.1. Further Maxwell-Fock embedding exclusions are

outside VII.3.7.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.29 VII.4. Operator product expansion and contact extensions

Part: VII, identification as Yang–Mills. **Lean:** `YM/OPE`, `YM/Contacts`. **Size:** about 10,200 lines, 2,035 declarations.

3.29.1 1. What this unit proves

For VII.4.1–4 assume (RG): grid approximants, simple gauge Lie algebra, per-step and cluster RG estimates, and the continuum extraction record. Fix any coherent prescription and its real covariant Reader on the final record; VIII.1 supplies these data. Its strata $E_j \subset F_j$ have bases $O_{j,\kappa}$ fixed by conjugation, with coefficients $c_{j,\kappa}$.

An (r, M, η)-cluster satisfies $|x_i - o| \le Mr$, $|x_i - x_{i'}| \ge \eta r$ for $i \ne i'$. Spectators are c-clear if $|y_j - o| \ge c$, $|y_j - y_{j'}| \ge c$ for $j \ne j'$. Small-radius thresholds are uniform in configurations and states.

Proposition VII.4.1 (separated OPE). Fix finite label lists a, b, $M \ge 0$, $\eta, c, \varepsilon > 0$, and $D \in \mathbb{N}$. For all sufficiently small r, every such cluster and clear spectator configuration gives separated (x, y) and (o, y) and

$$\|\mathcal{E}_D\|_{\mathcal{A}} \le r^{D+1-w(a)-\varepsilon}, \qquad \mathcal{E}_D = M(a, x; b, y) - \sum_{j \le D, \kappa} c_{j,\kappa}(a; x, o) M((j, O_{j,\kappa}), o; b, y).$$

The same bound holds in every phase and the mixed state. For each fixed channel (j, κ), $|c_{j,\kappa}| \le r^{j-w(a)-\varepsilon}$ uniformly on clusters for sufficiently small r. Coefficients are independent of spectators, state and D. *Status:* theorem. *Lean:* `Orbits.Foundation`.

Proposition VII.4.2 (integrated remainder). With VII.4.1's fixed data, let K be a measurable space, μ a positive measure, $q \in L^1(\mu; \mathbb{C})$, $\int |q|\, d\mu \le B$, $B \ge 0$. Let configurations be a.e.-measurable and satisfy the geometry a.e. on $\{q \ne 0\}$. For all sufficiently small r, for the mixed state or any a.e.-measurable phase map, the weighted correlation, truncation and remainder are integrable, and $|\int q\, s(\mathcal{E}_D)\, d\mu| \le Br^{D+1-w(a)-\varepsilon}$. The pointwise threshold works for every K, B; every probability mixture of phases satisfies the remainder bound of VII.4.1. *Status:* theorem. *Lean:* `estimates_weighted`.

Proposition VII.4.3 (nested OPE). Fix a finite rooted tree with inner root, leaf labels v, output (m, κ), inner truncations D_p, $M \ge 0$, $\eta, \varepsilon > 0$. A plot assigns leaf points and inner anchors z_p; positive radii satisfy $s_{\mathrm{root}} = r$, $s_p \le s_{\mathrm{parent}(p)}$. Every subtree lies within Ms_p of z_p; positions in different children are separated by ηs_p. Put $\rho_p = s_p / s_{\mathrm{parent}(p)}$, and let w_p be the degree below p. Over non-root inner nodes define

$$\mathrm{toll}_m = r^{m-w(v)-\varepsilon} \prod_p \rho_p^{-w_p-\varepsilon}, \qquad \mathrm{nest}_D = \sum_p \rho_p^{D_p+1}.$$

Writing $c_{m,\kappa} = c_{m,\kappa}(v; z_{\mathrm{leaves}}, z_{\mathrm{root}})$, for all sufficiently small r, $|c_{m,\kappa}| \le \mathrm{toll}_m$ and $|c_{m,\kappa} - c^{(D)}_{m,\kappa}| \le \mathrm{toll}_m \mathrm{nest}_D$. Here $c^{(D)}$ recursively multiplies node coefficients and sums inner-child channels through D_p, leaving leaves unchanged; the root output is fixed and D_{root} is unused. For fixed spectators b

and $\delta > 0$, separated mutually and from every plot point by δ, the degree-m remainder has norm, and absolute value in every state, at most toll_{m+1}.

For a.e.-measurable plots and spectators satisfying this geometry a.e. on $\{q \neq 0\}$, VII.4.2's integration conclusion holds: direct and composed coefficients are integrable after weighting, with $|\int qc\,d\mu| \leq B\mathrm{toll}_m$ and $|\int q(c - c^{(D)})\,d\mu| \leq B\mathrm{toll}_m\mathrm{nest}_D$. Weighted tested correlations, truncations and remainders are integrable, with remainder bound $B\mathrm{toll}_{m+1}$ for the mixed state or any a.e.-measurable phase map. Radii remain fixed during integration; thresholds are uniform in radii, measures, weights, B and configuration maps. *Status:* theorem. *Lean:* `Nesting.Uniform`.

Proposition VII.4.4 (contacts). Fix contact devices sharing the Reader's prescription and the final record's rotations. For k labels $a_i \in F_{d_i}$, any state s, and $0 < s_0 \leq B_0/2$, $B_0 = \min(1, \mathrm{cl}(w(a)))$, the forest kernel defines a distribution $\mathcal{D}^s_a$ of order at most $(2 + w(a))k$: on each compact K, its absolute integral is bounded by $C_K\|f\|_{C^{(2+w(a))k}}$. It equals $\int fS^s(a;x)\,dx$ for tests supported on distinct configurations. Distributions also exist on slices $o + P(E)$, with P a linear isometry whose range contains a group diagonal's normal space, and o on that diagonal.

Shell and iterated forest sums give the pairing, with integrable kernels and summable L^1 norms, under these stage-family hypotheses: a partition Q of g; affine isometric placements based on block diagonals and containing their normals; positive subtraction lengths s; $R_i \in (0, 1]$; common ceiling $2s \leq B \leq \min(1, \mathrm{cl}(\sum_i d_i))$. A common frame scales each block normal by R_i and preserves its orthogonal complement. Closed stages are bounded in frame coordinates and within normal distance B; nonsingleton blocks satisfy $R_i \leq C_1 s$. Original and spectator labels have uniform norm bounds; spectator positions and labels are continuous on stages and constant in block normal directions. All nonempty subgroup centroids of different blocks and all spectators are uniformly separated. Constants may depend on the fixed family.

For $Q = \{g\}$, $g \neq \emptyset$, pure normal dilation $R \in (0, 1]$, and $f_R(x) = R^{-4(|g|-1)}f(R^{-1}x_{K_g} + x_{K_g^\perp})$, assume rescaled-stage support and $\|f\|_{C^N} \leq H$, $N = (2 + \sum_i d_i)(|g| - 1)$. For every $\varepsilon > 0$, the absolute kernel integral is $\leq CHR^{u-w(a_g)-\varepsilon}$: $u = 0, n, n + 1$ for full, weight-n, and degree-n remainder. The latter two require stages within normal distance $\mathrm{cl}(n)$ of the g-diagonal. For $|g| > 1$, tests away from its diagonal agree with the expression retaining proper-subcluster subtractions.

Device strata have bases fixed by conjugation, containing the unit in degree zero. Every positive sequence f_n admits $0 < t_n \leq f_n$, $4t_{n+1} < t_n$, $t_n \to 0$, $2(n + 1)t_n \leq \mathrm{cl}(n)$. *Status:* theorem. *Lean:* `provided`.

Proposition VII.4.5 (slab bound). Assume construction properties, ordered generic point extraction, Euclidean hierarchy, IV.1's corrected field estimates, and the cell hypothesis: reflection positivity and chessboard inequalities at every cutoff, with centred one-field bounds $\leq Aa_m^{-d-\varepsilon} + \eta$ late in the extraction, as in IV.1. For every $D \in \mathbb{N}$, $R > 0$, there is $C_0 > 0$, with $P = 4(D + 1)$, such that every nonempty word w, degrees $\leq D$, coefficient norms $\leq R$, r-buffered in $(0, d)$, $d > 0$, $0 < r \leq 1$, admits a level b and width $0 < s \leq d$, $s \in a_b\mathbb{N}$, containing w in $(0, s)$. Every positive-time exterior word v then has B_v with

$$|S(\Theta v_n \cdot v_n)| \leq B_v\rho^{4n} \quad (n \in \mathbb{N}), \qquad \rho = C_0^{|w|}(|w|/r)^{P|w|},$$

where v_n precedes v by $2n$ consecutive width-s slabs holding alternately w and its reflection. Iterated Schwarz gives $\|T_s(w)\| \leq \rho$ in V.1. *Status:* theorem. *Lean:* `seats`.

Proposition VII.4.6 (prescription). A probe is a finite separated token configuration, anchored at its first point. The correlation maps sp_n of anchored fields against probes are injective and inclusion-compatible. For modes indexing bases of $F_n/\iota(F_{n-1})$, $F_{-1} = 0$, there exist vectors $u_i \in F_{\deg i}$ and finitely supported complex probe weights λ_i, such that for every n, $\lambda_i^{(n)}(\iota u_j) = \delta_{ij}$ for degrees $\le n$, and $\phi = \sum_{\deg i \le n} \lambda_i^{(n)}(\phi)\iota u_i$ for every $\phi \in F_n$. Here $\lambda_i^{(n)} = \sum_p \lambda_i(p)\mathrm{sp}_n(\cdot)(p)$. Higher-degree tests vanish on F_n; one prescription serves every truncation. *Status:* theorem. *Lean:* `Prescription.chosen`.

3.29.2 2. What it uses

- II.2, III.13: cutoff construction; III.11 and Part III: RG estimates, assembled in VIII.1 (`Services`).
- IV.1: corrected fields, extraction, centred chessboard bounds; V.1–V.3: ground algebra and phases; VI.1: derivatives, rotations, reality.
- Gelfand isometry, reflection positivity, iterated Schwarz, Haar averaging, Taylor subtraction and Zimmermann forests.

3.29.3 3. Strategy

Probes reconstruct finite-degree fields. RG replaces clusters by fields plus controlled remainders; translated chessboard moments give ground-algebra norm bounds, hence estimates in every phase. Exact reconstruction of replacement fields gives the grid OPE. Derivative induction extends it to arbitrary configurations, and bringing auxiliary scale heights together gives arbitrary ε-loss. Recursive replacements yield trees; weight-dependent normal Taylor subtraction makes contact kernels integrable, with partial collisions treated by finer partitions.

3.29.4 4. Steps

1. Extend biorthogonal probes degree by degree; average rotations and conjugations and remove lower degrees. `Reader`.
2. Fuse nearby RG groups, retaining separated groups; obtain a degree-t replacement with remainder gain $(a_m/a_k)^{t+1/16}$. `funding`.
3. Bound translated moments by chessboard roots; full phase support and Gelfand give norm control. `Cards.carrying`.
4. Descend in degree from crude high-degree bounds; grid distance compensates the derivative degree increase. `interpolated`.
5. Telescope replacements over trees, merging children at the parent's rate and expanding others to sufficiently high auxiliary degree. This controls each node's truncation. `Fusion.tariff`.
6. Subtract weight-dependent normal Taylor jets; finer-partition induction handles partial collisions and sums dyadic shells. With $d_h = \sum_{i \in h} d_i$, $q_h = 4(|h|-1)$, and $L_h = \max(0, d_h + 1 - q_h)$, subtraction leaves shell exponent $q_h + L_h - d_h \ge 1$. `glues_total`.
7. Apply two-scale chessboard bounds to reflection-positive slab trains; choose levels with $a_k \ge c(r/|w|)^4$ to obtain $P = 4(D+1)$. Iterated Schwarz removes the exterior cost. `replicas`.

3.29.5 5. Where the difficulty lies

The 1/16 gain margin comes from RG transport. Norm control ensures spectator and state uniformity; partial collisions require more than radial integrability. Estimates allow arbitrary ε-loss, without identifying logarithmic corrections.

3.29.6 6. Notation

$w(a) = \sum_i d_i$; F_d: degree-d fields; $\mathcal{A}$: ground algebra; M: compressed word; S^s: correlation for $s \in X \sqcup \{\mathrm{mix}\}$, with X the phase spectrum; a_k: lattice spacing; $\mathrm{cl}(n)$: device clearance; K_g: normal space of the g-diagonal.

3.29.7 7. What remains unexplained

The internal polymer estimates and their constants, rotated comparisons, high-horizon tree replacement and partial-collision sector induction need fuller accounts. Device projections are not identified with Reader coefficients. Uniqueness, extraction independence, infinite-order convergence, contact positivity, contact Ward identities and perturbative agreement are not asserted.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.30 VII.5. Asymptotic freedom of the running coupling

Part: VII, identification as Yang–Mills. **Lean:** `YM/Pert/Asymptotics`, `YM/Extra/WEAK_SCALING_ACTLAW`, `YM/TraceExtension/Foundation`, `YM/TraceExtension/Soot`. **Size:** 232 lines, 43 declarations.

3.30.1 1. What this unit proves

For VII.5.1–3, let $G \subset U(q)$ be compact, $q \geq 1$. Write $2b_0 = \kappa_G = 11C_{\mathfrak{g}}/(24\pi^2)$, $a_m = (\prod_{k<m} b_k)^{-1}$, $s_m = \log(1/a_m)$, with odd $b_k \geq 3$. The trajectory has inverse couplings $\beta_{N,p,m}$, bare endpoint $\beta_{N,p} = \beta_{N,p,N}$, reference value $\beta_{N,p,0} = \bar{\beta} > 0$, and $g_{N,p,m} = \beta_{N,p,m}^{-1/2}$. Simple means real simple and nonabelian. The running clause says that for every $\varepsilon > 0$, eventually in m,

$$|\beta_{N,p,m} - (\bar{\beta} + 2b_0 s_m)| \leq \varepsilon s_m$$

uniformly for $N \geq m$ and $p \in \mathbb{N}$.

Proposition VII.5.1. If the family satisfies the construction conditions (II.2), then for every $\varepsilon > 0$ there is $m_0 \geq 1$ such that

$$\left|\beta_{N,p,m}/s_m - 2b_0\right| < \varepsilon \qquad (m \geq m_0,\ N \geq m,\ p \in \mathbb{N}).$$

Status: theorem; only the running clause is used, without simplicity. *Lean:* `YM.Pert.Asymptotics.zeroUp_bound`.

Corollary VII.5.2. Under the construction conditions and simplicity of $\mathfrak{g}$, the Wilson family satisfies (Reg1): for every $\varepsilon > 0$ there is $N_0 \geq 1$ such that $|\beta_{N,p}/s_N - b_*| < \varepsilon$ for all $N \geq N_0$ and $p \in \mathbb{N}$, where $b_* = 2b_0 > 0$. *Status:* theorem in VIII.1. *Lean:* `YangMills.regular`.

Proposition VII.5.3. Under the same hypotheses, for any lattice-level and volume sequences $(N_i), (k_i)$ with $i \le N_i$ eventually,

$$g_{N_i,k_i,i}\sqrt{s_i} \longrightarrow (2b_0)^{-1/2}, \qquad g^2_{N_i,k_i,i}s_i \longrightarrow (2b_0)^{-1}.$$

Status: theorem; the second limit follows by squaring. *Lean:* `YM.Pert.Asymptotics.comparison`.

For VII.5.4–5, $\mathcal{F}_d, \mathcal{F}_d^\circ$ are finite-dimensional complex label spaces of degree at most d in Euclidean hierarchies on S and its free Maxwell comparison table, on the same reference labels. Derivative plans have inclusions $\iota_{d\to e}$ and derivatives $\partial_{d,a}$. Exact compatibility means preserving inclusions, signed permutations Γ_4, conjugation and units.

Proposition VII.5.4. Fix the free Maxwell data of VI.2, including the finite separating tests and boundary, polynomial and internal-rotation conditions. Its plan W satisfies the inclusion, derivative and orientation axioms; J° satisfies Ward identities with translation part $\kappa\partial^W$ and rotation part κ times the infinitesimal rotation action, where $\kappa \neq 0$ and $\partial_4^W \neq 0$. Let V satisfy the inclusion and derivative axioms. Suppose $J \in \mathcal{F}_4^{4\times4}$ is symmetric, real, Γ_4-covariant and satisfies $\sum_a \partial^V_{4,a} J_{ab} = 0$ for every b, and (J, p, Y) satisfies the local Ward identities. If some $c \in \mathbb{R}$ satisfies $p_{4,a} = c\partial^V_{4,a}$ for all $a = 1, \dots, 4$, there exist $D, N, M \in \mathbb{N}$, $R_0 \ge 1$ such that $c/\xi_k \to \kappa$ for every tracking scaling sequence at every level $(n, d, j, R) \ge (N, D, M, R_0)$ componentwise (VI.1). Also, for any hierarchy and local Ward identities, $p \neq 0$ implies $J \neq 0$. A scaling sequence has complex-linear isomorphisms $U_k(d) : \mathcal{F}_d^\circ \to \mathcal{F}_d$, positive strictly decreasing $r_k \to 0$ and $\xi_k > 0$. Tracking (V, W, J) at (n, d, j, R) requires exact compatibility through $\max(d, 4)$, convergence of transported correlations at $r_k x$ for each fixed list of at most n fields of degrees at most d, to the free correlations, with all Fréchet derivatives through order j, uniformly on $\{x : \|x\| \le R,\ \|x_a - x_b\|_\infty \ge R^{-1}\ (a \neq b)\}$, and $\xi_k^{-1} r_k^4 U_k(4)^{-1} J_{ab} \to J^\circ_{ab}$.

Status: theorems. *Lean:* `YM.Extra.WeakScaling.tracks_ratio`, `flux_nontrivial`.

Proposition VII.5.5. Let V, W satisfy the inclusion and orientation axioms. For every d there is a complex-linear, Γ_4-equivariant, conjugation-preserving retraction P_d of $\iota_d^W := \iota^W_{d\to d+1}$. For any complex-linear isomorphisms $A_d : \mathcal{F}_d^\circ \to \mathcal{F}_d$, define

$$X_0 = A_0, \qquad X_{d+1} = \iota_d^V X_d P_d + A_{d+1}(1 - \iota_d^W P_d).$$

Fix $D \in \mathbb{N}$. If A_d preserve symmetries and units for $d \le D$, then X_d are exactly compatible through degree D. For sequences of such isomorphisms $A_d^{(k)}$ satisfying $(A_{d+1}^{(k)})^{-1}\iota_d^V A_d^{(k)} v \to \iota_d^W v$ for every $d < D, v$, one has $(A_d^{(k)})^{-1} X_d^{(k)} \to 1$ in operator norm for every $d \le D$. These $X_d^{(k)}$ are eventually bijective; replacing nonbijective terms by $A_d^{(k)}$ gives isomorphisms $\widetilde{X}_d^{(k)}$ with $(A_d^{(k)})^{-1}\widetilde{X}_d^{(k)} v \to v$. If the original frames preserve symmetries and units through D, the replacements are eventually exactly compatible simultaneously through D. *Status:* theorem. *Lean:* `retraction`, `adjusted_exact`, `perturb_limit`, `relative`.

Proposition VII.5.6. Along any filter: (a) for an injective linear map $f : E \to X$ between finite-dimensional real normed spaces, $f(a_y) \to f(e)$ implies $a_y \to e$; (b) continuous real-linear maps $f_y, g : (\mathbb{R}^4)^n \to \mathbb{C}$ satisfy $f_y \to g$ in operator norm if $f_y(e_{i,a}) \to g(e_{i,a})$ at all $4n$ coordinate vectors. *Status:* theorem. *Lean:* `norm_fit`, `norm_fit_deriv`.

Definition VII.5.7. In gauge-invariant radial-jet polynomials, with curvature weight 2 and derivatives adding 1, for $t \in \mathbb{R}^4$ set

$$\mathfrak{s} = \tfrac{1}{4}\sum_{i,j}\langle F_{ij}, F_{ij}\rangle, \quad T^{(t)}_{ab} = \delta_{ab}(t_0\mathbf{1} + t_1\sum_i \langle F_{ai}, F_{ai}\rangle + 4t_2\mathfrak{s}) + (1-\delta_{ab})t_3\sum_i \langle F_{ai}, F_{bi}\rangle.$$

Here $F, \partial F$ are linear curls of degree-0 and degree-1 radial Taylor coefficients. These are polynomial identities before quotienting by null fields. Put $T^{\mathrm{can}} = T^{(0,1,-1/4,1)}$, $\mu_b^1 = \sum_i \langle F_{bi}, \partial_b F_{bi} \rangle$, $\mu_b^2 = \sum_{i,j} \langle F_{ij}, \partial_b F_{ij} \rangle$, $E_b = \sum_{a,i} \langle F_{bi}, \partial_a F_{ai} \rangle$, and, for $w \in \mathbb{R}^3$,

$$\beta(w) = w_0 \mu^1 + w_1 \mu^2 + w_2 E, \quad \delta(w) = \beta(2(w_0 - w_2), 2w_1 + w_2/2, w_2).$$

Both μ^1, μ^2 belong to the real space $\mathcal{R}_1$ of real, Γ_4-covariant vectors of degree at most 5. Every degree-0 radial jet is realised by a constant-curvature free Maxwell field with affine potential. *Status:* definitions and two lemmas. *Lean:* `raw`, `basisV`, `bareVector`, `shell_affine`.

Definition VII.5.8. Fix a family, nondecreasing $K : \mathbb{N} \to \mathbb{R}$ with $K \geq 1$, degree d, stages $\ell_d \leq k \leq N$, volume p, and $u \in \mathbb{R}$. The sharp read-out bound asserts $u\|c_k\|_1 \leq 4(1 + n_k)w$ for every $w \geq 0$ and every chain of insertion presentations A_i at the origin, with coefficient vectors c_i, remainder costs q_i and densities $\mathcal{D}_i$, satisfying the admissibility conditions of III.12 (regular and forced remainders, with square-root forced allowance), $c_{\ell_d} = 0$, and, for every $\ell_d \leq i < k$,

$$\begin{aligned} \mathcal{D}_i &= b_i^d \mathcal{T}^{(i)}_{\beta_{N,p,i+1}} \mathcal{D}_{i+1}, \\ \|c_i - B_i S_i c_{i+1}\|_1 &\leq b_i^{-3/8}(\|P c_{i+1}\|_1 + q_{i+1}), \\ q_i &\leq K_i \|P S_i c_{i+1}\|_1 + b_i^{-3/8} q_{i+1}. \end{aligned}$$

For each $\ell_d \leq i < k$, costs q_i, q_{i+1} are nonnegative and B_i is a complex-linear automorphism commuting with P, with $\|B_i^{\pm 1} c\|_1 \leq K_i \|c\|_1$ for every c. For $\ell_d \leq i \leq k$, the admissible remainder cost is at most q_i, and $q_k \leq w + n_k(w + \|c_k\|_1)$. *Status:* definition. *Lean:* `YM.MergeProof.Accent.SharpRead`.

3.30.2 2. What it uses

- I.1: Casimir positivity and divergent logarithmic scales (`Schedule.divergent`); II.2, III.1–2: construction conditions, selected histories and shooting (`YM.Family.Makes`).
- III.7, III.9: one-loop shift and coupling reset.
- IV.1, VI.1–2: hierarchies, derivative plans, Ward identities, A04–A05 and free Maxwell data (`MaxwellData`); III.12: insertion estimates.
- Standard results: Cesàro summation, continuity of square root and reciprocal, Maschke averaging, finite-dimensional continuity and openness of invertibility.

3.30.3 3. Strategy

The inverse coupling follows a linear reference trajectory in logarithmic scale. Uniform one-step errors sum to $o(s_m)$ around $\bar{\beta} + 2b_0 s_m$. Division by $s_m \to \infty$ removes the reference coupling. The auxiliary arguments compare two limits proportional to the same nonzero derivative, which determines the stress normalisation. Averaged retractions split each inclusion; correcting frames recursively on the included subspace makes compatibility exact, while the relative correction tends to the identity.

3.30.4 4. Steps

1. Given $\beta_{N,p,0} = \bar{\beta}$ and transfer errors bounded by δ_{m+1} with $\delta_{m+1}/\log b_m \leq J(m+1)^{-2} \to 0$, summation gives the uniform running clause. Split off a finite initial segment and bound the remaining error by a small multiple of s_m (III.2). `trajectory`.

2. Divide by s_m; specialise to $m = N$ for (Reg1), and use continuity of inverse square root for VII.5.3; positivity uses simplicity. `zeroUp_bound`, `comparison`.
3. A04–A05 give derivative and Ward limits; a linear functional nonzero on ∂_4^W extracts $c/\xi_k \to \kappa$. `tracks_ratio`.
4. Average retractions over Γ_4 and conjugation; the recursive correction preserves inclusions and tends to the original frames in relative operator norm, ensuring eventual invertibility. `retraction`, `perturb_limit`.
5. At the polynomial level, sign cancellation gives covariance of μ^1, μ^2; constant curvature realises every degree-0 jet on a Maxwell solution. `momentum1_mem`, `shell_affine`.

3.30.5 5. Where the difficulty lies

The analytic work lies upstream in the one-loop estimate, coupling reset and shooting. This unit establishes the leading lattice logarithm; the main theorem retains only a positive slope through (Reg1).

3.30.6 6. Notation

$C_{\mathfrak{g}}$ is the Casimir for $\langle X, Y\rangle = -q^{-1}\mathrm{Re}\,\mathrm{tr}(XY)$. For an orthonormal basis (e_i), $C_{\mathfrak{g}} = -(\dim \mathfrak{g})^{-1}\sum_i \mathrm{tr}_{\mathfrak{g}}((\mathrm{ad}\, e_i)^2)$, with value 0 if $\mathfrak{g} = 0$. In VII.5.8, ℓ_d is the family's reference stage, P removes the unit coordinate, $\|c\|_1 = \sum_\alpha |c_\alpha|$, $(S_i c)_\alpha = b_i^{d-d_\alpha} c_\alpha$, $K_i = K(J(i+1))$; n_k is the column number. Here d_α is label weight and J the scale-stage index. In VII.5.4, $\xi_k > 0$ is stress normalisation.

3.30.7 7. What remains unexplained

The one-loop integral belongs to III.7. A declaration establishing cutoff removal, infinite-volume limits and comparison with a coupling from continuum correlations has not been identified (tentative). Tracking-sequence existence belongs to VI.1 and VII.1; the read-out bound requires upstream one-step hypotheses.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.31 VII.6. Scaling variants and fixed-field estimates

Part: VII, identification as Yang–Mills. **Lean:** `YM/TraceExtension/Prom/`, with outputs in `Trace2`, `FREE_STRESS_CERTIFICATE` and `SharpGram`. **Size:** 4,205 lines in 374 declarations.

3.31.1 1. What this unit proves

Setting. Let G be a compact matrix group with simple Lie algebra, C a Wilson family with its construction record, and F an extraction order with ordered-tail filter Φ_F on cutoff pairs (N, p). Let S be its limit table, with ordered smeared and generic point limits and Euclidean hierarchy H; the free Maxwell reference has hierarchy H_0. Write $\mathcal{F}_d^S = \mathrm{Label}_d/\mathrm{Null}_d(S)$, projection π_d and unit $\mathbf{1}_d$, and similarly $\mathcal{F}_d^{\mathrm{free}}$. Fix the RG-history binding, Haar budget and chart ratio of III.11. Its one-step cost assumption is (S), and $c_d = 2(w_d + 1)$, where w_d is the degree-d label-frame polynomial degree plus 4.

A scaling consists of positive radii r_k strictly decreasing to zero, positive constants ξ_k, and complex-linear isomorphisms $U_k(d) : \mathcal{F}_d^{\mathrm{free}} \to \mathcal{F}_d^S$. It is *approaching* when, for every arity and free fields $v_i \in \mathcal{F}_{d_i}^{\mathrm{free}}$, the smooth kernels

$$K_k(x) = \langle U_k v_1(r_k x_1) \cdots U_k v_n(r_k x_n) \rangle^S - \langle v_1(x_1) \cdots v_n(x_n) \rangle^{\mathrm{free}}$$

satisfy: for every compact $K \subset \mathrm{Conf}_n$ and $M, \varepsilon > 0$, eventually $|\int t K_k| \le \varepsilon$ for every smooth t supported in K with $\|t\|_\infty \le M$.

Proposition VII.6.1 (prescribed null sequences). Assume (S), real $z_d \ge c_i$ for every $i \le d$, the uniform-band property $\mathrm{Fire}_d(z_d)$ for every d, and $\dim \mathcal{F}_d^S \le \dim \mathcal{F}_d^{\mathrm{free}}$ for every d. Let $\mathcal{L}$ be an ultrafilter on $\mathbb{N}^2$ finer than Φ_F. For every positive null sequence s_i, there exist strictly increasing $f : \mathbb{N} \to \mathbb{N}$ and an approaching scaling with $r_k = s_{f(k)}$ and $\xi_k = 1$. For each d, eventually $U_k(d)$ is real, covariant under signed coordinate permutations between H_0, H, and unit-preserving. There are depth-k vanes V_k, checked at level k, whose stages tend to infinity and whose radii equal r_k; for each d, eventually their full lift tuples through degree k converge along $\mathcal{L}$ to Y_k, with $U_k(d) = \pi_d \circ Y_{k,d}$.

Status: theorem conditional on the stated inputs. III.2 supplies the uniform-band property; VI.2 supplies the dimension cap from IV.1's corrected rank widths in the construction used by VII.2. *Lean:* `pays_sky`.

Proposition VII.6.2 (checked vanes). Assume (S), $z_d \ge c_i$ for $i \le d$, and $\mathrm{Fire}_d(z_d)$ for every d. Retain the ordered and generic point limits, smoothness of S, and H_0; the full hierarchy H is unnecessary. For every positive null sequence s, there exist strictly increasing $p : \mathbb{N} \to \mathbb{N}$, with $s \circ p$ strictly decreasing, and depth-d vanes V_d of stage $\ge d$, checked at level d, with radius $s_{p(d)}$.

Status: theorem. *Lean:* `wakesSky`.

Proposition VII.6.3 (moving-band chessboard bound). Assume (S). Fix $d \in \mathbb{N}$, $z \ge c_i$ for $i \le d$, Taylor order $\Delta \in \mathbb{N}$, real exponent e, real band ceiling u_0, and degree-$\le d$ stage packages P_k with order Δ, exponent e and bands $b_k \le u_0$. Fix $n, m \in \mathbb{N}$ with $n \ge 2$, $m + 1 \le n$, labels $g_i \in \mathrm{Label}_{d_i}$ with $d_i \le d$ for $0 \le i \le m$, and compact $K \subset \mathrm{Conf}_{m+1}$. For all sufficiently large k, take any:

- stage-k cost record for P_k with exponent -3, mark exponent z and arity bound n;
- cutoff $N > k$, period p, and compatible stage-k RG representative along the same history;
- configuration whose stage-k rounding, in units of $\ell(b_k, k)$, lies in K and fits the stopping stage;
- order-Δ polynomial lifts T_i for that representative and labels $f_i \in \mathrm{Label}_{d_i}$ whose images under T_i and the coupling dilation equal g_i weighted by $\ell(b_k, k)/a_k$.

For every nonempty $s \subset \{0, \dots, m\}$,

$$\rho_{\mathcal{T}}(\Phi_s) \le \mu_k^{-1/(16n)}, \qquad \Phi_s = [\eta_s] \left(e^{-D} \prod_{i=0}^{m} (1 + \eta_i O_i) \right).$$

Here the commuting markers satisfy $\eta_i^2 = 0$, $\eta_s = \prod_{i \in s} \eta_i$, O_i is the probe labelled f_i, and D is the free cumulant polynomial of g_i at the rounded rescaled points. The exponential is finite. The root uses coupling $\beta_{N,p}$ and width-stage cells. The threshold depends on $(d, z, \Delta, e, u_0, g, K, n)$, with band dependence only through u_0.

Status: theorem. *Lean:* `earningsSky`.

Proposition VII.6.4 (fixed-field supply). Let Y be a final record for (C, F, S), φ a filter finer than Φ_F, $d \in \mathbb{N}$ with $d \le 3$, and $x \in \mathrm{Label}_d$. There is $p \in \mathbb{N}$ such that, for all large k, there are $y \in \mathrm{Label}_d$ and cutoff labels q with $\pi_d y = \pi_d x$, $q \to y$ along φ, and $\mathrm{quota}_k^{N,p'}(q_{N,p'}) \le \widehat{\beta}_k^p$ eventually along φ. One can take $p = 1$ and q constant.

Status: theorem. *Lean:* `inletFirst`.

Proposition VII.6.5 (degrees below four). For every $d \in \mathbb{N}$ with $d < 4$, $\dim \mathcal{F}_d^{\mathrm{free}} = 1$. If Y is a final record for (C, F, S), every $u \in \mathcal{F}_d^S$ equals $t\mathbf{1}_d$ for some $t \in \mathbb{C}$.

Status: theorem. *Lean:* `dim_before`, `first_grade`.

3.31.2 2. What it uses

- III.2: uniform-band construction; III.9–III.11: RG histories and stage estimates, including (S).
- IV.1: extraction, limit tables, reflection positivity and corrected rank widths.
- VI.1–VI.2: free Maxwell comparison, stations, eventual injectivity and dimension cap.
- VIII.1: final record, comparator isomorphisms and nonzero unit.
- Intermediate value theorem, diagonal subsequences, finite-dimensional ultrafilter limits, Wilson chessboard estimates, and invariant theory for simple Lie algebras.

3.31.3 3. Strategy

Varying the reading band over a unit interval makes successive reading intervals overlap, so they cover every sufficiently small radius. Uniform stage estimates across one block-factor change then permit exact prescribed radii. Chessboard comparison and two auxiliary bands control coincidence strips and Riemann sums; diagonal selection and ultrafilter limits produce approaching maps. Eventual injectivity and the dimension cap make them isomorphisms. Low-degree supply follows separately from the comparator and unit coordinates.

3.31.4 4. Steps

1. Dynamic stopping windows allow two neighbouring price indices. Under the block-factor bound supplied by poised stages, continuity gives overlapping reading intervals and exact radii. Reading intervals.
2. Uniform probe and lift bounds propagate source expansions to the width stage; affordability there gives VII.6.3. Root estimate.
3. Two auxiliary bands make the strip and interpolation errors vanish; station admissibility follows. Balanced estimates.
4. Diagonal choice gives checked vanes; bounded lifts converge along the common ultrafilter. Projection preserves symmetry and the unit; eventual injectivity plus the dimension cap gives bijectivity. Lift limits, isomorphisms.
5. Below degree four, the comparator transfers one-dimensionality. Constant unit representatives have quota $|t| a_k^d \le \widehat{\beta}_k$ eventually. Unit supply.

3.31.5 5. Where the difficulty lies

The stage threshold must be uniform over the entire band interval. Two neighbouring price indices replace constant-factor plateaus. The dimension cap supplies surjectivity; weak convergence alone does not give the downstream trace argument. Scaling compatibility across degrees and stress-tensor normalization are not asserted.

3.31.6 6. Notation

$a_k \downarrow 0$ is the stage spacing, with block factors $a_k/a_{k+1} \geq 2$; $\hat{\beta}_k = \bar{\beta} + c_G \log a_k^{-1}$, $c_G > 0$; $\mu_k = \exp((\log\log \hat{\beta}_k)^2)$; $\ell(b,k) = a_k \mu_k^b$. Quota means stage-coordinate coefficient size; Conf_n is the configuration space of distinct points.

The coupling dilation multiplies a frame vector with q curvature factors by $s_{k,N,p}^q$, where $s_{k,N,p} = \beta_{N,p;k}^{-1/2}/(\hat{\beta}_k^{-1/2}\mu_k)$ and $\beta_{N,p;k}$ is the actual stage coupling. Position weighting multiplies an engineering-degree-h frame vector by the corresponding scale to power h.

A degree-D package has band $b \geq D+6$, Taylor order $\theta \geq D$, exponent e and price-index cost parameters with credit at least $bD + D + 2$. Poised stages have reached the reference stages, both neighbouring block factors are at most μ_k, and their tables preserve the unit, reality and signed covariance, with relative coordinate errors at most $1/2$ and $\mu_k^{-1/2}$. Each package has a cost record with exponent -3, the specified mark and arity bounds, and compatible representatives and polynomial lifts realizing those tables for every $N > k, p$.

The uniform-band property $\mathrm{Fire}_d(z)$ requires three package families with bands $b_i + s$, $s \in [0,1]$, fixed Taylor orders and exponents, and $\kappa < 0$ satisfying

$$b_1 + 6 < b_0, \quad b_2 + 5 < b_0 + \kappa, \quad (b_0 - b_1)d(d+2) + \kappa < 0, \quad (b_0 - b_2)d(d+2) - b_0 - \kappa < 0.$$

One threshold makes every triple poised for every $s \in [0,1]$, with arity bound $d+2$ and mark exponent z.

A depth-D vane records stage k, band b, strip exponent and complex-linear cutoff tables on labels through degree D. Checked at level j requires $5 < b \leq \log\log \hat{\beta}_k$, reference stages reached through degree D, and the table bounds above for every $N > k, p$. All generic rescaled lattice pairs of norm at most 100 fit; every degree-$\leq D$ free basis pair has recentred chessboard roots at most $1/(j+1)$ on the fitted radius-100 window. Its station is admissible through arity j, at window radius $j+1$ and tolerance $1/(j+1)$.

3.31.7 7. What remains unexplained

Constituent one-step estimates and eventual feasibility of the nonlinear small-field bounds remain in III.11. Paying all numerical requests belongs to III.2. The checked two-point bounds' role in deriving the trace anomaly belongs to VII.2.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.

3.32 VIII.1. Assembly of the main theorem

Part: VIII, Assembly of the main theorem. **Lean:** `Solution.lean`, `Construction.lean`, `ConcreteSU2.lean`, `YM/Nonperturbative`, and the public translations. **Size:** 12,060 lines in

1,674 declarations.

3.32.1 1. What this unit proves

A gauge group $(m, G, \mathfrak{g})$ has $m \geq 1$, $G \subset U(m)$ closed, and $\mathfrak{g} = \{X : \exp(tX) \in G \text{ for all } t \in \mathbb{R}\}$. Simplicity includes nonabelianness; Lie-algebra dimensions are real. Clause labels refer to statement.md.

Proposition VIII.1.1 (public theorem). For every gauge group with simple $\mathfrak{g}$, there exists a Wilson family $\mathcal{W}$ satisfying (Reg1)–(Reg4). For every strictly increasing $p : \mathbb{N} \to \mathbb{N}$, there exist a strictly increasing q and a hierarchy satisfying (P1)–(P4) along $p \circ q$:

- For every n, token list $v \in \mathcal{T}^n$ and $f \in C_c^\infty(\mathrm{Conf}_n)$, admissibility holds eventually along $p \circ q$. At every admissible volume p', $C_{N,p'}(v; f) \to R_{p'}$ along all N, and $R_{p(q(k))} \to \int f S^{\mathrm{mix}}(\pi(v); \cdot)$.
- The selected theory satisfies (W0)–(W3), has faithful fields, and satisfies (R1)–(R15). The mixed correlations are averages of continuous phase correlations against a full-support probability measure ν; the selected continuation is holomorphic in complex times at fixed real spatial positions, with selected Wightman boundary values on product Schwartz tests.
- (O1)–(O2) use common coefficients uniformly over all phases and the mixed state.
- Θ is centered and $r^8(\log 1/r)^2 G_\Theta(r) \to A_{\mathfrak{g}} = 3 \dim_{\mathbb{R}} \mathfrak{g}/\pi^4 > 0$ as $r \downarrow 0$. Every finite Wick profile differs from G_Θ somewhere in every $(0, \delta)$, $\delta > 0$.

Status: theorem. *Lean:* `YangMills.jointExistence`.

Proposition VIII.1.2 (detailed construction). For every such group and $b > 0$, there exists an internal family $\mathcal{C}$, depending only on (G, b), satisfying (D3), penalty b (D16), and Wilson regularity. For every extraction order σ, there exist τ, S, a final object $\mathfrak{F}$ on $(\mathcal{C}, \sigma \circ \tau, S)$, every-phase reconstruction $\mathfrak{L}$, reader Q, contact devices M and scaling data $\Sigma = (r_k, \xi_k, U_k)$ with $r_k \downarrow 0$, $\xi_k > 0$, satisfying (D1)–(D15) and (Core). These include (P1)–(P4) along $(\sigma \circ \tau)^{\mathrm{v}}$, distribution extensions, full-mesh ordered and positional limits, exact trajectory witnesses, every-phase Wightman axioms and barycentre identity, separated and nested OPE, trace and spectral laws, contact resolution with real bases and thresholds, Maxwell comparison and degree-four normalization. (Core) also includes integrated OPE remainder bounds for integrable complex densities against positive measures with bounded L^1 mass.

Status: theorem. *Lean:* `YangMills.constructed`.

Proposition VIII.1.3 (chosen run). For every such group and $b > 0$, the chosen funded run $\mathsf{R}_{G,b}$ has penalty b. For every σ, its family admits $\tau, S, \mathfrak{F}, \mathfrak{L}, Q, \mathsf{M}, \Sigma$ on $\sigma \circ \tau$ satisfying VIII.1.2's full-mesh ordered limit, positional convergence, (Core), contact conclusions, uniform nested OPE over admissible finite trees, and (D14)–(D15).

Status: theorem. *Lean:* `continuum_with_position_tests`.

Proposition VIII.1.4 (funded runs). Let $\mathfrak{g}$ be simple, R a run with campaign T and pillars W, and Δ local step estimates. If `Funded` (Fd) holds, then (Goal) holds: funding, (UV_∞), logarithmic OPE, dense sharp-Gram coverage, and the trace result for every final object on every $(\mathcal{C}_{\mathsf{R}}, \sigma, S)$. For every σ, one refinement carries ordered coupling limits B_m at every level, $S, \mathfrak{F}, \Sigma, \mathfrak{L}, Q$, full-mesh ordered and positional convergence, (D14), every-phase reconstruction, trace results, OPE facilities, foundation and uniform nested OPE. All-shapes OPE holds for every coherent prescription; every ultrafilter finer than the refined extraction filter has a compatible sharp-Gram allocation.

Status: theorem. *Lean:* `R2IGoals.fromFunded`.

Proposition VIII.1.5 (ultraviolet endpoint). For simple $\mathfrak{g}$ and a run with campaign T and pillars W, assume (Fd)'s stationary-point/amplification condition, delay condition and weight-and-tableau data (ϑ, v). Then (UV_∞) holds: (D3), weights, refinement to final objects with full-mesh ordered limits, trajectories competing in and issued by the fixed tuning recursion, successful tuning to level-zero target $\bar{\beta} + ws$ for every (p, N, s) with $s \in [-1, 1]$ and fixed $w > 0$, uniqueness of bare couplings within that recursion, analytic interpolation and coupling/activity convergence as $N \to \infty$ at every fixed (m, p) whenever level-zero targets converge. Complex-parameter convergence, with all parameter derivatives, is uniform on compact subsets of the unit disc in the fixed weighted activity norm. For every fixed (m, p) there is $c_{m,p} > 0$ with $e^{-\mathrm{scalar}_{N,p,m}} Z_{N,p} \geq c_{m,p}$ for $N \geq m$; scalar density tends to $-\infty$. Volume-uniform bounds and (D5) hold. For every fixed p, a successful parameter family with targets $\bar{\beta} + \lambda ws$, some $\lambda > 0$, has the chosen family as central member and the same step relations and insertion normalization. For every token list, compact $K \subset \mathrm{NoHit}_p$ and finite derivative order h, there are a neighbourhood Ω of K and a smaller positive parameter width independent of N, giving a limiting correlation holomorphic in the parameter and C^h in position, with convergence of coordinate differences of order at most h, their interpolants and admissible smearings, uniformly on compact parameter/position sets with all parameter derivatives.

Status: theorem. *Lean:* `FullVolume.Final.jewelOf17`.

Proposition VIII.1.6 (full spacing sequence). Under VIII.1.5's hypotheses, for every p, n and $v \in \mathcal{T}^n$ there is smooth $f_{p,v}$ on NoHit_p with $C_{N,p}(v; t) \to \int t f_{p,v}$ for every p-admissible t. Every fixed finite list of normalized forward, backward or centered coordinate differences with fixed integer widths converges, after interpolation, locally uniformly to the corresponding derivative; this also holds along approaching lattice configurations and arbitrary cutoff indices tending to infinity. Smeared differences converge to the integrable derivative pairing. Thresholds may depend on volume, tokens, compacts, tests and stencils. If a final object exists on $(\mathcal{C}, \sigma, S)$ and these positional limits hold, then $\mathrm{Ord}_{(\mathrm{id},\sigma^\vee)}(\mathcal{C}, S)$ holds.

Status: theorems. *Lean:* `Campaign.realUV`, `FullVolume.Final.from_spec`.

Proposition VIII.1.7 (public translation). For a final object on $(\mathcal{C}, \sigma, S)$ and reader Q, public and internal energies, measures and moments agree; $\mathrm{Ord}_{(\mathrm{id},s)}(\mathcal{C}, S)$ implies (P1) along s. Under (D3) and simplicity, regularity holds with $b_* = 2b_0$. Every-phase reconstruction implies (R1)–(R15); separated OPE implies (O1)–(O2); under simplicity, the seven headline trace clauses imply (T1)–(T4). Full contact resolution implies (D2).

Status: theorems. *Lean:* `fromWilson`, `regular`, `publicReconstruction`, `publicOPE`, `publicTrace`, `contactDistribution`.

Proposition VIII.1.8 (momentum Wick exclusion). For any final object with $s^8(\log 1/s)^2 G_\Theta(s) \to A > 0$ as $s \downarrow 0$, and every finite complex linear combination of products of mass-shell modes defining Wick profiles, there exist Schwartz f, g on $\mathbb{R}^4$ with $v_{f,g} = \overline{\mathcal{F}_+ f} \mathcal{F}_+ g$ compactly supported such that $\mathcal{W}_2^{\xi_*}(\Theta^*, \Theta)(\bar{f} \otimes g) \neq \sum_\alpha c_\alpha \mathrm{read}_\alpha(v_{f,g})$.

Status: theorem. *Lean:* `TraceExtension.Wick.polynomial_excludes`.

Proposition VIII.1.9. $\mathfrak{su}(2)$ is simple; VIII.1.1 holds for $(2, SU(2), \mathfrak{su}(2))$.

Status: theorem. *Lean:* `su2Simple`, `su2Theory`.

Proposition VIII.1.10. Final-object structures, construction input bundles and volume-uniform bounds are filed here and used in III.10–III.15, V.1–V.3, VI.1–VI.2 and VII.1–VII.4.

Status: definitions and upstream theorems. *Lean:* `Spec.Final`, `Org.Alloy`, `FixedBounds`.

3.32.2 2. What it uses

- I.1, II.1–II.2: gauge groups, families and running law; III.2, III.12–III.15: funded runs and ultraviolet estimates (`Independent.constructionRun`); IV.1: continuum seeds.
- V.1–V.3: phases and reconstruction; VI.1–VI.2: field algebra and stress tensor; VII.1–VII.5: Maxwell comparison, trace, OPE, contacts and asymptotic freedom (`EveryPhase.all_characters`, `Solstice.provided`).
- Countable-product compactness, diagonal extraction, equicontinuity, discrete Taylor estimates, dominated convergence, Fourier surjectivity and Källén–Lehmann representation. Final existence has no additional analytic hypotheses.

3.32.3 3. Strategy

Fix the run before extraction. Refine first for coupling limits, then for the continuum table and final object; attach all later structures on that refinement. Fixed-volume convergence and uniqueness of limits remove the mesh subsequence. Public translation then gives the stated hierarchy.

3.32.4 4. Steps

1. Choose the funded run for (G, b) using the run-selection theorem with the fixed dynamics, local estimates and component prescription. `constructionRun`.
2. Recover smooth full-mesh limits from dense-grid Cauchy convergence, equicontinuity and discrete Taylor estimates; smear by dominated convergence. `realUV`.
3. Extract coupling limits by countable-product compactness, then a final object on a further refinement. `Running.extraction`, `Pert.withFinal`.
4. Attach reconstruction, trace, OPE and contacts; subsequence stability retains earlier limits. `fromFunded`, `contacts_on`.
5. For Wick exclusion, realize compactly supported relative-momentum tests as Fourier products. Dominated convergence gives equality of Laplace transforms under a hypothetical Wick identity, contradicting the trace asymptotic. `polynomial_excludes`
6. Translate the detailed theorem and set $b = 1$, $\sigma = (\mathrm{id}, p)$, $q = \tau^{\mathrm{v}}$. `constructed`, `jointExistence`.

3.32.5 5. Where the difficulty lies

Every conclusion must concern the same previously chosen family. The Wilson limit is mixed-state; the trace test is selected-phase. The polynomial low-energy and heat bounds do not establish a mass gap. Volume-subsequence independence and agreement between penalties are not asserted.

3.32.6 6. Notation

$\sigma = (\sigma^{\mathrm{m}}, \sigma^{\mathrm{v}})$: increasing mesh/volume maps; refinement is componentwise composition. Ord: mesh-then-volume smeared limit. $\mathcal{C}$: internal family; $\mathfrak{F}$: trajectory witnesses, continuum seed, field algebra, phase selection and realizing continuations; Q: OPE reader; S: correlation table;

Θ: centered stress trace. NoHit_p: configurations distinct modulo $L_p\mathbb{Z}^4$. $\mathcal{F}_+$: unnormalized positive-sign Fourier transform. $b_0 = 11C_{\mathfrak{g}}/(48\pi^2)$.

3.32.7 7. What remains unexplained

Exact funding inequalities and renormalization comparison estimates remain in III.12–III.15; every-phase analytic arguments in V.3; normalized trace-scheme conditions in VII.3.

Detailed notes: the full account of this unit, with proof sketches for every step, is on the detailed notes page.